Swirl-String-Theory Canon-v0.8.23

Canonical Reference and Research Framework

Omar Iskandarani*

Independent Researcher
Groningen, The Netherlands

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Abstract

This document presents Swirl-String Theory (SST) as a structured canonical synthesis rather than as a claim that every phenomenological sector has already been derived to theorem level. The text records the impedance–resonance and free-to-bound-mode origin of the programme; consolidates the primitive constant chain linking the effective fluid density, characteristic swirl speed, Compton anchoring, and horn/circulation scale; assigns a minimal relational link-field action as the microscopic research candidate beneath the continuum torsion bridge; records the operational special-relativity postulate that stable vortex matter constructs local spacetime with clocks, rods, and luminal R-phase torsion pulses; and organizes delay selection, atomic bridges, spectroscopy, knot-geometry certification, relational time, and topology-based mass within one explicitly stratified framework. Throughout, historical provenance, calibrated identities, SST-internal derivations, predictions, and conjectural extensions are kept distinct so that the canon remains internally coherent, externally comparable, and open to falsification.

Keywords: Swirl String Theory, SST, Canon, Topological Hydrodynamics, Knot Theory, Genesis, Provenance, Impedance, Relational Link Field, Ridgerunner, Operational Spacetime, Vortex Atom Luminal Torsion, Quasinormal Swirl Spectroscopy, Pseudospectral Robustness

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Canon reading guide.

- [**ORTHODOX**] established physics or mathematics.
- [**DERIVED**] consequences obtained from definitions and assumptions of this SST layer.
- [**SPECULATIVE**] conjectural extensions or identifications not yet closed as theorems.

Optional *role* labels (*e.g.*, *Definition*, *Benchmark*, *Constraint*) qualify how to read a statement but do not replace the primary tag. Orthodox constraints—including quantum predictions, atomic spectroscopy, and relativistic consistency—are not overridden: bridge constructions, benchmark sections, and research-track material remain subordinate to the consistency stratification introduced later in the canon.

* Email: info@omariskandarani.com
ORCID: 0009-0006-1686-3961
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Author's Origin Note: From Impedance and Resonance to Vorticity

Swirl–String Theory did not begin as a fluid-mechanical model built from a preselected set of hydrodynamic constants. Its first written development arose from a different question: whether the apparent constants and discontinuities of atomic excitation could be understood as consequences of a classical, local, resonant interaction. The earliest notebooks from 2012 and the first synthesis document from July 2013 investigated stored field energy, capacitance, wavelength, oscillator stiffness, internal rotation, and the transition between a freely propagating excitation and a temporarily or permanently bound rotating configuration.

The guiding intuition was that the quantum jump should not be treated as a primitive discontinuity without mechanism. It should instead be examined as a finite dynamical transfer between a free field mode and a bound material mode. The early language was electrical and mechanical: capacitance, dielectric response, oscillation, characteristic speed, force, and impedance. The later language of circulation, vorticity, vortex tubes, toroidal geometry, knots, and relative vortex clocks was developed in order to provide a more coherent dynamical substrate for that earlier program.

This historical order matters. The present Canon must not imply that the current symbols Γ_0, ρ_f, and r_c were the original ontological starting point. They are candidate parameters of a later continuum representation. The older and more persistent commitment is structural: freely propagating radiation and bound matter are different dynamical regimes of one underlying medium, and their coupling is controlled by the response and impedance of that medium.

The historical documents are not treated as evidence that the proposed physics is correct. They are primary provenance records. Their role is to preserve the origin of the research questions, prevent later formulations from erasing the original logic, and identify which modern results must be derived independently rather than reconstructed algebraically from known constants.

> **Canonical origin statement.** SST originated as an impedance–resonance attempt to explain atomic excitation, the proportionality between energy and frequency, the emergence of mass from a bound rotating state, and the relation between internal material speed and the free propagation speed. Vorticity and knot topology are later dynamical languages introduced to make that program local, conservative, geometric, and falsifiable.

1 Genesis, Foundational Commitments, and Epistemic Architecture

1.1 Purpose and canonical scope

This section records three distinct layers that must not be conflated:

1. **historical origin**: what the notebooks and the 2013 proto-Canon actually proposed;
2. **modern SST translation**: how those ideas are now expressed using vortex, link-field, topological, and operational-spacetime language;
3. **current epistemic status**: which statements are definitions, hypotheses, calibrated bridges, derived consequences, predictions, or falsifiers.

No historical claim becomes canonical merely because it appeared early. No modern equation may be projected backward into the notebooks as if it had been explicitly present there. Conversely, a modern reformulation should not erase the earlier physical question that motivated it.

1.2 Primary provenance record

The present origin layer is anchored by the following source classes:

Notebook 2012. The earliest surviving page-level record of rotating charge, free versus bound wave structures, toroidal geometry, Lorentz transformations, oscillator-speed constructions, explicit circulation, vorticity, and the capacitance path.

`DOC2013-CANON-001`. *Een hypothese over de constantes in de kwantum mechanica, verklaard met de klassieke logica*, July 2013. This is the first surviving attempt to turn the notebook ideas into a single explanatory document. It explicitly introduced an excitation or internal circumferential speed, treated mass as a rotating state, interpreted the fine-structure constant as a speed ratio, and developed a capacitance-based route to the Planck relation.

Notebook 2013–2014. Subsequent geometric and algebraic development, including toroidal field structures, relative rotation, clock ideas, impedance reasoning, and attempts to interpret the earlier numerical closures.

Modern SST Canon. The current hydrodynamic and topological architecture, including link-field dynamics, operational spacetime, and falsification rules.

Modern computational provenance. The 2026 KnotPlot-to-Ridgerunner pipeline, geometry sidecars, seed-selection records, certified and candidate knot exports, and downstream SSTcore/VortexLab audits. These records test the translation from a closed-mode idea to resolved finite-thickness geometry.

Stable notebook identifiers such as `NB2012-P070` and `NB2013-P019` shall be used in the provenance appendix. Their purpose is traceability, not authority.

1.3 Historical development of the core idea

The source record supports the following developmental sequence:

$$
\begin{aligned}
&\text{stored field energy and capacitance}\\
&\quad\longrightarrow \text{resonant photon–matter interaction}\\
&\quad\longrightarrow \text{internal circumferential or excitation speed}\\
&\quad\longrightarrow \text{mass as a bound rotating state}\\
&\quad\longrightarrow \text{relative time and spatial deformation of rotating matter}\\
&\quad\longrightarrow \text{circulation, vorticity, toroidal cores, and vortex knots}\\
&\quad\longrightarrow \text{link-field propagation and operational spacetime.}
\end{aligned} \tag{1}
$$

The 2013 proto-Canon already contained an important epistemic caution. After rewriting the Planck relation from known electromagnetic quantities and the new excitation speed, it described the result as, in effect, a calculation trick unless the same structure could be recovered through another independent combination. That caution is retained here as a canonical methodological rule: algebraic closure is not yet dynamical explanation.

1.4 Modern computational provenance: from KnotPlot seeds to certified finite-thickness representatives

The 2012–2014 source record proposed closed recurrent field structures before SST possessed a controlled finite-thickness knot geometry. The modern KnotPlot-to-Ridgerunner project is the first systematic attempt to resolve that old bound-mode intuition as an auditable embedded tube rather than as a sketch, a knot diagram, or a scalar topological label. Its role in the Genesis record is therefore methodological: it connects the early claim “matter is a closed self-maintaining mode” to a reproducible geometry pipeline. It does not by itself establish that physical particles are ideal knots.

Three distinct computational roles. The project separates three layers which must remain distinct:

1. **KnotPlot exploration.** KnotPlot creates and relaxes isotopic candidate centerlines and supplies checkpoint families. It is a seed and topology-exploration tool, not a final smooth-thickness certificate.

2. **Ridgerunner certification.** Ridgerunner minimizes polygonal length subject to explicit finite-thickness constraints and reports a first-order constrained-gradient balance through active struts and kinks [63].

3. **SSTcore/VortexLab use.** A certified or candidate centerline is converted into a separately versioned, uniformly arc-length sampled copy for Biot–Savart, reach, clock, parity, or spectroscopy diagnostics. The original Ridgerunner polish remains the audit geometry.

A successful result in one layer is not silently promoted to the next.

Orthodox finite-thickness core. For a smooth embedded centerline γ, radius-convention thickness is

$$\mathrm{Thi}_{\mathrm{rad}}(\gamma) = \min\left(\inf_{s} \frac{1}{\kappa(s)}, \frac{1}{2} d_{\mathrm{dcsd}}(\gamma)\right), \tag{2}$$

where d_{dcsd} is the minimum doubly-critical self-distance. Thus a resolved tube is limited either by local curvature or by nonlocal self-contact. In a polygonal approximation these active constraints become the kink and strut sets. At constrained criticality their nonnegative multipliers balance the length or declared SST energy gradient,

$$-\nabla E + \sum_{(p,q)\in\mathrm{Strut}(V)} \lambda_{pq} \nabla\left(\frac{d(p,q)}{2}\right) + \sum_{i\in\mathrm{Kink}(V)} \mu_i \nabla\,\mathrm{MinRad}_i = 0, \qquad \lambda_{pq}, \mu_i \geq 0. \tag{3}$$

Ropelength minimizers exist in every tame knot type and are $C^{1,1}$, but they need not be C^2 or unique [61]. Therefore knot type alone does not select one unique physical centerline, and a scalar ropelength cannot determine every shape-sensitive SST observable.

Project conclusion 1: "ideal" is a claim to be certified. A file labelled *ideal*, a visually tight KnotPlot configuration, a Fourier atlas entry, or a small scalar ropelength excess is not by itself a geometric certificate. The project found that the same knot may pass a length comparison while still lacking strict residual convergence, resolution convergence, smooth-thickness verification, or a stable contact skeleton. Consequently the admissible labels are graded:

$$\begin{gathered}\texttt{seed} < \texttt{relaxed candidate} < \texttt{near-ideal candidate},\\ \texttt{near-ideal candidate} < \texttt{strict near-ideal} < \texttt{certified smooth upper bound}.\end{gathered} \tag{4}$$

No software or database name substitutes for these gates.

Project conclusion 2: topology and geometry require independent gates. The automatic pipeline records a topology sidecar for every checkpoint, including at least a safety flag, component count, vertices per component, and pairwise linking matrix where applicable. A geometrically plausible seed with missing or inconsistent topology metadata receives the status `geometry-qualified-topology-unverified`; automatic tightening stops rather than guessing. The specific trial number selected in a run is provenance metadata, not a canonical physical fact.

Project conclusion 3: checkpoint selection must not be "take the latest". KnotPlot checkpoint quality is not assumed to improve monotonically. The project therefore uses signed fractional length gain, per-component flatness, and a finite-thickness proxy

$$D_{\mathrm{proxy}} = \min\left(2R_{\min}, d_{\mathrm{self}}^{\mathrm{nonlocal}}, d_{\mathrm{inter}}\right), \tag{5}$$

$$\mathcal{R}_{\mathrm{proxy}} = \frac{L}{D_{\mathrm{proxy}}}, \tag{6}$$

rather than raw length alone. A signed plateau requires nonnegative but small improvements over successive checkpoints; a length regression is never called a plateau. If no plateau is found within the preregistered scan range, the pipeline may retain a best-so-far seed only with an explicit `best-so-far-no-plateau` warning.

Project conclusion 4: radius and diameter conventions must be explicit. The Brian Gilbert Fourier database records $D = 1$ and, for the benchmark entries,

$$L(3{:}1{:}1) = 16.371637, \qquad L(4{:}1{:}1) = 21.043322. \tag{7}$$

Under the radius-thickness convention used in the SST ropelength sections, these correspond to

$$\mathrm{Rop}_{\mathrm{rad}}(3_1) = 32.743274, \qquad \mathrm{Rop}_{\mathrm{rad}}(4_1) = 42.086644. \tag{8}$$

Any comparison that omits the factor-of-two convention is rejected before physical interpretation.

Project conclusion 5: uniform sampling and physical relaxation are different operations. The raw tightened polygon is retained unchanged for geometry audit. A separate copy is uniformly resampled in arc length for downstream field solvers. The uniform copy improves quadrature and removes mesh-density bias; it does not improve the physical optimum and does not erase the original residual or contact data. For strict resolution convergence, the current best Ridgerunner polish is used as the common parent, independently resampled to $N = 600$ and $N = 1200$, and each branch is re-relaxed and re-certified. The convergence study does not restart KnotPlot and does not treat the read-only VortexLab $N = 300$ export as a new canonical seed.

Current strict near-ideal profile. The present SST project profile requires, for a strict near-ideal label,

$$\chi_{\mathrm{KKT}} \le 5 \times 10^{-3}, \qquad \epsilon_{\mathrm{Rop}} \le 5 \times 10^{-4}, \tag{9}$$

$$\epsilon_{\mathrm{multistart}} \le 2 \times 10^{-4}, \qquad \epsilon_{600\to1200} \le 2 \times 10^{-4}, \tag{10}$$

$$\frac{\ell_{\max}}{\ell_{\min}} \le 1.02, \qquad \mathrm{CV}(\ell) \le 5 \times 10^{-3}. \tag{11}$$

These are project acceptance thresholds, not mathematical constants of nature. They may be tightened in later releases, but they must not be weakened after seeing a desired physical output.

Trefoil snapshot and status discipline. One dated $N = 300$ project snapshot derived from a Ridgerunner polish reports $L = 16.3763431828$ at $D = 1$, corresponding to $\mathrm{Rop}_{\mathrm{rad}} = 32.7526863657$. Relative to the Gilbert reference in Equation (8), the excess is about 2.87×10^{-4}, while the uniform export has edge ratio approximately 1.000693 and edge-length coefficient of variation approximately 8.24×10^{-5}. It passes topology, ropelength, uniform-mesh, and the looser $\chi_{\mathrm{KKT}} \le 10^{-2}$ candidate gate, but fails the strict residual gate and has not yet completed the $600 \to 1200$ convergence test. Its correct status is therefore `near-ideal-candidate`, not `strict-near-ideal`. This example is deliberately retained because it shows that excellent scalar length agreement is not equivalent to complete certification.

Project conclusion 6: the contact skeleton is the physically richer output. For SST, the active set

$$\mathcal{C}_K = (\mathrm{Strut}(V), \mathrm{Kink}(V), \{\lambda_{pq}\}, \{\mu_i\}, \mathrm{cond}\, A_F) \tag{12}$$

contains more physical information than the scalar ropelength. It identifies where the resolved tube is compression-limited, where curvature saturates, how loads are distributed, and whether the constraint system is well conditioned. Ropelength convergence without contact-measure convergence is insufficient for a shape-sensitive mass, magnetic, clock, or impedance claim.

Connection to the rediscovered impedance path. The KnotPlot/Ridgerunner project supplies a candidate geometry and contact skeleton for the *bound* side of the free–bound transition. It does not supply the free link-field impedance Z_ℓ, the transition action $\mathcal{J}_0$, or the source normalization Q_0. A future jump impedance Z_J may depend on the certified knot domain and its linear response, but it cannot be identified with ropelength alone. The non-circular program therefore requires the chain

$$\begin{aligned} &\text{certified finite-thickness geometry} \longrightarrow \text{bound-mode response operator}, \\ &\text{bound-mode response operator} \longrightarrow Z_\mathrm{J}(\omega) \longrightarrow \text{comparison with } Z_\ell(\omega). \end{aligned} \tag{13}$$

with all geometry errors propagated into the final coupling uncertainty.

Canonical status. The orthodox thickness, reach, ropelength, and constrained-criticality results belong in the MAIN Canon. The specific KnotPlot commands, checkpoint heuristics, thresholds, trial selections, and current candidate values are [**RESEARCH-TRACK / COMPUTATIONAL PROVENANCE**]. Downstream SST claims must expose the geometry source, topology certificate, convention, resolution, residual, contact map, and error envelope used.

1.5 Foundational physical commitments

The following commitments summarize the persistent physical content of the origin program without promoting unverified historical mechanisms to theorem status.

1. **One-substrate commitment.** Matter, radiation, and operational clocks are organized regimes of one underlying medium rather than unrelated primitive ontologies.
2. **Free/bound mode distinction.** A freely propagating radiation mode and a stable material particle are not the same solution. The former is an extended propagating excitation; the latter is a closed, recurrent, topologically or dynamically trapped state.
3. **Rotational matter commitment.** The inertia and clock structure of matter are associated with recurrent internal motion. In modern SST this is represented by stable vortex and link-field configurations, not by the literal rigid charged sphere of the earliest model.
4. **Local transition commitment.** Absorption and emission are local transfers between a free mode and a bound mode. The transition must possess an action, energy flux, coupling law, and finite response structure.
5. **Impedance commitment.** The strength of the free–bound coupling is controlled by the response of the propagating link sector relative to the response of the bound transition sector.
6. **Operational relativity commitment.** Stable matter constructs time, length, and simultaneity with its internal cycles and exchanged torsion pulses. Relative motion mixes the resulting temporal and spatial coordinates while preserving the local torsion cone.

1.6 The original capacitance and transition hypothesis

[ORIGIN / HISTORICAL BRIDGE]. The early construction combined the classical relations

$$E_{\text{cap}} = \frac{Q^2}{2C_{\text{cap}}}, \qquad C_{\text{cap}} = \varepsilon\frac{A}{D}, \tag{14}$$

$$v_{\text{ex}} = f\lambda, \qquad E = hf. \tag{15}$$

With the historical geometric ansatz $A = \lambda^2$ and $D = \lambda/2$, the capacity becomes proportional to $\varepsilon\lambda$. Substitution of $\lambda = v_{\text{ex}}/f$ then produces a coefficient multiplying f in the stored-energy expression. Under that specific normalization, the historical construction may be summarized as

$$h_{\text{bridge}} = \frac{Q^2}{4\varepsilon v_{\text{ex}}}. \tag{16}$$

Equation (16) is not a current derivation of Planck's constant. It depends on an assumed interaction geometry, on the identification of the relevant charge, on a pre-existing electromagnetic susceptibility, and on a speed that was itself calibrated from known particle relations. Moreover, its later impedance form is an exact reparameterization of this same algebraic construction, not an independent second route. Its canonical value is that it exposes the original physical question:

> Can the linear energy–frequency law arise from the static and dynamic response of a finite transition region rather than being inserted as an unexplained proportionality?

1.7 Impedance form of the origin hypothesis

The capacitance construction can be rewritten more transparently in terms of a propagation impedance and a transition impedance. This change of variables does not by itself add independent evidence or close the multiple-route requirement. For an isotropic lossless link-field sector, define

$$c_T = \frac{1}{\sqrt{\epsilon_\ell \mu_\ell}}, \qquad Z_\ell = \sqrt{\frac{\mu_\ell}{\epsilon_\ell}} = \frac{1}{\epsilon_\ell c_T}. \tag{17}$$

Here ϵ_ℓ and μ_ℓ are SST link-response coefficients. They are not assumed to equal the SI vacuum constants. Let Q_0 denote a candidate minimal topological source unit and let $\mathcal{J}_0$ denote a candidate minimal transition action. The corresponding transition-impedance scale is

$$Z_{\text{J}} := \frac{\mathcal{J}_0}{Q_0^2}. \tag{18}$$

A modern statement of the old fine-structure intuition is then

$$\alpha_{\text{SST}} = \kappa_Z \frac{Z_\ell}{Z_{\text{J}}} = \kappa_Z \frac{Q_0^2 Z_\ell}{\mathcal{J}_0}, \tag{19}$$

where κ_Z is a dimensionless geometric and normalization factor. The present electromagnetic bridge corresponds to the target $\kappa_Z = 1/2$ under one conventional normalization, but that value is not to be imposed in the zero-legacy derivation. It must follow from the transition geometry and source normalization.

Equation (19) is a **[RESEARCH HYPOTHESIS]**. In the historical bridge it is the impedance notation of the same capacitance identity, so the two forms count as one route. It becomes stronger than a numerical identity only if Z_ℓ, Q_0, $\mathcal{J}_0$, and κ_Z are independently obtained from the SST action and are then compared with a genuinely separate bound-mode, static-source, or transition-action calculation.

1.8 Modern link-field translation

The minimal link action introduced later in this Canon supplies the first controlled candidate for the propagation side of the origin program. On a regular network with link spacing a_ℓ, link inertia I_ℓ, and link holonomy stiffness K_ℓ, the transverse long-wavelength branch satisfies

$$\omega^2(\mathbf{k}) = c_T^2|\mathbf{k}|^2 + O(a_\ell^4|\mathbf{k}|^4), \tag{20}$$

$$c_T = a_\ell\sqrt{\frac{K_\ell}{I_\ell}}. \tag{21}$$

This is the modern action-level descendant of the early oscillator construction $v \sim \lambda\sqrt{k/m}$. The difference is decisive: Equation (21) follows from a declared set of degrees of freedom and an action, while the early formula substituted measured particle quantities directly into a harmonic-oscillator analogy.

The material Euler-vorticity sector and the link-torsion sector remain distinct:

$$\boldsymbol{\zeta} = \nabla \times \mathbf{v}, \qquad \boldsymbol{\mathcal{B}}_\ell = \nabla \times \boldsymbol{\mathcal{A}}_\ell. \tag{22}$$

A torsion pulse may therefore propagate by changing relational link states without creating permanent material vorticity in an initially irrotational material element.

1.9 The non-circular route to the propagation speed

[NON-CIRCULARITY GUARD]. The empirical value of c may be used in calibrated phenomenology, but it may not be used directly or indirectly in a derivation that is presented as a prediction of c.

A legitimate derivation has the logical structure

$$\text{microgeometry and action} \longrightarrow (a_\ell, I_\ell, K_\ell) \longrightarrow c_T \longrightarrow c_T \stackrel{?}{=} c_{\text{obs}}. \tag{23}$$

A circular construction has the structure

$$c_{\text{obs}} \longrightarrow \text{calibrated microparameter} \longrightarrow c_T \longrightarrow c_{\text{obs}}. \tag{24}$$

For a candidate dimensional basis $(\Gamma_\star, \rho_\star, r_\star)$, one may write

$$a_\ell = \xi_\ell r_\star, \tag{25}$$

$$I_\ell = \iota_\ell \rho_\star a_\ell^5, \tag{26}$$

$$K_\ell = \kappa_\ell \rho_\star \Gamma_\star^2 a_\ell, \tag{27}$$

which gives

$$c_T = \frac{\Gamma_\star}{a_\ell}\sqrt{\frac{\kappa_\ell}{\iota_\ell}} = \frac{\Gamma_\star}{r_\star}\frac{1}{\xi_\ell}\sqrt{\frac{\kappa_\ell}{\iota_\ell}}. \tag{28}$$

The set $(\Gamma_\star, \rho_\star, r_\star)$ is a candidate dimensional basis, not yet the asserted ontological core of SST. Its numerical values must remain symbolic in a zero-legacy derivation unless their provenance is independent of c, h, e, m_e, α, ϵ_0, μ_0, and any other validation target.

1.10 Three valid modes of use

The same equations may be used under three different epistemic modes:

Symbolic mode. All dimensional primitives remain symbolic. The outputs are dimensionless spectra, ratios, topology-dependent coefficients, and scaling laws. This is the cleanest theory-development mode.

Calibrated consistency mode. Current canonical numerical values may be inserted to test stability, dispersion, convergence, mode count, energy conservation, and order of magnitude. The result is labelled [**CALIBRATED CONSISTENCY TEST**], not [**PREDICTION**].

Prediction mode. A preregistered set of independent observables fixes the dimensional scale and any permitted coefficients. All remaining observables are then computed without retuning. Only those unused observables qualify as predictions.

The use of known numbers in simulation is therefore not concealed. What is forbidden is to use a known number to define an input and later count its recovery as an independent success.

1.11 From relative rotation to operational spacetime

The notebooks combined Lorentz transformations, moving frames, rotating matter, and internal clock ideas before the modern Swirl Clock notation existed. The persistent question was whether the time and length measured by matter could be properties of its internal recurrent dynamics rather than primitive properties of a detached coordinate system.

The modern formulation distinguishes substrate ordering from operational measurement. A stable carrier A constructs proper time τ_A, calibrated length ℓ_A, and radar coordinates with exchanged torsion pulses. The local statement is

$$c_A^{(2\mathrm{w})} = \frac{2\ell_A}{\Delta\tau_A} = c_T. \tag{29}$$

The target is not merely that clocks slow. The same link-field cone must govern clock response, rod equilibration, synchronization, and free propagation. Only then can Alice and Bob reconstruct the same local invariant speed while assigning different temporal and spatial coordinates to the same event.

The languages of relative vorticity and relativistic congruence kinematics both use expansion, shear, rotation, and acceleration. This is only a structural vocabulary, not an identification by analogy. SST must derive the observer-dependent kinematic decomposition from the actual vortex and link-field dynamics.

1.12 Canonical provenance and dependency rules

The following rules govern all claims developed from the origin program:

1. **Single-source rule.** A claim is not normatively part of SST merely because it occurs in a chat, notebook, old manuscript, code comment, or research-track file. Active foundational claims must appear in the MAIN Canon with status and dependencies.

2. **No-backprojection rule.** Modern vortex, gauge, knot, or link-field mathematics must not be described as literally present in an early notebook unless the source supports that statement.

3. **Dependency rule.** Every claimed prediction must have an acyclic dependency path from declared inputs to output.

4. **Calibration-exclusion rule.** A quantity used directly or indirectly to determine an input parameter may not also be counted as an independent prediction.

5. **Multiple-route rule.** A proposed structural constant should, where possible, be obtained through at least two genuinely independent sectors—for example a link-field propagation/constitutive calculation and a bound-mode, static-source, or transition-action calculation. Rewriting one capacitance identity in impedance notation does not count as a second route.

6. **Historical-preservation rule.** Incorrect or incomplete old mechanisms are retained in the provenance appendix under [**ORIGIN**] or [**REJECTED**], while the corrected modern formulation is placed beside them.

1.13 Extended epistemic labels

The origin and dependency architecture uses the following labels in addition to the primary Canon stratification:

Label	Meaning
`[ORIGIN]`	Historical idea or source statement; not a present derivation.
`[FOUNDATIONAL POSTULATE]`	Explicit starting commitment of the current model.
`[HISTORICAL BRIDGE]`	Algebraic or conceptual link from the old formulation to known physics.
`[DERIVED]`	Consequence of explicitly declared current assumptions.
`[CALIBRATED]`	Numerical quantity fixed from external data or a bridge relation.
`[PREDICTION]`	Output not used directly or indirectly in calibration.
`[THEOREM TARGET]`	Statement that must be obtained from the action or governing equations.
`[FALSIFIER]`	Preregistered result that would reject the stated hypothesis.
`[REJECTED]`	Historical mechanism retained for provenance but not active in the model.

1.14 Immediate theorem targets

The rediscovered impedance path yields the following immediate research targets:

1. **Link propagation theorem.** Derive I_ℓ, K_ℓ, and a_ℓ from the microscopic adjacency and interaction law, and prove the existence of an isotropic transverse branch with a controlled infrared and front velocity.

2. **Monometricity theorem.** Show that all physically relevant torsion polarizations and all stable vortex-matter species reconstruct the same local characteristic cone.

3. **Transition-action theorem.** Derive a minimal transition action $\mathcal{J}_0$ from a closed phase-space cycle or a topological transition, rather than setting $\mathcal{J}_0 = h$ or $\hbar$.

4. **Source-quantization theorem.** Derive the minimal source unit Q_0 from knot/link topology or a Noether charge, rather than inserting the elementary SI charge.

5. **Impedance-ratio theorem.** Derive κ_Z and test Equation (19) without using α in any input or normalization.

6. **Free–bound conversion theorem.** Construct a coupled action for a propagating link excitation and a bound vortex mode, and derive absorption, emission, selection rules, linewidth, and energy conservation.

7. **Clock–rod–cone closure theorem.** Derive the response of vortex clocks and rods from the same link-field cone that governs propagation, rather than imposing Lorentz kinematics as an additional unrelated rule.

8. **Resolved-knot certification target.** Across independent seeds and both the $N = 600$ and $N = 1200$ branches, demonstrate convergence of topology, thickness, residual, smooth-upper-bound, and contact measures for the selected bound-state geometry.

9. **Bound-mode impedance target.** Derive the frequency-dependent transition response on a certified finite- thickness knot domain and show which contact-skeleton observables enter $Z_{\mathrm{J}}(\omega)$; scalar ropelength alone is not an admissible closure.

1.15 Preregistered falsifiers

The impedance and non-circularity program fails in its strong form if any of the following persists after controlled numerical and analytical refinement:

1. no stable transverse link branch exists;
2. the inferred propagation speed depends irreducibly on polarization, lattice orientation, or vortex species at low energy;
3. the value of c_T can match experiment only by inserting a parameter whose provenance already contains c;
4. the transition action or source unit remains continuously tunable rather than dynamically or topologically quantized;
5. the impedance-ratio route reproduces α only after α-dependent calibration;
6. the free torsion pulse necessarily creates permanent material vorticity in an initially irrotational region, contrary to the declared Helmholtz sector separation;
7. clock slowing, rod response, synchronization, and propagation require mutually inconsistent characteristic speeds;
8. alternative microparameters fit c and α equally well while making incompatible predictions elsewhere, without a dynamical selection principle;
9. the selected knot geometry changes topology, component linking data, or strict near-ideal classification under the declared refinement pipeline;
10. scalar ropelength converges while the positive strut/kink contact measures or active rigidity rank fail to converge;
11. a downstream mass, impedance, or constant extraction quotes precision parametrically smaller than the certified geometry-error envelope without an independent cancellation proof.

1.16 Canonical dependency summary

The intended non-circular dependency graph is

$$\begin{aligned}&\text{micro-adjacency, symmetry, topology, and local action}\\&\quad\longrightarrow \{a_\ell, I_\ell, K_\ell, Q_0, \mathcal{J}_0, \kappa_Z\}\\&\quad\longrightarrow \{c_T, Z_\ell, Z_{\mathrm{J}}, \alpha_{\mathrm{SST}}\}\\&\quad\longrightarrow \text{clock, rod, spectral, atomic, and scattering observables}\\&\quad\longrightarrow \text{comparison with } \{c, h, e, \alpha, m_e, \ldots\}_{\mathrm{obs}}.\end{aligned} \tag{30}$$

The reverse arrows are permitted only in the explicitly calibrated phenomenology layer. They are prohibited in a claimed zero-legacy prediction.

1.17 Provenance references for this section

The principal source anchors are summarized in the Historical Notebook and Proto-Canon Provenance Register, Appendix A. The full page-level ledger remains a supporting archive; the MAIN Canon records its stable identifiers, active conclusions, and status. The principal source anchors for the present section are:

- `NB2012-P002`, `P006`, `P017`: free and bound wave structures, rotating matter, and the early unification aim;
- `NB2012-P021–P025`: Lorentz transformations, speed ratios, transition geometry, and oscillator dynamics;
- `NB2012-P044–P050`, `P055`: the turn to circulation, vortices, toroidal cores, and time/mass as properties of rotating matter;
- `NB2012-P069–P071`: oscillator propagation speed, capacitance, vacuum response, and the constants dependency network;
- `NB2013-P010`, `P019`, `P045`: transition dynamics, the distinction between numerical closure and physical meaning, and the early relative-rotation/time interpretation;
- `DOC2013-CANON-001`: the first synthesis of excitation speed, rotating mass, the fine-structure speed ratio, and the capacitance route to the Planck relation.
- `PROJ2026-KNOTPLOT-RR-001`: KnotPlot checkpoint families, topology sidecars, signed-plateau seed selection, Ridgerunner constrained-gradient polishes, uniform-arc-length exports, and strict near-ideal certification thresholds.

Reading guard. The origin archive establishes continuity of the research program. It does not validate the physical claims. Validation requires the independent action-level derivations and falsification tests stated above.

2 Philosophical Prelude

2.1 Why a philosophical prelude is necessary

The SST canon is not only a collection of equations but also a statement about what counts as fundamental. Standard quantum theory and general relativity are highly successful formalisms, yet they begin from different ontological primitives: operator-valued states in one case and spacetime geometry in the other. SST instead adopts a single material substrate as the underlying level of description and treats particles, radiation, and clock structure as organized states of that substrate.

2.2 The ontological shift

The canon therefore makes the following conceptual replacement:

$$\text{point particles} \longrightarrow \text{stable swirl-string configurations,} \tag{31}$$

$$\text{vacuum as background} \longrightarrow \text{vacuum as structured medium,} \tag{32}$$

$$\text{metric spacetime as primitive} \longrightarrow \text{operational Lorentz geometry of vortex clocks and torsion pulses,} \tag{33}$$

$$\text{quantization as postulate} \longrightarrow \text{discreteness from dynamics and topology.} \tag{34}$$

This is not a denial of effective particle language. Rather, it is a claim that particle language is secondary and emerges from a more primitive hydrodynamic and topological layer.

2.3 Relation to orthodox physics

SST is not introduced as a replacement for the empirical content of orthodox physics. Its canonical role is instead threefold:

1. to reproduce known low-energy structures in the appropriate limits,
2. to identify which apparently fundamental constants or rules may be structurally redundant,
3. to supply a dynamical mechanism for phenomena that are otherwise postulated axiomatically.

In this sense the canon follows a *Rosetta principle*: whenever a standard description already works, SST must map to it before claiming any deeper reinterpretation.

2.4 Special relativity as the operational starting point

[ORTHODOX]. Special relativity does not assert that a clock reading and a spatial distance are numerically identical. It states that temporal and spatial coordinates are components of one Lorentzian interval and are mixed by relative motion while the interval and the null propagation speed remain invariant [45]. For observers Alice and Bob, their separate assignments of time and distance therefore depend on their relative state of motion, although both reconstruct the same causal ordering and the same local value of the invariant speed c.

[CANONICAL FOUNDATIONAL POSTULATE]. This operational observation is a starting point of SST. The substrate is modeled ontologically as a three-dimensional incompressible medium with a global ordering parameter T, but a localized vortex atom does not directly read that parameter. A vortex atom constructs physical time from its internal cycle accumulation, physical length from equilibrated matter standards, and distant-event coordinates from exchanged R-phase torsion pulses. For every stable low-energy vortex carrier A, the rod-calibrated two-way torsion-pulse speed is postulated to satisfy

$$c_A^{(2\mathrm{w})} := \frac{2\ell_A}{\Delta\tau_A} = c, \tag{35}$$

where ℓ_A is an independently calibrated local separation and $\Delta\tau_A$ is the elapsed proper time on the carrier clock for the out-and-back exchange.

The corresponding local operational interval is

$$d\sigma_A^2 = -c^2 dt_A^2 + d\ell_A^2, \tag{36}$$

with an R-phase torsion pulse satisfying $d\sigma_A^2 = 0$. Hence Alice measures

$$\left|\frac{d\ell_A}{dt_A}\right| = c, \tag{37}$$

and Bob must obtain the same result in his own vortex-clock and vortex-ruler coordinates. In this precise sense, "space and time form one structure" means that relative motion mixes their operational coordinate components while preserving Eq. (36); it does not erase the distinction between substrate foliation time and local measured proper time.

[CANON / COORDINATE-SPEED GUARD]. The bare substrate-coordinate ratio dX/dT is not, by itself, the operational definition of the speed of light. It may depend on the chosen substrate coordinates and on the clock/ruler response. The canonically observable statement is Eq. (35). Deriving this universality from the torsional constitutive law, rather than using measured c as calibration input, remains a theorem target.

2.5 Why delay and topology matter

Two themes recur across the development of the canon. First, finite circulation delay on closed carriers provides a natural source of temporal nonlocality. Second, topological invariants constrain admissible configurations and stabilize structures that would otherwise collapse or disperse. Together they motivate the canonical SST slogan

delay selects modes, topology protects them.

This is the philosophical reason why the canon treats quantization neither as a primitive operator rule nor as a purely statistical statement.

2.6 Scope of the present canon

The present version of the canon should be read as a structured reference document rather than a claim that every sector has already been derived to theorem level. Some components are mature and reusable, such as the primitive constant chain, circulation quantization, and the Swirl-Clock. Other components are deliberately retained as research-track layers, such as full gauge emergence, topology-to-charge assignments, and complete many-body atomic closure. The point of the canon is therefore not maximal rhetorical closure, but explicit logical organization.

2.7 Interpretive summary

The philosophical stance of SST can be summarized in one sentence:

Physics is interpreted as the dynamics of a continuous medium whose stable, delayed, and topologically organized motions appear effectively as particles, clocks, fields, and spectra.

3 Formal Foundations

3.1 Primitive Structure of the Theory

We define the Swirl-String Theory (SST) as a continuum-based framework built on a minimal set of primitive quantities.

Primitive calibration set:

$$\mathcal{P}_{\text{cal}} = \{\rho_f, v_{\circlearrowleft}^{*}, \omega_c\} \tag{38}$$

where:

- ρ_f is the effective background fluid density,
- $v_{\circlearrowleft}^{*} = \|\mathbf{v}_{\circlearrowleft}\|$ is the canonical characteristic swirl-speed magnitude,
- ω_c is the Compton angular frequency.

[CANON / NOTATION]: $v_{\circlearrowleft}^{*}$ is a calibrated scalar speed. The local swirl-velocity field is denoted

$$\mathbf{u}_{\circlearrowleft}(\mathbf{x}, t), \qquad \boldsymbol{\omega}_{\circlearrowleft}(\mathbf{x}, t) = \nabla \times \mathbf{u}_{\circlearrowleft}(\mathbf{x}, t). \tag{39}$$

The symbol $\mathbf{v}_{\circlearrowleft}$ without explicit arguments is reserved for the canonical characteristic carrier velocity; it must not be used as the unrestricted local field in bridge equations.

[ORTHODOX]: $\omega_c = \frac{m_e c^2}{\hbar}$ is a standard physical constant.

[SPECULATIVE]: The identification of ω_c as the primitive Compton phase frequency of the SST carrier is an SST postulate; it is not a local vorticity variable.

3.2 Derived Horn/Circulation Radius

The quantity previously denoted as the characteristic "core radius" is fixed more precisely as a Compton-locked horn/circulation radius:

$$r_c \equiv R_{\mathrm{horn}} = \frac{v_{\circlearrowleft}^{*}}{\omega_c} = \frac{\hbar v_{\circlearrowleft}^{*}}{m_e c^2} = \frac{\alpha \hbar}{2 m_e c} = \frac{r_e}{2}. \tag{40}$$

Here

$$\omega_c \equiv \frac{m_e c^2}{\hbar} \tag{41}$$

is the electron Compton angular frequency. Thus r_c is the radius at which the canonical tangential swirl speed $v_{\circlearrowleft}^{*}$ is obtained at the Compton angular frequency.

[**CANON / NOTATION**]: ω_c denotes the Compton angular frequency. It must not be confused with the local vorticity scale. If $\omega_{\mathrm{vort},c}$ denotes the local vorticity of a rigid local rotation, then

$$\omega_{\mathrm{vort},c} = 2\omega_c, \qquad R_{\mathrm{horn}} = \frac{2 v_{\circlearrowleft}^{*}}{\omega_{\mathrm{vort},c}}. \tag{42}$$

[**CANON / SEMANTIC RULE**]: r_c is not the resolved physical vortex-tube radius. The local tube/core thickness is denoted by a_{core} and remains a separate geometric or variational quantity:

$$a_{\mathrm{core}} \neq r_c. \tag{43}$$

[**DERIVED**] **Consequence**: Eq. (40) removes r_c as an independent parameter while avoiding the earlier ambiguity between horn-envelope scale, tube radius, and vorticity.

[**CALIBRATED CHAIN GUARD**]: Eq. (40) is a Compton–electron-radius closure inside the primitive calibration chain. It must not be cited as an independent hydrodynamic derivation of either α, $v_{\circlearrowleft}^{*}$, or r_c. A future fluid-dynamical derivation would have to obtain the same radius from a resolved vortex-tube variational problem using a_{core} or an explicit profile, without inserting the electron-radius identity.

3.3 Circulation Quantization

The circulation is defined as:

$$\Gamma = \oint \mathbf{v} \cdot d\ell \tag{44}$$

We introduce the fundamental circulation quantum:

$$\Gamma_0 = 2\pi r_c \, v_{\circlearrowleft}^{*} \tag{45}$$

and impose:

$$\Gamma = n\Gamma_0, \quad n \in \mathbb{Z} \tag{46}$$

[**ORTHODOX**]: Circulation quantization appears in superfluid systems.

[**DERIVED**]: In SST, Γ_0 is no longer an input but follows from Eq. 40.

3.4 Bounded-Domain Biot–Savart Admissibility

[ORTHODOX OPERATOR FOUNDATION]. For a smooth vector field $\mathbf{V}$ supported in a bounded smooth domain $\Omega \subset \mathbb{R}^3$, define

$$\mathrm{BS}[\mathbf{V}](\mathbf{y}) := \frac{1}{4\pi} \int_\Omega \mathbf{V}(\mathbf{x}) \times \frac{\mathbf{y}-\mathbf{x}}{|\mathbf{y}-\mathbf{x}|^3}\, d^3x. \tag{47}$$

Its curl contains explicit source and boundary corrections:

$$\begin{aligned}\nabla_{\mathbf{y}} \times \mathrm{BS}[\mathbf{V}] = \mathbf{1}_\Omega(\mathbf{y})\mathbf{V}(\mathbf{y}) + \frac{1}{4\pi}\nabla_{\mathbf{y}} \int_\Omega \frac{\nabla_{\mathbf{x}} \cdot \mathbf{V}(\mathbf{x})}{|\mathbf{y}-\mathbf{x}|}\, d^3x \\ - \frac{1}{4\pi}\nabla_{\mathbf{y}} \int_{\partial\Omega} \frac{\mathbf{V}(\mathbf{x}) \cdot \mathbf{n}(\mathbf{x})}{|\mathbf{y}-\mathbf{x}|}\, dA_x.\end{aligned} \tag{48}$$

Consequently,

$$\nabla \times \mathrm{BS}[\mathbf{V}] = \mathbf{V} \quad \text{in } \Omega \quad \iff \quad \nabla \cdot \mathbf{V} = 0, \quad \mathbf{V} \cdot \mathbf{n} = 0 \text{ on } \partial\Omega. \tag{49}$$

These conditions are necessary as well as sufficient [59].

The same source establishes

$$\ker \mathrm{BS} = \{\nabla\phi : \nabla\phi \text{ is normal to } \partial\Omega\}, \tag{50}$$

$$\operatorname{im} \mathrm{BS} \subsetneq \operatorname{im}(\nabla\times), \tag{51}$$

and the L^2 bound

$$\|\mathrm{BS}[\mathbf{V}]\|_{L^2(\Omega)} \leq \frac{\mathrm{OS}(\Omega)}{4\pi} \|\mathbf{V}\|_{L^2(\Omega)}, \qquad \mathrm{OS}(\Omega) := \sup_{\mathbf{y}\in\mathbb{R}^3} \int_\Omega \frac{d^3x}{|\mathbf{y}-\mathbf{x}|^2}. \tag{52}$$

The extended operator is bounded, compact, and self-adjoint on its L^2 completion.

[CANON / FAR-FIELD GUARD]. A general compact source gives $\mathrm{BS}[\mathbf{V}] = O(r^{-2})$. If $\mathbf{V}$ is divergence-free and boundary-tangent, the leading moment cancels and

$$\mathrm{BS}[\mathbf{V}](r) = O(r^{-3}). \tag{53}$$

Therefore the direct Biot–Savart velocity of one compact resolved vortex knot is not a Newtonian monopole sector. Any r^{-2} force claim must be derived from a separately declared pressure/source, non-compact background, singular-core, or boundary-supported mechanism.

3.5 Hodge Decomposition and Topological Circulation Sectors

[ORTHODOX VECTOR-CALCULUS FOUNDATION]. On a smooth bounded domain $\Omega \subset \mathbb{R}^3$, the space of smooth vector fields decomposes orthogonally as

$$VF(\Omega) = FK \oplus HK \oplus CG \oplus HG \oplus GG, \tag{54}$$

where

$$FK = \{\nabla \cdot \mathbf{V} = 0,\ \mathbf{V} \cdot \mathbf{n} = 0, \tag{55}$$

$$\text{all interior fluxes} = 0\}, \tag{56}$$

$$HK = \{\nabla \cdot \mathbf{V} = 0,\ \nabla \times \mathbf{V} = 0, \tag{57}$$

$$\mathbf{V} \cdot \mathbf{n} = 0\}, \tag{58}$$

$$CG = \{\mathbf{V} = \nabla\phi,\ \nabla \cdot \mathbf{V} = 0, \tag{59}$$

$$\text{all boundary fluxes} = 0\}, \tag{60}$$

$$HG = \{\mathbf{V} = \nabla\phi,\ \nabla \cdot \mathbf{V} = 0, \tag{61}$$

$$\phi \text{ constant on each boundary component}\}, \tag{62}$$

$$GG = \{\mathbf{V} = \nabla\phi,\ \phi|_{\partial\Omega} = 0\}. \tag{63}$$

The incompressible boundary-tangent sector is

$$K(\Omega) = FK \oplus HK, \tag{64}$$

and

$$HK \cong H_1(\Omega;\mathbb{R}) \cong H_2(\Omega,\partial\Omega;\mathbb{R}). \tag{65}$$

Hence $\dim HK$ is the total genus of the boundary [60].

[CANON / INTERPRETATION GUARD]. Vorticity determines the fluxless/curl-bearing part but does not determine the harmonic-knot circulation sector. A ball has $HK = 0$; a solid torus has $\dim HK = 1$. This counts independent global circulation channels, not particle species or knot type.

3.6 Energy and Mass Relation

Mass is interpreted as stored rotational energy:

$$E = mc^2 \tag{66}$$

[**ORTHODOX**]: Standard relativistic energy relation.
[**SPECULATIVE**]: SST interprets this energy as localized vortex kinetic energy.

3.7 Swirl Velocity Constraint

We define the dimensionless ratio:

$$\eta_0 = \frac{v_{\circlearrowleft}^{*}}{c} \tag{67}$$

Within the present calibrated constant chain:

$$\eta_0 = \frac{\alpha}{2} \tag{68}$$

[**CALIBRATED ALGEBRAIC IDENTITY**]: Once the Compton anchoring relation $r_c = v_{\circlearrowleft}^{*}/\omega_c$ is combined with the electron-radius closure $r_c = \alpha\hbar/(2m_e c)$, the identity $v_{\circlearrowleft}^{*} = \alpha c/2$ follows by direct substitution.

[**CRITICAL NOTE**] This is not an independent fluid-dynamical derivation of α or of the characteristic speed. The fine-structure constant enters through the adopted electron-radius / horn-radius calibration, so the result is a calibrated algebraic identity rather than an emergent theorem.

3.8 Swirl-Clock Definition

For a localized carrier or observer A, define the Swirl-Clock as the conversion between substrate foliation time and accumulated local proper time:

$$S_{(t)}^{\circlearrowleft}[A;T] := \frac{d\tau_A}{dT} = \sqrt{1 - \frac{u_A^2(T)}{c^2}}, \tag{69}$$

where u_A is the resolved kinematic speed entering the clock response. The canonical internal swirl-speed evaluation is the special case

$$S_{(t)_c}^{\circlearrowleft} = \sqrt{1 - \frac{{v_{\circlearrowleft}^{*}}^2}{c^2}}. \tag{70}$$

[ORTHODOX FORM]. Equation (69) has the standard Lorentz time-dilation form. Operational comparison between Alice and Bob is governed by their relative velocity after clocks and rulers are synchronized by the common torsion cone.

[CANONICAL SST POSTULATE]. The clock response is part of the founding operational interpretation of SST: stable vortex matter measures local spacetime through the same luminal torsion sector that defines c. It is therefore stronger than a merely decorative analogy.

[OPEN DERIVATION]. A substrate-level derivation must still show that the torsion sector dresses all stable matter carriers isotropically and produces Eq. (69) without species-dependent retuning. This is the monometricity and core–torsion impedance problem developed in the research track.

3.9 Two-Speed Discipline for Clock Emergence

The canon distinguishes the canonical horn/circulation speed $v_{\circlearrowleft}^{*}$ from the transverse causal speed c:

$$v_{\circlearrowleft}^{*} = 1.09384563 \times 10^{6}\ \mathrm{m\,s^{-1}}, \qquad c = 2.99792458 \times 10^{8}\ \mathrm{m\,s^{-1}}, \qquad \frac{v_{\circlearrowleft}^{*}}{c} = 3.64867628 \times 10^{-3}. \tag{71}$$

Thus $v_{\circlearrowleft}^{*}$ is a calibrated circulation/horn speed, not the universal causal speed.

[CANON / SEMANTIC RULE] The calibrated identity $v_{\circlearrowleft}^{*} = \alpha c/2$ must not be inverted into a claim that the internal core sound speed equals c. In a Gross–Pitaevskii / NLSE core model with circulation quantum and healing length

$$\Gamma_q = \frac{h_*}{m_*}, \qquad \xi = \frac{\hbar_*}{\sqrt{2}\, m_* c_s}, \tag{72}$$

imposing $\Gamma_q = \Gamma_0$ and $\xi = r_c$ gives

$$c_s = \frac{\Gamma_0}{2\sqrt{2}\pi r_c} = \frac{v_{\circlearrowleft}^{*}}{\sqrt{2}} = 7.73465663 \times 10^{5}\ \mathrm{m\,s^{-1}}. \tag{73}$$

Therefore a calculation that closes an inertia ratio as u^2/c_s^2 has produced an internal core-acoustic closure, not yet the Lorentz-compatible ratio u^2/c^2.

[RESEARCH-TRACK GATE] A Lorentz-compatible derivation of the Swirl-Clock requires a separate transverse torsion/shear sector with propagation speed $c_T = c$, plus a core–torsion impedance identity of the form

$$\mathsf{M}_{\mathrm{torsion}}[K] \overset{?}{=} \frac{2E_0[K]}{c_T^2}\,\mathsf{I}. \tag{74}$$

Equation (69) and the operational identification $c_A^{(2\mathrm{w})} = c$ are retained as canonical founding postulates of SST, motivated by the empirical structure of special relativity. What remains open is their non-circular derivation from the substrate: Eq. (74), universal clock-and-ruler dressing, and low-energy monometricity must follow from a controlled field calculation. The NLSE core alone does not supply that proof.

3.10 Density Hierarchy

We distinguish the following densities:

$$\rho_f \quad \text{(effective background fluid density)}, \tag{75}$$

$$\rho_E = \tfrac{1}{2}\rho_f\, v_{\circlearrowleft}^{*\,2} \quad \text{(swirl energy density)}, \tag{76}$$

$$\rho_m = \rho_E/c^2 \quad \text{(mass-equivalent density)}, \tag{77}$$

$$\rho_{\mathrm{core}} \quad \text{(canonical saturated envelope-equivalent density)}. \tag{78}$$

[ORTHODOX] Definition: $\rho_m = \rho_E/c^2$ is the mass-equivalent density associated with the swirl kinetic accounting above; in orthodox limits, total energy E in a region still yields mass density E/c^2.

[SPECULATIVE] Interpretation: The distinction between background, energetic, and saturated envelope-equivalent sectors is elevated in SST to a structural ontology of the medium.

3.11 Horn-Envelope Density Closure

The density formerly denoted as a "core density" is now interpreted as an effective horn-envelope normalization density, not as a measured local material density of the resolved vortex tube:

$$\rho_{\rm horn}^{\rm eff} \equiv \frac{m_e c^2}{2\pi v_{\circlearrowleft}^{*2} R_{\rm horn}^3} = \frac{m_e c^2}{2\pi v_{\circlearrowleft}^{*2} {r_c}^3}. \tag{79}$$

[CANON / SEMANTIC RULE]: Equation (79) defines the canonical horn-envelope density closure used by the present SST calibration program. Legacy appearances of $\rho_{\rm core}$ in expressions proportional to $\rho_{\rm core}{r_c}^3$ should be read as $\rho_{\rm horn}^{\rm eff}$ unless a separate resolved-tube model is explicitly introduced.

[DERIVED] Consequence: Within that closure, $\rho_{\rm horn}^{\rm eff}$ is no longer an independent parameter. If a local tube mass density is required, it must be introduced through the separate tube radius $a_{\rm core}$ and a corresponding local tube-volume or profile model.

3.12 Chronos–Kelvin Invariant and Clock–Radius Transport

For a thin material loop of radius $R(t)$ carried by an incompressible, inviscid flow with no reconnection events, Kelvin's theorem implies conservation of circulation. In the near-solid-body core approximation this gives

$$\frac{D}{Dt}(R^2\omega) = 0, \tag{80}$$

where ω is the characteristic local vorticity on the loop. For rigid local rotation, $v_t = (\omega/2)R$; evaluated at the canonical horn scale this gives $v_t = (\omega/2)r_c$. Using this relation together with Eq. (69), one may rewrite

$$\omega = \frac{2c}{r_c}\sqrt{1 - S_{(t)}^2}, \tag{81}$$

so that Eq. (80) takes the equivalent form

$$\frac{D}{Dt}\left(\frac{2c}{r_c}R^2\sqrt{1 - S_{(t)}^2}\right) = 0. \tag{82}$$

[ORTHODOX]: Eq. (80) is the thin-loop form of Kelvin circulation conservation in inviscid barotropic flow [6, 30].

[CONDITIONAL DERIVED]: Eq. (82) is the SST rewriting of the same invariant in terms of the Swirl-Clock, conditional on evaluating the local tangential speed at the fixed horn/circulation scale r_c. It is not a generic radius-transport theorem for an arbitrary resolved tube radius $a_{\rm core}(t)$.

Differentiating Eq. (82) gives a useful transport identity:

$$\frac{dS_{(t)}}{dt} = \frac{2\big(1 - S_{(t)}^2\big)}{S_{(t)}} \frac{1}{R} \frac{dR}{dt}. \tag{83}$$

[CONDITIONAL DERIVED]: within the horn-anchored approximation, contraction of a material loop drives stronger clock suppression, while expansion relaxes it.

3.13 Reparametrized Zero-Parameter Corollary

Older SST canon layers often used the triplet (Γ_0, ρ_f, r_c) as the primitive parameterization. In the present v0.8.10 architecture, Eq. (40) and Eq. (45) show that this is a reparameterization of the present primitive calibration set $(\rho_f, v_{\circlearrowleft}^*, \omega_c)$ rather than a distinct axiom.

Explicitly,

$$\Gamma_0 = 2\pi r_c\, v_{\circlearrowleft}^* = 2\pi \frac{v_{\circlearrowleft}^{*\,2}}{\omega_c}. \tag{84}$$

Hence any dimensional SST quantity written in terms of (Γ_0, ρ_f, r_c) can be rewritten in terms of $(\rho_f, v_{\circlearrowleft}^*, \omega_c)$ with no new calibration input.

A coarse-grained density law retained from earlier canon layers is

$$\rho_f = K\,\Omega, \qquad K = \frac{\rho_{\text{core}}\, r_c}{v_{\circlearrowleft}^*}, \tag{85}$$

where Ω is a coarse-grained local rotation rate and K has dimensions $\mathrm{kg\,m^{-3}\,s}$. This relation records the canonical bridge between a sparse effective bulk density and a high-density horn-envelope normalization; it is not a resolved local-tube density law unless a_{core} is supplied.

[CRITICAL NOTE] Eq. (85) is not an additional zero-parameter theorem unless the averaging prescription that defines Ω is fixed independently of the data being fitted. If Ω is chosen phenomenologically, the relation is a bridge ansatz and K is a calibration transfer coefficient, not a new prediction.

[DERIVED COROLLARY]: the older "zero-parameter" language is retained only as a statement of reparameterization discipline: once one primitive triplet is fixed, no additional dimensional constants should be introduced without explicit justification.

3.14 Canonical Master Equation

The preceding subsections define the four ingredients that later SST papers often treat separately: a local mass-equivalent density, a local Swirl-Clock, a geometric gate, and a topological kernel. We therefore introduce a single canonical equation that organizes these sectors in one place.

First define the local swirl energy density and its mass-equivalent form:

$$\rho_E(x,t) = \frac{1}{2}\rho_f(x,t)\,\|\mathbf{u}_{\circlearrowleft}(x,t)\|^2 \tag{86}$$

$$\rho_m(x,t) = \frac{\rho_E(x,t)}{c^2} \tag{87}$$

Using Eq. 69, the local Swirl-Clock factor is

$$S_{(t)}^{\circlearrowleft}(x,t) = \sqrt{1 - \frac{\|\mathbf{u}_{\circlearrowleft}(x,t)\|^2}{c^2}} \tag{88}$$

We then define a binary geometric exposure gate

$$G(K) \in \{0,1\} \tag{89}$$

together with the canonical geometric factor

$$\Pi_K = \left(\frac{\lambda_c}{\pi r_c}\right)^{G(K)} = \left(\frac{4}{\alpha}\right)^{G(K)}. \tag{90}$$

[CANON / ALPHA-GATE GUARD] The equality $\lambda_c/(\pi r_c) = 4/\alpha$ is a calibrated geometric rewriting inside the present constant chain. It must not be cited as an independent derivation of the fine-structure constant. The finite-cell trefoil result is retained separately as an α-free sub-per-mille coincidence plus obstruction result in Subsection E.10.

Numerically,

$$\frac{\lambda_c}{\pi r_c} = \frac{4}{\alpha} \simeq 5.48144 \times 10^2. \tag{91}$$

This number is therefore an exposure or shielding gate inside the calibrated mass functional, not a new measurement or theorem for α.

The canonical topological kernel is taken to be

$$\Xi_K = [\alpha_C\, C(K) + \beta_L\, L(K) + \gamma_H\, \mathcal{H}(K)]\, \varphi^{-2k(K)}. \tag{92}$$

[CANON / GOLDEN-LAYER GUARD]. The golden-layer factor has an exact hyperbolic reparametrization,

$$\eta_\varphi \equiv \operatorname{asinh}\left(\frac{1}{2}\right) = \ln\varphi, \tag{93}$$

$$\varphi^{-2k(K)} = \exp[-2k(K)\eta_\varphi]\,. \tag{94}$$

This is a parametrization guard for the existing layer factor, not a derivation of the golden factor from Euler dynamics, ropelength criticality, or knot stability. It also does not assert that the electron trefoil is a hyperbolic knot; the trefoil remains a torus knot in the topology dictionary. Any future promotion of this notation must identify $k(K)$ with a mechanically defined discrete transfer, impedance, or relaxation step.

The canonical SST master equation is therefore

$$M_K(x,t) = \rho_m(x,t)\, {r_c}^3\, \Pi_K\, \Xi_K\, S_{(t)}^{\circlearrowleft}(x,t)^{-2} \tag{95}$$

$$= \left(\frac{\rho_f(x,t)\, \|\mathbf{u}_\circlearrowleft(x,t)\|^2}{2c^2}\right) {r_c}^3 \left(\frac{\lambda_c}{\pi r_c}\right)^{G(K)} [\alpha_C\, C(K) + \beta_L\, L(K) + \gamma_H\, \mathcal{H}(K)]\, \varphi^{-2k(K)} S_{(t)}^{\circlearrowleft}(x,t)^{-2} \tag{96}$$

[SPECULATIVE] Organizing principle: This equation factorizes SST into four sectors:

$$\text{medium scale} \times \text{geometric gate} \times \text{topological kernel} \times \text{clock impedance}. \tag{97}$$

[DERIVED] Dimensional check:

$$\left[\frac{\rho_f\, \|\mathbf{u}_\circlearrowleft\|^2}{c^2} {r_c}^3\right] = \mathrm{kg} \tag{98}$$

and all remaining factors are dimensionless, so Eq. 96 has the correct mass dimension.

[CANON / SEMANTIC RULE]: the factor ${r_c}^3$ in Eq. (96) is a canonical horn/envelope normalization volume scale. It is not a resolved physical tube volume unless a separate model sets $a_{\rm core} = r_c$. Resolved tube energies, cutoffs, and local vorticity estimates must use $a_{\rm core}$ or an explicit profile.

[SPECULATIVE] Interpretation: The same local kinematic state that slows the Swirl-Clock also enters the effective mass scale; mass and local proper-time modulation are therefore treated as two aspects of one medium configuration.

[CRITICAL NOTE] Equation (96) is presently a canonical synthesis formula, not yet a theorem derived from first-principles filament dynamics alone.

[NORMALIZATION TRANSPARENCY] With the canonical background-energy reading, $\rho_m = \rho_E/c^2 = 4.65949 \times 10^{-12}\,\mathrm{kg\,m^{-3}}$ and $\rho_m {r_c}^3 = 1.30330 \times 10^{-56}\,\mathrm{kg} = 1.43072 \times 10^{-26} m_e$. Therefore an exposed electron-scale benchmark with $\Pi_K = 4/\alpha$ requires a dimensionless kernel of order

$$\Xi^{(e)}_{3_1} \sim \frac{m_e}{\rho_m {r_c}^3 (4/\alpha)} = 1.28 \times 10^{23}. \tag{99}$$

Unless a branch explicitly re-anchors the medium scale to $\rho^{\rm eff}_{\rm horn}$, the coefficients inside Ξ_K are therefore calibrated matching coefficients carrying the medium-to-particle hierarchy, not order-unity topological invariants. This note is a scale-accounting guard; it does not alter the dimensional validity of Eq. (96).

In the stationary weak-swirl limit $S^{\circlearrowleft}_{(t)} \to 1$, it reduces to the static mass-functional form; in the local-kinematic limit at fixed K, it reduces to a pure clock-impedance correction. The hierarchy is therefore carried predominantly by the geometric and topological factors, while the Swirl-Clock supplies a coherent local correction rather than the primary source of mass separation.

3.15 Geometric Representation

A swirl-string is represented as a closed curve:

$$X(t,\sigma) = (x(t,\sigma), y(t,\sigma), z(t,\sigma)) \tag{100}$$

with $\sigma \in [0, 2\pi)$.

3.16 Summary of Dependency Structure

The full dependency chain is:

$$(\rho_f, v^*_{\circlearrowleft}, \omega_c) \to r_c \to \Gamma_0 \to (\rho_E, \rho_m, \rho_{\rm core}) \to S^{\circlearrowleft}_{(t)} \to M_K. \tag{101}$$

[DERIVED] Consistency statement: No quantity is defined circularly. The canonical closure is summarized by Eq. 96.

3.17 Consistency Statement

- All derived quantities depend on the stated calibration set, explicit orthodox constants, and explicitly labeled bridge assumptions.
- No *unlabeled* external parameters are introduced.
- Quantities fixed by α, m_e, $\hbar$, c, R_∞, or t_p are labeled **[CALIBRATED]** rather than independent predictions.

3.18 Interpretation

The Formal Foundations establish SST as:

- a continuum theory with minimal primitives
- a geometrically grounded framework
- a non-circular parameter system

3.19 Maximal Swirl Tension

3.19.1 Definition and canonical form

Within the primitive calibration set $\mathcal{P}_{\text{cal}} = \{\rho_f, v_{\circlearrowleft}^*, \omega_c\}$, a characteristic force scale emerges naturally from the Compton energy stored over the canonical circulation cross-section. We define the *maximal swirl tension* as

$$F_{\text{swirl}}^{\max} = \frac{v_{\circlearrowleft}^* \hbar}{2{r_c}^2} = \frac{\hbar\omega_c}{2r_c}. \tag{102}$$

[CALIBRATED ALGEBRAIC IDENTITY] Canonical form: substituting $v_{\circlearrowleft}^* = \omega_c r_c$ into Eq. (102) gives $F_{\text{swirl}}^{\max} = \hbar\omega_c/2r_c$, the Compton energy scale divided by twice the canonical circulation radius. Numerically, $F_{\text{swirl}}^{\max} = 29.053507$ N.

[CRITICAL NOTE] $F_{\text{swirl}}^{\max}$ is a mechanical reformulation of the Compton energy scale over the canonical circulation cross-section. It is *not* independent of $\hbar$ and m_e; its role in the canon is interpretive and organisational rather than that of an independent primitive.

3.19.2 Internal consistency identities

The following expressions are algebraically equivalent within the canonical constant chain and serve as coherence verifications rather than independent derivations:

$$\begin{aligned}
F_{\text{swirl}}^{\max} &= \frac{e^2}{16\pi\varepsilon_0 {r_c}^2}, && (103)\\
&= \frac{h\alpha c}{8\pi {r_c}^2}, && (104)\\
&= \pi {r_c}^2 \rho_{\text{calc}} {v_{\circlearrowleft}^*}^2, && (105)\\
&= \frac{c^4}{4G}\,\alpha \left(\frac{r_c}{\ell_P}\right)^{-2}. && (106)
\end{aligned}$$

where $\rho_{\text{calc}} = 4F_{\text{swirl}}^{\max}/(\pi\alpha^2 c^2 {r_c}^2)$ is the derived bulk density consistent with the canonical neutral-circulation closure.

[CALIBRATED ALGEBRAIC IDENTITY] Consistency check: Eqs. (103)–(106) are equivalent given $v_{\circlearrowleft}^* = \alpha c/2$, the standard identity $e^2/(4\pi\varepsilon_0) = \alpha\hbar c$, $r_c = \alpha\hbar/(2m_e c)$, and $\ell_P^2 = \hbar G/c^3$. The Planck-force form in Eq. (106) is a hierarchy reparameterization of the same calibrated tension scale, not an independent derivation from Planck physics. Their mutual agreement is an internal algebraic coherence test of the calibrated chain, not an independent dynamical prediction.

3.19.3 Rydberg consistency identity

[**CALIBRATED IDENTITY**] The Rydberg constant is recoverable from the three SST calibration variables without e, ε_0, or m_e appearing *explicitly*:

$$R_\infty = \frac{v_{\circlearrowleft}^{*\,3}}{\pi\, r_c\, c^3}. \tag{107}$$

[**DIMENSIONAL GUARD**] The direction of Eq. (107) is fixed: the numerator is $v_{\circlearrowleft}^{*\,3}$ and the denominator is $\pi r_c c^3$. Its units are m^{-1}. The reciprocal expression $\pi r_c c^3 / v_{\circlearrowleft}^{*\,3}$ has units of length and is not a Rydberg constant.

This is a benchmark for algebraic consistency of the calibration chain, not for independent non-redundancy of the primitive set. [**CRITICAL NOTE**] Substituting the canonical definitions $v_{\circlearrowleft}^* = \alpha c/2$ and $r_c = v_{\circlearrowleft}^*/\omega_c = \alpha\hbar/(2m_e c)$ reduces Eq. (107) *exactly* to the standard $R_\infty = \alpha^2 m_e c/(4\pi\hbar)$ (verified to machine precision): m_e is hidden inside r_c and α inside both $v_{\circlearrowleft}^*$ and r_c. The identity is therefore a consistency check, not an independent non-circular derivation of R_∞. Combining Eq. (107) with Eq. (102) yields the compact relation

$$\boxed{F_{\text{swirl}}^{\max} = \frac{32\pi^2 \hbar R_\infty^2 c}{\alpha^5},} \tag{108}$$

which expresses $F_{\text{swirl}}^{\max}$ in terms of spectroscopic observables alone, with no reference to m_e, r_c, or e as separate inputs. A further identity connecting atomic and force scales is

$$R_\infty = F_{\text{swirl}}^{\max} \cdot \frac{\alpha^3}{4\pi m_e c^2}, \tag{109}$$

which states that the Rydberg constant is the maximal swirl tension scaled by the Compton energy and a purely geometric factor $\alpha^3/4\pi$.

3.19.4 Threefold physical interpretation

The quantity $F_{\text{swirl}}^{\max}$ admits three distinct physical interpretations, each arising from a different sector of physics. [**CRITICAL NOTE**] These three values are algebraically equivalent expressions built from the same primitive set $\{\alpha, m_e, \hbar, c\}$; their numerical agreement is an *identity*, not an independent coincidence, and is not on its own confirmation of the theory.

(i) Gravitational analogue: the Planck force ceiling. In general relativity, the combination $c^4/4G$ sets the maximum force that any physical process can transmit without forming a causal horizon [40]. We denote this scale the maximal gravitational tension:

$$F_{\text{gr}}^{\max} = \frac{c^4}{4G}. \tag{110}$$

[**ORTHODOX**] $F_{\text{gr}}^{\max}$ is the Planck force, an invariant upper bound in classical general relativity [40].

[**CALIBRATED ALGEBRAIC IDENTITY**] The ratio of the two tension scales reduces exactly to

$$\boxed{\frac{F_{\text{swirl}}^{\max}}{F_{\text{gr}}^{\max}} = \frac{4\alpha_g}{\alpha},} \tag{111}$$

where $\alpha_g = Gm_e^2/\hbar c$ is the gravitational fine-structure constant of the electron and α is the electromagnetic fine-structure constant. Numerically, $4\alpha_g/\alpha \approx 9.6 \times 10^{-43}$, in agreement with the directly computed ratio 29.053507 N / 3.026×10^{43} N.

[SPECULATIVE] Interpretation: the hierarchy between the electromagnetic and gravitational force scales may be organized by the ratio of the two electron coupling constants. This is an interpretive compression of calibrated constants; by itself it does not prove a new medium-dynamical stiffness law.

(ii) Electrostatic interpretation: the Coulomb ceiling at the canonical circulation scale. When two protons are compressed to a centre-to-centre separation equal to the canonical circulation radius r_c, the Coulomb repulsion is

$$F_{pp}(r_c) \;=\; \frac{e^2}{4\pi\varepsilon_0 {r_c}^2} \;=\; 4\,F_{\text{swirl}}^{\max}. \tag{112}$$

[CALIBRATED] The maximal Coulomb repulsion at the canonical circulation scale is exactly four times $F_{\text{swirl}}^{\max}$. **[CRITICAL NOTE]** The factor of four is an arithmetic identity once $v_{\circlearrowleft}^{*} = \alpha c/2$ is adopted (the $4/\alpha$ gate); it is a self-consistency of the calibrated chain, not an independent prediction.

[SPECULATIVE] Interpretation: $F_{\text{swirl}}^{\max}$ is the quarter-Coulomb force at the canonical circulation scale. The medium sustains one quarter of the maximum proton–proton Coulomb repulsion before the neutral circulation envelope reaches its structural limit. This provides a physical ceiling on the compression of charged vortex configurations.

(iii) Mechanical interpretation: the spring constant of photon–electron interaction. The electron's internal vortex is modelled as a linear spring with stiffness

$$K_e \;=\; \frac{F_{\text{swirl}}^{\max}}{n\,r_c}, \qquad n = 2, \tag{113}$$

where n counts the number of half-wavelength segments on the core and is fixed to $n = 2$ by the photon–electron swirl matching condition. The natural oscillation frequency of the electron core under this spring is

$$\omega_{\text{spring}} \;=\; \sqrt{\frac{K_e}{m_e}} \;=\; \sqrt{\frac{F_{\text{swirl}}^{\max}}{2\,r_c\,m_e}} \;=\; \frac{\omega_c}{\alpha} \;=\; \frac{m_e c^2}{\alpha\hbar}, \tag{114}$$

a frequency intermediate between the atomic Rydberg scale ($\sim \alpha^2\omega_c$) and the bare Compton scale (ω_c).

[DERIVED] The energy stored in this spring at maximum compression $\delta = r_c$ is

$$E_{\text{spring}} \;=\; \tfrac{1}{2}\,K_e\,{r_c}^2 \;=\; \frac{F_{\text{swirl}}^{\max}\,r_c}{4} \;=\; \frac{\hbar\omega_c}{8} \;=\; \frac{m_e c^2}{8}. \tag{115}$$

One eighth of the electron rest energy is stored as internal vortex-spring potential at maximum compression.

[SPECULATIVE] Physical picture: during photon absorption, the electron vortex core is compressed for a duration of order one Planck time t_p into a transient mass-like configuration. The restoring force scale of this compression is $F_{\text{swirl}}^{\max}$, and the energy scale of the transient is $m_e c^2/8$. The spring releases at the Compton period, generating the recoil that initiates the Coulomb interaction with the proton.

[CRITICAL NOTE] Equation (114) assumes $n = 2$ from empirical matching. A first-principles derivation of the photon wrap number from the delay dynamics of Section 8 is an open derivational step.

3.19.5 Unified summary

The three interpretations are collected in Table 1. Together they organize $F_{\text{swirl}}^{\max}$ as a *calibrated structural tension scale*. The table is useful for cross-sector bookkeeping, but the equalities are algebraic consequences of the same calibrated electron-scale constants unless a separate medium-dynamical derivation is supplied.

Table 1: Threefold physical interpretation of $F_{\text{swirl}}^{\max}$ within the SST canonical framework.

Context	Expression	Physical meaning	Status
Gravitational ceiling	$F_{\text{gr}}^{\max}/F_{\text{swirl}}^{\max} = \alpha/(4\alpha_g)$	Calibrated electron-coupling ratio $\alpha/4\alpha_g$	[Calibrated identity]
Coulomb ceiling	$F_{pp}(r_c) = 4\,F_{\text{swirl}}^{\max}$	Quarter of max. Coulomb force at circulation scale	[Calibrated identity]
Vortex spring	$K_e = F_{\text{swirl}}^{\max}/2r_c$	Spring constant of photon–electron interaction	[Speculative]

[SPECULATIVE] Canonical interpretation: $F_{\text{swirl}}^{\max}$ is retained as a candidate tension scale for topological integrity of the structured medium. Its comparison with $F_{\text{gr}}^{\max} = c^4/4G$ is a calibrated electron-normalized analogy, not a theorem that gravitational and electromagnetic stiffness have already been derived from the substrate.

4 Axiomatic Framework

4.1 Preliminaries

We formalize Swirl-String Theory (SST) as an axiomatic continuum theory defined over a three-dimensional spatial domain $\mathcal{M} \subset \mathbb{R}^3$ equipped with a preferred time parameter $t \in \mathbb{R}$.

The theory is constructed from a minimal primitive set $\mathcal{P}$ (Section 3) and a set of axioms $\mathcal{A}$ governing admissible configurations and dynamics.

4.2 Axiom 1: Continuum Substrate

Statement. There exists a continuous, incompressible, inviscid medium filling $\mathcal{M}$, characterized by a constant background density ρ_f.

$$\nabla \cdot \mathbf{v} = 0 \tag{116}$$

[ORTHODOX]: This corresponds to the standard incompressible Euler framework.

Interpretation. All physical structures arise as configurations within this medium; no discrete particles are assumed at the fundamental level.

4.3 Axiom 2: Existence of Stable Vortex Filaments

Statement. The medium admits stable, localized, topologically nontrivial vortex filament solutions.

[ORTHODOX] Physics input: vortex filaments are known idealized structures in fluid dynamics.

[SPECULATIVE] Extension: such filaments are assumed here to be dynamically stable and persist as fundamental carriers.

Mathematical form. A filament is represented as a closed embedding:

$$X : S^1 \to \mathcal{M}, \quad \sigma \mapsto X(t, \sigma) \tag{117}$$

4.4 Axiom 3: Circulation Quantization

Statement. Circulation around any stable filament is quantized:

$$\Gamma = \oint \mathbf{v} \cdot d\ell = n\Gamma_0, \quad n \in \mathbb{Z} \tag{118}$$

[**ORTHODOX**]: Quantized circulation appears in superfluid systems.
[**SPECULATIVE**]: Elevated here to a universal invariant.

4.5 Axiom 4: Compton Anchoring

Statement. The Compton phase frequency of the canonical horn/circulation carrier satisfies:

$$\frac{v_{\circlearrowleft}^*}{r_c} = \omega_c \tag{119}$$

[**ORTHODOX**]: ω_c is the Compton frequency.
[**SPECULATIVE**]: Identified as the intrinsic Compton phase frequency of the SST carrier, not as the local vorticity of the resolved tube.

Consequence. The horn/circulation radius is not independent but derived (Eq. 40); the local tube radius remains a_{core}.

4.6 Axiom 5: Delay as a Constitutive Principle

Statement. Finite propagation time along closed vortex structures is a fundamental dynamical ingredient.

$$\tau_{\text{circ}}(K) = \frac{L_K}{v_{\circlearrowleft}^*} \tag{120}$$

[**DERIVED**] Kinematic consequence: the delay follows from finite signal speed along a closed carrier.

Axiomatic role. The delay is not treated as negligible, but as structurally relevant to the dynamics.

Interpretation. Temporal nonlocality is intrinsic to the dynamics.

4.7 Axiom 6: Energy Localization

Statement. Energy is localized in vortex structures and determines mass via:

$$E = mc^2 \tag{121}$$

[**ORTHODOX**]: Relativistic energy relation.
[**SPECULATIVE**]: Interpreted as rotational energy of the vortex configuration.

4.8 Axiom 7: Operational Spacetime of Vortex Matter

Statement. The global substrate parameter T orders the medium evolution, but physical observers are localized vortex structures. They construct measurable time, distance, and simultaneity using internal vortex-clock cycles, equilibrated matter standards, and exchanged R-phase torsion pulses.

For every stable low-energy carrier A, the locally measured two-way torsion-pulse speed is universal:

$$c_A^{(2\mathrm{w})} = \frac{2\ell_A}{\Delta\tau_A} = c. \tag{122}$$

The local clock rate is

$$\frac{d\tau_A}{dT} = S_{(t)}^{\circlearrowleft}[A;T] = \sqrt{1 - \frac{u_A^2}{c^2}}. \tag{123}$$

[ORTHODOX CONTENT]. The invariant interval and the relative-motion mixing of temporal and spatial coordinates are the standard content of special relativity [45].

[CANONICAL SST IDENTIFICATION]. In SST, this operational Lorentz structure is attributed to the common torsion cone and to the dynamical response of vortex clocks and vortex rulers, rather than to a primitive geometric spacetime ontology.

[THEOREM TARGET]. The substrate dynamics must derive that Alice and Bob recover the same c, that their observable transformation depends only on relative velocity, and that all stable matter and radiation species share the same infrared cone. Failure of any of these conditions is a preferred-frame or non-monometric falsifier of SST.

Interpretation. The substrate ordering parameter is primitive at the ontological level; operational spacetime is relational and is reconstructed by vortex matter.

4.9 Axiom 8: Density Saturation and Horn-Envelope Structure

Statement. The canonical horn-envelope exhibits an effective saturated density $\rho_{\mathrm{horn}}^{\mathrm{eff}}$ determined by the mechanical normalization constraints of the medium. A resolved local tube density is not specified until a_{core} and a tube profile are supplied.

[SPECULATIVE] Closure hypothesis: the effective saturated envelope density is fixed by the closure relation of Eq. (79).

[DERIVED] Consequence: once that closure is adopted, matter corresponds to configurations whose neutral circulation envelope reaches a structural normalization limit.

4.10 Logical Structure of the Axioms

The axioms form a dependency hierarchy:

$$\text{Continuum} \to \text{Vortex Structures} \to \text{Quantization} \to \text{Torsion Cone and Operational Spacetime} \to \text{Delay Dyna} \tag{124}$$

4.11 Minimality Principle

Statement. The axioms are minimal in the sense that removing any one of them destroys at least one of the following:

- existence of stable structures
- emergence of discrete spectra
- compatibility with known physics

4.12 Philosophical Interpretation

The axiomatic structure implies:

- Particles are not fundamental objects but stable dynamical configurations
- Discreteness is emergent, not postulated
- Time is relational, not absolute

4.13 Consistency Condition

A valid SST realization must satisfy:

- Compatibility with orthodox physics in appropriate limits
- Internal logical closure
- Explicit traceability from axioms to predictions

4.14 Canonical Statement

Swirl-String Theory is defined as the minimal continuum theory in which stable vortex structures, governed by quantized circulation and delay dynamics, give rise to discrete physical phenomena consistent with known physics.

5 Consistency and Classification Framework

5.1 Classification Scheme

All statements are labeled as:

- [**ORTHODOX**] — established physics
- [**DERIVED**] — logically deduced within SST
- [**SPECULATIVE**] — novel or unverified

5.2 Extended Labeling Convention

We distinguish between *epistemic status* and *editorial role.*

Epistemic status. The primary status labels used in the present Canon are:

- [**ORTHODOX**] — established in standard physics or mathematics
- [**DERIVED**] — obtained internally from the present SST assumptions, definitions, or closures, *without re-using the target quantity as an input*
- [**CALIBRATED**] — fixed by, or algebraically equivalent to, a measured constant (α, m_e, $\hbar$, R_∞, ...) that enters through the primitive set; numerically exact but not an independent prediction
- [**CONDITIONAL**] — follows only if a stated, not-yet-proven premise holds (e.g. an open lemma or selection rule)
- [**SPECULATIVE**] — conjectural SST extension, interpretation, or unverified identification

A [**CRITICAL NOTE**] editorial overlay flags a circularity risk, dimensional issue, or reviewer vulnerability on any of the above.

Editorial role. Additional descriptors such as *Definition*, *Constraint*, *Interpretation*, *Benchmark*, *Roadmap*, and *Dimensional check* are not independent epistemic classes. They indicate only the role played by the statement.

Usage rule. When both are needed, the Canon uses the format

$$\texttt{[STATUS] Role: statement.} \tag{125}$$

For example, one may write **[ORTHODOX] Constraint**, **[DERIVED] Consequence**, or **[SPECULATIVE] Interpretation**.

5.3 Orthodox Constraints

The following must remain intact:

- Quantum mechanics predictions
- Atomic spectroscopy
- Relativistic consistency

5.4 Internal Consistency Rules

- No circular definitions
- All primitives explicitly defined
- Derived quantities must follow from axioms

5.5 Conflict Resolution Strategy

Priority order:

1. Logical consistency
2. Experimental agreement
3. Canonical simplicity

5.6 Epistemic Status Tracking

Each major result must explicitly state:

- Assumptions
- Derivation status
- Empirical testability

In particular, a definitional choice must not be presented as a theorem, and an interpretive bridge must not be presented as an orthodox result unless the underlying mathematical structure is itself standard.

5.7 Canon Integrity Principle

A valid SST canon must be internally consistent, externally compatible, and explicitly stratified by epistemic status.

6 Geometric and Topological Sector

6.1 Role of geometry and topology in the canon

This sector collects those parts of SST in which the identity of a state depends not only on local field amplitudes but on the global organization of the carrier. Geometry enters through lengths, radii, and closure constraints; topology enters through linking, crossing, chirality, and other invariant features that survive smooth deformations. At canon level, the main role of this sector is selective rather than exhaustive: it provides the organizing variables that feed the topological kernel Ξ_K and the geometric gate Π_K in Eq. (96).

The canon therefore treats geometry and topology as follows:

1. geometry sets admissible scales and shielding ratios,
2. topology labels persistent identity classes,
3. only those quantities that survive smooth deformation are allowed to enter the long-lived state classification.

This is also why layer or dressing indices are not treated as primary identity labels: they may modify a state, but they do not replace the topology-first classification of the carrier itself.

6.2 Euler topological preservation and closure discipline

[ORTHODOX]. In a smooth incompressible inviscid Euler flow, with no boundaries, forcing, viscosity, singular core collapse, or phase-slip defects, vortex lines are materially transported by the flow. Consequently, linking and knot type of already closed vortex carriers are preserved. A closed unknot cannot become a trefoil by ordinary ideal-Euler evolution:

$$0_1 \not\longrightarrow 3_1 \qquad \text{under smooth ideal Euler flow.} \tag{126}$$

[DERIVED SST CONSTRAINT]. Trefoil creation must therefore not be described as reconnection of an already closed ideal-Euler vortex tube. It is admissible only as a boundary-supported closure, phase-slip, nucleation, or other non-Euler event occurring before the final carrier has acquired a protected closed knot type. After closure, smooth evolution may deform the embedding but must preserve the knot class.

[CANON RULE]. Topology protects existing closed carriers; it does not by itself provide an ideal-Euler mechanism for creating them.

6.3 Framed self-linking of the swirl string and the spinorial selection of the lepton ladder

This subsection records the framed-tube self-linking of a closed torus-knot swirl string and the boundary class that selects its physical winding. It is the first genuinely non-circular topological result in the canon, in deliberate contrast to the calibrated α-identities of the Formal Foundations (which re-express α rather than predict it).

[DERIVED] Torus-surface self-linking. Let $K = T(p, q)$ be a closed swirl string on the standard torus and let U_{tor} be the torus surface-normal framing. The framed self-linking is

$$SL_{\text{tor}}(T(p, q)) \;=\; pq, \tag{127}$$

verified numerically to machine precision for $p \in \{2, 3\}$ and $q = 3, \dots, 11$ with *no* target injected. For the lepton family $T(2, q)$, $SL_{\text{tor}} = 2q$.

[DEFINITIONAL] Physical-framing winding. For any single-valued physical framing $U_{\rm phys}$ of K (the direction of the local core swirl-velocity field $\mathbf{u}_{\circlearrowleft}$), the relative winding

$$\chi \;:=\; SL(K, U_{\rm phys}) - SL(K, U_{\rm tor}) \;=\; \frac{1}{2\pi}\oint_K \mathrm{d}[\angle_{U_{\rm tor}}(U_{\rm phys})] \;\in\; \mathbb{Z} \tag{128}$$

is an integer relative winding number, so $SL_{\rm phys}(T(p,q)) = pq + \chi$.

[CRITICAL NOTE] Any test that fixes the twist by $Tw = \text{target} - Wr$ and then confirms the target is circular: it passes for *any* integer target and proves only the Calugareanu identity $SL = Wr + Tw$. Only the target-free framing (127) and the selection argument below are admissible.

[DERIVED NEGATIVE] Ordinary $U(1)$ closure gives $\chi = 0$. With positive longitudinal phase stiffness $A_\parallel > 0$, the winding energy

$$E(\chi) \;=\; E_0 + \frac{2\pi^2 A_\parallel}{L}\,\chi^2 \tag{129}$$

is minimized over ordinary (periodic) $U(1)$ windings $\chi \in \mathbb{Z}$ at $\chi_{\rm ground} = 0$, hence $SL_{\rm phys} = pq$ (even). Pure framed-curve geometry with positive stiffness therefore yields a *boson* and cannot produce the lepton $pq + 1$ ladder.

[CONDITIONAL DERIVED] Spinorial anti-periodicity gives $\chi = \pm 1$. If the core phase is a section of a spinor (rather than a $U(1)$) bundle, then under one circuit of K the spin frame rotates by 2π and the phase is anti-periodic, so the admissible windings are $\chi \in 2\mathbb{Z} + 1$. Minimizing (129) over this sector gives the degenerate ground winding

$$\chi_{\rm ground} = \pm 1 \quad\Longrightarrow\quad SL_{\rm phys}(T(p,q)) = pq \pm 1, \qquad SL_{\rm phys}(T(2,q)) = 2q \pm 1, \tag{130}$$

the two signs being the matter and antimatter sectors. The selection is a *superselection* effect: the boson winding $\chi = 0$ is always energetically lower (by $2\pi^2 A_\parallel/L > 0$), so it is the boundary class, not the energetics, that renders the carrier fermionic. The result is independent of $(A_\parallel, L)$ for all positive values.

Why the scalar substrate is bosonic and $\theta = \pi$ is the fermionic $\mathbb{Z}_2$ choice. The two boundary classes are not arbitrary continuous tunings. In an $S^2 \simeq \mathbb{C}P^1$ order-parameter theory they can be identified with the two values of a discrete topological/statistical sector.

[DERIVED NEGATIVE] Scalar $U(1)$ substrate. A single-component order parameter $\psi = |\psi| e^{i\varphi}$ is single-valued only if

$$\oint \mathrm{d}\varphi \in 2\pi\mathbb{Z}, \tag{131}$$

so its scalar phase is periodic:

$$e^{i\oint \mathrm{d}\varphi} = +1 \tag{132}$$

on every closed loop. This condition allows integer windings, but it does not force the longitudinal framing class to be odd. With the positive stiffness energy (129),

$$E_\chi = E_0 + \frac{2\pi^2 A_\parallel}{L}\,\chi^2, \qquad A_\parallel > 0, \tag{133}$$

minimization over the ordinary scalar class $\chi \in \mathbb{Z}$ selects

$$\chi_{\rm ground} = 0. \tag{134}$$

Hence a purely scalar SST substrate has no intrinsic mechanism for the odd core-phase sector required by fermions. In this operational sense the scalar substrate is bosonic: it admits the trivial periodic carrier and does not generate the $pq \pm 1$ lepton ladder.

[DERIVED / ALLOWED] $S^2 \simeq \mathbb{C}P^1$ **admits a** $\mathbb{Z}_2$ **statistical sector.** The matter target is not a scalar $U(1)$ phase but the director/Hopfion target

$$M_{\text{SST}} = S^2 \simeq \mathbb{C}P^1. \tag{135}$$

The relevant homotopy data are

$$\pi_3(S^2) = \mathbb{Z}, \qquad \pi_4(S^2) = \mathbb{Z}_2. \tag{136}$$

The first class gives the static Hopf charge,

$$Q_H = SL_{\text{phys}}, \tag{137}$$

while the second allows a $\mathbb{Z}_2$ topological/statistical phase for spacetime loops of soliton configurations [41]. More carefully, the fermionic sign is not simply a local scalar factor $e^{i\theta Q_H}$ in the static Hopf charge. It is the sign of the relevant $\mathbb{Z}_2$ spacetime class,

$$\exp(iS_\theta) = (-1)^\nu, \qquad \nu \in \mathbb{Z}_2, \tag{138}$$

whose representative may be read, for the relevant 2π rotation or exchange process, as the parity class of the Hopf sector. Thus the two possible statistical sectors are

$$\theta = 0: \quad \exp(iS_\theta) = +1 \quad \Rightarrow \quad \text{bosonic / periodic sector}, \tag{139}$$

$$\theta = \pi: \quad \exp(iS_\theta) = (-1)^\nu \quad \Rightarrow \quad \text{fermionic / spinorial sector for odd Hopf parity}. \tag{140}$$

Identifying the spinorial sector with the longitudinal core-phase boundary class gives

$$\theta = \pi \quad \Longleftrightarrow \quad \chi \in 2\mathbb{Z} + 1. \tag{141}$$

Then positive stiffness no longer minimizes over all integers but over the odd class:

$$\chi \in 2\mathbb{Z} + 1 \quad \Rightarrow \quad \chi_{\text{ground}} = \pm 1. \tag{142}$$

Since

$$Q_H = SL_{\text{phys}} = SL_{\text{tor}} + \chi = pq + \chi, \tag{143}$$

and $pq = 2q$ is even for the lepton family $T(2, q)$, the $\theta = \pi$ sector gives

$$SL_{\text{phys}}(T(2, q)) = 2q \pm 1, \tag{144}$$

i.e. the odd Hopf/self-linking carrier required for fermionic leptons.

[POSIT, pinned] $\theta = \pi$ **is selected by observation, not derived.** The framed-curve geometry and the quadratic stiffness energy do not prefer $\theta = \pi$ over $\theta = 0$. The value $\theta = \pi$ is therefore a discrete superselection input pinned by the empirical existence of fermions. It is naturally identified with the same $\mathbb{Z}_2$ datum appearing in half-quantum vortices of multi-component condensates [42, 43] and in the half/full circulation assignment of the magnetic-moment sector (Route B). With that identification, fermion statistics, the $pq \pm 1$ ladder, and the Route B origin of $g = 2$ are no longer independent posits but three faces of one $\mathbb{Z}_2$ sector.

[CRITICAL NOTE] The statement "$\pi_4(S^2) = \mathbb{Z}_2$ guarantees a fermionic Hopfion" is too strong. The homotopy group shows that a $\mathbb{Z}_2$ spacetime class exists, but hopfion statistics are set by the presence and value of the corresponding topological/statistical term. In modern language this classification can involve bordism/TQFT data rather than only bare homotopy groups [44]. What is solid here is the reduction:

$$\theta = 0 \Rightarrow \text{periodic/bosonic substrate}, \qquad \theta = \pi \Rightarrow \text{spinorial/fermionic substrate}. \tag{145}$$

The remaining open problem is to derive why the SST substrate realizes the $\theta = \pi$ sector rather than the $\theta = 0$ sector.

[OPEN] Derivation of the spinorial boundary condition. The spinorial anti-periodicity of the core phase is not yet derived from the swirl-field dynamics; it is the single remaining premise of the $pq + 1$ ladder.

[CONDITIONAL] Unification with the magnetic moment. The spinor premise above is the *same* posit as the spin-$\frac{1}{2}$ assignment of the magnetic-moment sector (Route B): one spinorial lift of the core phase yields both $g = 2$ and $\chi = \pm 1 \Rightarrow pq \pm 1$. If established, fermion statistics, the gyromagnetic ratio, and the lepton self-linking ladder collapse into a single mechanism.

[DERIVED] Falsifier. An independently computed $T(2, q)$ Hopfion self-linking with $Q_H \neq 2q \pm 1$ (e.g. $T(2, 5) \neq 11$) falsifies the conditional chain irrespective of the variational argument.

6.4 Spectral Structure of Locked Closed Carriers (Secondary Result)

Scope and status. This subsection records a *secondary consequence* of the kinematics of rationally locked, closed carriers. It does not introduce new primitives. All results are conditional on the existence of a well-defined phase field and a regular mode spectrum. Status: **derived under spectral assumptions; experimentally unverified**.

Kinematic reduction. Consider a closed carrier with a single central driving phase $\phi(t)$ and N_s sector variables $\theta_i(t)$ $(i = 1, \ldots, N_s)$ subject to a rational locking relation

$$\theta_i = \sigma_i \kappa_i \phi + \delta_i, \qquad \sigma_i \in \{+1, -1\}, \quad \kappa_i \in \mathbb{Q}. \tag{146}$$

In the rigid-lock limit, the dynamics reduce to a single collective coordinate $\phi(t)$ with effective Lagrangian

$$\mathcal{L}_{\text{eff}} = \frac{I_{\text{eff}}}{2} \dot{\phi}^2 - V(\phi), \tag{147}$$

where I_{eff} is an effective inertia and $V(\phi)$ is a periodic locking potential.

Distributed phase field. On a closed carrier of length L, allow small phase fluctuations

$$\phi(t) \longrightarrow \phi_0 + \eta(s, t), \qquad s \in [0, L], \tag{148}$$

with periodic boundary condition

$$\eta(s + L, t) = \eta(s, t). \tag{149}$$

A minimal quadratic field model consistent with Eq. (147) is

$$\mathcal{L}_{\text{field}} = \int_0^L \left[\frac{\mu_{\text{eff}}}{2} (\partial_t \eta)^2 - \frac{T_{\text{eff}}}{2} (\partial_s \eta)^2 - \frac{\mu_{\text{eff}} \omega_0^2}{2} \eta^2 \right] ds, \tag{150}$$

where μ_{eff} is an effective phase inertia density, T_{eff} an effective stiffness, and ω_0 the small-oscillation frequency set by $V(\phi)$.

Dimensional consistency:

$$[\mu_{\text{eff}}] = \text{kg m}, \qquad [T_{\text{eff}}] = \text{N}, \qquad [\omega_0] = \text{s}^{-1}.$$

Mode spectrum. Expanding in normal modes

$$\eta(s,t) = \sum_{n\in\mathbb{Z}} a_n(t)\, e^{ik_n s}, \qquad k_n = \frac{2\pi n}{L}, \tag{151}$$

yields the dispersion relation

$$\boxed{\omega_n^2 = \omega_0^2 + c_\phi^2 k_n^2, \qquad c_\phi^2 = \frac{T_{\text{eff}}}{\mu_{\text{eff}}}.} \tag{152}$$

For large n,

$$\omega_n \sim \frac{2\pi c_\phi}{L}\, n. \tag{153}$$

Cubic spectral sums and $\zeta(3)$. Consider a cubic-weighted spectral contribution of the form

$$\Delta\mathcal{E}^{(3)} = A_3 \sum_{n=1}^{\infty} \frac{1}{\omega_n^3}, \qquad [A_3] = \mathrm{J\,s^{-3}}. \tag{154}$$

Using the asymptotic scaling (153),

$$\Delta\mathcal{E}^{(3)} \;\approx\; A_3 \left(\frac{L}{2\pi c_\phi}\right)^3 \sum_{n=1}^{\infty} \frac{1}{n^3}. \tag{155}$$

Hence

$$\boxed{\Delta\mathcal{E}^{(3)} \;\approx\; A_3 \left(\frac{L}{2\pi c_\phi}\right)^3 \zeta(3).} \tag{156}$$

Interpretation. Equation (156) shows that $\zeta(3)$ arises from the *high-mode spectral tail* of a closed carrier, not from the geometric threefold structure alone. The role of the locking relation (146) is indirect: it fixes I_{eff}, ω_0, and admissible boundary conditions, thereby determining the spectral scale (L, c_ϕ).

Universality. Different geometric realizations (e.g., distinct multi-sector carriers) that reduce to the same class of effective phase dynamics and share the asymptotic linear spectrum (153) will produce the same $\zeta(3)$ factor in Eq. (156), up to system-dependent prefactors.

$$\boxed{\zeta(3) \text{ is a spectral invariant of the mode tower, not a direct invariant of the carrier geometry.}} \tag{157}$$

Limitations and falsifiers. The result fails if:

1. the mode spectrum is not asymptotically linear in n,
2. strong damping or nonlocal coupling invalidates the expansion (150),
3. the system does not admit a well-defined periodic phase field on a closed domain.

Minimal experimental test. Measure mode frequencies ω_n for $n = 1, 2, 3, \ldots$ and verify

$$\frac{\omega_n}{n} \to \text{constant} \quad \text{as } n \to \infty. \tag{158}$$

Deviation from this scaling excludes Eq. (156).

Status. **Secondary derived result.** Valid under spectral assumptions; not a defining postulate. The cubic weighting $\sum_n 1/\omega_n^3$ with $\omega_n \sim n$ yields $\zeta(3)$ for *any* asymptotically linear spectrum; this is a generic calculus property [CALIBRATED], not an SST-specific fluid-dynamical or topological invariant.

Closed-mode versus open-resonance guard. Equations (150)–(152) describe a closed, conservative quadratic phase model with periodic boundary conditions. They do *not* by themselves derive damped resonances, outgoing-flux conditions, quasinormal modes, or a non-self-adjoint evolution generator. The distinct research programme for open or coarse-grained vortex response is therefore kept in the companion research track as *quasinormal swirl spectroscopy*; see Section P.6. That programme imports operator and resonance methodology only. It does not import Kerr geometry, event horizons, or Einstein dynamics into SST.

6.5 Bounded-Domain Curl-Spectrum Benchmarks

[ORTHODOX / SPECTRAL BENCHMARK]. Let $\Omega \subset \mathbb{R}^3$ be a smooth bounded domain and let $\mathbf{V}$ be divergence-free and tangent to $\partial\Omega$. In the bounded-domain curl eigenproblem

$$\nabla \times \mathbf{V} = \kappa_{\mathrm{curl}}\mathbf{V}, \qquad \nabla \cdot \mathbf{V} = 0, \qquad \mathbf{V} \cdot \mathbf{n}\big|_{\partial\Omega} = 0, \tag{159}$$

the smallest positive eigenvalue is a property of the complete domain and its boundary conditions, not of the centreline alone [56].

For a ball B_b^3 of radius b,

$$\kappa_1(B_b^3) = \frac{x_1}{b}, \qquad \tan x_1 = x_1, \qquad x_1 = 4.493409458\ldots, \tag{160}$$

with multiplicity three. For a concentric spherical shell $B^3(a,b) = \{a < r < b\}$, the first positive eigenvalue is the smallest positive root of

$$J_{3/2}(a\kappa)Y_{3/2}(b\kappa) - J_{3/2}(b\kappa)Y_{3/2}(a\kappa) = 0. \tag{161}$$

In the normalization

$$\mathscr{E}_V = \int_\Omega |\mathbf{V}|^2 \, d^3x, \qquad \mathscr{H}_V = \left\langle \mathbf{V}, \mathrm{BS}_{K(\Omega)} \mathbf{V} \right\rangle_{L^2}, \tag{162}$$

a curl eigenfield satisfies

$$\mathscr{E}_V = \kappa_{\mathrm{curl}}\mathscr{H}_V, \qquad \frac{\mathscr{H}_V}{\mathscr{E}_V} = \frac{1}{\kappa_{\mathrm{curl}}}. \tag{163}$$

[CANON / SCOPE GUARD]. Equations (160) and (161) are exact benchmarks only for the stated filled spherical domains. The constant $4.493409458\ldots$ is not a universal SST particle constant, the distinguished central orbit of the eigenfield is not an independently resolved vortex core, and neither formula determines a_{core} or r_{horn} for a knotted tube.

6.6 Symmetry-Sensitive Dark-Knot Classification

A compact classification retained from earlier SST topology layers distinguishes visible, quasi-dark, and dark sectors by chirality, helicity, and low-order instability content.

Let $\chi(K)$ denote a chirality indicator, let

$$\mathcal{H}[K] = \int \mathbf{v} \cdot (\nabla \times \mathbf{v}) \, dV \tag{164}$$

be the helicity functional, and let $N_u^{(1)}(K)$ count low-order unstable modes in a chosen perturbative stability analysis. Then the retained taxonomy is

$$\begin{aligned} \text{dark:} &\quad \chi(K) = 0, \quad \mathcal{H}[K] \approx 0, \quad N_u^{(1)}(K) = 0, && (165)\\ \text{quasi-dark:} &\quad |\chi(K)| \ll 1, \quad \mathcal{H}[K] \approx 0, \quad 0 < N_u^{(1)}(K) \le 2, && (166)\\ \text{visible:} &\quad \chi(K) \neq 0 \quad \text{or} \quad \mathcal{H}[K] \neq 0. && (167) \end{aligned}$$

[ORTHODOX]: helicity is a standard invariant of ideal vortex dynamics [36, 6].

[SPECULATIVE]: the interpretation of Eq. (167) as a matter-sector taxonomy is an SST model choice.

Heuristic balance diagnostic from the SST helicity archive. As a supporting diagnostic (not a stand-alone classifier), define

$$\Delta_H(K) = \left| \widehat{\mathcal{H}}_b(K) + \frac{1}{2} \right|, \tag{168}$$

where $\widehat{\mathcal{H}}_b(K)$ is the archived normalized helicity-like base indicator for class K. Near-balanced values with $\Delta_H(K) \ll 1$ are treated only as candidate evidence for dark or quasi-dark sectors.

[CRITICAL NOTE]: in canon usage, the diagnostic above is subordinate to Eq. (167); no single scalar is sufficient for dark-sector membership without the symmetry and instability filters.

[CANON WORKFLOW NOTE]: if SST-labelled and parallel legacy-labelled helicity exports are numerically identical, only the SST-labelled archive is required for canon workflows.

Symmetry-first refinement (v0.8.x). In the taxonomy program, candidate sectors are labeled symmetry-first, then filtered by helicity balance and low-order instability content. Amphichiral knot types furnish the preferred *candidate* pool for the dark bin in symmetry-first tables; assigning "dark" still requires the conjunction of near-zero chirality indicators, helicity balance in the sense of Eq. (167), and vanishing low-order instability count. The implication "amphichiral $\Rightarrow$ dark" is a classifier proposal tested against observables and archive diagnostics, not a closed theorem.

Ambiguity tie-break. If rule conjunctions overlap, precedence is

$$\text{symmetry gate} \;\to\; \text{helicity balance gate} \;\to\; \text{stability gate}.$$

7 Swirl–Electromagnetic Bridge

7.1 Minimal transverse field sector

A conservative bridge from the fluid sector to a radiation sector is obtained by introducing a divergence-free transverse field $\mathbf{A}_{\text{eff}}$ with

$$\nabla \cdot \mathbf{A}_{\text{eff}} = 0, \tag{169}$$

and effective Lagrangian density

$$\mathcal{L}_{\text{rad}} = \frac{\rho_f}{2} \|\partial_t \mathbf{A}_{\text{eff}}\|^2 - \frac{K_\gamma}{2} \|\nabla \times \mathbf{A}_{\text{eff}}\|^2. \tag{170}$$

Varying Eq. (170) and using Eq. (169) gives

$$\frac{1}{c_\gamma^2}\partial_t^2 \mathbf{A}_{\text{eff}} - \nabla^2 \mathbf{A}_{\text{eff}} = 0, \qquad c_\gamma^2 = \frac{K_\gamma}{\rho_f}. \tag{171}$$

[CANONICAL FOUNDATIONAL IDENTIFICATION]. The photon-sector R-phase excitation is identified with the transverse torsion pulse used by vortex matter to define the local invariant speed. Therefore the low-energy, locally measured propagation speed satisfies

$$c_\gamma = c_T = c. \tag{172}$$

Within the quadratic bridge model this fixes the target stiffness

$$K_\gamma = \rho_f c^2 = 6.29128625115772348 \times 10^{10}\ \text{Pa}. \tag{173}$$

Equation (173) is dimensionally a shear/torsion stiffness target. Substituting the measured value of c calibrates the bridge; it is not yet a microscopic derivation of K_γ.

[NO-ACOUSTIC-MODE GUARD]. The speed c_γ is not an ordinary sound speed. In the exact incompressible SST substrate, density compression is excluded and pressure enforces a constraint on each foliation slice. The luminal branch belongs to the independent transverse torsion/director sector represented by $\mathbf{A}_{\text{eff}}$ [48].

Define

$$\mathbf{E}_{\text{eff}} = -\partial_t \mathbf{A}_{\text{eff}}, \qquad \mathbf{B}_{\text{eff}} = \nabla \times \mathbf{A}_{\text{eff}}. \tag{174}$$

Then one immediately recovers the source-free identities

$$\nabla \cdot \mathbf{B}_{\text{eff}} = 0, \tag{175}$$

$$\nabla \times \mathbf{E}_{\text{eff}} = -\partial_t \mathbf{B}_{\text{eff}}. \tag{176}$$

[ORTHODOX FORM]: Eq. (171) and Eq. (176) are the standard transverse-wave and source-free Faraday identities in vector-potential form [23].

[SPECULATIVE IDENTIFICATION]: SST identifies $\mathbf{A}_{\text{eff}}$ with a coarse-grained torsional or director displacement field of the medium.

Relational link–field interpretation. [CANON / RESEARCH-TRACK INTERFACE]. The fundamental torsional degree of freedom is assigned to the dynamical connection between neighbouring Æther elements, rather than to a necessary net local rotation of either endpoint. At a microscopic level an oriented link $e = (i \to j)$ carries a relative phase or twist variable ϑ_e, with reversal of orientation giving $\vartheta_{\bar{e}} = -\vartheta_e$ modulo the phase period. The continuum field $\mathbf{A}_{\text{eff}}$ is retained as a normalized infrared representative of this link connection; it is not identified with the material Euler vorticity $\nabla \times \mathbf{v}$.

The quadratic action in Eq. (170) is therefore retained as the long-wavelength bridge, while its microscopic completion is assigned to the compact relational link action in Section R.63. Ordinary R-phase propagation changes link holonomy and transports torsional energy, but it does not by itself switch an Æther element between the material sectors $\boldsymbol{\zeta} = 0$ and $\boldsymbol{\zeta} \neq 0$.

7.2 Photon / unknot sector

The minimal retained radiation picture is that an R-phase excitation is an unknotted, finite-duration torsional packet propagating in the field $\mathbf{A}_{\text{eff}}$. At the Rosetta level this corresponds to a source-free transverse wave packet with dispersion inherited from Eq. (171).

[**SPECULATIVE**]: handedness of the local torsional packet is taken to encode polarization, while additional azimuthal phase winding encodes orbital angular momentum.

[**CANON / SEMANTIC RULE**]: a photon-sector R-phase torsion packet is not a loose end of a vortex filament and does not yet carry a protected closed matter-knot invariant. It is a phase/torsion support on a swirl-characteristic. A spacetime geodesic is only the orthodox comparison language for the coarse-grained propagation path.

Helmholtz-sector compatibility. Define material Euler vorticity by

$$\boldsymbol{\zeta} := \nabla \times \mathbf{v}. \tag{177}$$

The torsion/director curl $\nabla \times \mathbf{A}_{\text{eff}}$ is not identified with the creation of new material Euler vorticity. On a smooth ideal branch, an R-phase pulse may transport energy, phase, and polarization while preserving the circulation sector of every material loop $C(T)$:

$$\Delta\Gamma_C = \Delta \oint_{C(T)} \mathbf{v} \cdot d\boldsymbol{\ell} = 0. \tag{178}$$

Thus propagation through an initially irrotational material region must not leave behind a permanent transition from $\boldsymbol{\zeta} = 0$ to $\boldsymbol{\zeta} \neq 0$. Such a transition requires an explicitly non-ideal source, boundary, singularity, or Kairos event; it is not ordinary photon propagation.

7.3 R-to-T closure threshold and conservation bookkeeping

[**SPECULATIVE SST BRIDGE**]. If electron-trefoil creation is retained, the conservative bookkeeping form is

$$\gamma_{\text{R}} + \mathcal{B}_{\text{atom}} \xrightarrow{\mathcal{K}_{\text{Kairos}}} e^-_{3_1} + \mathcal{R}, \tag{179}$$

where γ_{R} is an extended R-phase torsion packet, $\mathcal{B}_{\text{atom}}$ is a local boundary or trapping condition, and $\mathcal{R}$ carries recoil, momentum balance, and any required topological or charge compensation. Equation (179) is not a pure Euler reconnection event.

The Compton anchoring gives the closure energy scale

$$\omega_c = \frac{v^*_{\circlearrowleft}}{r_c}, \qquad \hbar\omega_c = m_e c^2. \tag{180}$$

This scale may be used as a seed/closure threshold only in the presence of a boundary reservoir or pre-existing attachment support. Free creation of a matter carrier from radiation must respect charge, momentum, and topological compensation. A lone photon-to-lone-electron channel is therefore excluded in the canon-safe reading. The minimal free pair bookkeeping is instead

$$\gamma + \gamma \longrightarrow 3_1 + \overline{3_1}, \qquad E_{\text{pair}} = 2m_e c^2, \tag{181}$$

or a field-assisted version

$$\gamma + \mathcal{F} \longrightarrow 3_1 + \overline{3_1} + \mathcal{F}'. \tag{182}$$

7.4 Swirl pressure law

For steady axisymmetric swirl, Euler radial balance gives

$$\frac{1}{\rho_f}\frac{dp_{\text{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}. \tag{183}$$

For a locally rigid rotation profile $v_\theta(r) = \Omega r$, Eq. (183) integrates to

$$p_{\text{swirl}}(r) = p_0 + \frac{1}{2}\rho_f \Omega^2 r^2, \tag{184}$$

while for an exterior irrotational profile $v_\theta(r) = \Gamma/(2\pi r)$ one finds

$$p_{\text{swirl}}(r) = p_\infty - \frac{\rho_f \Gamma^2}{8\pi^2 r^2}. \tag{185}$$

In a flat rotation curve with asymptotically constant azimuthal speed $v_\theta(r) \to v_0$ as r grows in a window where the radial balance (183) remains valid, integration yields the logarithmic tail

$$p_{\text{swirl}}(r) = p_0 + \rho_f v_0^2 \ln\left(\frac{r}{r_0}\right), \tag{186}$$

for suitable reference radius r_0 and offset p_0 within that window.

[ORTHODOX] Assumption block: Eq. (183) is taken in the usual incompressible, inviscid, stationary, axisymmetric regime with azimuthal-dominant flow, so that radial drift, viscosity, and stratification corrections are subleading in the domain of validity.

[ORTHODOX] Dimensional check: $[\rho_f{}^{-1}\, dp/dr] = \mathrm{m\,s^{-2}} = [v_\theta^2/r]$.

[ORTHODOX]: Eq. (183) is the ordinary radial Euler balance for steady swirl [6, 30].

[DERIVED SST ROLE]: this pressure law is the canonical bridge between localized circulation and large-scale force analogies. Interpreting p_{swirl} as a dark-sector classifier or support layer is an SST model step, logically separate from the orthodox fluid derivation.

[DERIVED] Observables / falsifiers: (i) consistency of a pressure profile reconstructed from archived $v_\theta(r)$ with the radial gradient implied by (183); (ii) null test: systematic sign or scale mismatch of the reconstructed dp_{swirl}/dr relative to v_θ^2/r across validated radial windows excludes the applicability of the balance in those windows.

7.5 Lorentz-Type Force Density as Swirl Stress

Status labels. The vector identity used in this section is **[ORTHODOX]**. The definition of the swirl-force density is **[DERIVED]** as a hydrodynamic force-density structure. The numerical SST scales are **[CALIBRATED]** with respect to the canonical constants ρ_f, r_c, and $v_{\circlearrowleft}^*$. No microscopic electromagnetic element-force law is canonized here.

Hydrodynamic identity. For incompressible inviscid flow,

$$\nabla \cdot \mathbf{v} = 0, \qquad \boldsymbol{\omega} = \nabla \times \mathbf{v}.$$

The convective acceleration satisfies the exact identity

$$(\mathbf{v}\cdot\nabla)\mathbf{v} = \nabla\left(\frac{1}{2}|\mathbf{v}|^2\right) - \mathbf{v}\times\boldsymbol{\omega}.$$

Equivalently,

$$\boxed{\mathbf{v}\times\boldsymbol{\omega} = \nabla\left(\frac{1}{2}|\mathbf{v}|^2\right) - (\mathbf{v}\cdot\nabla)\mathbf{v}.}$$

This provides the hydrodynamic structure behind a Lorentz-type transverse force-density term.

Swirl-force density. SST defines the local swirl-force density

$$\boxed{\mathbf{f}_{\circlearrowleft} = \rho_f \mathbf{v} \times \boldsymbol{\omega}.}$$

Its dimensional consistency is

$$[\rho_f \mathbf{v} \times \boldsymbol{\omega}] = \mathrm{kg\,m^{-3}\,m\,s^{-1}\,s^{-1}} = \mathrm{N\,m^{-3}},$$

as required for a force density.

Bernoulli relation and sign convention. The scalar quantity

$$p_{\circlearrowleft} = \frac{1}{2}\rho_f |\mathbf{v}|^2$$

is the dynamic swirl-pressure scale. It should not be confused with the static pressure, which decreases in high-swirl regions in the usual Bernoulli sense. For stationary Euler flow with constant ρ_f and no external body force,

$$\rho_f (\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla p_{\mathrm{stat}}.$$

Using the identity above gives

$$\boxed{\rho_f \mathbf{v} \times \boldsymbol{\omega} = \nabla\left(p_{\mathrm{stat}} + \frac{1}{2}\rho_f |\mathbf{v}|^2\right).}$$

Thus the swirl-force density is canonically tied to the gradient of total Euler pressure, not to an independently postulated electromagnetic force law.

Closed-loop observables. SST assigns canonical status to integrated and topological observables rather than to a unique microscopic segment-to-segment force kernel. The canonical loop observables include circulation,

$$\Gamma = \oint_C \mathbf{v} \cdot d\boldsymbol{\ell},$$

helicity,

$$H = \int_V \mathbf{v} \cdot \boldsymbol{\omega}\, d^3x,$$

and the Gauss linking number,

$$Lk(C_1, C_2) = \frac{1}{4\pi} \oint_{C_1} \oint_{C_2} \frac{(d\mathbf{x}_1 \times d\mathbf{x}_2) \cdot (\mathbf{x}_1 - \mathbf{x}_2)}{|\mathbf{x}_1 - \mathbf{x}_2|^3}.$$

Distinct local force kernels may yield identical closed-loop observables. Therefore SST canonizes circulation, helicity, linking number, pressure differences, and net forces, but not a unique microscopic electromagnetic element-force law.

Canonical scales. Using the canonical SST constants

$$v_{\circlearrowleft}^{*} = 1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15}\,\mathrm{m}, \qquad \rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}},$$

the natural circulation scale is

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}^{*} = 9.68361920 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}.$$

The associated dynamic swirl-pressure scale is

$$p_{\circlearrowleft,0} = \frac{1}{2}\rho_f v_{\circlearrowleft}^{*2} = 4.1877439 \times 10^{5}\,\mathrm{Pa}.$$

This value is the canonical SST stress scale for Lorentz-type force-density analogies and coincides with the calibrated source-pressure scale used in the swirl-pressure sector.

Canonical conclusion. The canonized result is not a replacement of Maxwell–Lorentz electrodynamics. The canonized result is the hydrodynamic statement that a Lorentz-type transverse force-density structure has the exact SST form

$$\boxed{\mathbf{f}_{\circlearrowleft} = \rho_f \mathbf{v} \times \boldsymbol{\omega} = \rho_f \nabla\left(\frac{1}{2}|\mathbf{v}|^2\right) - \rho_f (\mathbf{v} \cdot \nabla)\mathbf{v}.}$$

Accordingly, magnetic-type force densities in SST are interpreted as effective projections of pressure, vorticity, and closed-loop topology in the substrate.

7.6 Minimal effective Lagrangian bridge

A compact field-theoretic summary retaining the clock, radiation, and matter sectors is

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_T + \mathcal{L}_{\text{rad}} + \sum_K \bar{\Psi}_K (i\gamma^\mu D_\mu - m_K^{(\text{sol})})\Psi_K + \cdots , \tag{187}$$

where $\mathcal{L}_T$ is the clock/foliation sector introduced in Section 11, $\mathcal{L}_{\text{rad}}$ is given by Eq. (170), and the ellipsis denotes higher-order self-interaction, topological, or medium-response operators.

[**SPECULATIVE BRIDGE**]: Eq. (187) is a bookkeeping layer, not yet a completed first-principles derivation.

8 Delay and Mode Selection Theory

8.1 Circulation Delay as a Physical Primitive

We introduce the circulation delay as a constitutive element of the theory. For a closed vortex filament of effective length L and characteristic propagation speed $v_{\circlearrowleft}^*$, the circulation time is defined as:

$$\tau_{\text{circ}}(K) = \frac{L_K}{v_{\circlearrowleft}^*} \tag{188}$$

In the special case of a minimal closed loop associated with the neutral circulation scale, this reduces to:

$$\tau_{\text{circ,c}} = \frac{2\pi r_c}{v_{\circlearrowleft}^*} = \frac{2\pi}{\omega_c} \tag{189}$$

This establishes a direct identification between the circulation delay and the inverse Compton frequency.

[**DERIVED**]: The intrinsic delay of the neutral circulation loop is identical to the Compton period.

8.2 Delayed Self-Interaction Dynamics

We model the phase $\phi(t)$ of a circulating excitation as a delayed self-interacting oscillator:

$$\dot{\phi}(t) = \omega_0 + \kappa \sin\left(\phi(t - \tau_{\text{circ}}) - \phi(t)\right) \tag{190}$$

This equation represents a minimal effective description of a circulating field with finite propagation time.

[**ORTHODOX**]: Delay-differential equations of this type are well-established in feedback systems [1].

8.3 Emergence of Discrete Stable Modes

We seek steady-state solutions of the form:

$$\phi(t) = \Omega t \tag{191}$$

Substituting into Eq. 190 yields $\phi(t - \tau_{\mathrm{circ}}) - \phi(t) = -\Omega\tau_{\mathrm{circ}}$, hence:

$$\Omega = \omega_0 - \kappa \sin(\Omega\tau_{\mathrm{circ}}) \tag{192}$$

This transcendental equation admits a discrete set of stable solutions Ω_n determined by the delay parameter τ_{circ}. The sign convention is not physically unique—one may absorb it into κ or reverse the delay argument—but the displayed form is the direct consequence of Eq. (190) as written.

[**CONDITIONAL DERIVED**]: Given the effective DDE ansatz of Eq. (190), the delayed feedback equation admits discrete locked branches. The claim that the SST medium obeys this specific DDE remains a bridge assumption, not a first-principles Euler theorem.

8.4 Delay-Induced Quantization Mechanism

Unlike conventional quantum mechanics, where discreteness is postulated, here it emerges dynamically:

- No boundary condition required
- No operator quantization assumed
- Discreteness arises from stability selection

[**SPECULATIVE**]: Particle spectra may be interpreted as delay-selected stable circulation modes.

8.5 Extension to Nonlinear Schrödinger Dynamics

To connect with field-theoretic descriptions, we extend the delay mechanism to a nonlinear Schrödinger framework:

$$i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + F_{\mathrm{delay}}[\psi(t - \tau_{\mathrm{circ}})] \tag{193}$$

where the delayed functional introduces temporal nonlocality into the field evolution.

[**SPECULATIVE**]: This provides a bridge between classical delay systems and quantum wave dynamics.

8.6 Physical Interpretation

The delay τ_{circ} acts as:

- a memory kernel
- a nonlocal temporal coupling
- a selection mechanism for stable modes

Thus:

$$\textit{Delay} \Rightarrow \textit{Stability} \Rightarrow \textit{Discreteness}$$

This mechanism forms one of the central pillars of the SST canon.

9 Atomic Bridge Model

9.1 Introduction and epistemic framing

This section records the minimal SST atomic bridge rather than a full derivation of atomic structure. Its canon role is deliberately narrow: recover the orthodox Coulomb limit when the SST-specific sector is switched off, define a branch-resolved circulation coupling fixed by canonical medium scales, and make explicit which many-electron and shell-structure claims remain open. It is therefore a benchmark layer, not yet a complete atomic theory.

9.2 Baseline Hamiltonian

We adopt the standard Coulomb Hamiltonian as the orthodox baseline:

$$H_0 = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \tag{194}$$

[ORTHODOX]: This reproduces the hydrogenic spectrum.

9.3 Legacy Soft-Core Regularization

Early canon layers introduced a finite-core regularization of the hydrogenic potential. In the present document this is retained only as a controlled bridge ansatz, not as a replacement for Eq. (194).

Define the effective coupling scale

$$\Lambda_{\mathrm{SST}} = 4\pi\,\rho_{\mathrm{core}}\,{v_{\circlearrowleft}^{*}}^{2}\,{r_c}^{4} \tag{195}$$

with dimensions

$$[\Lambda_{\mathrm{SST}}] = \mathrm{J\,m} = \mathrm{N\,m}^2. \tag{196}$$

A finite-core bridge potential is then

$$V_{\mathrm{soft}}(r) = -\frac{\Lambda_{\mathrm{SST}}}{\sqrt{r^2 + r_c^2}} \tag{197}$$

satisfying the large-radius limit

$$V_{\mathrm{soft}}(r) \xrightarrow{r \gg r_c} -\frac{\Lambda_{\mathrm{SST}}}{r}. \tag{198}$$

[ORTHODOX] Effective-model input: finite-core regularizations of singular central potentials are standard tools.
[DERIVED] Canonical consequence: the SST scale in Eq. (195) is fixed by canonical constants rather than introduced phenomenologically.
[ORTHODOX] Constraint: this sector is admissible only if the large-radius limit reproduces the standard Coulomb problem to spectroscopic accuracy.

In particular, Eq. (197) is retained as a bridge ansatz only. It is not promoted here to the primary atomic potential.

9.4 SST Circulation Correction

We introduce a circulation-dependent perturbation:

$$\delta H^{(\sigma)} = -2\sigma E_R \eta_0 \gamma + \frac{E_R \eta_0 \sigma}{2} \left\{ \frac{\hat{L}_z}{\hbar}, \gamma \right\} \tag{199}$$

[BRIDGE ANSATZ]: This is an orthodox-operator perturbation template whose coefficient depends on the SST circulation amplitude $\gamma(r,t)$. Because $\hat{L}_z$ is imported from the standard quantum Hamiltonian, Eq. (199) is not a first-principles derivation of quantum angular momentum from the substrate.

[CRITICAL NOTE]: The fractional shift scales as $\delta E_n/E_n \sim \eta_0\gamma = (\alpha/2)\gamma$. With $\gamma = \mathcal{O}(1)$ this is $\mathcal{O}(\alpha)$, far larger than the orthodox fine-structure splitting $\mathcal{O}(\alpha^2)$ (fractional). Compatibility with the atomic spectroscopy bounds (Sec. 7) therefore requires γ to be α-suppressed; this suppression is not yet derived.

9.5 Circulation source chain

The atomic perturbation is not introduced as an arbitrary extra scalar, but is intended to descend from a circulation-bearing source chain,

$$\xi \to \omega_z \to u_\theta \to \Gamma_{\mathrm{loc}} \to \gamma, \tag{200}$$

where an axial displacement field ξ generates local vorticity, the vorticity sources azimuthal swirl, and the loop circulation is normalized by the canonical quantum Γ_0. This makes the circulation amplitude γ the natural canon-level coupling variable rather than a free phenomenological insertion.

9.6 Branch Structure

Two chiral sectors exist:

$$\sigma = \pm 1 \tag{201}$$

These correspond to:

- ↺ (left-handed)
- ↻ (right-handed)

9.7 Electron trefoil identity under ordinary atomic binding

[DERIVED SST CONSTRAINT]. Ordinary propagation through vacuum and ordinary atomic binding do not change the electron carrier's knot class. In the trefoil assignment, the transition is

$$3_1^{\mathrm{free}} \longrightarrow 3_1^{\mathrm{bound}}, \tag{202}$$

not a breaking or unknotting of the carrier. Binding changes the embedding, phase envelope, branch coupling, and dressing of the carrier:

$$e^-_{\mathrm{atom}} = 3_1^{\mathrm{core}} \otimes \Psi^{\mathrm{envelope}}_{n\ell m\sigma}. \tag{203}$$

[CANON RULE]. Ordinary electron capture by an ion is a binding and phase-locking process. Destruction or conversion of the separate electron trefoil requires a non-Euler process such as annihilation with an anti-trefoil or weak-sector capture, with the necessary compensation carried by the final state.

9.8 Consistency Requirement

To remain compatible with atomic physics:

$$|\delta E_n| \ll |E_n| \tag{204}$$

[ORTHODOX] Constraint: SST corrections must not violate known spectroscopy.

9.9 Interpretation

[STATUS]: The atomic bridge is a controlled benchmark sector. The orthodox Coulomb problem is imported as the baseline; the circulation-dependent term is the SST addition; shell counting, exclusion-like behavior, and heavy-atom extensions remain programmatic rather than derived here. The purpose of the section is therefore structural compatibility, not yet a complete replacement of atomic quantum mechanics.

10 Spectroscopic Constraints

10.1 Motivation

Any extension of atomic structure must remain consistent with high-precision spectroscopic measurements.

[ORTHODOX]: Atomic energy levels are experimentally verified to extremely high precision.

Requirement:

$$|\delta E_n| \ll |E_n| \tag{205}$$

where δE_n represents any SST-induced correction.

10.2 Internal Mode Hypothesis

We consider the possibility that atomic bound states possess an additional internal excitation sector associated with vortex filament dynamics.

[SPECULATIVE]: Each bound state carries an internal mode spectrum $\{E_{m,n}\}$.

The effective energy becomes:

$$E_n^{\text{eff}} = E_n^{(0)} + \delta E_n \tag{206}$$

10.3 Generic Coupling Structure

We parameterize the coupling between internal modes and observable energy levels:

$$|\delta E_n| \le \lambda_n \sum_m E_{m,n}\, \nu_{m,n} \tag{207}$$

where:

- λ_n is the coupling efficiency
- $\nu_{m,n}$ is the occupation factor

[DERIVED]: This form captures a general upper bound independent of microscopic details.

10.4 Ungapped Spectrum: Instability

If the internal spectrum is ungapped:

- arbitrarily low-energy excitations exist
- no universal suppression mechanism applies

Consequence:

$$\delta E_n \not\to 0 \quad \text{generically} \tag{208}$$

[**CRITICAL**]: An ungapped internal sector is generically incompatible with spectroscopy unless:

- $\lambda_n \ll 1$ (extremely weak coupling), or
- the density of states is highly suppressed

10.5 Gapped Spectrum: Decoupling Mechanism

We introduce an excitation gap Δ_K:

$$E_{m,n} \geq \Delta_K \tag{209}$$

Then occupation factors follow Boltzmann suppression:

$$\nu_{m,n} \sim \exp\left(-\frac{\Delta_K}{k_B T_{\text{eff}}}\right) \tag{210}$$

Substituting into Eq. 207:

$$\delta E_n \propto \exp\left(-\frac{\Delta_K}{k_B T_{\text{eff}}}\right) \tag{211}$$

[**DERIVED**]: Exponential suppression ensures compatibility with spectroscopy.

10.6 Physical Origin of the Gap

[**SPECULATIVE**]: The gap Δ_K may arise from:

- geometric constraints on filament curvature
- torsional rigidity
- topological restrictions
- core structure of the vortex

10.7 Order-of-Magnitude Estimate

Using filament parameters:

$$\Delta_K \sim \frac{\Gamma^2}{4\pi\xi L} \tag{212}$$

where:

- ξ is the core scale
- L is the loop length

For typical SST scales:

$$\Delta_K = \mathcal{O}(10^2 - 10^3\,\mathrm{eV}) \tag{213}$$

[**DERIVED**]: This places the internal sector well above atomic energy scales ($\sim$ eV).

10.8 Consistency Condition

To preserve atomic physics:

$$\frac{\Delta_K}{k_B T_{\mathrm{eff}}} \gg 1 \tag{214}$$

Interpretation: The internal sector is effectively frozen at atomic energies.

10.9 Observable Consequences

[**SPECULATIVE**]: Residual effects may include:

- tiny level shifts
- suppressed line broadening
- weak temperature dependence

10.10 Falsifiability

The theory is falsifiable via:

- detection of unexplained spectral shifts
- absence of expected suppression
- deviation from Boltzmann scaling

10.11 Integration with SST Framework

The spectroscopic constraint enforces:

- compatibility with the Atomic Bridge (Section 9)
- constraints on delay-induced excitations (Section 8)
- bounds on allowable vortex configurations

10.12 Canonical Statement

Any viable SST extension must include a mechanism—such as an excitation gap—that ensures exponential suppression of internal modes at atomic energy scales.

10.13 Precision electroweak bridge: lessons from the CMS W-mass measurement

Status. [**ORTHODOX**] External input: the CMS measurement is a precision constraint on any SST effective weak-sector realization. [**DERIVED**] Methodological consequence: the result supports a forward-modelling strategy for SST phenomenology. [**CRITICAL NOTE**] The paper does *not* provide direct evidence for a structured-substrate ontology of the W boson.

Empirical anchor. Let the experimentally inferred resonance mass be denoted by

$$m_W^{\mathrm{exp}} = 80.3602 \pm 0.0099 \text{ GeV}. \tag{215}$$

In the CMS analysis this quantity is not read off from a single observable, but extracted from a profiled likelihood over finely binned charged-lepton kinematics. This motivates the following SST principle:

> *Collider observables constrain SST through forward maps from latent swirl-string states to detector-level distributions, not through direct identification of a single internal parameter with a measured mass.*

Latent-to-observable map. Introduce a latent SST parameter set for an effective spin-1 weak excitation,

$$\Theta_W^{\mathrm{SST}} = \Big(K_W,\, \chi_W,\, \mathcal{L}_W,\, \mathcal{H}_W,\, {S^{\circlearrowleft}_{(t)}}_W,\, {\rho_f}^{\mathrm{eff}},\, \mathbf{u}_{\circlearrowleft}^{\mathrm{eff}}\Big), \tag{216}$$

where:

- K_W is the topology class or excitation family,
- χ_W is chirality / handedness label,
- $\mathcal{L}_W$ is an effective geometric length scale,
- $\mathcal{H}_W$ is a helicity/linkage content,
- ${S^{\circlearrowleft}_{(t)}}_W$ is the local clock factor for the mode,
- ${\rho_f}^{\mathrm{eff}}$ and $\mathbf{u}_{\circlearrowleft}^{\mathrm{eff}}$ are the coarse-grained medium density and effective local swirl-velocity field seen by the excitation.

The experimentally accessible distribution is then not a bare function of m_W alone, but a projected density

$$\mathcal{P}_{\mathrm{meas}}(x) = \int dz\; R(x|z)\, \mathcal{P}_{\mathrm{prod}}(z \mid \Theta_W^{\mathrm{SST}}), \tag{217}$$

where z denotes truth-level production/decay variables and $R(x|z)$ is the detector response kernel.

Mass as an effective resonance parameter. In SST one should distinguish:

$$m_W^{\text{internal}} \neq m_W^{\text{eff}} \tag{218}$$

in general, where:

- m_W^{internal} is the internal excitation-energy scale of the structured mode,
- m_W^{eff} is the resonance parameter inferred from scattering distributions.

The experimentally constrained quantity is m_W^{eff}.

A minimal canon-safe matching condition is therefore

$$m_W^{\text{eff}}(\Theta_W^{\text{SST}}) = m_W^{\text{exp}} + \delta_W, \qquad |\delta_W| \lesssim 10 \text{ MeV}, \tag{219}$$

with the current precision scale set by experiment.

Distribution-level fit target. The correct SST comparison object is a likelihood built from binned observables:

$$\mathcal{L}(\Theta_W^{\text{SST}}, \nu) = \prod_{b=1}^{N_{\text{bins}}} \text{Poisson}\Big(n_b \,\Big|\, \mu_b(\Theta_W^{\text{SST}}, \nu)\Big)\; \Pi(\nu), \tag{220}$$

where:

- n_b is the observed count in bin b,
- μ_b is the SST prediction after production, decay, and detector folding,
- ν denotes nuisance parameters,
- $\Pi(\nu)$ is the nuisance prior / constraint factor.

The profile likelihood for the physically interesting SST parameters is then

$$\mathcal{L}_{\text{prof}}(\Theta_W^{\text{SST}}) = \max_{\nu}\; \mathcal{L}(\Theta_W^{\text{SST}}, \nu). \tag{221}$$

Helicity decomposition as a falsifier. A particularly sharp SST test is not the central value of m_W alone, but whether the effective spin-1 mode reproduces the standard angular basis. Write

$$\frac{d\sigma}{d\Omega} = \sum_{i=0}^{8} A_i\, f_i(\theta, \phi) + \epsilon_{\text{SST}}\, \Delta(\theta, \phi), \tag{222}$$

where the $A_i f_i$ are the standard helicity structures and $\epsilon_{\text{SST}}\Delta$ is the first SST correction. Then:

- if $\epsilon_{\text{SST}} = 0$ within experimental precision, SST survives this channel,
- if $\epsilon_{\text{SST}} \neq 0$, the deviation is directly falsifiable through angular distributions.

Canon interpretation. This leads to the following conservative reading:

1. The weak sector must be *SM-like at the effective distribution level.*
2. Large $\mathcal{O}(10^{-4})$–$\mathcal{O}(10^{-3})$ relative distortions in the effective W mass are disfavoured.
3. The best SST discovery channels are therefore likely to be

   $$\text{small angular deviations,} \quad \text{polarization asymmetries,} \quad \text{off-shell tails,} \quad \text{rare differential distortions,} \tag{223}$$

   rather than a large shift in the central value of m_W.

Minimal bridge claim. The CMS result does *not* force the W boson to be fundamental in the ontological sense. It does force any SST realization of the weak sector to satisfy the stronger requirement:

$$\mathcal{P}_{\text{prod}}^{\text{SST}}(z \mid \Theta_W^{\text{SST}}) \approx \mathcal{P}_{\text{prod}}^{\text{SM}}(z \mid m_W, \text{SM nuisances}) \tag{224}$$

to current experimental precision after detector folding and nuisance profiling.

Practical SST program. A precision-compatible SST weak-sector program should therefore proceed in three layers:

$$\text{(i) effective resonance matching:} \quad m_W^{\text{eff}} \to m_W^{\text{exp}}, \tag{225}$$

$$\text{(ii) helicity matching:} \quad A_i^{\text{SST}} \to A_i^{\text{SM}}, \tag{226}$$

$$\text{(iii) residual search:} \quad \Delta \neq 0 \text{ only in subleading differential structure.} \tag{227}$$

Conclusion. The CMS W-mass result is best understood in SST as a *precision filter*: it does not supply ontology, but it strongly constrains phenomenology. Any viable SST weak-sector realization must be distributionally SM-like to present precision, while leaving room only for small, structured residuals in higher-order observables.

11 Relational Time Framework

11.1 Canonical Time Ontology

Status. [CANON / SEMANTIC ORGANIZATION]. This subsection fixes the time-language of SST. It does not introduce a new dynamical postulate. Its purpose is to prevent the absolute foliation parameter, laboratory time, local proper time, carrier loop time, the Swirl-Clock factor, and Kairos transition events from being conflated.

No bare-τ convention. Because SST uses both relativistic local clock accumulation and finite circulation delay, the unqualified symbol τ is avoided in mixed contexts. Circulation delay is denoted by τ_{circ}, while local proper time carried by a localized process A is denoted by τ_A or τ_{loc}.

Absolute foliation time. The symbol T denotes the global foliation parameter of the incompressible SST substrate. Each value of T labels a constraint slice

$$\Sigma_T. \tag{228}$$

The pressure field and incompressibility constraint are reconstructed on Σ_T. The parameter T is bookkeeping time for the medium foliation; it is not by itself a local clock reading. This foliation T is distinct from the coarse-grained clock field $T(x)$ introduced later in relational observables from conserved currents.

External comparison time. The symbol t_∞ denotes asymptotic laboratory time or external comparison time. It is the time coordinate used when matching SST clock predictions to standard relativistic observables. In weakly perturbed laboratory settings one may identify t_∞ with T, but the distinction should be retained whenever local clock rates are being discussed.

Swirl-Clock factor. The local Swirl-Clock factor is the dimensionless rate-conversion field already defined in Eq. (69):

$$S_{(t)}^{\circlearrowleft}(x,T) = \sqrt{1 - \frac{\|\mathbf{u}_{\circlearrowleft}(x,T)\|^2}{c^2}}. \tag{229}$$

It is not a duration. It converts the foliation rate into the local proper-clock rate of a resolved carrier or process.

Local proper time. The symbol τ_A (or $\tau_{\rm loc}$) denotes local proper clock time. Along a localized carrier A,

$$d\tau_A = S_{(t)}^{\circlearrowleft}[A;T]\,dT. \tag{230}$$

Thus two localized carriers A and B may share the same foliation slice Σ_T while accumulating different proper times:

$$\frac{d\tau_A}{dT} = S_{(t)}^{\circlearrowleft}[A;T], \qquad \frac{d\tau_B}{dT} = S_{(t)}^{\circlearrowleft}[B;T], \qquad \frac{d\tau_C}{dT} = S_{(t)}^{\circlearrowleft}[C;T]. \tag{231}$$

Operational coordinates of a vortex observer. Let Alice be a localized vortex observer. Her coordinate time t_A is assembled from a network of clocks identical to her local clock, synchronized by R-phase torsion-pulse exchange; at Alice's reference worldline, $dt_A = d\tau_A$. Let ℓ_A denote distance measured by an independently equilibrated vortex-matter ruler. The directly testable two-way propagation speed is

$$c_A^{(2\mathrm{w})} = \frac{2\ell_A}{\tau_A^{\rm recv} - \tau_A^{\rm emit}} = c. \tag{232}$$

Once this local calibration is established, Alice assigns radar coordinates to a reflected event E by

$$t_A(E) = \frac{\tau_A^{\rm recv} + \tau_A^{\rm emit}}{2}, \tag{233}$$

$$\ell_A(E) = \frac{c}{2}\left(\tau_A^{\rm recv} - \tau_A^{\rm emit}\right). \tag{234}$$

Alice–Bob space–time mixing. Let Bob move uniformly relative to Alice with operationally measured speed u_{AB}, and define

$$\beta_{AB} = \frac{u_{AB}}{c}, \qquad \gamma_{AB} = \frac{1}{\sqrt{1-\beta_{AB}^2}}. \tag{235}$$

In the shared infrared torsion-cone limit, their operational coordinates obey

$$ct_B = \gamma_{AB}\left(ct_A - \beta_{AB}x_A\right), \tag{236}$$

$$x_B = \gamma_{AB}\left(x_A - \beta_{AB}ct_A\right), \tag{237}$$

with

$$-c^2dt_A^2 + dx_A^2 = -c^2dt_B^2 + dx_B^2. \tag{238}$$

Equations (237)–(238) formalize the statement that space and time form one operational Lorentz structure: a change of relative motion mixes their coordinate components. SST does not interpret this as proof that the substrate itself lacks a preferred ordering parameter; rather, it requires that the preferred foliation be operationally hidden at low energy when clocks, rulers, and signals are all dressed by the same torsion sector.

[PREFERRED-FRAME FALSIFIER]. If an Alice–Bob experiment depends separately on Alice's or Bob's velocity relative to the substrate, rather than only on u_{AB}, then exact operational Lorentz invariance fails. The allowed residual is constrained by preferred-frame experiments and must vanish in the exact monometric limit.

Proper loop or turnover time. The symbol τ_{circ} denotes a carrier-internal circulation or delay time. For a loop of resolved length L_K,

$$\tau_{\text{circ}}(K) = \frac{L_K}{v_{\mho}^{*}}. \tag{239}$$

At the canonical core scale this reduces to

$$\tau_{\text{circ,c}} = \frac{2\pi r_c}{v_{\mho}^{*}} = \frac{2\pi}{\omega_c}. \tag{240}$$

The turnover time τ_{circ} counts internal circulation cycles; it is therefore distinct from the dimensionless Swirl-Clock factor $S_{(t)}^{\mho}$.

Kairos transition event. A Kairos event is not a continuous clock variable. It is a discrete topological transition marker,

$$\mathcal{K}_{\text{Kairos}} : K_{-} \longrightarrow K_{+}, \tag{241}$$

used when a process changes topological sector, chirality class, closure state, or reconnection class. It may occur at some foliation value T, but it is not itself a time coordinate.

Summary schema.

Layer	Symbol	Role
Foliation slice	Σ_T	global incompressible constraint slice
Absolute foliation time	T	ordering parameter of the substrate
External comparison time	t_∞	laboratory/asymptotic comparison time
Swirl-Clock factor	$S_{(t)}^{\mho}$	dimensionless local rate conversion
Local proper time	τ_A, τ_{loc}	accumulated local clock time
Operational chart	(t_A, ℓ_A)	local radar coordinates
Loop/circulation delay	τ_{circ}	internal circulation period
Kairos event	$\mathcal{K}_{\text{Kairos}}$	discrete topological transition

Units guard. $S_{(t)}^{\mho}$ is dimensionless; T, t_∞, τ_A, and τ_{circ} have units of time; $\mathcal{K}_{\text{Kairos}}$ is an event, not a coordinate.

11.2 Swirl-Clock and local proper time

Section 11.1 fixes the time-language of SST. The canonical time variable is not introduced as an independent primitive. It is defined relationally from the local kinematic state of the medium through the Swirl-Clock already introduced in Eq. (69). In its coarse-grained form,

$$d\tau_{\text{loc}} = S_{(t)}^{\mho}(x)\, dt_\infty, \tag{242}$$

consistent with Eq. (230). Here t_∞ denotes the asymptotic laboratory time and τ_{loc} the effective proper time associated with a localized process in the medium. In SST, this factor is interpreted as a fluid-kinematic clock response. Operational special relativity requires the corresponding vortex-ruler response and R-phase torsion propagation to be governed by the same infrared cone; time dilation alone is insufficient to derive invariant local c.

No-signalling research-track note. The pressure field $p_{\mho}(\mathbf{x}, T)$ is a constraint field on Σ_T. Its elliptic reconstruction may be instantaneous in the T-bookkeeping sense, but this does not by itself define an operational superluminal signal channel. Observable clock and photon phases are read through the local Swirl–Clock projection $d\tau_A/dT = S_{(t)}^{\mho}[A;T]$. The no-signalling status of the pressure constraint is a research-track theorem target, not an assumed device capability.

11.3 Carrier-local clocking and the turnover scale

For localized carriers the natural microscopic rate is the turnover frequency

$$\Omega_0 = \frac{v_{\circlearrowleft}^*}{r_c} = \omega_c, \tag{243}$$

which coincides with the Compton anchoring scale and satisfies $\tau_{\mathrm{circ,c}} = 2\pi/\omega_c$ (Eq. (189)). A complete SST clock description therefore involves two linked but distinct levels:

- the *microscopic turnover scale* Ω_0, which counts circulation cycles of the carrier,
- the *coarse-grained Swirl-Clock* $S_{(t)}$, which relates local rates to asymptotic time.

This resolves a recurring ambiguity in earlier canon versions: local cycle counting and coarse-grained time dilation are related, but they are not identical observables.

11.4 Relational observables from conserved currents

A purely operator-based notion of time is avoided in SST by defining time observables through conserved currents. Let J^μ denote a matter current and let j^μ_{ev} denote a coarse-grained event current associated with structured activity of the medium. The relational clock field $T(x)$ is then recovered only after coarse graining,

$$\partial_\mu T(x) \approx \frac{1}{\nu_0} \langle j^{\mathrm{ev}}_\mu \rangle_\ell, \tag{244}$$

where ℓ is the coarse-graining scale and ν_0 a reference event frequency. The canon interpretation is that time is read from correlations between currents, not from an abstract external operator.

11.5 Time-of-arrival as a relational field observable

Let $\mathcal{W}$ be a detector world-tube with boundary $\Sigma = \partial\mathcal{W}$. Then the arrival-time distribution associated with a detector label Θ is written as

$$p(\Theta) = \frac{1}{\mathcal{N}} \int_\Sigma d\sigma_\mu \, J^\mu(x)\, \delta(T(x) - \Theta). \tag{245}$$

This formulation is canonically important because it bypasses the need for a self-adjoint time operator conjugate to a bounded Hamiltonian. The measured time is instead the value of a physical clock field at the point where matter flux crosses the detector boundary.

11.6 Clock broadening and continuum limits

If the event current has a finite variance, then the observed time-of-arrival distribution is broadened relative to the ideal classical prediction. Writing P_{cl} for the classical arrival distribution and σ_τ^2 for the effective clock variance, one expects a convolution structure of the form

$$P_{\mathrm{obs}}(\Theta) = \int dt\, P_{\mathrm{cl}}(t) \frac{1}{\sqrt{2\pi\sigma_\tau^2}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma_\tau^2}\right]. \tag{246}$$

In the limit $\ell \gg r_c$, the event structure is smoothed and one recovers a continuum clock field. In the opposite limit $\ell \to r_c$, the underlying discreteness of carrier events becomes operationally relevant.

11.7 Connection to gravity and geometry

In the canon, the relational-time sector and the geometric-response sector are not independent. Variations in the Swirl-Clock track variations in local swirl energy density and therefore provide the time component of the effective geometric response of the medium. At this level, the SST reading is:

gravity and time are two coarse-grained aspects of the same clock-bearing medium.

This statement should be read as a canon-level organizing principle, not yet as a completed field-theoretic proof.

11.8 Clock Field and Foliation Variable

[DERIVED] EFT bridge: the scalar Swirl-Clock field is promoted from the kinematic relation of Eq. (69) to a long-wavelength foliation variable

$$\chi(x), \tag{247}$$

whose normalized gradient defines a preferred local clock frame:

$$u_\mu = \frac{\nabla_\mu \chi}{\sqrt{g^{\alpha\beta}\nabla_\alpha \chi\, \nabla_\beta \chi}}. \tag{248}$$

[ORTHODOX] Equation (248) matches the standard hypersurface-orthogonal foliation construction used in khronometric / Einstein–Æther effective field theory [14, 15].

[DERIVED] Within SST, the interpretation is different: u_μ is not taken as a fundamental spacetime structure, but as the coarse-grained clock-flow direction associated with the medium.

11.9 Clock-Sector Effective Action

[ORTHODOX] EFT template: the most general second-order covariant clock-sector action compatible with hypersurface orthogonality is

$$\begin{aligned} S_\chi = \int d^4x\sqrt{-g}\,\Big[&c_1(\nabla_\mu u_\nu)(\nabla^\mu u^\nu) + c_2(\nabla_\mu u^\mu)^2 \\ &+ c_3(\nabla_\mu u_\nu)(\nabla^\nu u^\mu) + c_4 u^\mu u^\nu(\nabla_\mu u_\alpha)(\nabla_\nu u^\alpha)\Big]. \end{aligned} \tag{249}$$

[SPECULATIVE] SST interpretation: Equation (249) is not itself derived from first-principles filament dynamics in the present Canon. It is recorded as the minimal orthodox EFT bridge into a relativistic clock-field description.

11.10 Poisson Limit and Effective Gravity Bridge

[DERIVED] Bridge statement: in the weak-field, slowly varying limit, the scalar clock field is taken to obey a Poisson-type closure

$$\nabla^2\chi(x) = 4\pi G_{\text{eff}}\,\rho_{\text{matter}}(x), \tag{250}$$

so that outside localized sources

$$\chi(r) \propto \frac{1}{r}, \qquad \mathbf{g}_{\text{eff}} = -\nabla\chi. \tag{251}$$

[ORTHODOX] The $1/r$ scalar potential and $1/r^2$ force law are standard consequences of Eq. (250).

[SPECULATIVE] The identification of χ with the Swirl-Clock and the interpretation of Eq. (250) as an emergent gravity law are SST-specific.

11.11 Observational Constraint from Tensor-Mode Speed

[ORTHODOX] Constraint: If the clock-sector EFT is used, multimessenger gravitational-wave observations require [16]

$$c_{13} \equiv c_1 + c_3 = 0, \tag{252}$$

to enforce luminal tensor-mode propagation at observational precision.

[ORTHODOX] Constraint: any SST realization employing the EFT bridge of Eq. (249) must preserve Eq. (252).

11.12 Internal Tensor-Speed Naturalness

[DERIVED, conditional] In the clock-field / Einstein–Æther comparison layer, the spin-2 tensor-mode speed is conventionally written as

$$c_T^2 = \frac{c^2}{1 - c_{13}}, \qquad c_{13} := c_1 + c_3. \tag{253}$$

Multimessenger observations require $c_T = c$ to high precision and therefore impose $c_{13} \simeq 0$ observationally [16, 53]. In SST there is a stronger conditional interpretation: if the metric tensor mode of the coarse-grained clock-field sector is identified with the transverse excitation mode of the underlying medium, and if c is the canonical transverse propagation speed of that medium, then

$$c_T = c \quad \Longrightarrow \quad c_{13} = 0. \tag{254}$$

Status. Equation (254) is not an independent derivation of the Einstein–Hilbert dynamics. It is a conditional mode-identification result inside the orthodox Einstein–Æther comparison layer [51, 52]. Its canon role is to upgrade the luminal tensor-speed condition from a purely external observational constraint to an internal naturalness condition, provided the SST tensor-mode identification is valid.

[CRITICAL NOTE] The proposition applies only to the spin-2/tensor sector. It does not by itself constrain the scalar and vector Æ-modes, nor does it prove monometricity for all matter, gauge, and knot species. The full monometricity target and the Einstein-dynamics routes are developed in the research track (Relativity Emergence Ladder).

11.13 Interpretive Role in the Canon

This section has a deliberately limited role:

- it upgrades the Swirl-Clock from a purely local kinematic factor to a candidate field variable;
- it records a minimal orthodox EFT bridge for relativistic comparison;
- it supplies the internal reference point needed by Section 12 for the time–gravity link.

Status summary.

- Eq. (248): **[ORTHODOX]** mathematical form; **[DERIVED]** role as an SST bridge variable
- Eq. (249): **[ORTHODOX]** EFT template
- Eq. (250): **[DERIVED]** bridge closure
- gravity interpretation: **[SPECULATIVE]**

11.14 Relation to the Chronos–Kelvin law

Section 3.12 provides the local transport identity of the Swirl-Clock on a material loop. The present section supplies the long-wavelength field-theoretic and analogue-metric reading of the same clock sector. The two layers are therefore complementary:

- Eq. (83) is the loop-level transport law;
- Eq. (248)–Eq. (250) provide the coarse-grained field description.

11.15 Canonical Electromagnetic–Gravity Closure

[CANONICAL BRIDGE LAW]. The electromagnetic and gravitational sectors of SST are canonically linked by a common local swirl state, but they remain distinct projections. Electromagnetic response is carried by transverse phase/flux variables and topological piercing changes. Gravitational response is carried by pressure gradients, swirl stress, and the local clock factor. Thus SST does not identify the ordinary electromagnetic field with gravity. It identifies both as different observable projections of the same incompressible, vorticity-bearing medium.

The canonical source state for the coupled EM–gravity bridge is

$$\boxed{\mathcal{X}_{\circlearrowleft}^{\mathrm{EMG}} = \left(\mathbf{u}_{\circlearrowleft}, p_{\circlearrowleft}, \boldsymbol{\omega}_{\circlearrowleft}, \varrho_{\circlearrowleft}, \boldsymbol{\sigma}^{\mathrm{swirl}}, \rho_E, \rho_m\right), \qquad \rho_m = \frac{\rho_E}{c^2}.} \tag{255}$$

The topological piercing density $\varrho_{\circlearrowleft}$ belongs to the electromagnetic impulse channel, while $p_{\circlearrowleft}$, ρ_m, and $\boldsymbol{\sigma}^{\mathrm{swirl}}$ belong to the force and gravity channel.

[CANON / PREFACTOR GUARD] The SI-normalized electromagnetic bridge, when invoked, uses the electron mass-to-charge factor

$$\mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\,\mathbf{u}_{\circlearrowleft}(\mathbf{x}, t), \qquad \mathbf{B}_{\mathrm{SST}} = \nabla \times \mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\,\boldsymbol{\omega}. \tag{256}$$

The inverse factor e/m_e is rejected in this bridge because it has the wrong SI dimensions for mapping vorticity to magnetic flux density. At the Compton angular frequency this calibrated bridge gives

$$B_{\mathrm{core}}^{\mathrm{SST}} = \frac{m_e}{e}\omega_c = \frac{m_e^2 c^2}{e\hbar} = B_{\mathrm{QED}}, \tag{257}$$

while preserving the epistemic status of the result as a normalization bridge, not a derivation of QED from Euler dynamics.

Electromagnetic projection. The canonical topological electromagnetic impulse is

$$\boxed{\Delta\Phi_{\circlearrowleft} = \Phi_0\,\Delta N, \qquad \Phi_0 = \frac{h}{2e}, \qquad \Delta N \in \mathbb{Z}.} \tag{258}$$

In local differential notation, the canonical part of the effective Faraday projection is

$$\boxed{\nabla \times \mathbf{E}_{\mathrm{eff}} = -\frac{\partial \mathbf{B}_{\mathrm{eff}}}{\partial t} - \Phi_0 \frac{\partial \varrho_{\circlearrowleft}}{\partial t}\,\hat{\mathbf{n}} - \mathbf{b}_{\circlearrowleft}^{(\sigma)}.} \tag{259}$$

Here $\mathbf{b}_{\circlearrowleft}^{(\sigma)}$ is a calibrated stress-projection correction. It is not a primitive canonical constant. If no stress-projection closure has been derived for a given apparatus, set

$$\mathbf{b}_{\circlearrowleft}^{(\sigma)} = 0. \tag{260}$$

Equation (259) is an effective-medium projection. In the R-phase null limit,

$$\partial_t \varrho_{\circlearrowleft} \to 0, \qquad \mathbf{b}_{\circlearrowleft}^{(\sigma)} \to 0, \tag{261}$$

and the ordinary source-free Maxwell–Faraday identity is recovered.

Gravity projection. The canonical SST gravitational force density is

$$\boxed{\mathbf{f}_{\mathrm{grav},\circlearrowleft} = \rho_m \mathbf{g}_\circlearrowleft + \nabla\cdot\boldsymbol{\sigma}^{\mathrm{swirl}}.} \tag{262}$$

The bulk acceleration field is

$$\boxed{\mathbf{g}_\circlearrowleft = -\nabla\Phi_\circlearrowleft - \frac{\partial \mathbf{A}_\circlearrowleft}{\partial t}, \qquad \mathbf{B}_\circlearrowleft = \nabla\times\mathbf{A}_\circlearrowleft.} \tag{263}$$

In the weak, stationary, coarse-grained limit the scalar sector obeys the Poisson-type closure

$$\boxed{\nabla\cdot\mathbf{g}_\circlearrowleft = -4\pi G_{\mathrm{swirl}}\rho_m, \qquad \nabla^2\Phi_\circlearrowleft = 4\pi G_{\mathrm{swirl}}\rho_m.} \tag{264}$$

This closure is valid only after coarse graining over swirl-string microstructure and only in the weak-field, slow-flow regime where pressure reconstruction is consistent with Euler balance.

The canonical pressure anchor remains the incompressible Euler radial balance

$$\boxed{\frac{1}{\rho_f}\frac{dp_{\mathrm{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}.} \tag{265}$$

Therefore the Newtonian-looking gravitational acceleration is not fundamental curvature in SST; it is the coarse-grained acceleration produced by pressure/stress gradients in the swirl medium.

Clock and optical closure. The clock sector supplies the shared time-rate projection:

$$\boxed{\frac{d\tau_{\mathrm{loc}}}{dt_\infty} = S_{(t)}^{\circlearrowleft} = \sqrt{1-\frac{\|\mathbf{u}_\circlearrowleft\|^2}{c^2}}, \qquad n_\gamma = S_{(t)}^{\circlearrowleft\,-1}.} \tag{266}$$

Photon phase therefore probes n_γ, while bulk matter probes Eq. (262). These are coupled through $\mathbf{u}_\circlearrowleft$, but they need not have identical spatial gradients.

Electron-normalized coupling calibration. The weak gravitational coupling is written in electron-normalized SST variables as

$$\boxed{G_{\mathrm{swirl}} = \frac{v_\circlearrowleft^{*} c^3 t_p^2}{r_c M_e} = 6.6743013\times10^{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}.} \tag{267}$$

Thus the EM–gravity bridge is not an unlabeled free-parameter assertion. However, because t_p already contains G in orthodox units, Eq. (267) is a **[CALIBRATED]** electron-normalized consistency identity unless a future SST sector derives t_p or G_{swirl} independently from medium dynamics.

Compton–horn reduction guard. Substituting the calibrated horn closure $r_c = v_\circlearrowleft^{*}/\omega_c$ into Eq. (267) gives

$$G_{\mathrm{swirl}} = \frac{\omega_c c^3 t_p^2}{M_e}. \tag{268}$$

With the orthodox definitions

$$\omega_c = \frac{M_e c^2}{\hbar}, \qquad t_p^2 = \frac{\hbar G}{c^5}, \tag{269}$$

Eq. (268) reduces identically to G. This confirms that the electron-normalized bridge is algebraically closed under the v0.8.18 calibration chain, but it also fixes the epistemic status: the identity must not be cited as an independent derivation of Newton's constant unless t_p, $G_{\rm swirl}$, or the Compton–horn closure is derived without importing G through orthodox Planck units.

[CANON / STATUS GUARD] A dimensionless balance angle ϕ_G may be introduced only as a research-track diagnostic of Eq. (267). Since the calibrated chain gives $\tan^2\phi_G = 1$, the associated 45° angle is a visualization of equal radial-confinement and tangential-swirl weighting, not a physical direction of gravitational force and not an additional gravitational law. See Sec. R.56.

[ORTHODOX GR REPARAMETERIZATION]. In orthodox units the electron Schwarzschild radius may be written as

$$r_s(e) = \frac{2Gm_e}{c^2} = \frac{4\pi\ell_P^2}{\lambda_c}, \qquad \ell_P = \sqrt{\frac{\hbar G}{c^3}}. \tag{270}$$

This is a standard GR identity and notational bridge. It does not derive G from SST medium dynamics.

Dimensional checks. The electromagnetic source term has units

$$[\Phi_0\partial_t\varrho_\circlearrowleft] = (\mathrm{V\,s})(\mathrm{m^{-2}\,s^{-1}}) = \mathrm{V\,m^{-2}}, \tag{271}$$

matching $[\nabla\times\mathbf{E}_{\rm eff}]$. The gravitational force density has units

$$[\rho_m\mathbf{g}_\circlearrowleft] = \mathrm{kg\,m^{-3}\,m\,s^{-2}} = \mathrm{N\,m^{-3}}, \qquad [\nabla\cdot\boldsymbol{\sigma}^{\rm swirl}] = \mathrm{N\,m^{-3}}. \tag{272}$$

Finally,

$$\left[\frac{\|\mathbf{v}_\circlearrowleft\|c^3t_p^2}{r_cM_e}\right] = \frac{(\mathrm{m\,s^{-1}})(\mathrm{m^3\,s^{-3}})(\mathrm{s^2})}{\mathrm{m\,kg}} = \mathrm{m^3\,kg^{-1}\,s^{-2}}. \tag{273}$$

Numerical hierarchy. At the canonical speed $v_\circlearrowleft^* = 1.09384563\times10^6\,\mathrm{m\,s^{-1}}$,

$$\frac{1}{2}\rho_f v_\circlearrowleft^{*2} = 4.1877439\times10^5\,\mathrm{Pa}, \tag{274}$$

$$S_{(t)}^\circlearrowleft = 0.9999933436, \tag{275}$$

$$n_\gamma - 1 = 6.6564858\times10^{-6}. \tag{276}$$

Hence pressure/stress channels can be macroscopically large at canonical scale, while single-pass optical-index shifts are small unless a resonant optical path enhances them. This hierarchy is now part of the canonical interpretation of the EM–gravity bridge.

Canonical exclusion rules. The following statements are excluded from the canonical EM–gravity bridge:

1. Ordinary electromagnetic fields are not identified with gravity: $\mathbf{B}_{\rm eff}\neq\mathbf{B}_\circlearrowleft$.
2. Uncalibrated vorticity terms may not be inserted directly into Maxwell source equations.
3. Device-specific gains such as Q_γ, $\mathcal{G}_\circlearrowleft^{(\sigma)}$, and impedance gains are research-track unless derived from a resonator or stress-closure model.
4. Apparent laboratory forces must first be decomposed into electromagnetic, thermal, acoustic, electrohydrodynamic, mechanical, and ordinary vorticity-impulse channels before any residual is interpreted as SST-specific.

Canonical summary. The promoted law is therefore

$$\boxed{\text{EM response and gravity response are distinct sectoral projections of } \mathcal{X}_\circlearrowleft^{\mathrm{EMG}}.} \tag{277}$$

The electromagnetic projection is controlled by topological flux impulse $\Delta\Phi_\circlearrowleft = \Phi_0 \Delta N$. The gravitational projection is controlled by $\rho_m \mathbf{g}_\circlearrowleft + \nabla\cdot\boldsymbol{\sigma}^{\mathrm{swirl}}$ with weak closure $\nabla\cdot\mathbf{g}_\circlearrowleft = -4\pi G_{\mathrm{swirl}}\rho_m$. Both recover their orthodox limits when the SST projection channels vanish.

11.16 Triadic Gravity-Response Corollary

[DERIVED] Canonical corollary. Because the electromagnetic, gravitational, optical, and boundary-stress responses are distinct projections of the same SST state $\mathcal{X}_\circlearrowleft^{\mathrm{EMG}}$, a laboratory observation must not be classified as "gravity" from force response alone. The admissible coarse response separates into three limiting channels:

$$\mathcal{R}_{\mathrm{plume}} := (\Gamma_\circlearrowleft \neq 0, \quad \nabla n_\gamma \approx 0, \quad \mathbf{g}_\circlearrowleft \approx 0)\,, \tag{278}$$

$$\mathcal{R}_\gamma := (\nabla_\perp n_\gamma \neq 0, \quad \Gamma_\circlearrowleft \approx 0, \quad \mathbf{f}_{\mathrm{matter}} \approx 0)\,, \tag{279}$$

$$\mathcal{R}_\Pi := \left(\nabla\cdot\boldsymbol{\sigma}^{\mathrm{swirl}} \neq 0, \quad \nabla n_\gamma \approx 0\right). \tag{280}$$

Here $\mathcal{R}_{\mathrm{plume}}$ is an impedance or damping response of a plume or flame-like medium, $\mathcal{R}_\gamma$ is an optical phase/caustic response, and $\mathcal{R}_\Pi$ is a boundary-stress or elastic-shell response. The three modes may share a common source state but are not mutually identical.

Local Euler locking of pressure and optical projections. Under the additional assumptions of a passive, stationary, inviscid Euler region with constant ρ_f, the pressure and optical projections are not independent. Bernoulli reconstruction gives

$$\delta p_{\mathrm{swirl}} = -\frac{1}{2}\rho_f \|\mathbf{u}_\circlearrowleft\|^2, \tag{281}$$

while the clock/optical closure gives

$$n_\gamma = \left(1 - \frac{\|\mathbf{u}_\circlearrowleft\|^2}{c^2}\right)^{-1/2} = 1 + \frac{\|\mathbf{u}_\circlearrowleft\|^2}{2c^2} + O\left(\frac{\|\mathbf{u}_\circlearrowleft\|^4}{c^4}\right). \tag{282}$$

Eliminating $\|\mathbf{u}_\circlearrowleft\|^2$ yields the parameter-free leading-order locking relation

$$\boxed{\delta n_\gamma = -\frac{\delta p_{\mathrm{swirl}}}{\rho_f c^2}} \tag{283}$$

with validity restricted to the passive Euler regime. Forcing, damping, viscosity, thermal gradients, acoustic drive, plasma response, or explicit Mode-I terms such as $\Gamma_\circlearrowleft \mathbf{u}_{\mathrm{plume}}$ break the assumptions and move the prediction to the research-track transfer model. Thus the refined diagnostic statement is not three fully independent channels, but two locally locked projections plus one nonlocal Poisson/bulk response whenever Eq. (264) is invoked.

With $\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}}$,

$$\frac{1}{\rho_f c^2} = 1.58950008 \times 10^{-11}\,\mathrm{Pa}^{-1}, \tag{284}$$

$$\frac{1}{2}\rho_f v_\circlearrowleft^{*2} = 4.1877439 \times 10^5\,\mathrm{Pa}, \tag{285}$$

$$\delta n_\gamma = 6.6565 \times 10^{-6} \tag{286}$$

at canonical swirl speed, consistent with Eq. (276).

Therefore

$$\boxed{\text{matter stabilization} \neq \text{photon caustic} \neq \text{boundary shell force.}} \tag{287}$$

[CRITICAL NOTE]. This corollary canonizes only the diagnostic separation of response channels. It does not canonize any specific witness claim, device claim, or anti-gravity interpretation. Device-specific excitation factors, optical gains, and shell stiffness parameters remain research-track unless derived from an explicit resonator, stress-closure, or calibrated transfer model.

12 Unified Interpretation

12.1 Overview

Swirl-String Theory (SST) proposes a unified interpretation in which fundamental physical phenomena emerge from the dynamics of a continuous medium supporting stable vortex structures.

This section synthesizes the results of the previous sections into a coherent conceptual framework.

12.2 Ontology of the Theory

[SPECULATIVE] Ontological claim: The fundamental ontology consists of:

- A continuous incompressible medium (Section 4)
- Stable vortex filament configurations (Section 6)
- Circulation as a conserved topological invariant

Interpretation: Particles are not elementary objects, but persistent dynamical structures.

12.3 Mass as Trapped Circulation Energy

From Section 3, and in particular from Eq. (96), the mass sector is organized as the product of a medium scale, a geometric gate, a topological kernel, and a local clock-impedance factor. The canon-level reading is therefore not merely that "energy is stored in rotation," but that persistent inertial scale is assigned only after geometry, topology, and local clocking have all been specified.

$$M_K(x) = \rho_m(x)\, {r_c}^3\, \Pi_K\, \Xi_K\, S^{\circlearrowleft}_{(t)}(x)^{-2}. \tag{288}$$

[SPECULATIVE] Unified interpretation: mass corresponds to localized, self-sustained circulation energy in the medium, modulated by state geometry, topology, and local clock impedance.

12.4 Time as a Relational Quantity

From Section 11 and Section 3, time is not introduced as an independent primitive but as a relational field read from the local state of the medium. The same kinematic quantity that enters the Swirl-Clock also appears in the organizing equation of the mass sector, so the canon treats proper-time modulation and inertial scaling as two projections of one shared substrate state.

$$S_{(t)}^{\circlearrowleft}(x,t) = \sqrt{1 - \frac{\|\mathbf{u}_{\circlearrowleft}(x,t)\|^2}{c^2}}. \tag{289}$$

[SPECULATIVE] Unified interpretation: time is not fundamental, but emerges from the kinematic state of the medium and is read operationally through clock-bearing processes.

12.5 Discreteness without Quantization Postulates

From Section 8:

- Delay introduces temporal nonlocality
- Stability selects discrete modes

[DERIVED] Unified consequence:

> Discrete physical spectra arise dynamically from delay-induced mode selection.

This replaces operator-based quantization with a dynamical mechanism.

12.6 Charge and Circulation

[SPECULATIVE] Provisional bridge only. The canon retains circulation as a structurally important invariant, but it does *not* yet claim a completed derivation of observed gauge charges from circulation data alone. At most, the present document records a working hypothesis that orientation and magnitude of circulation may enter a later charge-assignment program.

12.7 Unification of Interactions

The canon records a *unification program*, not yet a completed unification theorem. In the present document:

- electromagnetic effects are represented by the minimal transverse-field bridge of Section 7,
- gravitational effects are represented by the clock-field and Poisson-limit bridge of Section 11,
- inertial mass is organized by the medium/topology/geometry/clock structure of Eq. (96).

[SPECULATIVE] Programmatic claim: these sectors may ultimately be read as different effective descriptions of one substrate, but their full common derivation remains open.

12.8 Atomic Structure as an Emergent Layer

From Section 9:

- Standard quantum mechanics appears as an effective description
- SST introduces circulation-dependent corrections

[SPECULATIVE] Unified interpretation:

> Quantum mechanics is an emergent effective theory of underlying vortex dynamics.

12.9 Consistency with Spectroscopy

From Section 10:

- Internal modes are suppressed by an excitation gap
- Low-energy physics remains unchanged

[**ORTHODOX**] **Constraint**: The unified picture does not violate known experimental constraints.

12.10 Hierarchy of Description

The theory admits multiple levels:

1. **Microscopic level:** vortex filament dynamics
2. **Mesoscopic level:** delay-induced mode structure
3. **Macroscopic level:** effective quantum description

12.11 Conceptual Summary

The SST framework can be summarized as:

Mass	$\rightarrow$ trapped circulation energy
Time	$\rightarrow$ relational swirl-clock field
Charge	$\rightarrow$ provisional circulation bridge
Quantization	$\rightarrow$ delay-induced stability

12.12 Canonical Statement

All observed physical structure emerges from the dynamics of a continuous medium in which stable vortex configurations, governed by quantized circulation and delay dynamics, produce the phenomena conventionally attributed to particles, fields, and quantum mechanics.

12.13 Scope and Limits

[**IMPORTANT**]:

- The framework is not yet a complete fundamental theory
- Several mechanisms remain to be derived rigorously
- Many results are conditional or phenomenological

12.14 Final Consistency Principle

A valid unified interpretation must preserve all experimentally verified results while providing a deeper structural explanation of their origin.

13 Integration of Prior Canon Versions

13.1 Integration policy (v0.5.12 reintegration)

This block records only material from earlier SST canon layers that can be retained inside the v0.8.1 architecture without weakening the explicit classification discipline of Section 5.

Older canon documents are therefore *not* cited as authorities. Transferable content is rewritten here as current-canon statements, each labeled according to its present status.

13.2 Retained v0.5.12 kernel sectors

The present draft now retains the following v0.5.12 kernel sectors in rewritten form:

- the Chronos–Kelvin invariant and clock–radius transport law (Section 3.12);
- the reparametrized zero-parameter corollary and coarse-graining rule (Section 3.13);
- a minimal Swirl–EM bridge and photon / unknot sector (Section 7);
- the swirl pressure law as the Euler-level force bridge (Section 7.4);
- the analogue-metric / clock-field bridge (Section 11).

13.3 Wave–particle phase rule and measurement bridge

A retained qualitative rule from earlier canon layers is that SST distinguishes two limiting phases:

$$\text{R-phase}: \quad \text{extended, unknotted, radiative,} \tag{290}$$

$$\text{T-phase}: \quad \text{localized, knotted, tangible.} \tag{291}$$

A minimal dynamical bookkeeping law for phase conversion is

$$\partial_t P_T = \Gamma_{R\to T}[\mathcal{K}]\, P_R - \Gamma_{T\to R}[\mathcal{K}]\, P_T, \qquad P_R + P_T = 1, \tag{292}$$

where $\mathcal{K}$ denotes an environment- and coherence-dependent kernel functional. The intended interpretation is that "measurement" corresponds not to an axiomatically imposed collapse, but to a medium-induced transfer of support from the extended R-phase to the localized T-phase.

[ORTHODOX LIMIT]: in weak coupling and after coarse graining, Eq. (292) should reduce to ordinary environment-induced decoherence models [38].

[SPECULATIVE]: the identification of decoherence with an actual R↔T phase transition is SST-specific.

13.4 Gauge and weak-mixing bridge

The retained gauge-level statement is that the effective low-energy algebra should reduce to

$$\mathfrak{g}_{\text{eff}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1), \tag{293}$$

with gauge potentials interpreted as holonomy data of coarse-grained multi-director transport.

A minimal bridge ansatz for the weak mixing angle is

$$\sin^2 \theta_W^{\text{eff}} = \frac{K_Y}{K_Y + K_W}, \tag{294}$$

where K_Y and K_W are effective Abelian and weak-sector stiffness scales. Eq. (294) is not a completed derivation, but it preserves the v0.5.12 idea that weak mixing should descend from a ratio of medium-response coefficients rather than appear as an unstructured free number.

[ORTHODOX]: θ_W is the standard electroweak mixing parameter of the SM [39, 3].

[SPECULATIVE]: Eq. (294) is an SST bridge ansatz only.

13.5 What the v0.5.12 reintegration does not import

The present reintegration does *not* import v0.5.12 cosmogony, engineering subsystems, cosmological-term closures, or highly application-specific protocols into the main canon body. Those remain suitable for appendices or standalone papers until they are more tightly tied to the minimal axiom system.

13.6 Purpose of this integration layer

Version 0.8.1 is not a fresh theory written from zero, but an explicit consolidation of earlier canon lines. This section records what has been retained, what has been renamed or reorganized, and what has been demoted from canon status to research-track status.

13.7 Retained from the early canon layers

From the v0.1.x–v0.3.x line, the present canon retains the primitive closure logic:

- a continuum substrate with effective density ρ_f,
- a core scale r_c fixed by Compton anchoring,
- circulation quantization through $\Gamma_0 = 2\pi r_c \, v_{\circlearrowleft}^{*}$,
- the interpretation of matter as stable organized structures rather than point primitives.

These elements remain non-negotiable because later sectors depend on them directly.

13.8 Retained from the middle canon layers

From the v0.4.x–v0.6.x line, the canon retains three durable structures:

1. the Rosetta strategy for mapping SST statements into orthodox language,
2. the photon/torsion sector as the minimal radiation bridge,
3. delay-induced mode selection as the main route to discreteness without operator postulates.

These are now distributed across Sections 8, 11, and 12 rather than treated as isolated notes.

13.9 Retained from the late canon layers

From the v0.7.x line, v0.8.1 retains the strongest structurally reusable sectors:

- the Swirl-Clock as the canonical relational-time field,
- the atomic bridge as a benchmark consistency layer rather than a complete atomic derivation,
- the spectroscopic gap logic as a hard consistency filter,
- the topology-first mass program, with topology separated from thermodynamic or radiative dressing,
- the EFT-style foliation and gauge-emergence program as a long-range development track.

13.10 Notation and density cleanup

Earlier canon versions sometimes conflated background density, mass-equivalent density, and saturated envelope-equivalent density. The present version adopts the cleaned hierarchy

$$\begin{aligned} \rho_f &\quad \text{effective fluid density,} \\ \rho_E &\quad \text{swirl energy density,} \\ \rho_m = \rho_E/c^2 &\quad \text{mass-equivalent density,} \\ \rho_{\text{horn}}^{\text{eff}} &\quad \text{canonical saturated envelope-equivalent density,} \\ \rho_{\text{core}} &\quad \text{legacy alias, deprecated unless explicitly defined.} \end{aligned} \tag{295}$$

Any older passage that identifies a single symbol with more than one of these roles is superseded by the present separation.

13.11 Topology-first reinterpretation of layers

A major consolidation from the recent lepton and knot-mass program is the separation between topology and layering. Knot type is retained as the primary identity label, while Golden-layer or thermal ladder indices are treated as dressing or coherence variables rather than species-defining labels. This prevents the canon from over-reading numerical layer fits as ontological identities.

13.12 Status of optional sectors

The following sectors are retained as important but non-primitive research layers:

- full gauge emergence and charge assignments,
- dual-vacuum phenomenology and Unruh-echo interpretation,
- complete many-electron atomic closure,
- full weak-sector and collider phenomenology,
- explicit topology-to-particle dictionary beyond the best-supported benchmark assignments.

They remain in the canon as organized programs, not as closed theorem-level results.

13.13 Integration principle

The purpose of v0.8.1 is therefore not to maximize novelty per page, but to minimize duplication and to expose the dependency graph of the theory. Whenever two earlier canon fragments make the same structural claim in different language, the present version keeps one canonical form and demotes the others to historical context.

13.14 Research-track companion

The sectors that are useful for continuity but too long or too speculative for a minimal main-canon reading are also collected in the companion file `canon-0.8.1-research-track.tex`. That excerpt includes the thermodynamic-radiation interface, the minimal-coupling QED analogy, the ribbon-invariant chirality refinement, the particle-candidate mapping layer, the hydrodynamic exchange / Pauli-barrier estimate, the numerical-benchmark policy block, the gauge-sector roadmap, the four topology/stability appendices on non-Abelian protection, reconnection decay, fractional torus-phase winding, and topological-fluid mechanics, and the *dark-sector and galactic canonization execution package* (frozen topology toolkit, benchmark protocol template, Euler-pressure anchor details, galactic branch gate, and probe–equation map). The integration subsections below retain the full in-context treatments with labels and cross-references; the companion file provides a portable standalone copy. Editorial separation does not reject these sectors; it prevents the core narrative from over-claiming closure.

13.15 Research-track knot-energy normal form

Status. [RESEARCH TRACK / NON-DERIVED SCREENING FUNCTIONAL]. The main canon does not assume a closed universal knot-energy functional beyond the calibrated constant chain and the explicitly derived hydrodynamic energy scales. For classification and numerical screening, the research track uses the dimensionless normal form

$$\frac{E_{\text{eff}}[K]}{E_0} = \alpha_C C(K) + \alpha_L L(K) + \alpha_H \widehat{\mathcal{H}}(K) + \alpha_G I_G(K) + \cdots, \tag{296}$$

where $C(K)$ denotes a crossing/contact-complexity proxy, $L(K)$ a ropelength or relaxed-length invariant, $\widehat{\mathcal{H}}(K)$ a nondimensional helicity measure, and $I_G(K)$ a scalar index extracted from a possible non-Abelian topological charge sector $Q_G(K)$. The map

$$I_G : Q_G(K) \longrightarrow \mathbb{R}_{\geq 0} \tag{297}$$

is required because $Q_G(K)$ is generally a sector label, not a real-valued energy contribution. Equation (296) is therefore a research-track screening functional, not a derived SST Hamiltonian or mass theorem.

13.16 Dark-sector and galactic canonization status (v0.8.x)

This block records promotion labels for the dark-sector / galactic program consistent with the internal canonization plan. Detailed freezes, the mandatory benchmark-protocol checklist, the galactic profile branch gate (Branch A: legacy-profile refine with benchmark traceability), and the probe falsification table live in `canon-0.8.1-research-track.tex` (Section titled `Dark-sector and galactic canonization execution package`).

Table 2: Master tracker: dark-sector and galactic program (v0.8.x).

Item	Prior state	Target	Status after Stage 0–4 freeze
Dark-sector Euler pressure anchor	BRIDGE	CANON	CANON-CANDIDATE (derivation, assumptions, limits, falsifiers frozen; see Section 7.4)
Dark taxonomy	HOLD	CANON	BRIDGE (symmetry-first classifier + filters; Section 6.6)
Galactic swirl law	BRIDGE	CANON	BRIDGE (Branch A closure; companion execution package)
Experimental probes	HOLD	CANON-BRIDGE	CANON-BRIDGE (equation / null-result map; companion)
Benchmark / reproducibility	BRIDGE	CANON	CANON-CANDIDATE (protocol template frozen; repository-wide reuse pending)
Topological reference identities	CANON-CANDIDATE	CANON	CANON-ready scope/notation (Appendix B)

[DERIVED] Stage 5 review: cross-item dependencies are respected (pressure anchor and benchmark layer precede full galactic closure; taxonomy strengthens other sectors). No sector is promoted on fit quality alone; each carries explicit falsifiability or reproducibility hooks as in Table 2 and the companion execution package.

13.17 Integration Policy

This section records only material from prior internal SST development that can be retained inside the v0.8.1 architecture *without* weakening the epistemic classification framework of Section 5.

Rule. Older canon documents are *not* cited as external authorities. Instead, transferable content is rewritten here as current-canon statements, each labeled as [**ORTHODOX**], [**DERIVED**], or [**SPECULATIVE**].

13.18 Selective Recovery from Pre-v0.7 Canon Layers

The present draft absorbs only those legacy ingredients that can be rewritten in the current epistemic format without appealing to earlier canon versions as authorities.

Analogue line-element bridge. Legacy SST drafts recorded a schematic analogue line element and a metric–clock identification of the form

$$ds^2 = -c^2\, S^{\circlearrowleft}_{(t)}(x)^2\, dt^2 + \delta_{ij}\, dx^i\, dx^j\,, \tag{298}$$

$$g_{00}(x) = -S^{\circlearrowleft}_{(t)}(x)^2. \tag{299}$$

Recovered legacy sectors include:

- the finite-core atomic bridge, Eqs. (195)–(197);
- the analogue line-element bridge, Eqs. (298)–(299);
- compact research-track interfaces to thermodynamic radiation and minimal-coupling QED analogies;
- topological chirality refinements based on standard ribbon invariants.

Legacy material not imported includes version history, cosmogonic narrative, and phenomenological galaxy-scale laws that are not yet sufficiently integrated with the present minimal axiom system.

13.19 Thermodynamic Radiation Interface

A recurring legacy idea is that a local swirl-energy density may define an effective temperature field and thereby an emergent radiation sector.

Let

$$T_{\text{eff}} = T_{\text{eff}}\Big(\rho_E,\, \omega,\, S_{(t)}\Big) \tag{300}$$

denote an unspecified constitutive relation. A minimal legacy interface then asks whether Planck-type spectra can emerge from a medium whose local excitation scale is set by T_{eff}.

[**ORTHODOX**]: blackbody spectra are governed by Planck's law once a valid temperature variable is defined.

[**SPECULATIVE**]: SST does not yet derive Eq. (300), so this sector remains a research program rather than canonically closed physics.

13.20 Minimal Coupling Interface to Quantum Electrodynamics

Early research notes repeatedly explored whether an effective vector potential could be built from circulation variables. The most conservative placeholder takes the form

$$\nabla \to \nabla - i\,\frac{m}{\hbar}\,\mathbf{A}_{\text{eff}}, \qquad \mathbf{A}_{\text{eff}} = \chi\,\mathbf{v}, \tag{301}$$

where χ is a scalar weight and $\mathbf{v}$ is the local carrier velocity field.

[ORTHODOX]: minimal coupling is the standard gateway from free to gauge-coupled wave dynamics.
[SPECULATIVE]: Eq. (301) is only an analogy template unless a full gauge-covariant derivation is supplied.

13.21 Ribbon-Invariant Chirality Refinement

The older canon layers also suggested using the Călugăreanu–White–Fuller relation

$$Lk = Tw + Wr \tag{302}$$

as a refinement of the knot-based matter/antimatter dictionary.

[ORTHODOX]: Eq. (302) is a standard geometric identity for framed curves and ribbons.
[DERIVED BOOKKEEPING]: once a framed curve is specified, the identity separates writhe and twist contributions.
[SPECULATIVE]: the claim that writhe-dominated and twist-dominated realizations are dynamically stable physical isomers, or map to particle/antiparticle sectors, remains a model hypothesis.

13.22 Framed-Tube Ontology and the Attachment Gate

The particle carrier in SST is not a bare centerline knot alone. The canonical object is a framed, phase-bearing tube,

$$\mathcal{K}_{\text{SST}} = (K,\, \mathcal{F},\, \Gamma,\, \theta_{\text{core}}), \tag{303}$$

where K is the centerline knot or link, $\mathcal{F}$ is a ribbon/framing, Γ is circulation, and θ_{core} is a material or order-parameter phase attached to the core.

The current Attachment Lemma is the statement that a local Compton/core cycle can be promoted to a globally closed framed-tube holonomy,

$$\frac{1}{2\pi}\oint_K d\theta_{\text{core}} = \chi, \qquad \chi \in \mathbb{Z}. \tag{304}$$

A compensated form allows local swirl-clock redistribution while preserving the exact global integer,

$$\chi_{\text{total}} = \chi_{\text{local}} + \chi_{\text{attachment}} = \chi, \qquad \delta Q = 0. \tag{305}$$

[CANON / OPEN GATE]: Eq. (304) is not yet a theorem of the Euler–SST dynamics. It is a load-bearing gate. Integer energy selection alone is insufficient; exact holonomy is required for canon status.

[CANON / NOTATION]: the framing phase follows the Compton angular cycle ω_c, while the associated local vorticity scale is $2\omega_c$. The factor of two must not be absorbed into χ.

Table 3: Minimal knot-class and analogue summary retained in v0.8.3, with current semantic corrections.

Carrier / composition	Candidate role	Status / interpretation
3_1 trefoil	electron / positron sector	baseline chiral lepton calibration; not used for fractional quark charge
$T(2,5)$ or equivalent higher torus class	muon	lepton hierarchy candidate
$T(2,7)$ or equivalent higher torus class	tau	lepton hierarchy candidate
twist/clasp knots, quasi-unknot loops	quark-like defects	research-track fractional boundary winding candidates
$5_2 + 5_2 + 6_1$ and nearby composites	baryon candidates	legacy mass-functional dictionary, still under audit
triple-gear locked analogue	proton-like charged baryon analogue	mechanical model for nonzero exterior collective rotation
Borromean-type analogue	neutron-like neutral baryon analogue	mechanical/topological model for zero exterior charge with internal triple linking

13.23 Particle-Candidate Mapping Layer

[**SPECULATIVE**] Working dictionary: a minimal knot/composition mapping retained from earlier SST work is summarized in Table 3. This table is organizational rather than definitive; it records current topological assignments and mechanical analogues used by the mass and taxonomy program.

[**CRITICAL NOTE**]: Table 3 is a working dictionary only. Electric charge is not identified with raw Hopf helicity or pairwise linking number. The safer current placeholder is a boundary-winding charge variable q_∂, while knot/link data supply mass, chirality, confinement, and internal topological structure.

[**CANON / DECISION**]: The trefoil assignment is retained for the electron/positron sector only. Fractional values such as 2/3 belong, if anywhere, to the twist-knot/quark-like boundary-winding research track rather than to the trefoil-electron calibration.

13.24 Hydrodynamic Exchange and the Pauli Barrier

[**ORTHODOX**] Fluid-mechanical input: for two thin filamentary loops C_1 and C_2 in an incompressible medium, the Biot–Savart interaction energy is [6]

$$E_{\text{int}} = \frac{\rho_f \Gamma_1 \Gamma_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|}. \tag{306}$$

[**ORTHODOX**] Regularized filament-energy template: if two identical electron-candidate loops are forced into the same spatial state, a short-distance cutoff a_{cut} gives the overlap penalty

$$V_{\text{Pauli}}(\mathbf{d}; a_{\text{cut}}) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \oint\oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{\sqrt{\|\mathbf{r}_1(\theta_1) - \mathbf{r}_2(\theta_2) + \mathbf{d}\|^2 + a_{\text{cut}}^2}}, \tag{307}$$

with maximal-overlap scale

$$V_{\text{Pauli}}^{\max}(a_{\text{cut}}) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \left(\frac{L}{a_{\text{cut}}}\right) \mathcal{O}(1). \tag{308}$$

Using the canonical values

$$\begin{aligned}\rho_f &= 7.0\times10^{-7}\ \mathrm{kg\,m^{-3}},\\ r_c &= 1.40897017\times10^{-15}\ \mathrm{m},\\ \Gamma_0 &= 2\pi r_c v_{\circlearrowleft}^{*} \approx 9.6836\times10^{-9}\ \mathrm{m^2\,s^{-1}},\end{aligned} \tag{309}$$

and $L\sim 2\pi a_0$ at the hydrogenic scale, the legacy benchmark corresponds to the cutoff calibration $a_{\mathrm{cut}}=r_c$:

$$V_{\mathrm{Pauli}}^{\max}(a_{\mathrm{cut}}=r_c)\sim 1.23\times10^{-18}\ \mathrm{J}\sim 7.6\ \mathrm{eV}. \tag{310}$$

Away from this cutoff calibration the scaling is

$$V_{\mathrm{Pauli}}^{\max}(a_{\mathrm{cut}})\sim V_{\mathrm{Pauli}}^{\max}(r_c)\left(\frac{r_c}{a_{\mathrm{cut}}}\right). \tag{311}$$

[CRITICAL NOTE] The cutoff a_{cut} is not the resolved physical tube radius a_{core}. The benchmark choice $a_{\mathrm{cut}}=r_c$ is retained only as a legacy ultraviolet regularization scale. It must not be read as $a_{\mathrm{core}}=r_c$, and it is not a theorem-level derivation of Pauli exclusion from ideal Euler dynamics.

As written, Eq. (310) is an order-of-magnitude, cutoff-calibrated benchmark attached to the canonical constants. A physical resolved-core calculation must provide a_{core}, the tube profile, and the relation between that profile and a_{cut} independently.

13.25 Numerical Benchmark Layer

[DERIVED] Computational-program requirement: earlier SST work contained explicit benchmark tables for particle and atomic masses. In v0.8.1, the benchmark layer is retained as a reproducibility requirement rather than as a standalone authority claim.

Table 4: Benchmark summary structure to be populated from canonical scripts and archived CSV output.

Particle	m_{exp} (MeV)	m_{SST} (MeV)	rel. error	mode	class
e	0.51099895	0.51099895	0	predictive/closure	3_1
μ	105.6583745	...	...	predictive	candidate torus class
τ	1776.86	...	...	predictive	candidate torus class
p	938.272088	...	...	predictive or closure	composite class
n	939.565420	...	...	predictive or closure	composite class

[CRITICAL NOTE] A future v0.8.x benchmark appendix should include:

- the exact script or package path used for generation,
- archived CSV outputs,
- mode definitions (predictive vs. exact-closure),
- uncertainty propagation on canonical constants and geometric factors,
- theorem-style pass/fail metadata (including tail-quality checks, postchecks, and extrapolation summaries),
- explicit separation between practical best-estimate closure and theorem-level closure,
- normalization of legacy constant names into SST notation (ρ_f, r_c, Γ_0) while preserving numerical values.

This requirement is included to keep the benchmark layer falsifiable and reproducible, rather than merely illustrative.

13.26 Gauge-Sector Roadmap

[**SPECULATIVE**] Structural roadmap: a retained minimal statement of the gauge program is:

$$\mathfrak{g}_{\text{eff}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1), \tag{312}$$

with effective gauge phases organized as holonomy data of multi-director transport.

A minimal phase-based representation is

$$A^a_\mu(x) = \partial_\mu \theta^a(x), \tag{313}$$

and clock coupling acts as a common phase reference,

$$\theta^a(x) \mapsto \theta^a(x) + q^a \chi(x) \quad \Rightarrow \quad A^a_\mu \mapsto A^a_\mu + q^a \partial_\mu \chi. \tag{314}$$

[**CRITICAL NOTE**] The gauge layer is retained in v0.8.1 only as a roadmap statement. Running couplings, anomaly cancellation, and realistic mixing matrices remain open work.

Research-track contact-channel guard. Any QCD-like factor-nine or confinement-sheet tension claim must be routed through the resolved colored contact-channel rank test in Sec. M.5.10. In the core canon this remains [**RESEARCH TRACK / MATCHING DIAGNOSTIC**]: the number is not inferred from $\dim SU(3) = 8$, nor from a fitted hadronic scale, but only from a future positive-rank condition on the finite-thickness contact-stress matrix.

14 Discussion and Limits

14.1 Status of the recovered v0.5.12 sectors

The material reintroduced here does not have uniform epistemic status.

- The Chronos–Kelvin and pressure-law sectors are orthodox fluid-mechanical structures with SST reinterpretations.
- The analogue-metric and clock-field EFT layers are orthodox comparison tools, but only partially derived within SST.
- The R/T measurement bridge and weak-mixing stiffness ansatz remain explicitly speculative.

14.2 What v0.8.1 now recovers from v0.5.12

After the present patch, v0.8.1 no longer omits the main v0.5.12 kernel sectors that are structurally compatible with the current architecture:

- Chronos–Kelvin / clock transport,
- zero-parameter reparameterization discipline,
- Swirl–EM and photon / unknot bridge,
- swirl pressure law,
- analogue metric and clock-field gravity bridge,
- R/T measurement bridge,
- weak-mixing bridge ansatz.

14.3 Open items that remain outside the present closure

The following remain open and are *not* upgraded to full canonical closure in this draft:

- a first-principles derivation of the full SM gauge sector;
- a completed EWSB derivation tied to the canon constants;
- a rigorous derivation of the R/T measurement kernel from continuum dynamics;
- full CANON promotion of the dark-sector taxonomy and galactic closure beyond the current BRIDGE labels (Section 13.16).

14.4 What is established at canon level

The present document supports the following as canon-level structures:

- the primitive constant chain $(\rho_f, v_{\circlearrowleft}, \omega_c) \to R_{\text{horn}} \equiv r_c \to \Gamma_0$,
- the Swirl-Clock as the coarse-grained relational time factor,
- delay-induced mode selection as the preferred discreteness mechanism,
- the atomic bridge and spectroscopic sectors as compatibility filters,
- the topology-first program for mass organization.

These claims are not equally proven, but they are the sectors on which the rest of the canon is structurally built.

14.5 What remains closure-level or conjectural

Several central SST ambitions remain open:

1. a rigorous derivation of full particle spectra from topology alone,
2. a first-principles derivation of charge assignments and gauge couplings,
3. a complete many-body derivation of shell filling and exclusion,
4. a full demonstration that delay plus topology reproduces the observed quantum rule set without importing it.

The canon records these as active programs rather than as settled conclusions.

14.6 Compatibility limits

Any viable SST realization must remain compatible with:

- atomic spectroscopy,
- relativistic effective kinematics,
- precision electroweak data where an SST bridge is claimed,
- the absence of large low-energy deviations from standard quantum mechanics.

This is why the canon repeatedly imposes suppression, weak-coupling, and low-energy recovery conditions.

14.7 Minimal falsifiers

The present canon would be materially weakened by any of the following:

- failure of the constant chain to remain dimensionally and numerically self-consistent,
- absence of any workable suppression mechanism for internal modes at atomic energies,
- inability to formulate a non-circular route from delay dynamics to discrete stable spectra,
- systematic contradiction between any SST atomic bridge and precision spectroscopy.

Conversely, strong support would come from a successful theorem ladder showing how discrete spectra, topological stabilization, and low-energy effective quantum structure arise from one shared substrate model.

14.8 Methodological limit

The canon is deliberately hybrid in style. Some sectors are written as exact closures, some as effective models, and some as programmatic roadmaps. This is a feature rather than a flaw, provided the epistemic status of each claim remains explicit. The document should therefore not be judged as though every section were intended to be a stand-alone journal theorem.

14.9 Near-term priorities

The next canon-strengthening tasks are clear:

1. complete the theorem ladder for the mass and clock sectors,
2. expand numerical validation blocks using the canonical constants,
3. refine the atomic bridge into sharper benchmark observables,
4. decide which optional sectors belong in the core canon and which should migrate to separate technical notes or the research-track companion.

14.10 Final limit statement

SST is presently best understood as a structured unification program with several strong internal bridges and several open derivational gaps. The canon is therefore mature enough to organize the theory coherently, but not yet so mature that every phenomenological sector should be presented as theorem-complete.

14.11 Status of Legacy Recoveries

The additional material imported here from older canon layers does not have uniform status.

- The analogue-metric bridge and ribbon identity are mathematically orthodox tools.
- Their SST interpretation is partially derived but not fully closed dynamically.
- The thermal-radiation and QED-interface sectors remain explicitly research-track.

This section therefore serves as a controlled retention layer, not as a blanket validation of all legacy material.

Accordingly, these recoveries should be read as structured extensions of the minimal core, not as fully promoted final doctrine.

14.12 What v0.8.1 Now Retains

After integration, the present Canon retains the following prior-development sectors in current-canon form:

- a particle-candidate topological dictionary (Section 13.23);
- a hydrodynamic exchange-energy estimate for fermionic exclusion (Section 13.24);
- a benchmark / reproducibility layer (Section 13.25);
- a minimal gauge-sector roadmap (Section 13.26);
- a clock-field EFT bridge and Poisson gravity limit (Section 11).

14.13 What is Deliberately Not Claimed

The present document still does *not* claim:

- a completed first-principles derivation of the Standard Model;
- a unique finalized knot-to-particle dictionary;
- a closed proof of the exclusion barrier beyond the current scaling benchmark;
- a fully renormalized relativistic field theory for the gauge and clock sectors.

These omissions are deliberate. v0.8.1 is intended to remain logically strict even when phenomenological sectors are reintroduced.

14.14 Canon Rule Going Forward

Any further reintegrated material must satisfy all of the following:

1. it can be written without citing older internal canon versions as authorities;
2. it can be assigned an explicit epistemic label;
3. it includes either an internal derivation path or a reproducible benchmark route;
4. it preserves compatibility with the orthodox limits recorded earlier in the document.

Appendices

A Historical Notebook and Proto-Canon Provenance Register

A.1 Purpose and authority

This register preserves the traceable origin of active SST questions without assigning evidential authority to age or priority. Notebook pages and early synthesis documents are primary historical sources; current definitions, status labels, and normative claims are determined only by the MAIN Canon. The complete page-level inventory is maintained as a supporting ledger, while this appendix records the stable source classes and the conclusions that have survived into the present architecture.

Source class	Stable identifier	Canonical use
Primary notebook	`NB2012-Pxxx`	Earliest surviving sketches, equations, and local reasoning. Read as `[ORIGIN]`, not as current derivation.
Partial later notebook	`NB2013-Pxxx`, `NB2014-Pxxx`	Development of transition geometry, toroidal fields, relative rotation, impedance, circulation, and vorticity.
Proto-Canon	`DOC2013-CANON-001`	First synthesis of excitation speed, rotating mass, the fine-structure speed ratio, and the capacitance route to the Planck relation.
Modern computational project	`PROJ2026-KNOTPLOT-RR-001`	KnotPlot seeds, topology sidecars, Ridgerunner constrained polishes, resampling, and near-ideal certification provenance.
Modern Canon	versioned `SST_CANON-vX.Y.Z`	Sole normative source for active definitions, theorem status, calibration, predictions, and falsifiers.

A.2 High-priority source concordance

Source anchor	Historical content	Current SST correspondence
NB2012-P002/P006/P017	Free and bound wave structures; rotating matter; unification aim.	Free link-field packet versus bound vortex/link mode; mass as inertial response of a closed carrier.
NB2012-P021-P025	Lorentz transformations, speed ratios, transition geometry, and oscillator dynamics.	Operational Alice–Bob spacetime; Swirl-Clock; action-level propagation-speed target.
NB2012-P044-P055	Circulation, vortices, toroidal cores, and time/mass associated with rotation.	Euler-vorticity sector, finite-thickness carrier geometry, and relational clock response.
NB2012-P069-P071	Oscillator propagation speed, capacitance, vacuum response, and constant dependencies.	Link inertia/stiffness, constitutive response, impedance, and non-circularity ledger.
NB2013-P010/P019/P045	Transition dynamics, caution about numerical closure, and relative-rotation/time interpretation.	Transition-action research programme, calibration guard, and observer-dependent kinematics.
DOC2013-CANON-001	First coherent constant-chain and excitation model.	Historical bridge only; inputs and algebra are audited before any prediction claim.
PROJ2026-KNOTPLOT-RR-001	Candidate generation, topology verification, constrained tightening, and mesh export.	Bound-mode geometry certification and input domain for a future response/impedance operator.

A.3 Preservation and correction policy

Historical formulas are preserved in their original role and may be quoted under `[ORIGIN]` or `[HISTORICAL BRIDGE]`. Algebraic, dimensional, or physical corrections are written beside them rather than silently substituted backward into the source. A historical capacitance formula and its impedance rewriting count as one dependency path. Promotion requires an independent action-level derivation, explicit assumptions, and a falsification test.

B Topological reference toolkit

The following orthodox items form the mandatory internal mathematical toolkit cited by topology-first SST sectors. Statements are standard; SST-specific mappings appear in the main text with explicit status labels.

1. Gauss linking integral and linking number for disjoint closed curves.
2. Helicity $\mathcal{H} = \int \mathbf{v} \cdot (\nabla \times \mathbf{v})\, dV$ in ideal (inviscid) fluid flow and its conservation in the appropriate limit.
3. Thin-tube reduction of helicity to flux–linkage form in the standard filament limit.
4. Hopf invariant for localized field configurations (degree–type integral) where used as a topological density.
5. Basic torus-knot invariants as needed in the mass program (e.g. minimal crossing number, braid index, genus) with fixed naming consistent with Rolfsen notation in the main text.

6. Călugăreanu–White–Fuller relation $Lk = Tw + Wr$ for framed ribbons (already recalled as Eq. (302) in the integration layer).

[ORTHODOX]: each item is classical mathematics [36, 29, 6]. **[DERIVED]**: scope and notation for v0.8.x are frozen to this list plus house symbols $(\rho_f, \Gamma, r_c, \mathcal{H})$ without parallel aliases.

C Full Derivations

C.1 Canonical horn-radius constant chain

The present canon uses the closure chain

$$r_e = \frac{\alpha\hbar}{m_e c}, \tag{315}$$
$$r_c = \frac{r_e}{2}, \tag{316}$$
$$\omega_c = \frac{m_e c^2}{\hbar}, \tag{317}$$
$$v_\circlearrowleft = \omega_c r_c, \tag{318}$$
$$\Gamma_0 = 2\pi r_c v_\circlearrowleft. \tag{319}$$

Substituting $r_c = \alpha\hbar/(2m_e c)$ into $v_\circlearrowleft = \omega_c r_c$ gives

$$v_\circlearrowleft = \left(\frac{m_e c^2}{\hbar}\right)\left(\frac{\alpha\hbar}{2m_e c}\right) = \frac{\alpha}{2}c, \tag{320}$$

hence the canonical ratio

$$\eta_0 = \frac{v_\circlearrowleft}{c} = \frac{\alpha}{2}. \tag{321}$$

C.2 Horn-Envelope Density Normalization

The electron rest-energy normalization may be written as an effective horn-envelope density relation,

$$\rho_{\text{horn}}^{\text{eff}} \equiv \frac{m_e c^2}{2\pi v_\circlearrowleft^{*\,2} R_{\text{horn}}^3}. \tag{322}$$

This relation is dimensionally exact,

$$\left[\frac{m_e c^2}{v^2 r^3}\right] = \mathrm{kg\,m^{-3}}, \tag{323}$$

but its interpretation is now restricted.

[CANON / SEMANTIC RULE]: Eq. (322) is an effective horn-envelope or normalization density. It is not a measured local material density of the resolved vortex tube. Any resolved-tube model must use a_{core}, a profile function, or ropelength/finite-thickness data rather than silently identifying $a_{\text{core}} = r_c$.

[CANON / MASS-FUNCTIONAL NOTE]: Existing mass-functionals that use $\rho_{\text{core}} {r_c}^3$ should be read as horn-envelope normalizations unless a separate local-core profile is explicitly introduced.

C.3 Geometric gate identity

Using the Compton wavelength $\lambda_c = h/(m_e c)$ and the horn-radius closure $r_c = R_{\text{horn}} = \alpha\hbar/(2m_e c)$, one obtains

$$\frac{\lambda_c}{\pi r_c} = \frac{h/(m_e c)}{\pi\alpha\hbar/(2m_e c)} = \frac{2h}{\pi\alpha\hbar} = \frac{4}{\alpha}. \tag{324}$$

This is the canonical geometric shielding factor used throughout the late mass-functional papers.

C.4 Useful numerical anchors

With the canonical values

$$v_{\circlearrowleft} \approx 1.09384563 \times 10^6 \ \mathrm{m\,s^{-1}}, \tag{325}$$

$$r_c \approx 1.40897017 \times 10^{-15} \ \mathrm{m}, \tag{326}$$

$$\Gamma_0 \approx 9.6836 \times 10^{-9} \ \mathrm{m^2\,s^{-1}}, \tag{327}$$

one has the Compton turnover time

$$\tau_{\text{circ,c}} = \frac{2\pi r_c}{v_{\circlearrowleft}^*} = \frac{2\pi}{\omega_c}, \tag{328}$$

and the geometric gate

$$\frac{\lambda_c}{\pi r_c} \approx 5.48 \times 10^2. \tag{329}$$

D Delay Mathematics

D.1 Fixed-point condition for locked modes

Starting from the delayed phase equation

$$\dot{\phi}(t) = \omega_0 + \kappa \sin\big(\phi(t - \tau_{\text{circ}}) - \phi(t)\big), \tag{330}$$

seek a rigidly rotating solution

$$\phi(t) = \Omega t + \phi_\star. \tag{331}$$

Substitution gives the algebraic mode condition

$$\Omega = \omega_0 - \kappa \sin(\Omega\tau_{\text{circ}}), \tag{332}$$

where the minus sign follows from $\sin[-\Omega\tau_{\text{circ}}] = -\sin(\Omega\tau_{\text{circ}})$. This admits multiple branches labelled by the effective phase winding across the delay interval.

D.2 Linear stability of a locked branch

Let

$$\phi(t) = \Omega t + \epsilon(t), \qquad |\epsilon| \ll 1. \tag{333}$$

Linearizing yields

$$\dot{\epsilon}(t) = \kappa\cos(\Omega\tau_{\text{circ}})\big(\epsilon(t - \tau_{\text{circ}}) - \epsilon(t)\big). \tag{334}$$

Using the exponential ansatz $\epsilon(t) \propto e^{\lambda t}$ gives the characteristic equation

$$\lambda = \kappa\cos(\Omega\tau_{\text{circ}})\left(e^{-\lambda\tau_{\text{circ}}} - 1\right). \tag{335}$$

Stability requires $\mathrm{Re}(\lambda) < 0$ for all nontrivial roots. Thus discrete spectra in SST are not arbitrary: only those branches satisfying both the fixed-point equation and the stability inequality survive dynamically.

D.3 Small-delay expansion

For $|\lambda\tau_{\mathrm{circ}}| \ll 1$,

$$e^{-\lambda\tau_{\mathrm{circ}}} = 1 - \lambda\tau_{\mathrm{circ}} + \mathcal{O}((\lambda\tau_{\mathrm{circ}})^2), \tag{336}$$

so the characteristic equation becomes

$$\lambda \approx -\kappa\tau_{\mathrm{circ}} \cos(\Omega\tau_{\mathrm{circ}})\, \lambda. \tag{337}$$

Nontrivial damping therefore requires the effective factor $\kappa\tau_{\mathrm{circ}} \cos(\Omega\tau_{\mathrm{circ}})$ to have the correct sign. This provides a simple branch-selection criterion at weak delay.

D.4 Interpretive consequence

The important canon point is that discreteness is produced in two stages:

$$\text{delay} \longrightarrow \text{multiple algebraic branches} \longrightarrow \text{stability filter}. \tag{338}$$

Quantization is therefore not inserted by hand; it is the residue of a dynamical selection problem on a delayed closed carrier.

E Conversation-Derived Insights

E.1 Role of this appendix

This appendix records recurrent insights that emerged during the project-level development of the canon and are useful for guiding future work, even when they do not yet qualify as theorem-level results.

E.2 Topology first, layers second

A central refinement of the recent canon is the separation

$$\text{particle identity} \longleftrightarrow \text{topology}, \qquad \text{excitation or dressing state} \longleftrightarrow \text{layer index}. \tag{339}$$

This prevents Golden-layer fits from being mistaken for a species ontology and keeps the topology program structurally cleaner.

E.3 Spectral origin of $\zeta(3)$

The project discussions repeatedly converged on the conclusion already encoded in Section 6.4: the appearance of $\zeta(3)$ is best interpreted as a spectral invariant of a cubic-weighted closed-carrier mode tower, not as a direct invariant of threefold geometry by itself.

E.4 Atomic bridge status discipline

Another recurring project insight is that the atomic bridge is strongest when treated as a benchmark layer with explicit status tags. Hydrogenic consistency, branch-sensitive couplings, and the spectroscopic gap form a robust bridge; shell filling, exclusion, and full heavy-atom structure remain open programs rather than completed derivations.

E.5 Master-equation viewpoint

A major organizational gain of the recent conversations is the recognition that many SST sectors can be read as different projections of one shared structure:

$$\text{medium scale} \times \text{topology} \times \text{geometry} \times \text{clocking}. \tag{340}$$

Even when this is not written as a single canonical equation inside the present draft, it remains a useful guide for reducing duplication across the mass, time, and thermodynamic sectors.

E.6 State-transition taxonomy from the trefoil-closure discussion

The present conversation also fixes a useful state-transition taxonomy:

$$\text{smooth deformation/binding} : K = 3_1 \text{ preserved}, \tag{341}$$

$$\text{R-to-T closure/nucleation} : K \text{ created at a boundary event}, \tag{342}$$

$$\text{annihilation or weak capture} : K \text{ converted with compensation}. \tag{343}$$

This taxonomy prevents ordinary atomic binding, non-Euler closure, and true particle conversion from being conflated.

E.7 Dual-vacuum and photon sector status

The project-level discussions also clarified a canon policy: the photon/torsion bridge and the dual-vacuum phenomenology are valuable and fertile, but they should not silently override the simpler core canon. They are best treated as phenomenological extensions built on top of the primitive constant chain and Swirl-Clock structure.

E.8 Canon edition notes

E.8.1 v0.8.23

v0.8.23 integrates the Author's Origin Note and the Genesis/provenance architecture, records the rediscovered capacitance–impedance path with an explicit same-route guard, adds the Historical Notebook and Proto-Canon Provenance Register (Appendix A), inserts the MAIN-Canon interface to relational link torsion (R.63), restores the minimal compact link-field action and its zero-legacy speed target in the research track, and records the KnotPlot–Ridgerunner–SSTcore/VortexLab provenance and finite-thickness certification ladder on top of v0.8.22.

E.8.2 v0.8.22

v0.8.22 elevates special relativity to an operational founding SST input via vortex clocks, rods, and luminal R-phase torsion pulses (2.4, 4.8, 36), records Alice–Bob radar/Lorentz maps and a non-circular c-emergence ladder in the research track (864, 866), and adds the Quasinormal Swirl Spectroscopy (QSS) spectral-robustness programme (P.6, 724, 735) on top of v0.8.21.

E.8.3 v0.8.21

v0.8.21 adds the eight-source research appendices and matching main-canon guards: spherical/shell curl spectrum and helicity capacity (6.5, M.7), coarse writhe/helicity upper bounds (M.8), Gehring link-criticality and continuum contact measure (M.9), bounded-domain Biot–Savart operator admissibility (3.4, M.10), five-sector Hodge topology and harmonic circulation (3.5, M.11), cohomological helicity and mapping-class/Dehn-twist jumps (E.9, M.4.2), isoperimetric domain-spectrum and enstrophy guards (E.9, M.4.4), and Ridgerunner polygonal-to-smooth certification (E.9, M.5.8) on top of v0.8.20. The operational-spacetime clarification further records special relativity as a founding SST input: vortex clocks, vortex rulers, and R-phase torsion pulses reconstruct one local Lorentz structure, while the derivation of universal monometricity from substrate microdynamics remains open.

E.8.4 v0.8.20

v0.8.20 adds Chronos–Kelvin topological hitting conditions with a first-hitting kink-versus-reconnection split and Kairos boundary ledger (M.2.5, 488), a contact-pressure saturation and Swirl-Clock kinematic shielding block (M.5.3, 557), a main-text research-track guard and colored rank-nine contact-channel diagnostic (M.5.10, 602), a Separatrix Surface-Density Lift monopole DtN research-track route (R.60) with audit v0.2 numeric support, and a canon-safe Research-Track particle/clock ansatz with preregistered mass falsifiers (N, O) on top of v0.8.19.

E.8.5 v0.8.19

v0.8.19 adds an orthodox radius-convention ropelength/thickness foundation, a main-text guard that knot type alone does not determine a unique physical representative, a scalarized research-track screening functional E_{screen} with Q_G retained only as a sector label unless mapped to I_G, a derived leading thin-filament length/log energy anchor, a density and dimensional audit for geometric mass branches, a geometric impedance bridge demoted to research-track pointer form with an explicit T_{core} free-symbol guard, stricter reintegration rules for derived Hamiltonian terms versus screening surrogates, a consolidated audit bundle (unified $S_{(t)}^{\circlearrowleft}$ discipline, F_max hygiene, Poisson–clock bridge, EMG guards, trefoil precision, pure-geometric mass collapse), a Foucault-pendulum rotation-to-clock inference protocol in the research track, projected swirl-vorticity frequency diagnostics, symmetry-metadata guards for dark-knot taxonomy, benchmark CSV schema hardening, operator-valued Swirl-Clock visibility diagnostics, a cosmological-impedance route for ρ_f (§R.58, Eq. 901) with horn-effective density label hygiene, and Taylor-column / rotating-fluid Swirl-Clock parity benchmarks (§R.70, §R.71) on top of v0.8.18.

E.8.6 v0.8.18

v0.8.18 adds calibration-chain guardrails (rc/α gates, kernel normalization, EM prefactor, $\rho_{\text{horn}}^{\text{eff}}$ baseline fix), research-track observational falsifier bounds (GW170817, pulsar preferred-frame limits), orthodox resolved-tube reach/thickness definitions, the four topology/stability research appendices (non-Abelian protection, reconnection decay, fractional torus-phase winding, and topological-fluid mechanics), the research-track knot-energy normal form with I_G scalarization, Euler/Magnus probe-transport polish, canonical time ontology with no-bare-τ policy, golden-layer hyperbolic rapidity guards, Planck-suppressed $F_{\text{swirl}}^{\text{max}}$ reparameterization, Compton–horn G_{swirl} reduction guard and balance-angle diagnostic, hybrid density-source Swirl-Clock benchmark, and the contact-stress geometry research appendix on top of v0.8.17.

E.8.7 v0.8.17

v0.8.17 adds the T-foliation versus local Swirl-Clock remark (Eq. 231), the ropelength/trefoil-α convention guard, Route-I SST-63/23/56 integration in the research track, and aligned updates in SST-73, SST-63, SST-64 v2, and SST-34 on top of v0.8.16. RC2 release-clean polish: p.119 summary quote, ‘Æ‘-modes fix, ‘tabularx‘ particle-candidate tables, and targeted version-phrase cleanup.

E.8.8 v0.8.16

v0.8.16 adds the pressure–optical locking relation (Eq. 283) in the Triadic Gravity-Response Corollary, the research-track no-monopole transport-stress audit (§R.34), and the relativity falsifier ladder on top of v0.8.15. SST-73 receives matching critical notes on Candidate C and $[\alpha_{\mathrm{grav}}] = \mathrm{s}^2$.

E.8.9 v0.8.15

v0.8.15 adds the canon hydrodynamic Lorentz-type swirl-force density (§7.5: Bernoulli tie, loop observables, calibrated scales) and the research-track EM-to-swirl correspondence with $\lambda_{\mathrm{EM}\to\circlearrowleft}$, SST-44 stress closure, and pending $\mathbf{f}_{\mathrm{link}}$ (§R.69.1) on top of v0.8.14.

E.8.10 v0.8.14

v0.8.14 adds the two-speed clock discipline (§3.9: $v^*_{\circlearrowleft}$ vs. c, NLSE core sound speed $c_s = v^*_{\circlearrowleft}/\sqrt{2}$, research-track gate), the α-gate guard on $\lambda_c/(\pi r_c) = 4/\alpha$, and the research-track core–torsion impedance-matching lemma (§R.64) on top of v0.8.13.

E.8.11 v0.8.13

v0.8.13 adds the relativity-emergence audit: the internal tensor-speed naturalness proposition (§11.12, conditional $c_{13} = 0$) in the main canon, the research-track Relativity Emergence Ladder (§R.44) separating derivable SR/GR kinematics from the open Einstein-dynamics program, and relativity/induced-gravity/cosmology bibliography on top of v0.8.12.

E.8.12 v0.8.12

v0.8.12 applies the Gemini round-3 epistemic audit: $\mathcal{P}_{\mathrm{cal}}$-aligned identity relabels, coarse-grain density caveat, Pauli a_{cut} disambiguation, galaxy-scale ρ_f note, and research-track EM-bridge limits on top of v0.8.11.

E.8.13 v0.8.11

v0.8.11 completes the final hygiene pass: consistent $\mathcal{P}_{\mathrm{cal}}$, $v^*_{\circlearrowleft}$ vs. $\mathbf{u}_{\circlearrowleft}$ usage in EMG, framed self-linking, W-boson sector, and research-track numerical bridges on top of v0.8.10.

E.8.14 v0.8.10

v0.8.10 applies the Gemini round-2 audit: canonical speed $v^*_{\circlearrowleft}$ vs. local $\mathbf{u}_{\circlearrowleft}(x,t)$ discipline, η_0 algebraic-within-calibration status, delay-equation sign convention, Rydberg/finite-cell relabeling, and research-track v_K disambiguation on top of v0.8.9.

E.8.15 v0.8.9

v0.8.9 adds the triadic gravity-response corollary (§11.16) and research-track flame/caustic/shell diagnostics plus atomic condensate closure notes on top of v0.8.8.

E.8.16 v0.8.8

v0.8.8 applies the Gemini audit: primitive calibration notation ($\mathcal{P}_{\mathrm{cal}}$, $v^*_{\circlearrowleft}$, $\mathbf{u}_{\circlearrowleft}$), conditional Kelvin/DDE labels, Pauli a_{cut} disambiguation, and research-track epistemic relabeling on top of v0.8.7.

E.8.17 v0.8.7

v0.8.7 adds the scalar-vs-$\mathbb{C}P^1$ substrate argument, the $\theta = \pi$ fermionic $\mathbb{Z}_2$ sector, and supporting bibliography on top of v0.8.6.

E.8.18 v0.8.6

v0.8.6 adds framed-tube self-linking ($SL = pq$), spinorial $\chi = \pm 1$ selection, and the lepton ladder (§6.3) on top of v0.8.5.

E.8.19 v0.8.5

v0.8.5 adds the highres conversation audit (Euler discipline, R-to-T bookkeeping, twist-ladder research track, horn/envelope wording) and the CALIBRATED circularity-honesty relabeling on top of v0.8.4.

E.8.20 v0.8.4

v0.8.4 integrates the canonical EM–gravity bridge, finite-cell α obstruction summary, and extended research-track taxonomy blocks.

E.8.21 v0.8.3

v0.8.3 integrates framed-tube ontology, attachment gate, trefoil closure discipline, updated particle dictionary, and Pauli a_{core} regularization per conversation, highres, and trefoil diffs.

E.8.22 v0.8.2

v0.8.2 integrates radius/density semantic correction: $r_c \equiv R_{\text{horn}}$, separate a_{core}, and horn-envelope density normalization per conversation and highres diffs.

E.9 Trefoil closure and persistence discipline

The project-level conclusion is that electron-trefoil creation, if retained, must be formulated as a boundary-supported R-to-T closure event rather than as ordinary reconnection of an already closed ideal-Euler vortex tube. The canon distinction is therefore

$$\text{pre-closure R-phase support} \xrightarrow{\mathcal{K}_{\text{Kairos}}} \text{closed } 3_1 \text{ carrier}, \tag{344}$$

$$\text{closed } 3_1 \text{ carrier} \xrightarrow{\text{smooth Euler}} \text{same knot class } 3_1. \tag{345}$$

This keeps the orthodox frozen-in topology constraint intact while preserving the SST closure program as a clearly labeled bridge.

The radius notation is correspondingly split into the local tube radius a_{core}, the neutral circulation radius r_c, and the classical electron envelope radius $r_e = 2r_c$:

$$a_{\text{core}} \neq r_c, \qquad r_e = 2r_c. \tag{346}$$

Ordinary atomic binding changes the phase envelope of the carrier, not its knot class:

$$e^-_{\text{atom}} = 3_1^{\text{core}} \otimes \Psi^{\text{envelope}}_{n\ell m\sigma}. \tag{347}$$

Helicity mapping-class and framing guard. [ORTHODOX / CANON SEMANTIC GUARD]. Let $\Omega \subset \mathbb{R}^3$ be a compact resolved-tube domain and let $\mathbf{V}$ be divergence free and tangent to $\partial\Omega$. The dual two-form $\alpha = \iota_{\mathbf{V}} d^3x$ is closed and vanishes when pulled back to the boundary. Helicity may then be written using the Biot–Savart/Hodge-selected primitive,

$$\mathcal{H}(\alpha) = \int_\Omega \alpha \wedge \mathrm{BS}(\alpha), \tag{348}$$

but an arbitrary primitive is not a gauge-complete replacement on a multiply connected domain [37].

For an orientation-preserving diffeomorphism $f : \Omega \to \Omega'$, the general mapping-class correction is a quadratic form in the independent fluxes,

$$\Delta\mathcal{H} = \mathcal{H}\Big((f^{-1})^*\alpha\Big) - \mathcal{H}(\alpha) = \sum_{i,j} c_{ij}\Phi_i\Phi_j. \tag{349}$$

For a solid torus subjected to j Dehn twists this reduces to

$$\Delta\mathcal{H} = j\Phi^2. \tag{350}$$

In the hydrodynamic identification $\mathbf{V} = \boldsymbol{\omega}$, $\Phi = \Gamma$ and hence $\Delta\mathcal{H} = j\Gamma^2$. A smooth ideal material evolution generated from the identity remains in the identity mapping class and preserves helicity; a nonzero jump in Eq. (349) is therefore not ordinary Euler advection. It requires a declared boundary operation, framing change, phase slip, reconnection, or other Kairos transition.

[CANON CONSEQUENCE]. Knot type, centerline, and ropelength do not by themselves fix the helicity sector. Any SST state that uses helicity or twist energetically must expose at least

$$(K, \{\Gamma_i\}, \text{framing/mapping class})\,, \tag{351}$$

and must not fit Tw after the fact to force a desired value of $\mathcal{H}$.

Ropelength representative and thickness-constrained ground state. For a tame knot or link type K, orthodox ropelength theory proves the existence of a ropelength-minimizing representative [61, 64]. With the radius convention for thickness,

$$\mathrm{Rop}_{\mathrm{rad}}(\gamma) = \frac{\mathrm{Len}(\gamma)}{\mathrm{Thi}_{\mathrm{rad}}(\gamma)}, \tag{352}$$

and the minimizer is $C^{1,1}$, with bounded curvature, but need not be C^2 [61].

[ORTHODOX GEOMETRY / RESOLVED-TUBE DEFINITION]. For a C^1 embedded centerline, orthodox ropelength theory identifies thickness with both reach and normal injectivity radius [61]. Therefore the SST resolved-tube radius is, when an ideal-tube representative is being used,

$$a_{\mathrm{core}}(K) \equiv \mathrm{Thi}_{\mathrm{rad}}(\gamma_K) = \mathrm{reach}(\gamma_K) = \mathrm{NIR}(\gamma_K), \tag{353}$$

$$\mathrm{Thi}_{\mathrm{rad}}(\gamma_K) = \min\left(\inf_s \frac{1}{\kappa(s)}, \frac{1}{2} d_{\mathrm{dcsd}}(\gamma_K)\right), \tag{354}$$

where d_{dcsd} is the doubly-critical self-distance. Equations (353)–(354) are geometric definitions of the resolved tube; they do not license the identification $a_{\mathrm{core}} = r_c$. In an SST sector where the resolved vortex-string energy is tension dominated, with an orthodox knotted flux-tube analogue in the glueball literature [65],

$$E_{\mathrm{SST}}[\gamma] = T_{\mathrm{eff}}\,\mathrm{Len}(\gamma) + E_{\mathrm{sub}}[\gamma], \qquad \mathrm{Thi}_{\mathrm{rad}}(\gamma) \geq r_c, \tag{355}$$

the leading-order ground-state problem reduces to ropelength minimization:

$$E_{\mathrm{SST}}^{(0)}(K) \approx T_{\mathrm{eff}} r_c \, \mathrm{Rop}_{\mathrm{rad}}(K). \tag{356}$$

This is a derived-conditional reduction, not an independent proof that all shape-dependent observables are ropelength functions.

Critical note: degeneracy and subleading splitting. Ropelength minimizers need not be unique. Therefore, while the scalar mass proxy $M_K \propto \mathrm{Rop}(K)$ may remain well-defined, shape-sensitive observables such as dipole structure, magnetic moment, moment of inertia, writhe distribution, or local swirl-stress maps may require subleading splitting terms:

$$E_{\mathrm{sub}} = E_{\mathrm{bend}} + E_{\mathrm{twist}} + E_{\text{Biot-Savart}} + E_{\mathrm{core}} + \cdots . \tag{357}$$

Writhe and helicity upper-bound guard. [ORTHODOX / NON-SHARP BOUNDS]. For a smooth knot or link of length L carrying an embedded tube of radius R,

$$|\,\mathrm{Wr}(K)| < \frac{1}{4}\left(\frac{L}{R}\right)^{4/3} = \frac{1}{4}\mathrm{Rop}_{\mathrm{rad}}(K)^{4/3}. \tag{358}$$

For a smooth vector field $\mathbf{V}$ on a bounded domain Ω, the corresponding rough geometric helicity bound is

$$|\mathscr{H}_V| \leq R_{\mathrm{vol}}(\Omega)\, \mathscr{E}_V, \tag{359}$$

where $R_{\mathrm{vol}}(\Omega)$ is the radius of a ball with the same volume as Ω and $\mathscr{E}_V = \int_\Omega |\mathbf{V}|^2 d^3x$ [57].

[CANON / PRECISION GUARD]. Equations (358) and (359) are deliberately non-sharp. They are admissibility and regression bounds, not estimators of particle mass, chirality strength, or equilibrium helicity. Sharp helicity bounds require the domain-specific projected Biot–Savart/curl spectrum.

Multi-component Gehring-thickness guard. [ORTHODOX / DISTINCT CONSTRAINT PROBLEM]. For a link $L = L_1 \cup \cdots \cup L_N$, Gehring link-thickness is

$$\mathrm{LThi}(L) := \min_{i \neq j} \mathrm{Dist}(L_i, L_j), \qquad \mathrm{LRop}(L) := \frac{\mathrm{Len}(L)}{\mathrm{LThi}(L)}. \tag{360}$$

It constrains distances between distinct components but does not constrain self-distance or curvature within one component [58].

Therefore an SST composite made from finite-radius components requires the hybrid admissibility conditions

$$\mathrm{Thi}_{\mathrm{rad}}(\gamma_i) \geq a_i, \qquad\qquad \mathrm{Dist}(\gamma_i, \gamma_j) \geq a_i + a_j \quad (i \neq j), \tag{361}$$

together with preservation of each component's intended isotopy class.

[CANON / TOPOLOGY GUARD]. Link-homotopy permits self-intersections of an individual component and is therefore too weak, by itself, to protect an SST particle component from changing its knot type. Gehring criticality may diagnose intercomponent exclusion, but it does not replace ordinary reach/thickness constraints.

Domain-spectral helicity optimization guard. [ORTHODOX / CANON SEMANTIC GUARD]. On a fixed smooth domain Ω, let $K(\Omega) = \{\mathbf{V} : \nabla \cdot \mathbf{V} = 0, \mathbf{V} \cdot \mathbf{n} = 0\}$ and define the boundary-compatible Biot–Savart operator

$$\mathrm{BS}_K = P_{K(\Omega)}\,\mathrm{BS}\,. \tag{362}$$

Its largest positive eigenvalue is the sharp helicity-to-L^2 ratio and is the inverse of the smallest positive curl eigenvalue on its image [62]:

$$\lambda_{\max}(\Omega) = \max_{\mathbf{V}\in K(\Omega)\setminus\{0\}} \frac{\mathcal{H}(\mathbf{V})}{\int_\Omega |\mathbf{V}|^2 d^3x} = \frac{1}{\kappa_1^{\mathrm{curl}}(\Omega)}. \tag{363}$$

If $\mathbf{V} = \boldsymbol{\omega}$, the denominator in Eq. (363) is enstrophy, not the SST kinetic energy $\frac{1}{2}\rho_f \int |\mathbf{u}_\circlearrowleft|^2 d^3x$. The two optimization problems must not be identified without an explicit constitutive relation.

A smooth global optimizer over all fixed-volume domains would have nonzero constant $|\mathbf{V}|$ on each boundary component, toroidal boundary components, and boundary field lines that are intrinsic geodesics. These are necessary isoperimetric conditions, not a proof that an SST particle is a horn torus, not a derivation of r_c or a_{core}, and not a proof that ideal Euler evolution relaxes to the extremal eigenfield. Ropelength minimization of a centerline and spectral optimization of a filled three-dimensional domain are distinct variational layers.

Convention guard for trefoil alpha comparisons. The mathematical literature uses both radius- and diameter-based thickness conventions. If

$$\mathrm{Thi}_{\mathrm{diam}} = 2\,\mathrm{Thi}_{\mathrm{rad}}, \tag{364}$$

then

$$\mathrm{Rop}_{\mathrm{diam}} = \frac{1}{2}\,\mathrm{Rop}_{\mathrm{rad}}\,. \tag{365}$$

For the trefoil, using certified ideal-knot data [63],

$$\mathrm{Rop}_{\mathrm{rad}}(3_1) \approx 32.7436, \qquad \mathrm{Rop}_{\mathrm{diam}}(3_1) \approx 16.3718. \tag{366}$$

The leading coincidence

$$\alpha_{\mathrm{lead}}^{-1} = \frac{8\pi}{3}\,\mathrm{Rop}_{\mathrm{diam}}(3_1) \approx 137.1561 \tag{367}$$

is therefore a robust sub-promille numerical coincidence, not an exact or ppm-certified derivation of the fine-structure constant.

Numerical certification guard for ideal-knot inputs. [ORTHODOX NUMERICAL GEOMETRY / CANON VALIDATION RULE]. A Ridgerunner-style output is an approximately thickness-critical polygon and a candidate smooth ropelength upper bound, not automatically a unique smooth global minimizer [63]. The equivalence between constraint thickness and polygonal thickness is established for equilateral polygons. Every imported ideal-knot input must therefore expose at least

$$\mathcal{D}_K^{\mathrm{cert}} = \left(N, \epsilon_{\mathrm{eq}}, \mathrm{Thi}_p, \mathrm{CThi}, \mathrm{Rop}_p, \mathrm{Rop}_{\mathrm{smooth}}^{\mathrm{ub}}, \min_i \mathrm{MinRad}(v_i), \frac{1}{2}d_{\mathrm{dcsd}}, \chi_{\mathrm{KKT}}, N_{\mathrm{seed}}\right), \tag{368}$$

with a declared rigidity-matrix sign convention and

$$\chi_{\mathrm{KKT}} = \frac{\|-\nabla E + A\Lambda\|_2}{\|\nabla E\|_2}, \qquad \Lambda \geq 0. \tag{369}$$

The smooth bound must be recomputed after controlled $C^{1,1}$ reconstruction. The label "ideal" on a Fourier or atlas file is not a certificate.

Scalar ropelength convergence is insufficient: active strut/kink support, positive multiplier measures, and arclength contact maps must also converge under refinement and across independent initial embeddings. A small Eq. (369) establishes only first-order constrained criticality. It does not prove global minimality, positive second variation, or dynamical SST stability. No downstream mass, coupling, or constant claim may quote relative precision substantially finer than the worst certified mesh, thickness, smoothing, contact-map, and KKT error of its knot input.

E.10 Finite-cell α coincidence and obstruction summary

[BRIDGE ANSATZ / SPECULATIVE FIT]. A parameter-light finite-cell variational construction for an ideal trefoil filament, containing partially open modelling choices, yields a leading dimensionless scale

$$\alpha_{\text{lead}}^{-1} = \frac{8\pi}{3}\,\mathcal{L}_{3_1} \approx 137.15471, \tag{370}$$

using the numerically computed ideal trefoil ropelength $\mathcal{L}_{3_1} = 16.371637$ [63] as a *benchmark* topological-geometric input. This lies within 866 ppm of CODATA $\alpha^{-1} = 137.035999$ and is an α-free sub-per-mille coincidence, not a part-per-million derivation.

The prefactor $8\pi/3$ contains a mode-count factor $N_p = 4$ derived from the soft sector of the spherical finite-cell surface spectrum $k_\ell = \ell(\ell+1) - 2$ (zero and unstable modes only). The sector volume $4\pi/3$ and overall normalization remain partially open modelling choices.

[NEGATIVE / OBSTRUCTION RESULT]. Closing the residual to ppm accuracy is underdetermined: at fixed cell-radius gate the required transverse shell weight is $w_\perp = 1.0675$, not the round value 1 (which reaches only 10.5 ppm). An independent Gross–Pitaevskii core-profile proxy returns $w_\perp \approx 2.076$, overshooting to -157 ppm. The 10.5 ppm residual can be absorbed independently by any of five sub-percent coefficients (shell weight, cell-radius gate, sector volume, ropelength value, or a third-order term), so ppm closure carries no discriminating power without pinning all inputs to $\sim 10^{-6}$ relative precision—beyond present ideal-knot ropelength certification [63, 61].

The gate hierarchy is therefore

$$G_0\ (\text{input provenance}) \longrightarrow G_1\ (\text{identifiability}) \longrightarrow G_2\ (\text{operator Hessians}) \longrightarrow G_3\ (\text{ppm closure}), \tag{371}$$

with G_0 and G_1 blocking closure before Hessian-level work becomes relevant.

The ring-constant proximity $\alpha_{\text{ring}} \approx \varphi$ is treated in Subsection I.14 of the research track. It is not a canon-level trefoil curvature closure.

[CANON / NON-CLAIM]. This block does not assert a first-principles derivation of the fine-structure constant. The defensible content is the α-free sub-per-mille leading coincidence with partially derived $N_p = 4$, plus a localized obstruction to ppm closure. Full scripts, appendices, and audit details are in `finite_cell_obstruction.tex` and `Derive_Constants/`.

E.11 Open theorem ladder

The most important unfinished task identified across the project is not adding new sectors, but tightening the derivational ladder between existing ones. In practice this means:

1. turning heuristic closures into explicit lemmas,
2. attaching numerical validation to each nontrivial scaling law,

3. separating benchmark results from full ontological claims.

This appendix therefore serves as a reminder that canon quality increases more by sharpening structure than by increasing conceptual breadth.

F Explicit Topological Mass Closures

F.1 Purpose and status

This appendix supplies an explicit closure layer for the mass program without replacing the canonical master equation of Eq. (96). Its role is to make concrete those invariant choices that were historically used in pre-v0.8 layers whenever actual particle-mass calculations were attempted.

[DERIVED] Canon rule: the abstract kernel Ξ_K of the main text may be specialized to more explicit invariant combinations provided the resulting branch remains dimensionless, benchmarkable, and subordinate to the topology-first hierarchy.

[CRITICAL NOTE] Nothing in this appendix upgrades the explicit kernel to a theorem of first-principles filament dynamics. The present role is closure discipline and reproducible benchmarking.

F.2 Pure geometric baseline branch

[DERIVED-CONDITIONAL / NORMALIZATION BRANCH]. A pure geometric baseline mass scale for topology class T is

$$M_0(T) = 2\pi^3\, \rho_{\rm horn}^{\rm eff}\, \frac{{r_c}^5}{\lambda_c^2}\, \mathcal{L}_{\rm tot}(T). \tag{372}$$

This branch is subordinate to Eq. (96). It records a normalization-level geometric closure used in historical mass-program benchmarks; it does not replace the canonical master functional or the explicit kernel branch below. The density in Eq. (372) is the horn-envelope normalization density of Eq. (79); using the background mass-equivalent density ρ_m here would move the benchmark down by the factor $\rho_m/\rho_{\rm horn}^{\rm eff} \simeq 1.20 \times 10^{-30}$ and is therefore a symbol-collision error, not a physical branch.

F.3 Explicit invariant branch of the canonical kernel

A conservative explicit specialization is obtained by resolving the topological kernel into ropelength, braid complexity, and Seifert-genus data:

$$\Xi_K^{(\rm exp)} = \left[\alpha_{\mathcal{L}} \frac{\mathcal{L}_{\rm tot}(K)}{\mathcal{L}_{\rm tot}(3_1)} + \alpha_b\big(b(K)-1\big) + \alpha_g\, g(K)\right] \varphi^{-2k(K)}. \tag{373}$$

Here:

- $\mathcal{L}_{\rm tot}(K)$ is the total ropelength-like invariant used for the chosen embedding or ideal representative,
- $b(K)$ is the braid index,
- $g(K)$ is the Seifert genus,
- $k(K)$ is the canonical layer / dressing index already used in the main text.

[ORTHODOX] The invariants $\mathcal{L}_{\rm tot}(K)$, $b(K)$, and $g(K)$ are standard topological or geometric knot descriptors.

[DERIVED / MATCHING-ANSATZ GUARD] The dimensional admissibility of Eq. (373) is derived. The displayed three-coefficient form and the $\varphi^{-2k(K)}$ dressing are a benchmark matching ansatz; the coefficients $\alpha_{\mathcal{L}}$, α_b, and α_g remain calibrated unless a separate variational or statistical-mechanical calculation derives them.

Substituting Eq. (373) into Eq. (96) yields the explicit benchmark mass branch

$$M_K^{(\rm exp)}(x) = \rho_m(x)\, {r_c}^3\, \Pi_K\, \Xi_K^{(\rm exp)}\, S_{(t)}^{\circlearrowleft}(x)^{-2}. \tag{374}$$

F.4 Baryon benchmark closure

For composite baryons built from constituent topology classes K_u and K_d, a conservative additive benchmark branch is

$$M_B^{(\mathrm{add})} = \Pi_B\, M_0 \left(n_u\, \Xi_{K_u}^{(\mathrm{exp})} + n_d\, \Xi_{K_d}^{(\mathrm{exp})} \right), \tag{375}$$

where n_u and n_d are constituent counts and M_0 is the common medium scale extracted from the main mass functional.

A retained legacy golden-closure benchmark is

$$M_p^{(\varphi)} = \varphi^{-2} 3^{-1/\varphi}\, (2M_u + M_d), \tag{376}$$

$$M_n^{(\varphi)} = \varphi^{-2} 3^{-1/\varphi}\, (M_u + 2M_d). \tag{377}$$

[**SPECULATIVE**] Equations (376)–(377) are retained as baryon-sector closure benchmarks only. They are not promoted to universal mass laws.

F.5 Canonical use policy

The explicit closures above are intended for:

1. theorem-ladder benchmarking,
2. archived CSV reproduction,
3. side-by-side comparison of topology branches,
4. explicit baryon fits.

They are *not* intended to replace the abstract organizing form of the main text.

G Effective Quantum Bridge

G.1 Purpose and status

The main canon states that quantum mechanics should appear as an effective low-energy description of the medium. The present appendix supplies the minimal bridge formulas needed to make that statement mathematically legible without claiming a completed derivation of the full quantum rule set.

[**ORTHODOX**] The Madelung decomposition and Schrödinger-form continuum limit are standard.

[**DERIVED**] SST may use these formulas as a coarse-grained field representation once a local amplitude density and phase have been specified.

[**SPECULATIVE**] The identification of the resulting fields with branch-resolved swirl carriers remains an SST interpretation.

[**CRITICAL NOTE**]: ρ_ψ is a coarse-grained mode-support density of the emergent carrier field, *not* a compression of the incompressible substrate ρ_f (Axiom 1, $\nabla \cdot \mathbf{v} = 0$). The two densities are distinct objects, so the Madelung amplitude does not violate substrate incompressibility.

G.2 Amplitude–phase decomposition

Introduce an effective complex mode field

$$\psi(x,t) = \sqrt{\rho_\psi(x,t)}\, e^{i\theta(x,t)}, \tag{378}$$

with $\rho_\psi \geq 0$ and real phase θ. The associated effective transport velocity is

$$\mathbf{u}_\psi = \frac{\hbar}{m_{\text{eff}}}\nabla\theta. \tag{379}$$

Assuming a Schrödinger-form bridge equation

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m_{\text{eff}}}\nabla^2 + V_{\text{eff}}\right)\psi, \tag{380}$$

one obtains the equivalent real system

$$\partial_t\rho_\psi + \nabla\cdot(\rho_\psi\mathbf{u}_\psi) = 0, \tag{381}$$

$$\partial_t\theta + \frac{m_{\text{eff}}}{2}\|\mathbf{u}_\psi\|^2 + V_{\text{eff}} + Q_{\text{eff}} = 0, \tag{382}$$

with effective quantum potential

$$Q_{\text{eff}} = -\frac{\hbar^2}{2m_{\text{eff}}}\frac{\nabla^2\sqrt{\rho_\psi}}{\sqrt{\rho_\psi}}. \tag{383}$$

[ORTHODOX] Equations (381)–(383) are the standard Madelung system associated with Eq. (380) [21, 22].

G.3 SST reading of the bridge

The conservative SST interpretation is:

$$\rho_\psi \leftrightarrow \text{coarse-grained mode-support density}, \tag{384}$$

$$\theta \leftrightarrow \text{effective circulation phase}, \tag{385}$$

$$\mathbf{u}_\psi \leftrightarrow \text{phase-transport velocity of the low-energy envelope}. \tag{386}$$

At this level, the branch-resolved atomic sector of Section 9 is read not as a replacement of the orthodox wavefunction, but as a candidate microscopic underpinning for the effective fields ρ_ψ and θ.

G.4 Two-component branch bookkeeping

A minimal two-branch extension consistent with the $\sigma = \pm 1$ structure of the atomic sector is

$$\Psi = \begin{pmatrix}\psi_+\\ \psi_-\end{pmatrix}, \qquad \psi_\pm = \sqrt{\rho_\pm}\,e^{i\theta_\pm}, \tag{387}$$

with total coarse-grained density

$$\rho_{\text{tot}} = \rho_+ + \rho_-. \tag{388}$$

[SPECULATIVE] This two-component structure is the minimal SST bridge to spinor-like bookkeeping. It is not yet a first-principles derivation of fermionic spin statistics.

G.5 Topological reading of the bridge

The most conservative topology statement retained at appendix level is that nontrivial phase maps may encode nontrivial carrier organization. In particular, the effective phase field may carry nontrivial winding or linking data even when the coarse-grained description is written in wave language. This is the appendix-level reason why the canon treats topology and wave structure as compatible rather than as competing descriptions.

H Dark-Sector and Galactic Continuum Law

H.1 Purpose and status

The main canon already retains the dark-knot taxonomy and the radial swirl pressure law. This appendix collects the next phenomenological layer: the galaxy-scale force law and the simplest rotation-curve ansatz inherited from earlier SST development.

[**ORTHODOX**] The underlying force law is ordinary Euler balance in steady swirl.

[**DERIVED**] SST reinterprets that balance as a candidate large-scale acceleration source.

[**SPECULATIVE**] The explicit galaxy-fit profile below is a phenomenological closure ansatz, not a theorem.

[**CRITICAL NOTE**] The canonical $\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}}$ is an effective medium parameter, not an ordinary gravitating baryonic/cosmological rest-mass density. Applying it on galactic scales requires a separate coupling and screening law. Without that law, the galaxy-scale pressure profile is a fit ansatz only and must not be interpreted as a literal dense fluid through which stars experience ordinary hydrodynamic drag.

H.2 Effective dark acceleration law

Starting from the steady radial pressure law of Eq. (183),

$$\frac{1}{\rho_f}\frac{dp_{\mathrm{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}, \tag{389}$$

define the effective SST acceleration by

$$a_{\mathrm{SST}}(r) := \frac{1}{\rho_f}\frac{dp_{\mathrm{swirl}}}{dr}. \tag{390}$$

Then

$$a_{\mathrm{SST}}(r) = \frac{v_\theta(r)^2}{r}. \tag{391}$$

[**DERIVED**] Equations (390)–(391) are immediate consequences of the retained pressure law.

H.3 Two-component galactic rotation ansatz

A retained phenomenological velocity profile is

$$v_{\mathrm{swirl}}(r) = \frac{C_{\mathrm{core}}}{\sqrt{1 + \left(r_c^{\mathrm{gal}}/r\right)^2}} + C_{\mathrm{tail}}\left(1 - e^{-r/r_c^{\mathrm{gal}}}\right), \tag{392}$$

with corresponding total circular speed

$$v_{\mathrm{tot}}^2(r) = v_{\mathrm{bar}}^2(r) + v_{\mathrm{swirl}}^2(r). \tag{393}$$

[**SPECULATIVE**] Equation (392) is a phenomenological fit ansatz and should be used only at the appendix / research-track level.

H.4 Relation to dark-knot taxonomy

The taxonomy of Eq. (167) classifies low-helicity, low-instability knot sectors as dark or quasi-dark candidates. The present galactic law does *not* assume that all galactic anomalies are produced by localized dark knots alone. Rather, it provides the complementary continuum statement: long-range swirl pressure gradients may produce effective extra acceleration even when the underlying microscopic carriers are topologically sparse or weakly luminous.

H.5 Use policy

The galactic law belongs in the canon only as:

1. an appendix-level phenomenological sector,
2. a separate astrophysics / dark-sector track,
3. a fit target for future rotation-curve studies.

It is deliberately not promoted in v0.8.18 to the same status as the primitive constant chain, the Swirl-Clock, or the delay-selection sector.

I Research-track sectors migrated from the main canon

This companion document collects those sectors that remain useful for continuity and future development but are too long, too provisional, or too phenomenology-specific for the core body of *Swirl-String-Theory_Canon-v0.8.23*. Their migration out of the main canon is editorial rather than negative: each subsection remains part of the broader SST research program, but none is required for the minimal logical closure of the core canon.

Editorial note (v0.8.23): the maximal swirl tension sector ($F^{\max}_{\text{swirl}}$, Rydberg bridge identities, Gibbons-class maximum-tension comparison, and the threefold interpretation table) is developed in the core document `SST_CANON-v0.8.23.tex`, subsection `Maximal Swirl Tension` (label `sec:Fmax`); it is not duplicated here.

I.1 Thermodynamic Radiation Interface

A recurring legacy idea is that a local swirl-energy density may define an effective temperature field and thereby an emergent radiation sector. Let

$$T_{\text{eff}} = T_{\text{eff}}\big(\rho_E,\, \omega,\, S^{\circlearrowleft}_{(t)}\big) \tag{394}$$

denote an unspecified constitutive relation. A minimal legacy interface then asks whether Planck-type spectra can emerge from a medium whose local excitation scale is set by T_{eff}.

[**ORTHODOX**]: blackbody spectra are governed by Planck's law once a valid temperature variable is defined.

[**SPECULATIVE**]: SST does not yet derive the constitutive relation above, so this sector remains a research program rather than canonically closed physics.

I.2 Minimal Coupling Interface to QED

Early research notes repeatedly explored whether an effective vector potential could be built from circulation variables. The most conservative placeholder takes the form

$$\nabla \to \nabla - i\frac{m}{\hbar}A_{\text{eff}}, \qquad A_{\text{eff}} = \chi\, v, \tag{395}$$

where χ is a scalar weight and v is the local carrier velocity field.

[**ORTHODOX**]: minimal coupling is the standard gateway from free to gauge-coupled wave dynamics.

[**SPECULATIVE**]: the above substitution is only an analogy template unless a full gauge-covariant derivation is supplied.

I.3 Ribbon-Invariant Chirality Refinement

The older canon layers also suggested using the Calugareanu–White–Fuller relation

$$Lk = Tw + Wr \tag{396}$$

as a refinement of the knot-based matter/antimatter dictionary.

[**ORTHODOX**]: this is a standard geometric identity for framed curves and ribbons.

[**DERIVED BOOKKEEPING**]: once a framed curve is specified, the identity separates writhe and twist contributions.

[**SPECULATIVE**]: the dynamical stability of writhe-dominated or twist-dominated realizations, and the mapping from sign(Tw) or related chirality measures to particle/antiparticle sectors, remains a model hypothesis.

I.4 Particle-Candidate Mapping Layer

[SPECULATIVE] Working dictionary: a minimal knot/composition mapping retained from earlier SST work is summarized in Table 5. This table is organizational rather than definitive; it records the current topological assignments used by the mass and taxonomy program.

Knot class / composition	Candidate	Status / role
3_1 trefoil	electron / positron sector	baseline chiral lepton calibration
$T(2,5)$ or equivalent higher torus class	muon	lepton hierarchy candidate
$T(2,7)$ or equivalent higher torus class	tau	lepton hierarchy candidate
twist/clasp knots, quasi-unknot loops	quark-like defects	research-track fractional boundary winding candidates
$5_2 + 5_2 + 6_1$ and nearby composites	baryon candidates	legacy mass-functional dictionary, still under audit
triple-gear locked analogue	proton-like charged baryon analogue	mechanical model for nonzero exterior collective rotation
Borromean-type analogue	neutron-like neutral baryon analogue	mechanical/topological model for zero exterior charge with internal triple linking

Table 5: Minimal knot-class and analogue summary (v0.8.23 research-track), aligned with main canon semantic corrections.

[CRITICAL NOTE]: Table 5 is a working dictionary only. A full canonical assignment requires an explicit invariant set and a mass-functional comparison program.

I.5 Quark-Like Twist Knots as Quasi-Unknot Defects

[RESEARCH TRACK / CANON CANDIDATE] Twist knots and related clasp knots may be interpreted as quasi-unknot carrier loops with a localized twist/clasp defect. Geometrically such a knot can be stretched so that it resembles an unknot with a central aperture, while topologically remaining nontrivial until a crossing change or reconnection occurs.

A useful diagnostic decomposition is

$$K_{\text{twist}} \simeq 0_1 + \Delta_{\text{clasp}}, \tag{397}$$

where Δ_{clasp} denotes a localized topological defect region. Candidate audit quantities are

$$\mathcal{U}_{\text{quasi}}, \qquad A_{\text{aperture}}, \qquad f_{\text{clasp}}, \qquad f_{\kappa,\text{localized}}, \tag{398}$$

standing for quasi-unknot score, central aperture, clasp-localization fraction, and curvature-concentration fraction.

[CHARGE PLACEHOLDER] Fractional quark-like charge should not be assigned to the trefoil. If retained, it is represented as fractional exterior boundary winding,

$$\frac{Q}{e} = q_\partial, \qquad q_u = +\frac{2}{3}, \qquad q_d = -\frac{1}{3}. \tag{399}$$

The topological knot supplies the defect carrier; q_∂ supplies the charge-sector bookkeeping.

I.6 Twist-Ladder Closure from a Folded Clasp

[ORTHODOX / RESEARCH TRACK]. The rope experiment described in the project discussion is naturally modeled as a twist box plus a single clasp closure. If m denotes the number of signed half-twists retained in the folded region at closure, the resulting twist-knot ladder is

$$K_m = \text{clasp} + m\text{-half-twist box}, \qquad m \in \mathbb{Z}. \tag{400}$$

For the positive convention used here,

$$K_1 = 3_1, \qquad K_2 = 4_1, \qquad K_3 = 5_2, \qquad K_4 = 6_1, \qquad K_5 = 7_2, \qquad K_6 = 8_1. \tag{401}$$

The standard invariant checks are

$$c_{\min}(K_m) = m + 2, \qquad \det(K_m) = 2m + 1 \quad (m \geq 1). \tag{402}$$

Changing the twist sign gives the mirror sector,

$$K_{-m} = \overline{K_m}. \tag{403}$$

[CANON-COMPATIBLE INTERPRETATION]. End-rotation and plectonemic folding redistribute twist and writhe according to the framed-ribbon bookkeeping $Lk = Tw + Wr$, but they do not change the topological class of an already closed ideal carrier. The knot class is fixed only when the clasp/boundary closure is performed. This module is therefore a research-track generator for twist/clasp candidates and for the $m = 1$ trefoil closure picture; it is not yet a dynamical theorem of Euler–SST.

I.7 Triple-Gear and Borromean Baryon Analogues

[**MECHANICAL ANALOGUE / NOT A PROOF**] The triple-gear print and Borromean-type links are retained as baryon analogues, not as final particle assignments.

The proton-like analogue is a three-component locked system with nonzero exterior collective rotation,

$$\mathcal{B}_p^{\text{gear}} = (C_1, C_2, C_3;\ q_\partial = +1,\ \Theta_{\text{ext}} \neq 0), \tag{404}$$

where the charge proxy is the boundary winding q_∂, not raw pairwise helicity.

The neutron-like analogue is a Borromean or Borromean-adjacent system with no net exterior boundary winding but nontrivial three-component topology,

$$\mathcal{B}_n^{\text{Bor}} = (C_1, C_2, C_3;\ q_\partial = 0,\ \mu_{123} \neq 0). \tag{405}$$

This preserves the intuition that the neutron is neutral externally while still carrying internal topological structure.

[**FALSIFIERS**] The proton analogue fails if the locked triple system has no stable exterior collective phase. The neutron analogue fails if the Borromean sector necessarily induces nonzero exterior boundary winding.

I.8 Trefoil–Gear Kinematic Non-Equivalence

[**RESEARCH TRACK**] The triple gear and segmented trefoil are not kinematically identical even if they use related printed parts. The gear model measures rotation about a straight axle; the trefoil model measures phase transport along a curved framed tube.

For the gear analogue,

$$\omega_{\text{gear}} = \frac{1}{5}\,\omega_{\text{core}} \tag{406}$$

if five full core rotations produce one full gear rotation.

For the segmented trefoil Blender model, the shell/blade winding budget is

$$N_{\text{shell,total}} = 10, \qquad N_{\text{shell,segment}} = \frac{10}{3}. \tag{407}$$

If the same core-shaft mechanism gives five core rotations per segment, then

$$\frac{N_{\text{shell,segment}}}{N_{\text{core,segment}}} = \frac{10/3}{5} = \frac{2}{3}. \tag{408}$$

This 2/3 is therefore a conditional shell/core kinematic ratio, not a trefoil-electron charge assignment.

I.9 Higgs-as-Trefoil High-Temperature Hypothesis

[**PARKED / SPECULATIVE**] The idea that a Higgs-like state is a thermally activated trefoil phase is not canon-ready. A single trefoil is chiral and lepton-like in the current taxonomy, whereas a Higgs-like excitation must behave effectively as a scalar sector.

A viable future version would need at least one of the following mechanisms:

1. a trefoil–anti-trefoil paired condensate with scalar effective symmetry,
2. a high-temperature unframed phase where chirality averages out,
3. an independent scalar amplitude mode coupled to trefoil attachment without identifying the Higgs with the trefoil itself.

Until such a mechanism is derived, the hypothesis remains outside the canon.

I.10 Hydrodynamic Exchange and the Pauli Barrier

[**ORTHODOX**]: for two thin filamentary loops C_1 and C_2 in an incompressible medium, the Biot–Savart interaction energy is

$$E_{\text{int}} = \frac{\rho_f \Gamma_1 \Gamma_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\ell_1 \cdot d\ell_2}{\|r_1 - r_2\|}. \tag{409}$$

[**DERIVED**]: if two identical electron-candidate loops are forced into the same spatial state, regularization at a short-distance cutoff a_{cut} yields the overlap penalty

$$V_{\text{Pauli}}(d) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \iint \frac{d\ell_1 \cdot d\ell_2}{\sqrt{\|r_1(\theta_1) - r_2(\theta_2) + d\|^2 + a_{\text{cut}}^2}}, \tag{410}$$

with maximal-overlap scale

$$V_{\text{Pauli}}^{\max} \approx \frac{\rho_f \Gamma_0^2}{4\pi} \left(\frac{L}{a_{\text{cut}}}\right) \mathcal{O}(1). \tag{411}$$

Using the canonical values

$$\rho_f = 7.0 \times 10^{-7}\ \text{kg}\,\text{m}^{-3}, \tag{412}$$

$$r_c = 1.40897017 \times 10^{-15}\ \text{m}, \tag{413}$$

$$\Gamma_0 = 2\pi r_c \,\|\mathbf{v}_{\circlearrowleft}\| \approx 9.6836 \times 10^{-9}\ \text{m}^2\,\text{s}^{-1}, \tag{414}$$

and $L \sim 2\pi a_0$ at the hydrogenic scale, the legacy benchmark corresponds to the cutoff calibration $a_{\rm cut} = r_c$. This does *not* identify the resolved tube radius $a_{\rm core}$ with r_c; it only fixes the ultraviolet cutoff used in this benchmark. One obtains

$$V_{\rm Pauli}^{\rm max} \sim 1.23 \times 10^{-18}\ {\rm J} \sim 7.6\ {\rm eV}. \tag{415}$$

[**CRITICAL NOTE**]: this is retained as a cutoff-calibrated scale benchmark, not yet as a closed theorem. Its role is to show that the exclusion-energy estimate falls in an atomically relevant window rather than at obviously pathological scales. A physical resolved-core calculation must use an independently specified $a_{\rm core}$ or profile model.

I.11 Numerical Benchmark Layer

[**DERIVED**] **Computational-program requirement:** earlier SST work contained explicit benchmark tables for particle and atomic masses. In the present canon, the benchmark layer is retained as a reproducibility requirement rather than as a stand-alone authority claim.

Particle	$m_{\rm exp}$ (MeV)	$m_{\rm SST}$ (MeV)	rel. error	mode class
e	0.51099895	0.51099895	0	predictive/closure
μ	105.6583745	...	...	predictive candidate
τ	1776.86	...	...	predictive candidate
p	938.272088	...	...	predictive or closure
n	939.565420	...	...	predictive or closure

Table 6: Benchmark summary structure to be populated from canonical scripts and archived CSV output.

[**CRITICAL NOTE**]: a future v0.8.x benchmark appendix should include the exact script or package path used for generation, archived CSV outputs, mode definitions (predictive vs. exact-closure), and uncertainty propagation on canonical constants and geometric factors.

I.12 Helicity-by-base archive (diagnostic layer)

[**DERIVED**] **Archive role:** the SST helicity-by-base outputs are retained as a measurable balance diagnostic for taxonomy support, not as an independent law. Define the midpoint-distance diagnostic

$$\Delta_H(K) = \left| \widehat{\mathcal{H}}_b(K) + \frac{1}{2} \right|, \tag{416}$$

where $\widehat{\mathcal{H}}_b(K)$ is the archived normalized helicity-like base indicator for class K.

Near-balanced values with $\Delta_H(K) \ll 1$ indicate candidate dark or quasi-dark sectors only when symmetry and instability filters are also satisfied.

[**CRITICAL NOTE**]: no single helicity-like scalar is sufficient for sector assignment; classifier decisions remain conjunction rules over symmetry, helicity balance, and low-order stability content.

[**CANON WORKFLOW NOTE**]: if SST-labelled and parallel VAM-labelled helicity outputs are numerically identical, the SST-labelled archive is the only required source for canon workflows.

I.13 Trefoil-closure archive (benchmark discipline)

[**DERIVED**] **Benchmark role:** ideal-trefoil robustness sweeps, final-best-estimate tables, and theorem-style reports are retained as benchmark-discipline infrastructure.

[CRITICAL NOTE]: practical best-estimate closure and theorem-level closure are distinct. A result may remain useful for calibration while still failing strict theorem-level tail or postcheck requirements.

[REQUIRED METADATA]: benchmark reports should include explicit pass/fail logic, postchecks, extrapolation summaries, and local-curve diagnostics so that closure claims are auditable.

I.14 G7: Ring Constant versus Trefoil-Geometric Closure

[NUMERICAL OBSERVATION / OPEN GATE / NOT CANONICAL]. Track-B calculations give the numerical vortex-ring constant

$$\alpha_{\text{ring}} = 1.61935\ldots = \varphi + 0.00132\ldots, \tag{417}$$

with

$$\frac{\alpha_{\text{ring}} - \varphi}{\varphi} \approx 8.16 \times 10^{-4} = 0.0816\%. \tag{418}$$

This is retained as a genuine sub-per-mille numerical proximity between the GP/NLSE ring-energy constant and the golden ratio.

However, the identification

$$\alpha_{\text{ring}} \stackrel{?}{=} \mathcal{C}_{\text{trefoil}} = \frac{R_{\text{eff}}}{a_{\text{eff}}} \tag{419}$$

is not derived. The quantity α_{ring} is a vortex-ring energy constant, whereas $\mathcal{C}_{\text{trefoil}}$ is a geometric centerline-thickness observable of an ideal trefoil.

The recursive closure relation

$$\mathcal{C}_{\text{cl}} = 1 + \frac{1}{\mathcal{C}_{\text{cl}}} \tag{420}$$

implies

$$\mathcal{C}_{\text{cl}} = \varphi, \tag{421}$$

but this proves only the consequence of the ansatz. It does not prove that the ideal trefoil realizes the ansatz.

We therefore split G7 into three independent subgates:

$$\text{G7a}: \quad \alpha_{\text{ring}} \to \varphi, \tag{422}$$

$$\text{G7b}: \quad \mathcal{C}_{\text{trefoil}} = \frac{R_{\text{eff}}}{a_{\text{eff}}} \to \varphi, \tag{423}$$

$$\text{G7c}: \quad \xi = r_c. \tag{424}$$

At present all three remain open. In particular, $\xi = r_c$ is an imposed core-matching condition in the NLSE/GP diagnostic layer, not a derived identity.

Canon rule. The canon may cite

$$\alpha_{\text{ring}} \approx \varphi \tag{425}$$

as a numerical observation, but it must not cite this as evidence that

$$\mathcal{C}_{\text{trefoil}} \approx \varphi \tag{426}$$

without an independent geometric convergence test on ideal-trefoil data.

I.15 ϕ_{3_1} transform archive (deferred)

[**SPECULATIVE**] **Status:** retained as research-track spectral evidence for later zeta/spectral programs.

[**SCOPE RULE**]: this archive is not part of the current dark-sector or galactic canonization closure path in v0.8.x.

I.16 Gauge-Sector Roadmap

[**SPECULATIVE**] **Structural roadmap:** a retained minimal statement of the gauge program is

$$g_{\text{eff}} = su(3) \oplus su(2) \oplus u(1), \tag{427}$$

with effective gauge phases organized as holonomy data of multi-director transport.

A minimal phase-based representation is

$$A^a_\mu(x) = \partial_\mu \theta^a(x), \tag{428}$$

and clock coupling acts as a common phase reference,

$$\theta^a(x) \mapsto \theta^a(x) + q^a \chi(x) \quad \Rightarrow \quad A^a_\mu \mapsto A^a_\mu + q^a \partial_\mu \chi. \tag{429}$$

[**CRITICAL NOTE**]: the gauge layer is retained only as a roadmap statement. Running couplings, anomaly cancellation, and realistic mixing matrices remain open work.

J Dark-sector and galactic canonization execution package (v0.8.x)

J.1 Stage-0 freeze: topology toolkit and benchmark protocol

[**DERIVED**] **Scope freeze:** the mandatory topology toolkit for v0.8.x canonization is fixed to:

1. Gauss linking integral,
2. helicity in ideal fluids,
3. thin-tube helicity reduction,
4. Hopf invariant,
5. basic torus-knot invariants (c, b, g),
6. Calugareanu–White–Fuller relation $Lk = Tw + Wr$.

[**DERIVED**] **Notation freeze:** one notation family is used in this package:

$$\rho_f, \quad v_\theta(r), \quad p_{\text{swirl}}(r), \quad \mathcal{H}(K), \quad N_u^{(1)}(K), \quad \Gamma, \quad \kappa, \quad S^{\circlearrowleft}_{(t)}.$$

No parallel symbol aliases are introduced for the same quantity.

[**DERIVED**] **Mandatory benchmark protocol template:**

1. **Inputs:** canonical constants, topology identifiers, profile parameters, and dataset/version tags.
2. **Procedure:** exact script or package path, run command, and branch/commit hash.
3. **Outputs:** archived CSV files with fixed column schema and an attached metadata note.

4. **Uncertainty:** first-order propagation for calibrated constants and sensitivity sweep on profile parameters.

5. **Rerun:** deterministic rerun steps and acceptance tolerances.

[CANON WORKFLOW NOTE]: every promoted quantitative claim in this package is invalid without an explicit input-to-CSV trace.

J.2 Stage-1 freeze: dark-sector Euler pressure anchor

[ORTHODOX] Euler radial balance (steady, axisymmetric, azimuthal-dominant):

$$\frac{1}{\rho_f}\frac{dp_{\rm swirl}}{dr} = \frac{v_\theta(r)^2}{r}. \tag{430}$$

[DERIVED] Flat-tail limit (constant tail speed):

$$v_\theta(r) \to v_0 \quad \Longrightarrow \quad p_{\rm swirl}(r) = p_0 + \rho_f v_0^2 \ln\left(\frac{r}{r_0}\right). \tag{431}$$

[DERIVED] Assumption block: incompressible, inviscid, stationary, axisymmetric, azimuthal-dominant flow, with validity restricted to regions where radial drift and stratification corrections are subleading.

[DERIVED] Dimensional check:

$$\left[\frac{1}{\rho_f}\frac{dp}{dr}\right] = \frac{\mathrm{kg\,m^{-2}\,s^{-2}}}{\mathrm{kg\,m^{-3}}} = \mathrm{m\,s^{-2}} = \left[\frac{v^2}{r}\right].$$

[SPECULATIVE] SST interpretation: Eq. (430) is interpreted as a classifier/driver layer for dark-sector support. This interpretation is model-level and separate from the orthodox derivation.

[CRITICAL NOTE]: Applying Eq. (430) unscreened with the microscopic $\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}}$ at galactic scales ($v_\theta \sim 200\,\mathrm{km\,s^{-1}}$, $r \sim 10\,\mathrm{kpc}$) gives $dp/dr \sim 10^{-16}\,\mathrm{Pa\,m^{-1}}$ and $\Delta p \sim 2.8 \times 10^3\,\mathrm{Pa}$ over 1 kpc, far above the observed ISM. The swirl pressure is not the thermal ISM pressure, but no screening/scaling law for ρ_f across scales is yet derived; this sector stays research-track and is falsifiable on exactly this point.

[DERIVED] Attached observables/falsifiers:

1. Rotation-curve consistency after pressure reconstruction from archived $v_\theta(r)$ data.

2. Null falsifier: systematic mismatch in the reconstructed pressure gradient sign or scale across validated radial windows.

J.3 Stage-2 freeze: measurable dark taxonomy

[SPECULATIVE] Classifier variables: symmetry class (including amphichirality status), chirality sign/state, helicity-balance indicator $\Delta_H(K)$, and low-order instability count $N_u^{(1)}(K)$.

[SPECULATIVE] Decision rules (symmetry-first, then stability-filtered):

$$\text{Dark}: \ \text{amphichiral candidate} \ \wedge \ \Delta_H(K) \ll 1 \ \wedge \ N_u^{(1)}(K) = 0, \tag{432}$$

$$\text{Quasi-dark}: \ \text{near-balanced symmetry/helicity} \ \wedge \ 0 < N_u^{(1)}(K) \le 2, \tag{433}$$

$$\text{Visible}: \ \text{chiral-dominant and/or helicity-unbalanced sectors.} \tag{434}$$

[CANON WORKFLOW NOTE] **Ambiguity tie-break:** if classes overlap, choose the class with the strongest conjunction score in the order

$$\text{symmetry gate} \to \text{helicity balance gate} \to \text{stability gate}.$$

[SPECULATIVE] Baseline templates and falsifiers:

- **Visible baseline:** trefoil-like chiral template; falsifier = persistent amphichiral-plus-balanced signature under validated data updates.
- **Quasi-dark baseline:** Borromean-type weak-coupling template; falsifier = stable zero-instability assignment across independent scans.
- **Dark baseline:** figure-eight/amphichiral template; falsifier = robust nonzero chirality or high-instability assignment.

[**CANON WORKFLOW NOTE**]: amphichiral ⇒ dark remains a classifier proposal tested against observables, not a closed theorem.

J.4 Stage-3 freeze: galactic law branch gate and closure route

[**CANON WORKFLOW NOTE**] **Branch-gate decision:** Branch A (legacy profile refine) is selected for current v0.8.x closure because it preserves tested limits while allowing benchmark traceability.

Working profile:

$$v_\theta(r) = \frac{C_{\text{core}}}{\sqrt{1 + (r_c/r)^2}} + C_{\text{tail}} \left(1 - e^{-r/r_c}\right). \tag{435}$$

[**DERIVED**] **Required checks:** small-r behavior, flat-tail behavior, parameter semantics, and pressure reconstruction via Eq. (430).

[**DERIVED**] **Failure modes:**

1. non-physical parameter combinations (wrong sign or unstable growth),
2. failure to satisfy tail-limit behavior in benchmark windows,
3. inability to reconstruct a consistent pressure profile from the same archived inputs.

J.5 Stage-4 freeze: probe falsification map

[**CANON WORKFLOW NOTE**] probe classes are fixed as astrophysical, laboratory pressure/swirl analog, HV-coil/thrust-inspired, optical/torsional coupling, and null dark-sector tests.

[**CANON WORKFLOW NOTE**]: probe outcomes can support or reject already-stated equations, but they do not replace derivation-level closure.

J.6 Promotion snapshot after execution

[**DERIVED**] post-freeze status in this package:

- Item 6 (topology toolkit): CANON-ready in scope and notation.
- Item 5 (benchmark protocol): CANON-CANDIDATE pending repository-wide protocol reuse.
- Item 1 (Euler pressure anchor): CANON-CANDIDATE with explicit assumptions, limits, and falsifier path.
- Item 2 (dark taxonomy): BRIDGE with measurable classifier gates and falsifiers.
- Item 3 (galactic swirl law): BRIDGE with branch gate closed and benchmark route fixed.
- Item 4 (probe map): CANON-BRIDGE with equation/null-result mapping frozen.

Probe class	Equation under test	Null-result meaning	Primary confounder	Expected sign/trend
Astrophysical rotation curves	Eq. (430) + profile closure	weakens pressure-law applicability window	baryonic-model bias	reconstructed dp/dr consistent with v_θ^2/r
Lab swirl analog	Euler radial balance scaling	weakens scale-transfer claim	viscosity/edge forcing	monotone pressure rise with imposed swirl
HV-coil/thrust-inspired	circulation-pressure bridge claims	kills quantized-bridge claim if no scaling	EM/thermal artifacts	response tracks circulation proxy
Optical/torsional coupling	swirl-clock bridge signatures	weakens coupling interpretation	detector drift/noise	phase/timing shift with controlled swirl proxy
Null dark tests	taxonomy conjunction rules	kills classifier branch if conjunction fails	mislabelled topology class	class score decreases under contradictory diagnostics

Table 7: Frozen equation-to-probe map for falsification-first workflow in the v0.8.x dark-sector program.

K Candidate CANON-v0.8.x Modules from Swirl-Flux, Envelope, and Reconnection Analysis

K.1 Status Classification

The following modules are proposed as CANON-v0.8.x candidates.

1. The Schrödinger kinetic coefficient is retained as an effective envelope stiffness,
$$\mathcal{S}_e = \frac{\hbar^2}{2M_e}.$$
The replacement $-\mathbf{v}_\circlearrowleft^2\nabla^2/2$ is rejected on dimensional grounds.

2. The Meissner effect is interpreted as macroscopic swirl-flux phase locking in a coherent paired condensate.

3. R/T wave–string interaction is represented by conservation of radiative pseudomomentum plus vorticity impulse.

4. Reconnection is excluded in the ideal sector but allowed in measurement, decay, and dissipative sectors, where helicity is redistributed between topology, writhe, twist, radiation, and dissipation.

K.2 Envelope Stiffness and the Rejection of the $\mathbf{v}_{\circlearrowleft}^2\nabla^2$ Kinetic Term

For an effective R/T envelope amplitude $\Psi(\mathbf{r},t)$, the kinetic coefficient must have dimension $\mathrm{J\,m^2}$. The correct envelope stiffness is therefore

$$\mathcal{S}_e \equiv \frac{\hbar^2}{2M_e}, \qquad [\mathcal{S}_e] = \mathrm{J\,m^2}.$$

The effective low-energy envelope equation is

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[-\mathcal{S}_e\nabla^2 + V_{\mathrm{SST}}(\mathbf{r},t)\right]\Psi.$$

The operator $-\mathbf{v}_{\circlearrowleft}^2\nabla^2/2$ is not a Schrödinger kinetic energy operator because

$$\left[\mathbf{v}_{\circlearrowleft}^2\nabla^2\right] = \frac{\mathrm{m}^2}{\mathrm{s}^2}\frac{1}{\mathrm{m}^2} = \mathrm{s}^{-2},$$

which is not an energy.

For tunneling through a one-dimensional effective SST barrier,

$$T \sim \exp\left(-2\int_{x_1}^{x_2}\kappa_{\mathrm{SST}}(x)\,dx\right),$$

where

$$\kappa_{\mathrm{SST}}(x) = \sqrt{\frac{V_{\mathrm{SST}}(x)-E}{\mathcal{S}_e}} = \sqrt{\frac{2M_e\left[V_{\mathrm{SST}}(x)-E\right]}{\hbar^2}}.$$

This is dimensionally consistent because

$$\left[\frac{V-E}{\mathcal{S}_e}\right] = \mathrm{m}^{-2}, \qquad [\kappa_{\mathrm{SST}}] = \mathrm{m}^{-1}.$$

K.3 Numerical Bridge Anomaly

A numerically close but dimensionally non-identical bridge is

$$\mathcal{B}_e = \frac{F_{\mathrm{swirl}}^{\max}{r_c}^3}{5\lambda_c v_{\circlearrowleft}^*}.$$

Using

$$\lambda_c = \frac{h}{M_e c},$$

one obtains

$$\frac{\hbar^2}{2M_e} = 6.10426431\times10^{-39}\,\mathrm{J\,m^2},$$

whereas

$$\mathcal{B}_e = 6.12394931\times10^{-39}\,\mathrm{J\,s}.$$

The numerical ratio is

$$\left.\frac{\mathcal{B}_e}{\mathcal{S}_e}\right|_{\text{SI numbers}} = 1.00322479.$$

This is classified as a Research Track numerical bridge, not a canonical identity, because

$$[\mathcal{B}_e] = \mathrm{J\,s} \neq \mathrm{J\,m^2} = [\mathcal{S}_e].$$

The dimensionally meaningful relation is instead

$$5\lambda_c v_{\circlearrowleft}^* = 5\alpha\frac{h}{2M_e},$$

with both sides having dimension $\mathrm{m^2\,s^{-1}}$.

K.4 Swirl-Flux Locking and the Meissner Effect

[**CRITICAL NOTE**]: The values $2M_e$ and $q_{\text{pair}} = -2e$ are imported from the orthodox Cooper-pair description. Per Sec. 8.6 (Charge and Circulation), SST does not yet derive gauge charges from circulation alone, so this module is a [CALIBRATED] bridge using empirical pairing inputs, not a first-principles flux-exclusion derivation.

For a coherent paired electronic condensate of mass $2M_e$ and charge $q_{\text{pair}} = -2e$, phase single-valuedness gives

$$\oint_{\partial\Sigma} (2M_e\mathbf{v}_{\text{pair}} + q_{\text{pair}}\mathbf{A}) \cdot d\boldsymbol{\ell} = nh.$$

Since $q_{\text{pair}} = -2e$,

$$2M_e\Gamma_{\text{pair}} - 2e\Phi_B = nh.$$

Thus

$$\Phi_B = \frac{M_e}{e}\Gamma_{\text{pair}} - n\frac{h}{2e}.$$

In magnitude,

$$|\Phi_B| = \left|n\Phi_0 - \frac{M_e}{e}\Gamma_{\text{pair}}\right|, \qquad \Phi_0 = \frac{h}{2e}.$$

For a non-rotating coherent bulk,

$$\Gamma_{\text{pair}} \to 0,$$

and therefore

$$|\Phi_B| = n\Phi_0.$$

The Meissner effect is then interpreted in SST as exclusion of non-quantized magnetic flux from a phase-locked coherent R/T condensate.

K.5 R/T Pseudomomentum–Impulse Exchange

For a closed R/T interaction domain,

$$\frac{d}{dt}(\mathbf{P}_R + \mathbf{I}_T) = 0.$$

Here $\mathbf{P}_R$ denotes radiative R-phase pseudomomentum and $\mathbf{I}_T$ denotes T-phase vorticity impulse,

$$\mathbf{I}_T = \frac{\rho_f}{2}\int_\Omega \mathbf{x} \times \boldsymbol{\omega}\, d^3x.$$

The dimension is

$$[\mathbf{I}_T] = \mathrm{kg\,m\,s^{-1}},$$

so $\mathbf{I}_T$ is a true momentum-like quantity.

For a thin circular swirl string of circulation Γ and radius R,

$$\mathbf{I}_T = \rho_f\Gamma\pi R^2\hat{\mathbf{n}}.$$

The core circulation scale is

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}^* = 9.68361920 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}.$$

K.6 Reconnection and Helicity Redistribution

In the ideal sector, topology is conserved:

$$\frac{D\mathcal{K}}{Dt} = 0,$$

under the assumptions

$$\nabla \cdot \mathbf{v} = 0, \qquad \nu = 0, \qquad \text{barotropic}, \qquad \text{no reconnection}.$$

In the dissipative, measurement, or decay sector, reconnection is allowed only as a non-ideal event:

$$\frac{D\mathcal{K}}{Dt} \neq 0.$$

This is not a free creation mechanism. A reconnection claim must state the trigger class, e.g. $\nu_{\text{eff}} > 0$, a boundary injection, a phase-slip singularity, a resolved-core collapse, or an explicit external forcing term. The helicity budget is written

$$\Delta H_{\text{topology}} + \Delta H_{\text{writhe}} + \Delta H_{\text{twist}} + \Delta H_{\text{radiative}} = -\Delta H_{\text{diss}}.$$

In the weak-dissipation limit,

$$\Delta H_{\text{topology}} + \Delta H_{\text{writhe}} + \Delta H_{\text{twist}} + \Delta H_{\text{radiative}} \approx 0.$$

For filamentary structures,

$$H_{\text{centerline}} = \sum_{i\neq j} \Gamma_i \Gamma_j \mathrm{Lk}_{ij} + \sum_i \Gamma_i^2 \left(\mathrm{Wr}_i + \mathrm{Tw}_i\right).$$

K.7 Spherical Pressure Envelopes

For inviscid incompressible SST flow,

$$\nabla p_{\text{SST}} = \rho_f (\mathbf{v} \cdot \nabla)\mathbf{v}.$$

For an azimuthal swirl profile $v_\theta(r)$,

$$\frac{dp}{dr} = \rho_f \frac{v_\theta^2(r)}{r}.$$

If

$$v_\theta(r) = \frac{\Gamma}{2\pi r},$$

then

$$\frac{dp}{dr} = \rho_f \frac{\Gamma^2}{4\pi^2 r^3},$$

and hence

$$p(r) = p_\infty - \frac{\rho_f \Gamma^2}{8\pi^2 r^2}.$$

Therefore

$$\Delta p(r) = p_\infty - p(r) = \frac{\rho_f \Gamma^2}{8\pi^2 r^2} = \frac{1}{2} \rho_f v_\theta^2(r).$$

At the canonical swirl speed,

$$\frac{1}{2} \rho_f {v_{\circlearrowleft}^{*}}^2 = 4.18774392 \times 10^5 \,\text{Pa}.$$

At the core-density stress scale,

$$\rho_{\text{core}} {v_{\circlearrowleft}^{*}}^2 = 4.65848920 \times 10^{30} \,\text{Pa}.$$

The canonical maximum swirl force is recovered as

$$F_{\text{swirl}}^{\max} = \rho_{\text{core}} {v_{\circlearrowleft}^{*}}^2 \pi {r_c}^2 = 29.053507 \,\text{N}.$$

K.8 Golden-Layer Scaling

The universal rule

$$E_n = \varphi^n E_0$$

is rejected as too strong. The weaker Research Track ansatz is

$$E_{K,k} = E_K^{(0)} \varphi^{-2k}, \qquad k \in \mathbb{Z}_{\geq 0}.$$

Here K labels a knot family and k labels a relaxation layer inside that family.

K.9 Non-Holonomic Gravity Claims

Podkletnov-type anti-gravity claims are not adopted as Canon. At most, the formal analogy

$$\text{non-holonomic twist} \longleftrightarrow \text{foliation twist/vorticity}$$

may be retained in a historical or speculative appendix. No experimental anti-gravity claim is inferred.

L Mode-Locked Swirl-Coil Excitation and $v_{\circlearrowleft}^* = f\Delta x$ Metrology

Status. [RESEARCH TRACK / CANON-CANDIDATE]. This appendix formulates a falsifiable experimental protocol for testing whether externally driven multi-phase coil geometries can inject coherent swirl modes into a bounded measurement region. The construction is not assumed as a new force law. It is a mode-selection and metrology rule for identifying when the canonical characteristic swirl speed

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^6\,\mathrm{m\,s^{-1}} \tag{436}$$

is sampled by a laboratory-scale frequency–length product.

L.1 Motivation

The basic dimensional identity

$$\|\mathbf{v}_{\circlearrowleft}\| = f\,\Delta x \tag{437}$$

has the units of speed and is therefore not, by itself, a dynamical claim. Its SST use is metrological: a driven system is said to probe a swirl-resonant channel only when a temporal frequency f and a geometrically defined spatial period Δx satisfy Eq. (437) within experimental bandwidth.

For a circular or quasi-circular coil, the relevant spatial period is generally not an arbitrary external distance. It is the azimuthal wavelength selected by the coil radius and spatial mode number. Hence the physical coil radius enters the resonance condition directly.

L.2 Ring-mode closure condition

Let R_{coil} be the effective radius of the active winding, f_0 the electrical base frequency, $n \in \mathbb{N}$ the temporal harmonic, and $m \in \mathbb{Z} \setminus \{0\}$ the azimuthal spatial mode number around the coil. The spatial wavelength of mode m along the coil circumference is

$$\Delta x_m = \frac{2\pi R_{\mathrm{coil}}}{|m|}. \tag{438}$$

The (n)-th temporal harmonic has frequency

$$f_n = n f_0. \tag{439}$$

The dimensionless swirl-lock parameter is therefore

$$\boxed{\eta_{n,m} = \frac{f_n \Delta x_m}{\|\mathbf{v}_{\circlearrowleft}\|} = \frac{n f_0}{\|\mathbf{v}_{\circlearrowleft}\|} \frac{2\pi R_{\text{coil}}}{|m|}}. \tag{440}$$

A ring mode is swirl-locked when

$$\boxed{\eta_{n,m} = 1.} \tag{441}$$

Equivalently, the resonant coil radius for a given (n, m, f_0) is

$$\boxed{R_{\text{coil}}^{(n,m)} = \frac{|m| \, \|\mathbf{v}_{\circlearrowleft}\|}{2\pi n f_0}}. \tag{442}$$

Dimensional check.

$$[R_{\text{coil}}] = \frac{\mathrm{m\,s^{-1}}}{\mathrm{s^{-1}}} = \mathrm{m}.$$

Thus Eq. (442) is dimensionally well-defined.

L.3 Numerical calibration examples

For the canonical value

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^6 \,\mathrm{m\,s^{-1}},$$

and a base drive frequency

$$f_0 = 150\,\mathrm{kHz},$$

the first three $m = 1$ radius locks are

$$R_{\text{coil}}^{(1,1)} = \frac{1.09384563 \times 10^6}{2\pi(1)(150000)} = 1.1606\,\mathrm{m}, \tag{443}$$

$$R_{\text{coil}}^{(2,1)} = 0.5803\,\mathrm{m}, \tag{444}$$

$$R_{\text{coil}}^{(3,1)} = 0.3869\,\mathrm{m}. \tag{445}$$

For a smaller experimental coil with

$$R_{\text{coil}} = 0.150\,\mathrm{m},$$

the $(n = 3,\ m = 1)$ lock frequency is

$$f_0 = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{2\pi n R_{\text{coil}}} = \frac{1.09384563 \times 10^6}{2\pi(3)(0.150)} = 3.8687 \times 10^5\,\mathrm{Hz}. \tag{446}$$

Hence a 0.150 m coil driven at 150 kHz does not satisfy the $(n = 3, m = 1)$ ring-lock condition. Its lock parameter is

$$\eta_{3,1} = \frac{3(150000)\,2\pi(0.150)}{1.09384563 \times 10^6} = 0.3877. \tag{447}$$

This is sub-resonant for the $(3, 1)$ ring mode, although it may still couple to different (n, m) channels.

L.4 Three-phase selection rule

Consider a three-phase coil drive with phase index $q = 0, 1, 2$. If the q-th phase is shifted by $2\pi q/3$, the discrete phase sum for temporal harmonic n and spatial azimuthal mode m is

$$T^{(3\phi)}_{n,m} = \frac{1}{3}\left|\sum_{q=0}^{2} \exp!\left[iq\frac{2\pi}{3}(m-n)\right]\right|. \tag{448}$$

The exact selection rule is

$$\boxed{T^{(3\phi)}_{n,m} = \begin{cases} 1, & m \equiv n \pmod 3, \\ 0, & m \not\equiv n \pmod 3. \end{cases}} \tag{449}$$

Thus a three-phase winding does not merely "add power"; it filters the allowed temporal–spatial mode pairs.

L.5 Double-stack phase-shift filter

Let the coil be built as two vertically separated stacks, with the second stack rotated azimuthally by

$$\delta = 30^\circ = \frac{\pi}{6}. \tag{450}$$

Let ψ denote the electrical inter-stack phase offset. The two-stack spatial filter is

$$L_m(\delta, \psi) = |1 + \exp[i(m\delta + \psi)]|. \tag{451}$$

For equal stack polarity and zero additional electrical offset ($\psi = 0$), the destructive condition is

$$m\delta = (2q+1)\pi, \qquad q \in \mathbb{Z}. \tag{452}$$

With $\delta = \pi/6$, this gives

$$\boxed{m = 6, 18, 30, \ldots} \tag{453}$$

Therefore a 30° double-stack geometry exactly suppresses the sixth azimuthal spatial harmonic and its odd multiples in this idealized equal-amplitude model.

L.6 Golden-duty PWM spectrum

Let the duty cycle be the inverse golden-square fraction

$$d = \varphi^{-2} = \left(\frac{1+\sqrt{5}}{2}\right)^{-2} = 0.381966011\ldots. \tag{454}$$

For an ideal unipolar rectangular pulse train, the magnitude of the n-th Fourier coefficient, ignoring the DC normalization convention, is proportional to

$$P_n(d) = \left|\frac{\sin(\pi n d)}{\pi n}\right|. \tag{455}$$

For $d = \varphi^{-2}$, the first seven relative amplitudes normalized to P_1 are approximately

n	$P_n(d)$	$P_n(d)/P_1(d)$
1	0.296675	1.000
2	0.107508	0.362
3	0.046948	0.158
4	0.079273	0.267
5	0.017794	0.060
6	0.042102	0.142
7	0.038864	0.131

Thus golden-duty PWM strongly suppresses the fifth harmonic, but does not by itself null the sixth and seventh harmonics. In the proposed three-phase double-stack geometry, the suppression of the fifth, sixth, and seventh bands must be understood as a composite filter effect:

$$\boxed{\mathcal{C}_{n,m} = P_n(d)\, T^{(3\phi)}_{n,m}\, S_m(\mathrm{P40}, +11, -9)\, L_m(\delta, \psi)\, G_m(k_n R_{\mathrm{coil}})}. \tag{456}$$

Here:

- $P_n(d)$ is the PWM harmonic amplitude;
- $T^{(3\phi)}_{n,m}$ is the three-phase selection factor;
- $S_m(\mathrm{P40}, +11, -9)$ is the spatial winding factor of the SawShape path;
- $L_m(\delta, \psi)$ is the double-stack filter;
- $G_m(k_n R_{\mathrm{coil}})$ is the geometric near-field coupling factor.

The wave number of the n-th temporal harmonic is

$$k_n = \frac{2\pi n f_0}{\|\mathbf{v}_{\circlearrowleft}\|}. \tag{457}$$

L.7 SawShape winding factor for $\mathrm{P40}, +11, -9$

For a discrete winding on $N = 40$ angular sites, define the node angle

$$\theta_j = \frac{2\pi p_j}{40}, \qquad p_j \in \mathbb{Z}/40\mathbb{Z}. \tag{458}$$

A SawShape path with alternating skip sequence $(+11, -9)$ can be represented by

$$p_{j+1} = p_j + s_j \pmod{40}, \qquad s_j \in \{+11, -9\}. \tag{459}$$

The normalized spatial Fourier factor is then

$$\boxed{S_m(\mathrm{P40}, +11, -9) = \left| \frac{1}{N_s} \sum_{j=0}^{N_s - 1} w_j \exp(-im\theta_j) \right|}. \tag{460}$$

Here w_j encodes segment weight, orientation, current direction, layer index, and local segment length. This factor is experimentally accessible: it can be inferred by measuring the spatial FFT of the near-field magnetic map over a circular sampling path.

L.8 Canonical interpretation

The proposed coil does not directly generate a canonical force. Instead, it prepares a structured, rotating, mode-filtered boundary condition for the local field. In SST language, the coil acts as a laboratory-scale swirl-mode injector.

The relevant energy-density comparison is

$$\rho_E = \frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2. \tag{461}$$

Using

$$\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}}, \qquad \|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}},$$

one obtains

$$\rho_E = \frac{1}{2}(7.0 \times 10^{-7})(1.09384563 \times 10^{6})^2 = 4.187 \times 10^{5}\,\mathrm{J\,m^{-3}}. \tag{462}$$

The corresponding mass-equivalent density is

$$\rho_m = \frac{\rho_E}{c^2} = \frac{4.187 \times 10^5}{(2.99792458 \times 10^8)^2} = 4.659 \times 10^{-12}\,\mathrm{kg\,m^{-3}}. \tag{463}$$

These values should be understood as canonical scale references, not as a claim that a laboratory coil reaches the full canonical swirl speed in ordinary electromagnetic matter.

L.9 Experimental falsification protocol

A coil experiment is SST-relevant only if it passes the following null hierarchy.

1. **Electrical null:** repeat with identical RMS current and power but randomized phase ordering.
2. **Geometric null:** replace SawShape $(\mathrm{P40}, +11, -9)$ by a symmetry-preserving circular winding with the same resistance and inductance.
3. **Duty-cycle null:** compare $d = 0.381966$ against $d = 0.500$, $d = 0.333$, and $d = 0.250$ at fixed RMS current.
4. **Stack null:** compare 30° double-stack shift against 0°, 15°, 45°, and 60°.
5. **Mode-lock scan:** sweep f_0 and test whether anomalous coherence peaks occur at

$$f_0^{(n,m)} = \frac{|m|\,\|\mathbf{v}_{\circlearrowleft}\|}{2\pi n R_{\mathrm{coil}}}.$$

6. **Environmental null:** repeat in still air, shielded enclosure, and, where safely possible, reduced pressure to remove ion wind and acoustic coupling.
7. **Force-decomposition null:** decompose any measured force into electromagnetic, thermal, acoustic, EHD, and mechanical vibration contributions before assigning a residual.

Only a residual signal satisfying both

$$\eta_{n,m} \approx 1 \tag{464}$$

and

$$\mathcal{C}_{n,m} \neq 0 \tag{465}$$

is admissible as an SST research-track candidate.

L.10 Vorticity-force decomposition layer

Any measured force must first be written schematically as

$$\mathbf{F}_{\mathrm{meas}} = \mathbf{F}_{\mathrm{EM}} + \mathbf{F}_{\mathrm{EHD}} + \mathbf{F}_{\mathrm{thermal}} + \mathbf{F}_{\mathrm{acoustic}} + \mathbf{F}_{\mathrm{mechanical}} + \mathbf{F}_{\omega} + \mathbf{F}_{\mathrm{res}}. \tag{466}$$

Here $\mathbf{F}_{\omega}$ denotes force reconstructed from ordinary vorticity transport and impulse balance. Only the remaining term $\mathbf{F}_{\mathrm{res}}$, after conventional reconstruction and uncertainty propagation, can be tested against the SST mode-locking rule. The canonical research criterion is therefore

$$\boxed{\mathbf{F}_{\mathrm{res}} = \mathbf{F}_{\mathrm{res}}(\eta_{n,m}, \mathcal{C}_{n,m}) \quad \text{with a peak near} \quad \eta_{n,m} = 1.} \tag{467}$$

L.11 Recommended measured observables

The minimum observable set is:

1. drive current $I_A(t), I_B(t), I_C(t)$;
2. voltage $V_A(t), V_B(t), V_C(t)$;
3. real input power $P_{\text{in}}(t)$;
4. local magnetic field map $\mathbf{B}(\mathbf{x}, t)$;
5. pickup-coil FFT around the active circumference;
6. accelerometer spectrum on the coil frame;
7. air temperature and pressure field if operated in air;
8. force sensor output with synchronized phase reference.

A positive SST candidate requires a phase-stable spectral feature at a predicted (n, m) lock that disappears under at least one of the symmetry-breaking null tests.

L.12 Conclusion

The 3-phase SawShape $(\text{P40}, +11, -9)$ double-stack coil with 30° inter-stack rotation and $d = \varphi^{-2}$ PWM is canonically useful as a mode-filtered swirl-excitation apparatus. Its SST value is not that it assumes a new force, but that it makes the frequency–length criterion

$$\|\mathbf{v}_{\circlearrowleft}\| = f\Delta x$$

experimentally precise.

The essential design rule is

$$\eta_{n,m} = \frac{n f_0}{\|\mathbf{v}_{\circlearrowleft}\|} \frac{2\pi R_{\text{coil}}}{|m|} \approx 1,$$

with observable coupling only when

$$\mathcal{C}_{n,m} = P_n(d)\, T^{(3\phi)}_{n,m}\, S_m(\text{P40}, +11, -9)\, L_m(\delta, \psi)\, G_m(k_n R_{\text{coil}})$$

is nonzero. This converts the coil concept into a falsifiable SST metrology protocol.

M Research Appendices: Topology, Stability, and Fluid Mechanics

M.1 Research Appendix: Non-Abelian Topological Charge for Protected Swirl Strings

Canon status. [SPECULATIVE] Research Track. This appendix does not promote a new particle assignment to canon. It records a canon-compatible extension of the SST topology sector: knot type and helicity may be insufficient to determine dynamical stability when reconnection and strand-crossing moves are allowed. A non-Abelian coloring invariant may supply an additional obstruction to decay.

M.1.1 Motivation

The current SST knot sector uses geometric and topological descriptors such as crossing number, ropelength, helicity, chirality, and hyperbolic-volume proxies. A minimal research-track screening form is

$$\frac{E_{\rm eff}[K]}{E_0} = \alpha_C C(K) + \alpha_L L(K) + \alpha_H \widehat{\mathcal{H}}(K) + \cdots, \tag{468}$$

where $C(K)$ is a crossing-complexity proxy, $L(K)$ is a ropelength or length proxy, and $\widehat{\mathcal{H}}(K)$ denotes nondimensional helicity or helicity-derived topological data. This is a scoring functional, not a derived Hamiltonian.

Recent work on non-Abelian topological defects shows that some linked structures cannot be removed by local reconnections and strand crossings because the strands carry non-commuting elements of an order-parameter fundamental group [71]. This suggests that SST should distinguish two layers:

1. the geometric knot type K of the swirl string centerline;
2. an internal topological coloring q valued in a discrete or continuous group G.

A concrete benchmark choice is the quaternion group

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}, \qquad ij = k, \qquad ji = -k, \tag{469}$$

so that $ij \neq ji$. The general symbol G below is retained because SST does not yet select a unique internal order-parameter group.

M.1.2 Group-colored swirl-string sector

Let a colored swirl-string configuration be denoted

$$\mathcal{K}_G = (K, q), \tag{470}$$

where

$$q : \pi_1\Big(\mathbb{R}^3 \setminus K\Big) \longrightarrow G \tag{471}$$

assigns a group element to each Wirtinger generator of the complement. For a diagram with arc generators a_i, the coloring must satisfy the usual crossing consistency relations. In SST language, $q(a_i)$ labels an internal director or phase-sector attached to a local swirl-string strand.

[ORTHODOX] Constraint. For a non-Abelian group G, strand exchange is obstructed when the corresponding labels do not commute:

$$[q(a_i), q(a_j)] = q(a_i)q(a_j)q(a_i)^{-1}q(a_j)^{-1} \neq e_G. \tag{472}$$

[SPECULATIVE] SST interpretation. Equation (472) is interpreted as a local reconnection barrier in the swirl medium. If two strands belong to non-commuting internal sectors, a direct crossing or reconnection requires an additional defect cord or phase bridge. In SST terms this is a candidate mechanism for topological protection beyond ordinary helicity conservation.

M.1.3 Protected-sector invariant

Define a protection marker

$$\mathcal{P}_G(\mathcal{K}_G) = \begin{cases} 1, & \text{if no allowed local reconnection/crossing sequence reduces } \mathcal{K}_G \text{ to the trivial sector}, \\ 0, & \text{otherwise.} \end{cases} \tag{473}$$

The marker $\mathcal{P}_G$ is not an energy term by itself. It is a sector label. Energetically, it enters as a barrier rule:

$$\Delta E_{\mathrm{rec}}(\mathcal{K}_G \to \mathcal{K}'_G) = \begin{cases} < \infty, & Q_G(\mathcal{K}_G) = Q_G(\mathcal{K}'_G), \\ \infty \text{ or macroscopically large}, & Q_G(\mathcal{K}_G) \neq Q_G(\mathcal{K}'_G), \end{cases} \tag{474}$$

where Q_G is a conserved coloring invariant under the permitted move set.

A useful research-track extension of Eq. (468) is therefore the scalarized screening form

$$\frac{E_{\mathrm{eff}}^{(G)}[\mathcal{K}_G]}{E_0} = \alpha_C C(K) + \alpha_L L(K) + \alpha_H \widehat{\mathcal{H}}(K) + \alpha_G I_G(\mathcal{K}_G), \tag{475}$$

where

$$I_G : Q_G(\mathcal{K}_G) \longrightarrow \mathbb{R}_{\geq 0} \tag{476}$$

extracts a real-valued index from the conserved coloring sector. The barrier scale Δ_G belongs to reconnection dynamics through Eq. (474); it is not promoted to a fitted mass term.

M.1.4 Dimensional check

The quantities $C(K)$, $L(K)$ after ropelength normalization, $\widehat{\mathcal{H}}(K)$, and I_G are dimensionless. Therefore the coefficients in Eq. (475) are dimensionless if the left-hand side is E_{eff}/E_0. A dimensional energy is recovered only after selecting an SST energy scale E_0, for example a calibrated tube energy, a core line-tension scale, or a controlled numerical relaxation scale. The group object Q_G itself is not added to the energy; only a scalar index extracted from it may enter a screening functional.

M.1.5 Minimal falsifier

If two SST candidate particles have the same $(C, L, \mathcal{H})$ but different observed stability, the model predicts that a missing internal sector label exists:

$$(C_1, L_1, \mathcal{H}_1) = (C_2, L_2, \mathcal{H}_2), \qquad \tau_1 \neq \tau_2 \quad \Rightarrow \quad Q_G(\mathcal{K}_1) \neq Q_G(\mathcal{K}_2). \tag{477}$$

Failure to find such a sector in simulation or experiment would demote this appendix from Research Track to rejected model branch.

M.2 Research Appendix: Reconnection Decay as a Stability Falsifier

Canon status. [ORTHODOX] Constraint + [SPECULATIVE] SST interpretation. This appendix records a negative constraint on SST. A single-component Gross–Pitaevskii-like medium with permitted reconnections does not automatically protect knotted filaments [72]. Therefore, SST particle stability requires additional structure beyond generic phase-defect topology.

M.2.1 Baseline statement

In a dissipative or reconnection-permitting medium, knotted filament topology can decay. The relevant orthodox lesson is not that SST is false, but that topology alone is insufficient unless the move set is restricted by the dynamics.

Let $K(t)$ denote the topology of a swirl-string centerline. A reconnection path is a sequence

$$K_0 \longrightarrow K_1 \longrightarrow \cdots \longrightarrow K_N, \tag{478}$$

where each arrow is a local reconnection or strand-crossing event. If K_N is an unlink or collection of unknotted loops, then K_0 is dynamically unprotected under that move set.

M.2.2 SST stability criterion

SST should not claim that every closed filament is stable. A canon-compatible stability criterion is

$$\tau_{\mathrm{life}}(K) \sim \tau_0 \exp\left(\frac{\Delta E_{\mathrm{rec}}(K)}{E_{\mathrm{drive}}}\right), \tag{479}$$

where $\Delta E_{\mathrm{rec}}(K)$ is the lowest reconnection barrier out of the topological sector, and E_{drive} is the effective environmental or internal driving energy available to trigger reconnection.

The stable-particle condition is therefore not merely

$$K \neq \bigcirc, \tag{480}$$

but rather

$$\Delta E_{\mathrm{rec}}(K) \gg E_{\mathrm{drive}} \quad \text{and/or} \quad Q_G(K) \neq Q_G(\bigcirc). \tag{481}$$

M.2.3 Relation to Chronos–Kelvin transport

For ideal SST evolution without reconnection or external source, the Chronos–Kelvin invariant remains

$$\frac{D}{Dt}\left(R^2\omega\right) = 0. \tag{482}$$

Reconnection is precisely a non-ideal event where this transport law is locally invalid or discontinuous. Thus a κ-type transition in the canon may be interpreted as the event layer at which topology can change.

[SPECULATIVE] Identification. A reconnection event is an SST Kairos event when it changes the topological sector:

$$\kappa: \quad K_i \to K_{i+1}, \qquad Q_G(K_i) \neq Q_G(K_{i+1}) \quad \text{or} \quad \mathcal{H}(K_i) \neq \mathcal{H}(K_{i+1}). \tag{483}$$

M.2.4 Dimensionless diagnostic

M.2.5 Topological Hitting Conditions and Chronos–Kelvin Phase Slips

Canon status ledger.

[ORTHODOX]	Geometric thickness	Curvature radius and doubly-critical self-distance control finite tube contact.
[ORTHODOX]	Phase-slip/reconnection layer	He-II/GP-like media permit non-ideal reconnection events.
[CANON-COMPATIBLE]	Ideal SST transport	Without reconnection/source, Chronos–Kelvin transport remains conserved.
[RESEARCH TRACK]	First-hitting boundary	A Kairos event is modeled as a distributional boundary condition, not a smooth correction.
[SPECULATIVE]	SST core scale	The identification $a_{\mathrm{rec}} = r_c$ requires a resolved-core SST move-set proof.
[FALSIFIER]	Two-scale test	He-II coherence length and SST core radius define distinct reconnection thresholds.

This block is Research Track. It does not modify the ideal SST evolution law. It defines the non-ideal boundary surface at which the ideal Chronos–Kelvin branch ceases to be a valid smooth continuation and the state must be represented in the Kairos event layer. Orthodox superfluid reconnections and phase-slip phenomenology provide the analogue event class, not the SST core proof [33, 34, 35, 72].

Orthodox geometric premise. In the orthodox ropelength setting, finite tube contact is controlled by curvature and doubly-critical self-distance [61, 63]. Let $\gamma_K : S^1 \times \mathbb{R} \to \mathbb{R}^3$ denote the centerline of a finite-thickness vortex tube in knot class K, with curvature $\kappa(s,t)$. Its geometric thickness is

$$\tau_K(t) = \min\left[\frac{1}{\kappa_{\max}(t)}, \frac{1}{2} d_{\text{dcsd}}(t)\right], \tag{484}$$

where

$$d_{\text{dcsd}}(t) = \min_{(s_1,s_2)\in\text{dcsd}} \|\gamma_K(s_1,t) - \gamma_K(s_2,t)\|. \tag{485}$$

The min-structure in Eq. (484) is the key point: a loss of thickness may occur either by excessive curvature or by self-contact. These are physically distinct events.

Derived split: kink versus reconnection. Define $\inf\emptyset := +\infty$. The first curvature-contact event is

$$t_{\text{kink}} = \inf\left\{t : \frac{1}{\kappa_{\max}(t)} \le a_{\text{rec}}, \quad D_t^+\left(\frac{1}{\kappa_{\max}}\right) < 0\right\}. \tag{486}$$

The first topological reconnection event is instead

$$t_{\text{rec}} = \inf\left\{t : d_{\text{dcsd}}(t) \le 2a_{\text{rec}}, \quad D_t^+ d_{\text{dcsd}}(t) < 0, \quad E_{\text{avail}}(t) \ge E_{\text{b}}, \quad \Delta w \in \mathbb{Z}\setminus\{0\}\right\}. \tag{487}$$

Here D_t^+ denotes the right-sided time derivative, and Δw denotes the integer winding or phase-slip jump across the local core-contact surface. The first breakdown of ideal smooth evolution is

$$\boxed{t_* = \min(t_{\text{kink}}, t_{\text{rec}}).} \tag{488}$$

Only t_{rec} is a topology-changing reconnection event. A curvature event t_{kink} is a local core-compression or kink-barrier event and must not be automatically labeled as a topological defect.

Reconnection scale. The reconnection radius is medium-dependent:

$$a_{\text{rec}} = \begin{cases} r_c, & \text{SST resolved-core hypothesis,} \\ \xi, & \text{orthodox He-II coherence-length falsifier layer.} \end{cases} \tag{489}$$

With $r_c = 1.40897017 \times 10^{-15}$ m and a representative He-II coherence length $\xi \sim 10^{-10}$ m,

$$\frac{\xi}{r_c} \simeq 7.09738 \times 10^4, \qquad d_{\text{rec}}^{\text{SST}} = 2r_c = 2.81794034 \times 10^{-15}\,\text{m}, \qquad d_{\text{rec}}^{\text{He-II}} \sim 2 \times 10^{-10}\,\text{m}. \tag{490}$$

Thus He-II reconnection experiments primarily test the allowed topological move-set of a superfluid medium; they do not directly probe the SST core radius unless an additional scale-bridge is derived.

Chronos–Kelvin branch and distributional failure. On the ideal branch $t < t_*$, define the scalar Chronos–Kelvin quantity

$$\Lambda_{\text{CK}} = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_s} R^2 \sqrt{1 - (S_t^{\circlearrowleft})^2}. \tag{491}$$

Then

$$\frac{D\Lambda_{\text{CK}}}{Dt_{ae}} = 0, \qquad t < t_*. \tag{492}$$

At a Kairos event the same quantity is not smoothly corrected. It acquires a distributional jump:

$$\frac{D\Lambda_{\text{CK}}}{Dt_{ae}} = \sum_j \Delta\Lambda_j\, \delta(t - t_j), \qquad t_j = t_{\text{rec},j}. \tag{493}$$

Consequently, the Swirl–Clock branch

$$S_t^{\circlearrowleft}(R)=\sqrt{1-\left(\frac{\Lambda_{\mathrm{CK}}r_s}{\|\mathbf{v}_{\circlearrowleft}\|R^2}\right)^2} \tag{494}$$

exists only while

$$R\geq R_{\mathrm{CK}}=\left(\frac{\Lambda_{\mathrm{CK}}r_s}{\|\mathbf{v}_{\circlearrowleft}\|}\right)^{1/2}. \tag{495}$$

The operative transition radius is therefore

$$\boxed{R_{\mathrm{trans}}=\max(a_{\mathrm{rec}},R_{\mathrm{CK}}).} \tag{496}$$

Inside R_{trans}, the ideal smooth Swirl–Clock continuation is no longer defined; the state must be represented in the non-ideal radiative defect sector.

Energetic jump condition. The available reconnection energy must be non-negative. For a proposed move $K_i\to K_{i+1}$ define

$$E_{\mathrm{avail}}^{ij}=W_{\mathrm{ext}}+[E_{\mathrm{tube}}(K_i)-E_{\mathrm{tube}}(K_{i+1})]_+\,,\qquad [x]_+:=\max(x,0). \tag{497}$$

The reconnection gate opens only if

$$\boxed{E_{\mathrm{avail}}^{ij}\geq E_{\mathrm{b}}^{ij}\,,} \tag{498}$$

where E_{b}^{ij} is the local core-contact or phase-slip barrier. A useful leading slender-tube scale is

$$E_{\mathrm{tube}}(K)\simeq\frac{\rho_f\Gamma_0^2}{4\pi}\,\ell_K\ln\left(\frac{R_{\mathrm{out}}}{a_{\mathrm{rec}}}\right). \tag{499}$$

The units are

$$[\rho_f\Gamma_0^2\ell_K]=\left(\frac{\mathrm{kg}}{\mathrm{m}^3}\right)\left(\frac{\mathrm{m}^4}{\mathrm{s}^2}\right)(\mathrm{m})=\mathrm{J}. \tag{500}$$

Canonical transition matrix. Let

$$N_{\mathrm{slip}}(t)=\sum_j n_jH(t-t_j),\qquad \dot{N}_{\mathrm{slip}}(t)=\sum_j n_j\delta(t-t_j). \tag{501}$$

Then the state split is

$$\boxed{\begin{aligned}\text{Stable T-phase / no-slip state:}\quad&\tau_K>a_{\mathrm{rec}},\quad N_{\mathrm{slip}}=\mathrm{const.},\quad E_{\mathrm{avail}}<E_{\mathrm{b}},\quad \frac{D\Lambda_{\mathrm{CK}}}{Dt_{ae}}=0;\\ \text{Dissipative Kairos event:}\quad&d_{\mathrm{dcsd}}\leq 2a_{\mathrm{rec}},\quad D_t^+d_{\mathrm{dcsd}}<0,\quad \Delta w\in\mathbb{Z}\setminus\{0\},\quad E_{\mathrm{avail}}\geq E_{\mathrm{b}},\\ &\frac{D\Lambda_{\mathrm{CK}}}{Dt_{ae}}=\sum_j\Delta\Lambda_j\delta(t-t_j)\neq 0.\end{aligned}} \tag{502}$$

Falsification rule. A simulation or experiment which labels a curvature-only event t_{kink} as a topological reconnection violates the status separation in Eq. (488). Conversely, a claimed reconnection without $d_{\mathrm{dcsd}}\leq 2a_{\mathrm{rec}}$, nonzero phase-slip index, and a positive energy gate violates Eq. (487). This makes the conclusion operational:

$$\boxed{\text{ideal SST conserves }\Lambda_{\mathrm{CK}};\quad\text{Kairos events inject distributional jumps.}} \tag{503}$$

M.2.6 Dimensionless diagnostic

Define the stability ratio

$$\Xi_{\rm rec}(K) = \frac{\Delta E_{\rm rec}(K)}{E_{\rm tube}(K)}. \tag{504}$$

For a slender swirl tube, the standard leading envelope energy scale is

$$E_{\rm tube}(K) \simeq \frac{\rho_f \Gamma^2}{4\pi}\, \ell_K \ln\!\left(\frac{R}{r_c}\right), \tag{505}$$

where Γ is circulation, ℓ_K is centerline length, and R is the outer cutoff. The units check:

$$[\rho_f \Gamma^2 \ell_K] = \frac{\rm kg}{\rm m^3}\frac{\rm m^4}{\rm s^2}\,{\rm m} = {\rm kg\, m^2 s^{-2}} = {\rm J}. \tag{506}$$

A practical SST classification rule is

$$\Xi_{\rm rec}(K) \lesssim 1 \Rightarrow \text{hydrodynamically fragile}, \qquad \Xi_{\rm rec}(K) \gg 1 \Rightarrow \text{candidate stable T-phase state}. \tag{507}$$

M.2.7 Minimal simulation protocol

A candidate SST knot assignment should be tested under three levels of dynamics:

1. single-component phase-defect evolution;
2. constrained incompressible evolution without reconnection;
3. multi-director or non-Abelian colored evolution with reconnection rules.

Only states stable under the third protocol should be promoted as particle candidates.

M.3 Research Appendix: Fractional Torus-Phase Winding and Swirl-Clock Phase

Canon status. [SPECULATIVE] Research Track. This appendix records a bridge between coordinated rotational phase mechanics and SST R-phase torsional packets, using fractional-order optical torus knots as the mathematical template [73]. It does not assert that optical polarization knots are literal SST particles. It supplies a useful mathematical template for phase winding, internal rotation, and torus-knot closure.

M.3.1 Coordinated phase generator

Let θ be an external azimuthal coordinate and let ϕ be an internal director angle. A coordinated rotation acts as

$$\theta \mapsto \theta + \alpha, \qquad \phi \mapsto \phi + \gamma\alpha. \tag{508}$$

The generator has the schematic form

$$J_\gamma = L_{\rm orb} + \gamma S_{\rm int}, \tag{509}$$

where $L_{\rm orb}$ shifts external azimuthal phase and $S_{\rm int}$ shifts internal swirl-director phase.

For two commensurate internal frequencies $p\Omega$ and $q\Omega$ carrying azimuthal mode numbers m_1 and m_2, the coordinated-winding parameter is

$$\gamma = \frac{m_2 p - m_1 q}{p+q}. \tag{510}$$

The corresponding torus-winding pair is

$$(m, n) = (m_2 p - m_1 q,\ p+q)\,. \tag{511}$$

If $\gcd(m,n) = 1$, the internal lobe trajectory is a single (m,n) torus knot. If $d = \gcd(m,n) > 1$, the trajectory splits into d linked components.

M.3.2 SST phase-action translation

To avoid notation conflict with the Swirl-Clock factor $S_{(t)}^{\circlearrowleft}$, define a phase-action field $\mathscr{S}(\vec{x},t)$. In an R-phase torsional packet, write the reduced phase on a toroidal carrier as

$$\Theta(\theta,\phi,t) = n\phi - m\theta - \Omega t. \tag{512}$$

The closure condition is

$$\oint d\Theta = 2\pi N, \qquad N \in \mathbb{Z}. \tag{513}$$

The phase-action field may be represented locally as

$$\mathscr{S}(\vec{x},t) = \hbar_{\mathrm{SST}}\,\Theta(\vec{x},t), \tag{514}$$

so that the circulation quantization condition becomes

$$\oint \nabla\mathscr{S}\cdot d\vec{\ell} = 2\pi N\hbar_{\mathrm{SST}}. \tag{515}$$

M.3.3 Connection to Swirl-Clock modulation

The local Swirl-Clock factor remains the kinematic clock factor

$$S_{(t)}^{\circlearrowleft} = \sqrt{1 - \frac{\|\mathbf{u}_{\circlearrowleft}(\mathbf{x},t)\|^2}{c^2}}. \tag{516}$$

The phase-action field controls wave-like periodicity, while $S_{(t)}^{\circlearrowleft}$ controls local clock rate. The research-track bridge is that a stable R-phase packet may require simultaneous closure of both:

$$\oint \nabla\mathscr{S}\cdot d\vec{\ell} = 2\pi N\hbar_{\mathrm{SST}}, \tag{517}$$

$$\frac{D}{Dt}\left(R^2\omega\right) = 0. \tag{518}$$

[SPECULATIVE] Interpretation. Equation (510) suggests a route to fractional internal winding without fractional circulation. The circulation quantum remains integer-valued, but the internal phase orientation can return only after multiple external turns. This may be useful for modeling R-phase torsional photons, spin-orbit-like packet structure, or generation-family phase layering.

M.3.4 Dimensional check

The variables m, n, p, q, N and γ are dimensionless. The phase Θ is dimensionless. Therefore $\mathscr{S} = \hbar_{\mathrm{SST}}\Theta$ has units of action, and Eq. (515) has units of action because $\nabla\mathscr{S}\cdot d\vec{\ell}$ has units $(\mathrm{J\,s\,m^{-1}})\mathrm{m} = \mathrm{J\,s}$.

M.4 Research Appendix: Topological Fluid Mechanics and the SST Energy Functional

Canon status. [ORTHODOX] Mathematical background + [SPECULATIVE] SST synthesis. This appendix records the topological-fluid mechanics background behind the SST knot-energy program [36, 74]. The orthodox part is that ideal incompressible flow preserves topology and helicity under the usual assumptions. The speculative part is the identification of selected relaxed knot classes with particle sectors.

M.4.1 Frozen-field topology

In an ideal incompressible medium, topology is transported by the flow map. For SST this supports the use of closed swirl strings as dynamically persistent structures under non-reconnecting evolution.

The primary invariants are

$$\Gamma = \oint_C \mathbf{u}_{\circlearrowleft} \cdot d\vec{\ell}, \tag{519}$$

$$\mathcal{H} = \int_V \mathbf{u}_{\circlearrowleft} \cdot \vec{\omega}\, d^3x, \tag{520}$$

$$\vec{\omega} = \nabla \times \mathbf{u}_{\circlearrowleft}. \tag{521}$$

For linked tubes with circulations Γ_i, the leading topological helicity decomposition is

$$\mathcal{H} \simeq \sum_i \Gamma_i^2\, SL_i + 2\sum_{i<j} Lk_{ij}\Gamma_i\Gamma_j, \tag{522}$$

where SL_i is self-linking and Lk_{ij} is Gauss linking number.

M.4.2 Cohomological helicity and mapping-class sectors

Status. [ORTHODOX DIFFERENTIAL-TOPOLOGICAL INPUT / RESEARCH-TRACK SST STATE LABEL]. Let $\alpha = \iota_{\mathbf{V}} d^3x$ be the two-form dual to a divergence-free field $\mathbf{V}$ tangent to the boundary of a compact domain $\Omega \subset \mathbb{R}^3$. Cantarella and Parsley express the same helicity both as a configuration-space cohomology pairing and as a wedge product with a Biot–Savart-selected primitive [37]:

$$\mathcal{H}(\alpha) = \int_{C_2[\Omega]} \alpha_x \wedge \alpha_y \wedge g^*(d\Omega_{S^2}), \tag{523}$$

$$= \int_\Omega \alpha \wedge \mathrm{BS}(\alpha). \tag{524}$$

The compactified configuration space $C_2[\Omega]$ resolves the pair diagonal $x = y$ by retaining the limiting direction of approach. This is a geometric regularization of the Gauss kernel; it is not an SST core cutoff and does not set a_{core} or r_c.

Primitive and gauge discipline. On a multiply connected domain, the value of $\int_\Omega \alpha \wedge \beta$ need not be independent of an arbitrary primitive $d\beta = \alpha$. The Biot–Savart/Hodge choice in Eq. (524), together with the independent flux and boundary data, is part of the observable definition. A numerical export of $\mathcal{H}$ is therefore incomplete unless it declares

$$(\boldsymbol{\omega}, \{\Gamma_i\}, [\beta|_{\partial\Omega}], \text{Hodge/Biot–Savart gauge})\,. \tag{525}$$

For the kinetic-helicity mapping one sets $\mathbf{V} = \boldsymbol{\omega}$ and $\mathrm{BS}(\boldsymbol{\omega}) = \mathbf{u}_{\circlearrowleft}$ up to the separately fixed harmonic sector.

Mapping-class jump law. For an orientation-preserving diffeomorphism $f : \Omega \to \Omega'$, the exact change is

$$\Delta\mathcal{H} = \mathcal{H}\Big((f^{-1})^*\alpha\Big) - \mathcal{H}(\alpha) = \sum_{i,j} c_{ij}\Phi_i\Phi_j = \mathbf{\Phi}^T C \mathbf{\Phi}, \tag{526}$$

where the coefficients c_{ij} are fixed by the induced action on boundary homology. For a solid torus and $j \in \mathbb{Z}$ Dehn twists,

$$\Delta\mathcal{H} = j\Phi^2. \tag{527}$$

With $\mathbf{V} = \boldsymbol{\omega}$, the spanning-disk flux is the circulation, $\Phi = \Gamma$. One canonical circulation unit therefore gives

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}^* = 9.68361920 \times 10^{-9}\,\mathrm{m}^2\,\mathrm{s}^{-1}, \tag{528}$$

$$\Delta\mathcal{H}_{j=1} = \Gamma_0^2 = 9.37724809 \times 10^{-17}\,\mathrm{m}^4\,\mathrm{s}^{-2}. \tag{529}$$

The dimensions are those of kinetic helicity. Equation (529) does not quantize circulation by itself; integer circulation requires an independent phase or Onsager closure.

Relation to writhe and twist. In the thin single-tube limit,

$$\mathcal{H} \simeq \Gamma^2(Wr + Tw), \tag{530}$$

and a Dehn twist changes the framing by $\Delta Tw = j$ while leaving the centerline embedding potentially unchanged. Thus two states with the same K, $\mathrm{Rop}(K)$, Wr, and Γ may still differ by $j\Gamma^2$ in helicity. The SST state label must therefore be enlarged to

$$\mathfrak{T}_K = (K, \{\Gamma_i\}, Wr, Tw, [f]_{\mathrm{map}})\,. \tag{531}$$

The twist variable is measured state data, not a residual parameter chosen to close a target helicity.

Chronos–Kairos separation. A smooth ideal material flow f_t with $f_0 = \mathrm{id}$ is homotopic to the identity and conserves helicity. A transition between distinct mapping classes is therefore entered in the Kairos ledger and must name its physical trigger. Fluxless sectors, $\Phi_i = 0$, are invariant under the quadratic jump law even when the mapping class is nontrivial.

Falsifier. If a numerical ideal-Euler evolution without boundary work, phase slip, or reconnection changes the tuple $(\mathcal{H}, \{\Gamma_i\}, [f]_{\mathrm{map}})$ beyond discretization error, the implementation violates the frozen-field topology gate.

M.4.3 Energy relaxation under topological constraints

For a slender swirl tube, the leading hydrodynamic energy scale is

$$E_{\mathrm{env}} \simeq \frac{\rho_f \Gamma^2}{4\pi}\,\ell_K \ln\!\left(\frac{R}{r_c}\right), \tag{532}$$

where ℓ_K is the tube centerline length. The core contribution can be represented by a line-tension term

$$E_{\mathrm{core}} \simeq T_{\mathrm{core}}\ell_K. \tag{533}$$

Near contacts and curvature corrections may be encoded by

$$E_{\mathrm{geom}}[K] = T_{\mathrm{core}}\ell_K + \frac{\rho_f \Gamma^2}{4\pi}\ell_K \ln\!\left(\frac{R}{r_c}\right) + \beta_\kappa \int_K \kappa_s^2\,ds + \beta_{\mathrm{nc}}\,C_{\mathrm{near}}(K). \tag{534}$$

Here κ_s is centerline curvature and C_{near} is a near-contact or crossing-complexity proxy.

M.4.4 Isoperimetric domain-shape and curl-spectrum gate

Status. [ORTHODOX OPERATOR/SHAPE CALCULUS / RESEARCH-TRACK SST DOMAIN OPTIMIZATION]. For a smooth compact domain $\Omega \subset \mathbb{R}^3$, define

$$K(\Omega) = \{\mathbf{V} : \nabla \cdot \mathbf{V} = 0,\ \mathbf{V} \cdot \mathbf{n} = 0\}, \tag{535}$$

$$\mathrm{BS}_K = P_{K(\Omega)}\,\mathrm{BS}. \tag{536}$$

The modified operator is compact and self-adjoint on the L^2 completion of $K(\Omega)$. Its largest positive eigenvalue, the maximal helicity-to-L^2 ratio, and the smallest positive curl eigenvalue on $\operatorname{im}\mathrm{BS}_K$ satisfy

$$\lambda_{\max}(\Omega) = \max_{\mathbf{V}\in K(\Omega)\setminus\{0\}} \frac{\langle \mathbf{V}, \mathrm{BS}_K\,\mathbf{V}\rangle}{\langle \mathbf{V}, \mathbf{V}\rangle} = \frac{1}{\kappa_1^{\mathrm{curl}}(\Omega)}, \tag{537}$$

$$\nabla \times \mathbf{V}_1 = \kappa_1^{\mathrm{curl}}\mathbf{V}_1. \tag{538}$$

Thus centerline ropelength and the spectrum of the filled tube domain are independent geometric data [62].

Energy-symbol guard. The operator paper uses

$$\mathscr{E}_V = \int_\Omega |\mathbf{V}|^2 d^3x. \tag{539}$$

For the hydrodynamic choice $\mathbf{V} = \boldsymbol{\omega}$, $\mathscr{E}_V$ is enstrophy while

$$E_{\mathrm{kin}} = \frac{\rho_f}{2}\int_\Omega |\mathbf{u}_{\circlearrowleft}|^2 d^3x \tag{540}$$

is kinetic energy. Maximizing helicity per enstrophy is not automatically the same problem as minimizing SST rest energy at fixed helicity. Any use of κ_1^{curl} in a mass functional must state which field occupies the operator slot and which physical energy is being extremized.

First variation under a volume-preserving shape deformation. Let $\mathbf{W}$ be the initial velocity of a volume-preserving deformation of Ω and carry $\mathbf{V}$ with the domain. The orthodox shape derivative is

$$\delta\mathscr{E}_V = -2\int_\Omega [\mathbf{V} \times (\nabla \times \mathbf{V})] \cdot \mathbf{W}\, d^3x - \int_{\partial\Omega} |\mathbf{V}|^2\, \mathbf{W} \cdot \mathbf{n}\, dA. \tag{541}$$

For a curl eigenfield the volume term vanishes, leaving a boundary-only shape sensitivity. For a simple eigenvalue the corresponding exact Rayleigh-quotient derivative is

$$\delta\lambda_1 = \lambda_1 \frac{\int_{\partial\Omega} |\mathbf{V}_1|^2\, \mathbf{W} \cdot \mathbf{n}\, dA}{\int_\Omega |\mathbf{V}_1|^2\, d^3x}. \tag{542}$$

Necessary boundary conditions for a smooth global optimum. At fixed domain volume, any smooth global maximizer must satisfy

$$|\mathbf{V}_1||_{\partial\Omega_a} = C_a > 0 \quad \text{on each boundary component}, \tag{543}$$

$$\chi(\partial\Omega_a) = 0, \tag{544}$$

$$k_g(\text{boundary orbit}) = 0. \tag{545}$$

Hence each smooth closed boundary component is a torus and the boundary field orbits are intrinsic geodesics. No smooth simply connected domain can be a global optimizer in this unrestricted fixed-volume problem.

Horn-envelope non-claim. The proposed singular "extreme apple" or pinched-torus optimizer in the source is conjectural. It does not identify a trefoil, does not determine R_{horn}, and does not resolve a finite SST vortex core. A canon-compatible research problem is instead the regularized, embedding-restricted program

$$\min_{\gamma_K,\Omega_K} \quad \kappa_1^{\text{curl}}(\Omega_K) \quad \text{subject to} \quad [\gamma_K]=K, \qquad \Omega_K=\text{Tube}_{a_{\text{core}}}(\gamma_K), \tag{546}$$

$$\text{Vol}(\Omega_K)=V_0, \qquad \text{reach}(\gamma_K)\geq a_{\text{core}}. \tag{547}$$

The knot/link embedding, flux data, and harmonic circulations are held fixed within each comparison. This target is a Research-Track shape optimization, not an existing SST theorem.

Operational diagnostics. A domain-shape solver should report

$$\mathcal{D}_{\text{iso}}=\left(\kappa_1^{\text{curl}},\lambda_{\max},\frac{\text{std}_{\partial\Omega}|\mathbf{V}_1|}{\text{mean}_{\partial\Omega}|\mathbf{V}_1|},\|k_g\|_{L^2(\partial\Omega)},\text{reach}(\partial\Omega)\right). \tag{548}$$

The extremal eigenfield is a variational benchmark or stationary candidate. In an ideal inviscid SST evolution, conservation of energy and helicity does not make it a dynamical attractor without a separately declared relaxation mechanism.

M.4.5 Reduction to the SST normal form

After nondimensionalization by a reference energy E_0, Eq. (534) reduces to the canon-style scoring form

$$\frac{E_{\text{geom}}[K]}{E_0}\rightsquigarrow \alpha_C C(K)+\alpha_L L(K)+\alpha_H\widehat{\mathcal{H}}(K)+\alpha_G I_G(K)+\cdots. \tag{549}$$

Here $I_G(K)$ is included only when a non-Abelian coloring sector $Q_G(K)$ has been specified and scalarized by a map $I_G:Q_G\to\mathbb{R}_{\geq 0}$. This justifies the SST normal form as a coarse-grained energy-ordering tool, not as a finished mass theorem.

M.4.6 Canon discipline

The functional in Eq. (549) should be used as follows:

1. **[ORTHODOX]** Use Γ, $\mathcal{H}$, Lk, writhe, twist, and ropelength as established geometric/topological descriptors.
2. **[DERIVED]** Within a chosen tube model, derive the units and leading energy scaling from ρ_f, Γ, ℓ_K, and r_c.
3. **[SPECULATIVE]** Only after this, map a relaxed topological class to an SST particle sector.

M.5 Research Appendix: Resolved-Tube Criticality and Contact-Stress Geometry

Canon status. [ORTHODOX GEOMETRIC BACKGROUND / RESEARCH-TRACK SST INTERPRETATION]. The orthodox input is the ropelength theory of finite-thickness embedded curves and the constrained-gradient formulation of polygonal knot tightening. The SST interpretation is that the active finite-thickness constraints of an ideal swirl-string may be read as a discrete contact-stress skeleton for the resolved tube. This appendix does not canonize a new mass formula; it canonizes a validation discipline for any future profile-resolved tube model.

M.5.1 Resolved tube radius as reach

For a resolved centerline γ_K, the physical tube radius is not the horn radius r_c. The resolved geometric radius is the reach/thickness of the centerline,

$$a_{\text{core}}(K) \equiv \text{Thi}_{\text{rad}}(\gamma_K) = \text{reach}(\gamma_K) = \text{NIR}(\gamma_K), \tag{550}$$

with radius-convention ropelength

$$\text{Rop}_{\text{rad}}(K) = \frac{\text{Len}(\gamma_K)}{a_{\text{core}}(K)}. \tag{551}$$

For smooth centerlines the controlling geometric obstructions are curvature and doubly-critical self-distance,

$$a_{\text{core}}(K) = \min\left(\inf_s \frac{1}{\kappa(s)}, \frac{1}{2}d_{\text{dcsd}}(\gamma_K)\right). \tag{552}$$

Equations (550)–(552) are **[ORTHODOX]** geometric definitions. They preserve the canon rule $a_{\text{core}} \neq r_c$ unless a separate profile calculation proves otherwise.

M.5.2 Struts, kinks, and contact-stress balance

In a polygonal approximation V of the centerline, finite thickness is controlled by two active constraint classes:

$$\text{MinRad}(v_i) \geq a_{\text{core}}, \tag{553}$$

$$\frac{1}{2}d(p,q) \geq a_{\text{core}}. \tag{554}$$

Vertices saturating Eq. (553) form the kink set $\text{Kink}(V)$; pairs saturating Eq. (554) form the strut set $\text{Strut}(V)$. The constrained-gradient criterion for a thickness-critical state has the schematic KKT form

$$-\nabla E[V] + \sum_{(p,q)\in\text{Strut}(V)} \lambda_{pq}\,\nabla\left(\frac{d(p,q)}{2}\right) + \sum_{i\in\text{Kink}(V)} \mu_i\,\nabla\,\text{MinRad}(v_i) = 0, \qquad \lambda_{pq}, \mu_i \geq 0. \tag{555}$$

For the mathematical ropelength problem, $E[V] = \text{Len}(V)$. For a profile-resolved SST tube one should instead test a functional of the form

$$E_{\text{SST}}[\gamma] = T_{\text{eff}}\,\text{Len}(\gamma) + E_{\text{bend}}[\gamma] + E_{\text{twist}}[\gamma] + E_{\text{swirl}}[\gamma] + E_{\text{core}}[\gamma]. \tag{556}$$

The multipliers λ_{pq} and μ_i are then research-track candidates for a discrete swirl-stress/contact-force map, not yet canonical physical forces.

M.5.3 Contact-pressure saturation and Swirl-Clock shielding

Status. [ORTHODOX INPUT / CANON SEMANTIC GUARD / CONDITIONAL DERIVED / RESEARCH-TRACK CONTACT PHYSICS]. The orthodox input is the finite-thickness constraint structure of ropelength criticality: active self-distance constraints are struts, active curvature constraints are kinks, and their critical balance is expressed by non-negative KKT multipliers. The canon semantic guard is that the Swirl-Clock is a kinematic factor determined by the resolved local swirl speed, not by an unrestricted contact multiplier. The algebraic saturation result below is **[CONDITIONAL DERIVED]** once the declared Bernoulli/Biot–Savart pressure closure and speed cap are imposed. The interpretation of the remaining hard multiplier as a physical contact-stress pressure is **[RESEARCH TRACK]**.

Pressure split. At an active self-contact patch, decompose the normal contact pressure into a kinematic Bernoulli/Biot–Savart part and a hard geometric constraint part:

$$\Pi_c = \Pi_{\mathrm{BS}} + \Pi_{\mathrm{hard}}, \tag{557}$$

$$\Pi_{\mathrm{BS}} = \frac{1}{2}\rho_\star u_\theta^2. \tag{558}$$

Here $\rho_\star$ is the declared load-bearing density of the modeled layer. The term Π_{hard} is the continuum counterpart of the active strut/kink multiplier sector enforcing the finite-thickness exclusion

$$d(s,t) \geq 2a_{\mathrm{core}}. \tag{559}$$

Thus Π_{BS} carries the kinematic swirl load, while Π_{hard} carries any additional geometric incompressibility required by contact.

Swirl-Clock coupling. The local Swirl-Clock remains the Lorentz-compatible kinematic factor

$$S_{(t)}^{\circlearrowleft}(\mathbf{x},t) = \sqrt{1 - \frac{\|\mathbf{u}_{\circlearrowleft}(\mathbf{x},t)\|^2}{c^2}}. \tag{560}$$

For a local azimuthal contact model with speed u_θ, Eq. (558) gives

$$S_{(t)}^{\circlearrowleft}(\Pi_{\mathrm{BS}}) = \sqrt{1 - \frac{2\Pi_{\mathrm{BS}}}{\rho_\star c^2}}. \tag{561}$$

Consequently,

$$\nabla S_{(t)}^{\circlearrowleft} = -\frac{1}{\rho_\star c^2 S_{(t)}^{\circlearrowleft}}\,\nabla\Pi_{\mathrm{BS}} \tag{562}$$

inside the declared Bernoulli/Biot–Savart layer. No corresponding rule is assigned to Π_{hard}; a hard-contact multiplier may diverge without forcing $S_{(t)}^{\circlearrowleft} \to 0$.

Canonical speed cap. For one canonical circulation unit,

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}^{*}. \tag{563}$$

The canonical contact-saturation discipline imposes

$$u_\theta \leq v_{\circlearrowleft}^{*}, \tag{564}$$

$$\Pi_{\mathrm{BS}} \leq \Pi_{\mathrm{cap}} = \frac{1}{2}\rho_\star v_{\circlearrowleft}^{*2}. \tag{565}$$

The Swirl-Clock therefore has the finite lower bound

$$\boxed{S_{(t)}^{\circlearrowleft} \geq S_{(t)\,\mathrm{min}}^{\circlearrowleft\,(c)} = \sqrt{1 - \frac{v_{\circlearrowleft}^{*2}}{c^2}}} \tag{566}$$

rather than collapsing to zero at contact. With $v_{\circlearrowleft}^{*} = 1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}}$,

$$S_{(t)\,\mathrm{min}}^{\circlearrowleft\,(c)} = 0.999993343558553, \tag{567}$$

$$1 - S_{(t)\,\mathrm{min}}^{\circlearrowleft\,(c)} = 6.65644145 \times 10^{-6}. \tag{568}$$

If the core-density layer is used, the corresponding kinematic pressure cap is

$$\Pi_{\mathrm{cap}}(\rho_{\mathrm{core}}) = \frac{1}{2}\rho_{\mathrm{core}} v_{\circlearrowleft}^{*2} = 2.32924460 \times 10^{30}\,\mathrm{Pa}. \tag{569}$$

Clock-shear shell. An active strut can still generate a steep transition layer in the clock field. For a characteristic transition thickness ℓ,

$$\|\nabla S_{(t)}^{\circlearrowleft}\| \sim \frac{1 - S_{(t)\,\min}^{\circlearrowleft\,(c)}}{\ell}. \tag{570}$$

At the limiting microscopic scale $\ell \sim r_c$,

$$\|\nabla S_{(t)}^{\circlearrowleft}\| \sim 4.724 \times 10^9 \, \mathrm{m}^{-1}. \tag{571}$$

This large gradient is a finite clock-shear shell, not an infinite time-dilation singularity. Its integrated clock deficit across the shell is only of order 6.66×10^{-6}.

Contact-Constraint Theorem. In any profile-resolved SST tube model that obeys the canonical speed cap,

$$\boxed{\Pi_c \to \infty \quad \Longrightarrow \quad S_{(t)}^{\circlearrowleft} \to S_{(t)\,\min}^{\circlearrowleft\,(c)} = \sqrt{1 - v_{\circlearrowleft}^{*\,2}/c^2}} \tag{572}$$

provided the divergent part of Π_c is assigned to Π_{hard}. Equivalently, the singularity is carried by the contact-stress sector

$$\nabla \cdot \mathbf{T}_{\mathrm{contact}}, \tag{573}$$

not by an unbounded collapse of the Swirl-Clock. The tube becomes geometrically hard before the local clock becomes infinitely slow.

Guard against denominator drift. A factor of the form $\sqrt{1 - u_\theta^2/v_{\circlearrowleft}^{*\,2}}$ may be used as a local topological saturation or transport-efficiency factor inside a declared model, but it is not the Lorentz-compatible Swirl-Clock. The operational clock factor in this canon retains the denominator c^2, as in Eq. (560).

Falsification protocol. A coupled knot-tightening/Biot–Savart audit must report, for each active strut,

$$\Pi_{\mathrm{BS}}, \qquad \Pi_{\mathrm{hard}}, \qquad \lambda_{pq}, \qquad S_{(t)}^{\circlearrowleft}. \tag{574}$$

The saturation guard fails if Π_{BS} reproducibly exceeds $\Pi_{\mathrm{cap}} = \frac{1}{2}\rho_\star v_{\circlearrowleft}^{*\,2}$ without activating a compensating contact-stress multiplier or without reducing the local speed field back below $v_{\circlearrowleft}^{*}$. A standalone ropelength minimizer does not compute Π_{BS} by itself; it supplies the contact skeleton that must be coupled to a Biot–Savart pressure postprocessor.

M.5.4 Contact-map refinement of the topological kernel

A scalar ropelength is not enough to determine all shape-sensitive SST observables. A resolved ideal tube carries a contact skeleton

$$\mathcal{C}_K = \{(p, q, \lambda_{pq})\}_{\mathrm{Strut}} \cup \{(i, \mu_i)\}_{\mathrm{Kink}}. \tag{575}$$

A future profile-resolved version of the dimensionless topology kernel may therefore have the schematic expansion

$$\Xi_K = \Xi_K^{(0)} + \epsilon_s \sum_{(p,q)} \lambda_{pq} + \epsilon_k \sum_i \mu_i + \epsilon_H \mathcal{H}_K + \cdots . \tag{576}$$

Equation (576) is a **[MATCHING ANSATZ]**. Its value is diagnostic: if the large calibrated normalization hidden in Ξ_K can be decomposed into reproducible contact, curvature, and helicity contributions, the mass kernel becomes mechanically inspectable rather than a single opaque coefficient.

M.5.5 Golden-layer factor as hyperbolic transfer parametrization

Status. [RESEARCH-TRACK / PARAMETRIC REWRITE]. This subsection records an exact algebraic reparametrization of the existing golden-layer factor in the SST kernel. It does not derive the factor from Euler dynamics, ropelength criticality, or knot stability.

The identity

$$\operatorname{asinh}\left(\frac{1}{2}\right) = \ln\left(\frac{1}{2} + \sqrt{1 + \frac{1}{4}}\right) = \ln\left(\frac{1+\sqrt{5}}{2}\right) = \ln\varphi \tag{577}$$

allows the layer term to be written as

$$\eta_\varphi \equiv \operatorname{asinh}\left(\frac{1}{2}\right) = \ln\varphi \simeq 0.4812118251, \tag{578}$$

$$\varphi^{-2k(K)} = \exp[-2k(K)\eta_\varphi]\,. \tag{579}$$

Equivalently,

$$\sinh\eta_\varphi = \frac{1}{2}, \qquad \cosh\eta_\varphi = \frac{\sqrt{5}}{2}, \qquad \tanh\eta_\varphi = \frac{1}{\sqrt{5}}. \tag{580}$$

The associated symmetric transfer matrix

$$B_\varphi = \begin{pmatrix} \sqrt{5}/2 & 1/2 \\ 1/2 & \sqrt{5}/2 \end{pmatrix}, \qquad \operatorname{spec}(B_\varphi) = \{\varphi, \varphi^{-1}\}, \tag{581}$$

therefore supplies a compact notation for one elementary layer-transfer step. Iteration gives attenuation eigenvalue

$$\lambda_-^{2k(K)} = \varphi^{-2k(K)}. \tag{582}$$

Canon discipline. This is not a hyperbolic-knot claim. In particular, the electron-sector assignment to the trefoil is not made hyperbolic by Eq. (579); the trefoil remains a torus knot. The useful content is only that the existing golden dressing factor can be written as a discrete exponential attenuation. To promote this from notation to a derived SST mechanism, a future calculation must identify $k(K)$ with a definite filament-transfer, impedance, phase-locking, or relaxation step and derive the elementary rapidity $\eta_\varphi = \operatorname{asinh}(1/2)$ from that model.

Dimensional check. The quantities φ, $k(K)$, η_φ, and all entries of B_φ are dimensionless. Therefore Eq. (579) can modify only a dimensionless kernel or impedance factor; it cannot by itself introduce a mass, length, energy, or force scale.

M.5.6 Geometric lower-bound gates for SSTcore

For any nontrivial knot in the radius-thickness convention, orthodox ropelength bounds give

$$\mathrm{Rop}_{\mathrm{rad}}(K_{\mathrm{nontrivial}}) \geq 4\pi + 2\pi\sqrt{2} \simeq 21.45213649. \tag{583}$$

Equivalently,

$$\mathrm{Rop}_{\mathrm{diam}}(K_{\mathrm{nontrivial}}) \geq 2\pi + \pi\sqrt{2} \simeq 10.72606825. \tag{584}$$

Using the canonical radius $r_c = 1.40897017 \times 10^{-15}$ m, the radius-convention bound corresponds to

$$L_{\mathrm{min}} = r_c\left(4\pi + 2\pi\sqrt{2}\right) \simeq 3.02254 \times 10^{-14}\,\mathrm{m}. \tag{585}$$

For the trefoil benchmark used elsewhere in the canon,

$$\mathrm{Rop}_{\mathrm{rad}}(3_1) \approx 2 \times 16.371637 = 32.743274, \tag{586}$$

$$L_{3_1} \approx 32.743274\, r_c \simeq 4.61343 \times 10^{-14}\,\mathrm{m}. \tag{587}$$

An SSTcore ideal-knot routine that reports a nontrivial knot below Eq. (583), or mixes the radius and diameter conventions without an explicit factor of two, fails the geometry gate.

M.5.7 Composite-link non-additivity warning

For composite links and connect-sum-like configurations, ideal ropelength is not generally additive. The ropelength literature reports a typical deficit of order

$$\Delta\mathcal{L}_{\mathrm{comp}} \sim 4\pi - 4 \simeq 8.56637. \tag{588}$$

Therefore any SST baryon or multi-component particle branch using an additive length proxy must expose a correction term,

$$\mathcal{L}_{\mathrm{comp}} = \sum_i \mathcal{L}_{K_i} - \Delta_{\mathrm{link}} + \delta_{\mathrm{stress}}, \tag{589}$$

rather than assuming $\mathcal{L}_{\mathrm{comp}} = \sum_i \mathcal{L}_{K_i}$. This is a research-track guard, not a universal law for all SST composites.

M.5.8 Polygonal-to-smooth certification ladder

Status. [ORTHODOX NUMERICAL GEOMETRY / RESEARCH-TRACK SST INPUT GATE]. This subsection isolates the numerical implications of the Ridgerunner constrained-gradient method [63]. It certifies an approximately thickness-critical polygon and a smooth ropelength upper bound; it does not certify a unique global minimizer or a physical SST energy ground state.

Equilateral constraint gate. For an equilateral polygon of edge length ℓ, Ridgerunner's constraint thickness and Rawdon polygonal thickness define the same feasible set,

$$\mathrm{CThi}(\ell, V) \geq 1 \quad \Longleftrightarrow \quad \mathrm{Thi}_p(V) \geq 1. \tag{590}$$

This equivalence must not be transferred silently to a strongly non-equilateral simulation mesh. Define

$$\bar{\ell} = \frac{1}{N}\sum_{i=1}^{N} \ell_i, \tag{591}$$

$$\epsilon_{\mathrm{eq}} = \max_i \left| \frac{\ell_i}{\bar{\ell}} - 1 \right|. \tag{592}$$

A pipeline must control ϵ_{eq} by arc-length remeshing or use a thickness formulation proved for its non-equilateral discretization.

Curvature-stiffness scaling. At normalized MinRad $= 1$, an equilateral polygon obeys

$$\left\|\nabla\, \mathrm{MinRad}^{\pm}\right\|_2 \geq \frac{2}{\ell^2}. \tag{593}$$

Thus halving the edge length can increase the curvature-constraint stiffness by a factor of four. Timestep, conditioning, and constraint-correction tolerances must scale with resolution; increasing the vertex count is not by itself a stability improvement.

Polygonal versus smooth ropelength. The polygonal value

$$\mathrm{Rop}_p(V) = \frac{\mathrm{Len}(V)}{\mathrm{Thi}_p(V)} \tag{594}$$

is not the final smooth certification. After replacing vertices by controlled circular or spline arcs, all newly introduced self-distances must be checked and one must report

$$\mathrm{Rop}^{\mathrm{ub}}_{\mathrm{smooth}} = \frac{\mathrm{Len}(\widetilde{V})}{\mathrm{Thi}(\widetilde{V})}. \tag{595}$$

The label "ideal" attached to a Fourier, KnotPlot, or atlas file is not a geometric certificate. The Ridgerunner authors could not reproduce every published Fourier/Gilbert ropelength claim under their polygonal thickness measurement, which makes independent reconstruction and smooth-thickness verification mandatory.

Constrained-gradient certificate. With rigidity matrix A assembled from active strut and kink gradients, the nonnegative least-squares problem is

$$\min_{\Lambda \geq 0} \|A\Lambda - (-\nabla E_{\mathrm{SST}})\|_2 \,, \tag{596}$$

and the relative first-order residual is

$$\chi_{\mathrm{KKT}} = \frac{\| -\nabla E_{\mathrm{SST}} + A\Lambda\|_2}{\|\nabla E_{\mathrm{SST}}\|_2}, \qquad \Lambda \geq 0. \tag{597}$$

The published dataset had mean residual approximately 2.99×10^{-3}, with specially refined examples reaching approximately 2.54×10^{-5}. These values are historical benchmarks, not universal acceptance thresholds. A small χ_{KKT} establishes only first-order criticality and must be interpreted together with the stability hierarchy in Eq. (641).

Contact-map and refinement convergence. Ropelength convergence is insufficient. The positive strut measure of Eq. (640), the analogous kink measure, their supports, total masses, and representative test integrals must converge under refinement. The arclength contact map should be compared across resolutions; new or disappearing contact branches are a geometry change even when the scalar ropelength changes little.

Multistart protocol. For each knot/link class use at least five independent initial embeddings, as in the published computation, and a resolution ladder. Suitable starts include analytic/Fourier, diagram/KnotPlot, previously tightened atlas, and independent isotopic perturbations. Retain the best certified smooth upper bound, not merely the shortest polygon encountered, and preserve seed, remeshing, active-set, correction-step, and stopping-residual metadata.

Precision ceiling. Define a conservative geometry-certification envelope

$$\epsilon_{\mathrm{geom}} = \max(\epsilon_{\mathrm{eq}}, \epsilon_{\mathrm{thi}}, \chi_{\mathrm{KKT}}, \epsilon_{\mathrm{smooth}}, \epsilon_{\mathrm{contact}}) \,, \tag{598}$$

where, for the normalized unit-thickness problem,

$$\epsilon_{\mathrm{thi}} := \max\Big(0, 1 - \mathrm{CThi}(\bar{\ell}, V)\Big) \,, \tag{599}$$

$$\epsilon_{\mathrm{smooth}} := \frac{\left|\mathrm{Rop}^{\mathrm{ub}}_{\mathrm{smooth}} - \mathrm{Rop}_p\right|}{\mathrm{Rop}^{\mathrm{ub}}_{\mathrm{smooth}}}. \tag{600}$$

Here $\epsilon_{\mathrm{contact}}$ is a declared resolution-to-resolution norm on the contact measures. A downstream SST mass, coupling, or constant extraction must not quote relative precision parametrically smaller than ϵ_{geom} unless an independent error-propagation analysis proves a cancellation.

M.5.9 Operational SSTcore diagnostics

A profile-resolved SSTcore tightening or imported ideal-knot pipeline should report at least

$$\mathcal{D}_K^{\mathrm{num}} = \Big(N, \bar{\ell}, \epsilon_{\mathrm{eq}}, \mathrm{Thi}_p, \mathrm{CThi}, \mathrm{Rop}_p, \mathrm{Rop}^{\mathrm{ub}}_{\mathrm{smooth}}, \min_i \mathrm{MinRad}(v_i), \frac{1}{2} d_{\mathrm{dcsd}}, |\,\mathrm{Strut}\,|, |\,\mathrm{Kink}\,|, \Lambda, \chi_{\mathrm{KKT}}, \mathrm{cond}_2(A_F), \tag{601}$$

where the rigidity-matrix column-sign convention is declared and A_F is the free-column submatrix in the final NNLS active set. This tuple supplements, rather than replaces, the continuum contact-measure audit of Sec. M.9.

M.5.10 Colored contact-channel rank test for the factor nine

Status. [ORTHODOX OPTIMIZATION BACKGROUND / RESEARCH-TRACK SST–QCD ROSETTA / NOT CANONICAL QCD]. The orthodox input is the finite-thickness ropelength/KKT framework summarized in Eqs. (552)–(555): active struts and kinks define a contact/rigidity matrix with nonnegative multipliers. The SST interpretation below is a confinement-sheet diagnostic for a $\mathbb{Z}_3$-locked colored trefoil triad. It is not a derivation of the QCD beta function, the full $SU(3)$ Yang–Mills action, running couplings, anomaly cancellation, or the observed hadron spectrum.

Claim discipline. A QCD-like numerical factor of nine must not be introduced as a fitted prefactor or as the statement "eight gluons plus one". In this research-track bridge it is instead the target positive rank of a resolved colored contact-stress matrix,

$$9 = 3_{\mathrm{color}} \times 3_{\mathrm{mech}}, \tag{602}$$

$$3_{\mathrm{color}} := \#\{RG, GB, BR\}, \tag{603}$$

$$3_{\mathrm{mech}} := \#\{A, S, T\}. \tag{604}$$

Here RG, GB, BR denote the three $\mathbb{Z}_3$ color root-planes of the ideal triad. The mechanical labels denote, respectively,

$$A : \text{antisymmetric-vector / torsional contact mode}, \tag{605}$$

$$S : \text{symmetric-traceless in-sheet shear mode}, \tag{606}$$

$$T : \text{transverse director-compression mode}. \tag{607}$$

Only the color triad is a discrete sector label. The three mechanical modes are finite-thickness contact modes and must be verified by the solver.

Resolved contact decomposition. Let Σ_Y be the putative confinement sheet joining a separated trefoil sector to the Steiner node of a $\mathbb{Z}_3$-locked triad. The active strut set is decomposed as

$$\mathcal{S}_\Sigma = \bigcup_{c \in \{RG, GB, BR\}} \left(\mathcal{S}_c^{(A)} \cup \mathcal{S}_c^{(S)} \cup \mathcal{S}_c^{(T)} \right), \tag{608}$$

where a strut is active only when the resolved finite-thickness condition is saturated,

$$d_s = 2a_{\mathrm{core}}, \qquad \lambda_s \geq 0. \tag{609}$$

The radius entering this condition is the resolved tube radius a_{core}, not automatically the canonical horn radius r_c.

For each channel $I=(c,m)$, with $c \in \{RG, GB, BR\}$ and $m \in \{A, S, T\}$, define the corresponding block of the rigidity/contact matrix by the gradients of its active constraints,

$$\mathcal{R}_I := \left[\nabla\left(\frac{d_s}{2}\right), \nabla \operatorname{MinRad}(v_k) \right]_{s,k \in I}. \tag{610}$$

The full reduced matrix is

$$\mathcal{R}_\Sigma^{\text{reduced}} := \left[\mathcal{R}_{RG}^{(A)}, \mathcal{R}_{RG}^{(S)}, \mathcal{R}_{RG}^{(T)}, \mathcal{R}_{GB}^{(A)}, \dots, \mathcal{R}_{BR}^{(T)} \right]_{\text{reduced}}. \tag{611}$$

Reduction removes rigid translations, rotations, reparametrization modes, and numerically duplicate constraints before the rank is computed.

Positive-rank criterion. Define rank_+ as the number of independent contact channels that survive the KKT balance with nonnegative strut/kink multipliers. The factor nine is promoted from a matching diagnostic to a derived contact-channel count only if

$$\boxed{N_+ := \text{rank}_+ \mathcal{R}_\Sigma^{\text{reduced}} = 9.} \tag{612}$$

This is a rank-stability condition, not exact numerical orthogonality.

The ideal symmetry limit is block diagonal,

$$\mathcal{R}_\Sigma = \bigoplus_{c \in \{RG, GB, BR\}} \left(\mathcal{R}_c^{(A)} \oplus \mathcal{R}_c^{(S)} \oplus \mathcal{R}_c^{(T)} \right). \tag{613}$$

A physical finite-thickness relaxation is expected to have controlled mixing,

$$\mathcal{R}_\Sigma = \mathcal{R}_0 + \delta\mathcal{R}, \tag{614}$$

where the leading cross-talk is expected between the S and T modes within the same color root-plane. The integer plateau survives if

$$\sigma_{\min}^+(\mathcal{R}_0) > \|\delta\mathcal{R}\|, \tag{615}$$

where $\sigma_{\min}^+$ denotes the smallest nonzero positive-channel singular scale after reduction. Thus the claim is not that cross-talk vanishes; the claim is that cross-talk does not collapse the positive rank.

A useful orthogonality diagnostic is the normalized channel Gram matrix

$$G_{IJ} = \frac{\langle \mathcal{R}_I, \mathcal{R}_J \rangle}{\|\mathcal{R}_I\| \, \|\mathcal{R}_J\|} = \delta_{IJ} + \varepsilon_{IJ}, \tag{616}$$

with reported mixing score

$$\eta_{\text{orth}} := \frac{\sum_{I \neq J} |G_{IJ}|}{\sum_I |G_{II}|}. \tag{617}$$

A small η_{orth} is supporting evidence, but Eq. (612) is the decisive test.

Energetic decoupling diagnostic. For each channel I, define the channel tension proxy

$$\sigma_I = \frac{1}{\ell_Y} \sum_{s \in \mathcal{S}_I} \lambda_s d_s, \tag{618}$$

and the full numerical sheet tension

$$\sigma_{\text{num}} = \frac{1}{\ell_Y} \sum_{s \in \mathcal{S}_\Sigma} \lambda_s d_s. \tag{619}$$

In the additive limit,

$$\sigma_{\text{num}} = \sum_{I=1}^{9} \sigma_I. \tag{620}$$

A finite-thickness sheet may instead produce

$$\sigma_{\text{num}} = \sum_{I=1}^{9} \sigma_I + \Delta\sigma_{\text{mix}}, \qquad \left| \frac{\Delta\sigma_{\text{mix}}}{\sum_I \sigma_I} \right| \ll 1. \tag{621}$$

The factor-nine interpretation survives only if $N_+ = 9$ and the mixing fraction is stable or decreases under mesh refinement.

Connection to the geometric shielding gate. If the rank test succeeds, the candidate confinement-sheet scale may be compared with the canon-level geometric shielding gate,

$$\frac{\lambda_c}{\pi r_c} = \frac{4}{\alpha}, \tag{622}$$

through the diagnostic estimate

$$\sigma_{\text{rank9}} := 9 \left(\frac{\lambda_c}{\pi r_c} \right) F_{\text{swirl}}^{\text{max}} = 9 \left(\frac{4}{\alpha} \right) F_{\text{swirl}}^{\text{max}}. \tag{623}$$

Equation (622) is not derived by this contact-rank lemma; it is imported from the calibrated geometric impedance chain. The lemma tests whether there are nine independent positive contact channels to which that gate could be applied.

Falsifiers. This research-track bridge fails, or remains a matching diagnostic only, if any of the following occurs:

1. the reduced positive rank satisfies $N_+ \neq 9$;
2. the KKT balance requires any negative strut or kink multiplier;
3. the S–T cross-talk collapses two channels into one under refinement;
4. the reported σ_{num} is not stable under mesh refinement, symmetry-preserving perturbation, and independent initialization;
5. the color root-plane assignment cannot be maintained as a $\mathbb{Z}_3$-locked triad around the central unknot.

Canon-safe summary.

$$\boxed{\text{The integer 9 is not postulated; it is the target positive rank of the resolved colored contact-stress matrix.}} \tag{624}$$

Until Eq. (612) is numerically and geometrically verified, Eq. (623) remains [**MATCHING DIAGNOSTIC, NOT DERIVED**].

Minimal falsifier for this appendix. If two independent high-resolution implementations, using the same thickness convention and topology, converge to incompatible contact skeletons $\mathcal{C}_K$ or incompatible χ_{contact} under the same declared E_{SST}, the proposed contact-kernel refinement Eq. (576) must be demoted from Research Track to numerical diagnostic only.

M.6 Minimal falsifier

If two distinct knots have the same SST particle assignment but relax to incompatible energy minima or different protected sectors under the same admissible dynamics, the assignment must be split or rejected:

$$K_1 \sim_{\text{SST}} K_2 \quad \text{but} \quad \min E[K_1] \neq \min E[K_2] \quad \Rightarrow \quad \text{taxonomy refinement required.} \tag{625}$$

M.7 Research Appendix: Spherical Curl Spectrum and Helicity Capacity

Status. [ORTHODOX SOURCE RESULT / SST RESEARCH-TRACK APPLICATION]. This appendix records the bounded-domain curl-spectrum consequences of Cantarella, DeTurck, Gluck, and Teytel without identifying a spheromak with an SST particle state [56].

Helicity capacity. For any domain on which the projected Biot–Savart/curl inverse is defined, set

$$C_H(\Omega) := \frac{1}{\kappa_1(\Omega)}, \qquad [C_H] = \text{m}. \tag{626}$$

This is the maximum helicity-to-L^2-energy ratio in the source convention. A scale-invariant resolved-tube diagnostic is

$$\chi_{\text{curl}}(K) := a_{\text{core}}\, \kappa_1(\Omega_K), \qquad [\chi_{\text{curl}}] = 1, \tag{627}$$

where Ω_K is the actual finite-thickness knot domain. A pure ropelength mass branch requires $\chi_{\text{curl}}(K)$ and the normalized field profile to be sufficiently universal across the compared knot classes.

Thin active-shell cost. Let $\delta = b - a \ll b$. Expanding the spherical-shell root condition in the thin-shell regime gives the leading radial scaling

$$\kappa_1(B^3(a,b)) \sim \frac{\pi}{\delta}. \tag{628}$$

Thus a progressively thinner *vorticity- or flux-carrying filled shell* acquires a large spectral cost. This result must not be transferred to an irrotational horn envelope surrounding a much smaller vorticity-bearing core; those are different field sectors.

Field-line geometry. The minimum-eigenvalue ball and shell fields are axisymmetric and are organized by nested toroidal invariant surfaces. This supports their use as qualitative benchmarks for a bent coarse-grained flux bundle. It does not imply that a remote background flow must be toroidal, nor does it establish the geometry of a knotted SST carrier.

Beltrami residual. For a numerical candidate $\mathbf{V}$, define

$$\mathcal{R}_{\text{CK}}(\kappa) := \frac{\|\nabla \times \mathbf{V} - \kappa \mathbf{V}\|_{L^2(\Omega)}}{\|\nabla \times \mathbf{V}\|_{L^2(\Omega)} + |\kappa|\, \|\mathbf{V}\|_{L^2(\Omega)}}. \tag{629}$$

The fitted κ, $\mathcal{R}_{\text{CK}}$, and $a_{\text{core}}\kappa$ must be reported together.

No automatic ideal-Euler relaxation. A lowest curl eigenfield is a stationary variational benchmark. In the ideal, incompressible, unforced Euler limit, kinetic energy and helicity are conserved; the dynamics therefore do not automatically dissipate toward the lowest mode. Any relaxation claim requires a separately declared dissipative, radiative, or coarse-graining mechanism. Hence

$$\text{small } \mathcal{R}_{\mathrm{CK}} \not\Rightarrow \text{dynamical attractor or particle stability.} \tag{630}$$

Minimal falsifier. Compute κ_1 on resolved unknot, trefoil, figure-eight, and selected higher-knot tubes at common a_{core} and common volume. Strong knot-dependent variation in χ_{curl} falsifies any claim that the entire field-energy splitting is determined by scalar ropelength alone.

M.8 Research Appendix: Writhe Bounds, Twist Guards, and Flat-Torus Benchmark

Status. [ORTHODOX SOURCE RESULTS / SST DIAGNOSTIC USE]. This appendix imports only the bounds and explicit model-domain results of Cantarella, DeTurck, and Gluck [57].

Writhe sanity gate. A computed centerline with radius-convention ropelength $\mathrm{Rop}_{\mathrm{rad}}$ must satisfy

$$|\,\mathrm{Wr}\,| < \frac{1}{4}\,\mathrm{Rop}_{\mathrm{rad}}^{4/3}\,. \tag{631}$$

Violation indicates a convention, discretization, or integration error. Passing the bound does not certify ideality because the inequality is intentionally broad.

Special zero-internal-twist tube. For the particular tube field used in the source—normal to each cross-sectional disk and with magnitude depending only on distance from the centerline—the internal twist contribution is absent and

$$\mathscr{H}_V = \Phi^2\,\mathrm{Wr}(K). \tag{632}$$

This equality is not a general SST identity. It may be used only after the selected framing and cross-sectional field have independently established zero internal twist. In particular,

$$\mathscr{H}_V/\Phi^2 = \mathrm{Wr} \quad\Longrightarrow\quad \text{verified zero-twist model,} \tag{633}$$

not a post-hoc choice of twist to match a target helicity.

Flat solid-torus benchmark. On the product domain $D^2(a)\times S^1$, the first positive curl eigenvalue is

$$\kappa_1 = \frac{j_{0,1}}{a}, \qquad j_{0,1} = 2.404825558\ldots, \tag{634}$$

with an eigenfield containing both longitudinal and azimuthal components. This supplies a local slender-tube benchmark for $a\kappa_1$. The product torus is not an embedded curved horn torus in $\mathbb{R}^3$; major-radius curvature, knotting, and self-contact are absent and must be recomputed in the physical domain.

Research requirement. For every candidate particle tube, report the tuple

$$\Big(\mathrm{Rop}_{\mathrm{rad}}, \mathrm{Wr}, \text{framing/twist prescription}, \mathscr{H}_V/\Phi^2, a_{\mathrm{core}}\kappa_1\Big)\,. \tag{635}$$

A scalar ropelength or writhe value alone is insufficient to specify the volumetric field state.

M.9 Research Appendix: Gehring-Link Criticality and Contact Measures

Status. [ORTHODOX VARIATIONAL GEOMETRY / SST CONTACT APPLICATION]. This appendix records the continuum balance theorem and its limitations from the Gehring link problem [58].

Continuum strut measure. Let $\mathrm{Strut}(L)$ be the set of intercomponent pairs realizing $\mathrm{LThi}(L)$. For a variation field $\boldsymbol{\xi}$, define

$$(A_S\boldsymbol{\xi})(x,y) := \frac{\langle \boldsymbol{\xi}(y) - \boldsymbol{\xi}(x), y - x\rangle}{|y-x|}. \tag{636}$$

Strong criticality is equivalent to the existence of a positive Radon measure μ on $\mathrm{Strut}(L)$ satisfying

$$\delta_{\boldsymbol{\xi}} \operatorname{Len}(L) = \int_{\mathrm{Strut}(L)} A_S\boldsymbol{\xi}\, d\mu, \qquad\qquad A_S^*\mu = -\mathbf{K}, \tag{637}$$

where $\mathbf{K}$ is the vector-valued curvature-force measure.

For a resolved SST energy with both self-contact and curvature constraints, the corresponding first-order structure is

$$-\frac{\delta E_{\mathrm{SST}}}{\delta\gamma} + A_S^*\mu + A_K^*\nu = 0, \qquad \mu,\nu \geq 0, \tag{638}$$

where ν is the kink/curvature multiplier measure.

Geometric contact versus load-bearing contact. The multiplier measure need not have support on every active geometric strut:

$$\operatorname{supp}\mu \subseteq \mathrm{Strut}(L), \tag{639}$$

with possible strict inclusion. Hence $d(x,y) = a_i + a_j$ does not by itself imply nonzero contact pressure. A solver must distinguish the active geometric strut set from the positive-multiplier load-bearing set.

Discrete-to-continuum convergence. A polygonal contact solution should be represented as

$$\mu_N = \sum_{q\in\mathrm{Strut}(V_N)} \lambda_q^{(N)}\delta_q. \tag{640}$$

Refinement requires weak-* convergence $\mu_N \overset{*}{\rightharpoonup} \mu$, not merely convergence of the number of struts or total ropelength. The total measure, support geometry, and selected test integrals $\int f\, d\mu_N$ must be reported.

Criticality is not stability. Equation (637) is a first-order KKT balance. It does not establish a nonnegative constrained Hessian and does not establish stability under Euler or Biot–Savart time evolution. The required hierarchy is

$$\text{balanced} \quad \not\Rightarrow \quad \text{variationally stable} \quad \not\Rightarrow \quad \text{dynamically stable}. \tag{641}$$

Clasp singularity and finite-core regularization. In the critical unit clasp, the tip curvature behaves as

$$\kappa(s) \sim |s|^{-1/3}, \tag{642}$$

so the curve is $C^{1,2/3}$, not $C^{1,1}$, while the central inter-tip distance is approximately 1.05639 for unit link-thickness. This demonstrates that intercomponent exclusion alone permits unbounded curvature and a non-contact gap at the visually closest tips. A physical SST tube therefore requires the ordinary curvature/reach gate or a resolved bending/core regularization in addition to Gehring separation.

Unequal component tensions. For component-dependent effective tensions T_i, the curvature-force side is weighted:

$$A_S^*\mu = -\sum_i T_i \mathbf{K}_i. \tag{643}$$

A geometrically symmetric multi-component state is not automatically critical when its components have unequal circulation, profile, or tension data.

Higher link-homotopy data. Pairwise linking numbers do not classify three-component Brunnian links. The Borromean sector therefore requires a higher invariant such as Milnor's μ_{123} in addition to Lk_{ij}. This is a topology label; it does not by itself establish the mechanical balance of Eq. (638).

Minimal falsifier. A claimed multi-component SST equilibrium fails this gate if it preserves only intercomponent distance while allowing a component self-crossing, if any required multiplier is negative, if the load-bearing support does not converge under refinement, or if KKT balance is reported as dynamical stability without a separate perturbation spectrum.

M.10 Research Appendix: Biot–Savart Realizability and Far-Field Gate

Status. [ORTHODOX OPERATOR RESULTS / SST IMPLEMENTATION REQUIREMENT]. This appendix maps the bounded-domain Biot–Savart theorems to SSTcore and VortexLab diagnostics [59].

Exact vorticity-to-velocity inversion. With $\mathbf{V} = \boldsymbol{\omega}$, the raw reconstruction $\mathbf{u} = \mathrm{BS}[\boldsymbol{\omega}]$ is a left inverse of curl only when

$$\nabla \cdot \boldsymbol{\omega} = 0, \qquad \boldsymbol{\omega} \cdot \mathbf{n} = 0 \quad \text{on } \partial\Omega. \tag{644}$$

Boundary leakage or discrete divergence produces the explicit gradient terms in Eq. (48); it is not a harmless numerical detail.

Potential-sector non-reconstruction. Since boundary-normal gradients lie in $\ker \mathrm{BS}$, a vorticity solver does not recover the complete pressure or potential-gradient sector. Therefore

$$\boldsymbol{\omega} \text{ and } \mathrm{BS}[\boldsymbol{\omega}] \quad \not\Rightarrow \quad \text{unique pressure, contact stress, or gravity closure.} \tag{645}$$

The pressure Poisson problem and its boundary data remain independent parts of the SST state.

Realizability gate. Because $\operatorname{im} \mathrm{BS} \subsetneq \operatorname{im} \nabla\times$, a smooth divergence-free target field need not be generated by sources confined to the same domain. Proposed analytic swirl profiles must therefore be tested against the projected Biot–Savart image rather than accepted from $\nabla \cdot \mathbf{u} = 0$ alone.

Optical-size diagnostic. Define the dimensionless geometry factor

$$\chi_{\mathrm{OS}}(K) := \frac{\mathrm{OS}(\Omega_K)}{a_{\mathrm{core}}}, \qquad [\chi_{\mathrm{OS}}] = 1. \tag{646}$$

It supplies an operator-norm bound independent of scalar ropelength and should be reported when comparing domains with different compactness or concavity.

Far-field regression test. For an isolated compact fluid-knot source, fit

$$\|\mathbf{u}(r)\| \propto r^{-p} \quad (r \gg \operatorname{diam} \Omega). \tag{647}$$

The direct Biot–Savart sector requires $p \simeq 3$. A measured numerical exponent near 1 or 2 indicates a non-compact source, boundary leakage, an added background/potential sector, or an implementation error; it must not be silently attributed to the compact knot.

Required residuals. A bounded-domain reconstruction must report

$$\epsilon_{\mathrm{div}} := \frac{\|\nabla \cdot \boldsymbol{\omega}\|_{L^2}}{\|\nabla \boldsymbol{\omega}\|_{L^2}}, \qquad \epsilon_{\mathrm{wall}} := \frac{\|\boldsymbol{\omega} \cdot \mathbf{n}\|_{L^2(\partial\Omega)}}{\|\boldsymbol{\omega}\|_{L^2(\Omega)}},$$
$$\epsilon_{\mathrm{curl}} := \frac{\|\nabla \times \mathbf{u} - \boldsymbol{\omega}\|_{L^2}}{\|\boldsymbol{\omega}\|_{L^2}}. \tag{648}$$

The denominators and treatment of exact-zero cases must be declared in code.

M.11 Research Appendix: Hodge-Complete SST Velocity Reconstruction

Status. [ORTHODOX HODGE DECOMPOSITION / DERIVED SST RECONSTRUCTION]. The following reconstruction uses the domain topology explicitly [60].

Harmonic basis and circulation coordinates. Let

$$g := \dim H_1(\Omega; \mathbb{R}), \tag{649}$$

and choose harmonic-knot fields $\mathbf{h}_1, \ldots, \mathbf{h}_g$ normalized by

$$\oint_{\gamma_i} \mathbf{h}_j \cdot d\boldsymbol{\ell} = \delta_{ij}. \tag{650}$$

A Hodge-complete bounded-domain velocity reconstruction has the form

$$\boxed{\mathbf{u} = P_{K(\Omega)}\, \mathrm{BS}[\boldsymbol{\omega}] + \sum_{j=1}^{g} \Gamma_j \mathbf{h}_j} \tag{651}$$

after the flux and gauge conventions defining the projection have been fixed. The coefficients Γ_j are independent topological circulation data; they cannot be reconstructed from $\boldsymbol{\omega} = \nabla \times \mathbf{u}$.

Solid-torus sector. For a solid torus, $g = 1$. In the standard torus of revolution an illustrative harmonic-knot field is locally

$$\mathbf{h} = \frac{1}{r}\,\widehat{\boldsymbol{\phi}}, \qquad \nabla \cdot \mathbf{h} = 0, \qquad \nabla \times \mathbf{h} = 0. \tag{652}$$

Thus local curl-free flow may carry nonzero circulation around a non-contractible loop. For the SST circulation quantum, the local slender-core scaling is

$$u_{\mathrm{HK}}(r) = \frac{\Gamma_0}{2\pi r} = v_{\circlearrowleft}^{*}\frac{r_c}{r}. \tag{653}$$

This is a local benchmark outside an excluded core, not a determination of r_{horn} or of the full curved-knot velocity profile.

Irrotational envelope guard. Equation (651) supplies a mathematically valid separation between a vorticity-bearing core sector and a curl-free global circulation sector. It therefore permits an irrotational horn/envelope flow around a smaller core. It does not prove that every SST horn is harmonic, nor does it fix the separatrix, pressure profile, or envelope radius.

Hodge data are not particle identity. Every tubular neighborhood of a single knot is topologically a solid torus, so

$$\dim HK(\Omega_{3_1}) = \dim HK(\Omega_{4_1}) = 1. \tag{654}$$

Hodge homology alone cannot distinguish a trefoil from a figure-eight knot. Particle classification still requires embedding-sensitive and knot-sensitive data such as knot group, chirality, writhe/twist, ropelength, contact map, and spectral geometry.

Numerical circulation residual. A bounded-domain solver must report

$$\epsilon_\Gamma := \max_j \frac{\left|\oint_{\gamma_j} \mathbf{u}\cdot d\boldsymbol{\ell} - \Gamma_j\right|}{\max(|\Gamma_j|, \Gamma_{\mathrm{floor}})}, \tag{655}$$

with a declared nonzero Γ_{floor}. A solver that matches $\nabla \times \mathbf{u}$ but fails this circulation test is not Hodge-complete.

Minimal research requirement. For each resolved knot or link domain, compute a homology basis, normalize the harmonic fields, project the raw Biot–Savart field into $K(\Omega)$, and verify both curl recovery and circulation recovery. The stored state must include

$$\Big(\boldsymbol{\omega}, \{\Gamma_j\}_{j=1}^{g}, \text{boundary conditions, projection/gauge convention}\Big). \tag{656}$$

N Research-Track particle and clock ansatz

SUBSTRATE INTERPRETATION / NON-ORTHODOX RESEARCH TRACK

Status. This section introduces candidate SST closures that are not consequences of orthodox Euler hydrodynamics, Gross–Pitaevskii theory, or established particle physics. The purpose is to formulate a constrained and falsifiable research programme while preserving the distinction between geometric definitions, canonically calibrated SST quantities, and unverified particle assignments.

Suppose, as an additional hypothesis, that microscopic matter structures are finite-core vortex knots of an incompressible underlying substrate. A predictive implementation must derive its equilibrium states from a declared continuum energy and finite-thickness constraint. Arbitrary pairwise short-range potentials introduced only to prevent strand crossings are excluded.

N.1 Resolved finite-core energy functional

Let

$$\gamma_K : S^1 \to \mathbb{R}^3, \qquad s \mapsto \boldsymbol{X}_K(s) \tag{657}$$

be an arclength-parametrized centreline of knot type K. Its resolved tube radius is defined geometrically by

$$a_{\text{core}}(K) = \text{reach}(\gamma_K) = \min\left[\inf_s \frac{1}{\kappa(s)}, \frac{1}{2} d_{\text{dcsd}}(\gamma_K)\right], \tag{658}$$

where d_{dcsd} is the minimum doubly-critical self-distance [61].

A candidate resolved-tube energy is

$$\mathcal{E}_{\text{eff}}[\gamma_K] = E_{\text{SI}}^{(a)}[\gamma_K, \Gamma_K] + T_{\text{eff}} L[\gamma_K] + B_{\text{eff}} \oint_{\gamma_K} \kappa^2 \, \mathrm{d}s + E_{\text{twist}}[\gamma_K], \tag{659}$$

$$L[\gamma_K] = \oint_{\gamma_K} \mathrm{d}s. \tag{660}$$

Here $E_{\text{SI}}^{(a)}$ is a finite-core regularization of the kinetic self-induction energy. It may be represented generally as

$$E_{\text{SI}}^{(a)} = \frac{\rho_f \Gamma_K^2}{8\pi} \oint_{\gamma_K} \oint_{\gamma_K} \boldsymbol{t}(s) \cdot \boldsymbol{t}(s') \, \mathcal{K}_{a_{\text{core}}} \left(|\boldsymbol{X}_K(s) - \boldsymbol{X}_K(s')|\right) \mathrm{d}s \, \mathrm{d}s', \tag{661}$$

where

$$\mathcal{K}_a(r) \longrightarrow \frac{1}{r} \qquad (r \gg a) \tag{662}$$

and $\mathcal{K}_a(0)$ is finite. The regularization kernel must be derived from, or fixed by, a declared core profile before mass predictions are evaluated.

The coefficients have dimensions

$$[T_{\text{eff}}] = \mathrm{J\,m^{-1}}, \qquad [B_{\text{eff}}] = \mathrm{J\,m}. \tag{663}$$

A finite-core scaling ansatz may take the form

$$B_{\text{eff}} = c_B T_{\text{eff}} a_{\text{core}}^2, \tag{664}$$

where c_B is dimensionless. Dimensional analysis does not determine $c_B = 1$; it must follow from a resolved cross-sectional calculation.

The minimization problem is therefore

$$\min_{\gamma_K \in K} \mathcal{E}_{\text{eff}}[\gamma_K] \qquad \text{subject to} \qquad \text{reach}(\gamma_K) \geq a_{\text{core}}. \tag{665}$$

For a polygonal discretization V, this constraint becomes

$$\text{MinRad}(v_i) \geq a_{\text{core}}, \tag{666}$$

$$\frac{1}{2} d(p, q) \geq a_{\text{core}}. \tag{667}$$

The active curvature constraints are kinks and the active self-distance constraints are struts. Their non-negative KKT multipliers supply a contact-stress skeleton for the resolved tube [63].

The hard reach constraint is a geometric admissibility condition, not an auxiliary Lennard–Jones interaction. Bending stiffness can oppose homothetic collapse, but it cannot by itself exclude a finite-curvature self-crossing.

N.2 Quantized circulation and characteristic velocity

The circulation is retained in the canonical SST form

$$\Gamma_K = n_K \Gamma_0, \qquad n_K \in \mathbb{Z}, \tag{668}$$

with

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}^{*} = 9.68361920 \times 10^{-9}\,\mathrm{m}^2\,\mathrm{s}^{-1}. \tag{669}$$

Thus the conserved circulation must not vary continuously with the knot radius R_K. Instead, the characteristic resolved velocity may depend on geometry through

$$U_K = \frac{\Gamma_K}{2\pi R_K}\,\mathcal{G}\left(\frac{a_{\mathrm{core}}}{R_K}, \mathcal{T}_K, \mathfrak{q}_K\right), \tag{670}$$

where $\mathcal{G}$ is dimensionless and $\mathfrak{q}_K$ denotes any independently specified internal coloring data.

Equation (670) separates the canonical circulation quantum from the geometry-dependent relation between circulation and a chosen characteristic velocity.

N.3 Topology-first particle assignment

A predictive particle dictionary must be fixed before the observed mass spectrum is used. Let

$$\mathsf{A}(P) = (K_P, n_P, \mathfrak{q}_P, \text{framing data}, \text{component structure}) \tag{671}$$

denote the preregistered assignment for particle P.

If a non-Abelian internal coloring is introduced, the mathematically appropriate structure is a group homomorphism

$$\mathfrak{q}_K : \pi_1\left(\mathbb{R}^3 \setminus K\right) \longrightarrow G, \tag{672}$$

whose values on Wirtinger generators satisfy the knot-group crossing relations.

A tuple such as

$$(L \bmod 3, S \bmod 2, \chi) \tag{673}$$

may serve as a collection of discrete labels, but it is not by itself a homomorphism into $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$.

The trefoil $K_{2,3} = 3_1$ may be retained as the preregistered electron benchmark of the current SST taxonomy. Baryon assignments remain under audit. A torus-knot candidate $K_{3,q}$, a three-component colored composite, or another explicitly specified topology may be tested, but threefold winding must not be identified with QCD color solely because both involve the integer three.

N.4 Preregistered falsifiability test

Before fitting any mass, the following objects must be fixed:

1. the particle-to-topology map A;
2. the core regularization kernel $\mathcal{K}_a$;
3. the circulation numbers n_P;
4. the dimensionless coefficient c_B;
5. the twist-energy law, if retained;
6. the admissible embedding class and numerical convergence criteria.

Only one overall energy normalization may then be calibrated using the electron mass. Define the dimensionless minimized energy

$$\widehat{\mathcal{E}}_P = \min_{\gamma \in \mathcal{C}_P} \widehat{\mathcal{E}}_{\text{eff}} [\gamma; \mathsf{A}(P)] , \tag{674}$$

where $\mathcal{C}_P$ is the preregistered admissible class.

The out-of-sample proton prediction is

$$\widehat{R}_{p/e} = \frac{\widehat{\mathcal{E}}_p}{\widehat{\mathcal{E}}_e}. \tag{675}$$

The model must reproduce

$$\widehat{R}_{p/e} \simeq 1836.15267 \tag{676}$$

within a preregistered numerical and theoretical uncertainty, without changing the topology assignment, the kernel, or any dimensionless coefficient after inspecting the result [54].

The same fixed model must then predict additional ratios, including the neutron-to-electron ratio and at least one further particle sector. Failure of any preregistered primary target falsifies the corresponding assignment or the energy closure.

N.5 Clock-rate discipline

Define

$$\beta_K = \frac{U_K}{c}, \qquad \Pi_a = \frac{a_{\text{core}}}{R_K}, \qquad \Pi_\kappa = a_{\text{core}}^2 \langle \kappa^2 \rangle. \tag{677}$$

The canon-compatible local clock factor is

$$\frac{\mathrm{d}\tau_K}{\mathrm{d}T} = \sqrt{1 - \beta_K^2}. \tag{678}$$

A Research-Track deviation may only be introduced in the explicitly factorized form

$$\frac{\mathrm{d}\tau_K}{\mathrm{d}T} = \sqrt{1 - \beta_K^2} \left[1 + \varepsilon_{\text{RT}} \Delta_K \left(\Pi_a, \Pi_\kappa, \mathcal{T}_K, \mathfrak{q}_K, \ldots\right)\right], \tag{679}$$

subject to

$$\Delta_K(0, 0, \ldots) = 0. \tag{680}$$

The correction must be local, universal, independently derived, and preregistered before experimental comparison. A ratio involving the galactic averaging radius R_{gal} is not admitted into the local clock law unless a specific non-local coupling mechanism is derived.

In the absence of such a derivation,

$$\varepsilon_{\text{RT}} = 0 \tag{681}$$

and Eq. (678) remains the operative clock law.

N.6 Reconnection, framing, and twist

A smooth filled Euler vortex tube has frozen topology and does not possess the Gross–Pitaevskii reconnection mechanism. The model must therefore choose among:

1. exact topological conservation in the ideal limit;
2. a declared finite-scale nonideal reconnection law;
3. a compressible or order-parameter core theory.

A framed finite-core tube may carry twist. Any twist term E_{twist} must specify the material frame, its transport equation, and the partition of helicity between linking, writhe, and twist [36]. Candidate falsifiers include Kelvin-wave dispersion, reconnection scaling, and helicity-transfer measurements.

O Minimal numerical programme

A staged numerical test must separate geometry, hydrodynamics, topology assignment, and clock interpretation.

1. Preregister the candidate particle topologies, circulation integers, core profile, regularization kernel, and all dimensionless coefficients.

2. Generate centreline discretizations for the preregistered knot and link classes. Measure

$$L, \qquad \mathrm{MinRad}, \qquad d_{\mathrm{dcsd}}, \qquad \mathrm{reach}, \qquad \int \kappa^2 \, \mathrm{d}s.$$

3. Minimize Eq. (659) subject to the hard finite-thickness conditions

$$\mathrm{MinRad} \geq a_{\mathrm{core}}, \qquad \frac{1}{2} d_{\mathrm{dcsd}} \geq a_{\mathrm{core}}.$$

   Record the active struts, kinks, KKT multipliers, and constrained-gradient residual.

4. Perform core-resolution and discretization convergence tests. The predicted energy ratios must be stable under changes in vertex count, quadrature order, kernel resolution, and optimizer tolerance.

5. Calibrate only the overall energy scale on the electron. Evaluate all preregistered baryon and additional particle ratios without further fitting.

6. For a fully contained closed vortex loop, verify the exact global identity

$$\int_V \boldsymbol{\omega} \, \mathrm{d}V = \Gamma_K \oint_{\gamma_K} \mathrm{d}\boldsymbol{X} = \mathbf{0}. \tag{682}$$

7. Test the local coarse-grained relation

$$\overline{\omega}_z(\boldsymbol{x}) \simeq \Gamma_K \mathcal{L}(\boldsymbol{x}) p_z(\boldsymbol{x}) \tag{683}$$

   only as a local segment average. For fully contained closed loops, verify that positive axial-vorticity regions are balanced by return-vorticity regions.

8. Compare separately:

   (a) straight vortex lines crossing the averaging boundary;
   (b) boundary-anchored or periodic vortex lines;
   (c) fully closed vortex knots;
   (d) space-filling vortex-cell models.

   Only the first two classes can support a nonzero global vector-vorticity monopole without an explicit return region.

9. Measure finite-size and homogenization errors as

$$\frac{L_{\mathrm{cg}}}{R_{\mathrm{gal}}} \to 0, \qquad \frac{\ell_v}{L_{\mathrm{cg}}} \to 0.$$

10. Use Gross–Pitaevskii simulations for depleted-core quantum superfluids and Euler, vortex-blob, or resolved finite-core methods for filled classical tubes. Do not transfer reconnection laws between these models without a declared bridge.

11. Do not introduce a modified clock law until the circulation, velocity, energy, and topology closures have passed their independent tests.

P Swirl-String Response and Resonance Framework

P.1 Resonance Selection Principle

Status: [RESEARCH-TRACK; candidate Canon principle].

A swirl-string state K is assigned a causal response function $\chi_K(\omega)$. An external perturbation $F(t)$, with Fourier spectrum $\widetilde{F}(\omega)$, excites the state K with amplitude

$$\mathcal{A}_K[F] = \int_0^\infty \widetilde{F}(\omega)\,\chi_K(\omega)\,d\omega. \tag{684}$$

The excitation probability is

$$P_K[F] = |\mathcal{A}_K[F]|^2. \tag{685}$$

For isolated narrow resonances, the response function may be written as

$$\chi_K(\omega) = \sum_n \frac{B_{Kn}\gamma_{Kn}}{\omega - \omega_{Kn} + i\gamma_{Kn}}, \tag{686}$$

where ω_{Kn} are internal swirl-string eigenfrequencies, γ_{Kn} are linewidths, and B_{Kn} are coupling weights.

The natural SST frequency scale is

$$\Omega_{\mathrm{SST}} = \frac{v_{\circlearrowleft}^{*}}{r_c}, \tag{687}$$

so that internal modes may be parametrized by

$$\omega_{Kn} = \eta_{Kn}\Omega_{\mathrm{SST}}, \tag{688}$$

with dimensionless mode factors η_{Kn}.

This principle unifies delay-loop selection, Swirl-Clock phase-locking, mode-locking, and knot-state excitation as instances of one spectral overlap law.

P.2 Resonance Selection Principle for Swirl–String Excitation

Status and purpose

Status: [RESEARCH-TRACK]. This appendix promotes the frequency-overlap idea from a specific beam–swirl model to a general SST excitation principle. It is not a proof of nuclear activation or fusion enhancement. It is a controlled response ansatz compatible with the present canon language of swirl strings, mode selection, circulation, helicity, and Kelvin-compatible dynamics.

Claim

A swirl-string state K is excited most efficiently when the spectrum of an external perturbation overlaps the internal response spectrum of K. Define a normalized external spectral density

$$p_{\mathrm{ext}}(\omega) \geq 0, \qquad \int_0^\infty p_{\mathrm{ext}}(\omega)\,d\omega = 1, \tag{689}$$

where p_{ext} has units of seconds, because $d\omega$ has units s^{-1}. Let the dimensionless response function of the swirl-string state be

$$R_K(\omega) = \sum_n B_{Kn}\frac{\gamma_{Kn}^2}{(\omega - \omega_{Kn})^2 + \gamma_{Kn}^2}, \tag{690}$$

where:

- ω_{Kn} is the n-th internal mode frequency,
- γ_{Kn} is the corresponding linewidth,
- B_{Kn} is a dimensionless coupling weight.

The dimensionless excitation yield is then

$$\boxed{\mathcal{Y}_K = \int_0^\infty p_{\rm ext}(\omega)\, R_K(\omega)\, d\omega} \qquad \text{[RESEARCH-TRACK].} \tag{691}$$

Dimensional check

Since

$$[p_{\rm ext}] = \mathrm{s}, \qquad [d\omega] = \mathrm{s}^{-1}, \qquad [R_K] = 1,$$

we obtain

$$[\mathcal{Y}_K] = \mathrm{s} \cdot 1 \cdot \mathrm{s}^{-1} = 1.$$

Thus $\mathcal{Y}_K$ is a dimensionless excitation efficiency.

Canonical scale

The natural SST angular-frequency scale is

$$\Omega_{\rm SST} = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}. \tag{692}$$

Using the Canon values

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^6\ \mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15}\ \mathrm{m},$$

gives

$$\Omega_{\rm SST} = 7.76344066 \times 10^{20}\ \mathrm{s}^{-1}. \tag{693}$$

The corresponding cycle period is

$$T_{\rm SST} = \frac{2\pi}{\Omega_{\rm SST}} = 8.09329985 \times 10^{-21}\ \mathrm{s}. \tag{694}$$

Interpretation

Equation (691) does not assert that all nuclear, atomic, or particle transitions are already derived in SST. It states a weaker and safer rule:

> External excitation is strongest when the perturbing spectrum overlaps an allowed internal swirl-string response spectrum.

This principle can later be specialized to:

- radiative transitions,
- qubit-like R/T phase transitions,
- knot-mode excitation,
- delayed mode selection,
- controlled response experiments in optical, plasma, or condensed-matter analogues.

Falsifiability

For a fixed target state K, SST predicts that changing the spectral center ω_0 of a narrow-band perturbation should change the activation yield according to $R_K(\omega_0)$, up to normalization and detector response. If no resonance-like dependence appears after controlling for beam power, fluence, and thermal effects, then Eq. (691) is falsified as a useful excitation law.

P.3 Time-Domain Kernel Form of Swirl–String Response

Status and purpose

Status: [RESEARCH-TRACK]. This appendix converts the frequency-domain selection rule into a time-domain response law. It is intended to connect SST mode selection to pulsed experiments, delay-loop systems, and Swirl-Clock phase locking.

Pulse-response formulation

Let $f(t)$ be a real external driving pulse. A Gaussian-windowed monochromatic pulse may be written as

$$f(t) = f_0 \exp\left(-\frac{t^2}{2\tau_p^2}\right)\cos(\omega_0 t + \phi_0), \tag{695}$$

where τ_p is the pulse duration and ω_0 is the carrier angular frequency.

Let $k_K(t)$ be the causal impulse-response kernel of the swirl-string state K, with

$$k_K(t < 0) = 0. \tag{696}$$

The induced response is

$$\boxed{s_K(t) = \int_{-\infty}^{t} f(t')\, k_K(t-t')\, dt'} \qquad \text{[ORTHODOX form; RESEARCH-TRACK SST interpretation]}. \tag{697}$$

Frequency-domain equivalence

With the Fourier convention

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{i\omega t}\, dt,$$

the convolution theorem gives

$$\hat{s}_K(\omega) = \hat{f}(\omega)\, \chi_K(\omega), \tag{698}$$

where $\chi_K(\omega)$ is the complex response function of the swirl-string state. The measurable spectral response is proportional to

$$R_K(\omega) \propto |\chi_K(\omega)|^2. \tag{699}$$

Bandwidth–duration relation

A Gaussian pulse obeys the scaling

$$\Delta\omega\, \tau_p \sim 1. \tag{700}$$

Therefore:

- short pulses have broad spectra and can excite many nearby modes;
- long pulses have narrow spectra and selectively excite one mode;
- coherent excitation requires phase stability over at least one internal response time.

For the canonical SST scale,

$$T_{\text{SST}} = 8.09329985 \times 10^{-21}\ \text{s}.$$

Thus, a pulse with duration $\tau_p \gg T_{\text{SST}}$ is narrow relative to the core-scale carrier, whereas $\tau_p \lesssim T_{\text{SST}}$ is broadband at the canonical core scale.

Connection to Swirl Clock

If the internal phase of a swirl string is denoted by $\theta_K(t)$, then a resonant perturbation can be represented as a phase-locking term

$$\frac{d\theta_K}{dt} = \omega_K + \epsilon\, f(t) \sin(\omega_0 t - \theta_K), \tag{701}$$

where ϵ is a coupling parameter. Phase locking occurs when

$$|\omega_0 - \omega_K| \lesssim \epsilon |f_0|. \tag{702}$$

This gives a direct bridge between frequency-selective excitation and Swirl-Clock synchronization.

Falsifiability

For a given K, changing only the pulse duration τ_p while keeping total pulse energy fixed should alter the response through the bandwidth $\Delta\omega$. A null result after controlling for heating and total fluence would falsify this time-domain response model.

P.4 Swirl Impedance and Mode Linewidth

Status and critical correction

Status: [RESEARCH-TRACK]. The earlier expression

$$Z_{\text{eff}} \sim \frac{P_{\text{core}} r_c}{\|\mathbf{v}_{\circlearrowleft}\| \hbar}$$

should *not* be canonised as written. Its dimensions are

$$\left[\frac{P_{\text{core}} r_c}{\|\mathbf{v}_{\circlearrowleft}\| \hbar}\right] = \text{m}^{-3},$$

so it is not an impedance and not dimensionless.

A Canon-compatible replacement is to define either:

1. a dimensionless rigidity ratio,
2. or an acoustic-like pressure-to-speed impedance.

Dimensionless swirl-rigidity ratio

Let V_K be the effective active volume of a swirl-string configuration and P_K the pressure scale associated with its deformation. Define

$$\boxed{\Pi_K(\omega) = \frac{P_K V_K}{\hbar\omega}} \qquad \text{[RESEARCH-TRACK]}. \tag{703}$$

This compares stored mechanical energy $P_K V_K$ to the quantum energy scale $\hbar\omega$. It is dimensionless:

$$[P_K V_K] = \text{J}, \qquad [\hbar\omega] = \text{J}.$$

Acoustic-like swirl impedance

An alternative intensive impedance is

$$\boxed{Z_{\text{swirl}} = \frac{P_K}{v_K}} \tag{704}$$

with units

$$[Z_{\text{swirl}}] = \frac{\text{kg}\,\text{m}^{-1}\,\text{s}^{-2}}{\text{m}\,\text{s}^{-1}} = \text{kg}\,\text{m}^{-2}\,\text{s}^{-1},$$

the same units as acoustic impedance. Here v_K denotes the coarse-grained velocity amplitude of the same mode family K used to define P_K; it may be instantiated as a phase-transport speed, local carrier-speed amplitude, or resonant surface-speed amplitude, but the choice must be declared in any benchmark. It is not the canonical scalar speed $v^*_{\circlearrowleft}$ unless the benchmark explicitly specializes to the canonical carrier limit. This object measures resistance to velocity-driven pressure excitation.

Connection to linewidth and quality factor

A resonant mode with angular frequency ω_K and quality factor Q_K has linewidth

$$\gamma_K = \frac{\omega_K}{2Q_K}. \tag{705}$$

A minimal phenomenological relation is

$$Q_K = Q_0\,\Pi_K^{\eta}, \qquad \eta > 0, \tag{706}$$

so that larger stored-energy rigidity produces narrower response lines.

Numerical check using canonical scales

Using

$$F_{\text{swirl}}^{\max} = 29.053507\ \text{N}, \qquad r_c = 1.40897017 \times 10^{-15}\ \text{m},$$

define the core pressure scale

$$P_{\max} = \frac{F_{\text{swirl}}^{\max}}{\pi {r_c}^2} = 4.65848920 \times 10^{30}\ \text{Pa}. \tag{707}$$

[CRITICAL NOTE]: $V_{\text{horn}} = 2\pi^2 {r_c}^3$ is the r_c-scale envelope (horn-torus) volume, *not* the resolved tube cross-section, for which Sec. 2.2 stipulates $a_{\text{core}} \neq r_c$. The resulting $\Pi_K \approx \pi$ is hence a geometric normalization at the envelope scale, not a stiffness theorem for the physical tube; a profile-resolved a_{core} model is required for the latter. For a horn-torus active volume

$$V_{\text{horn}} = 2\pi^2 {r_c}^3 = 5.52122107 \times 10^{-44}\ \text{m}^3, \tag{708}$$

the stored pressure energy is

$$P_{\max} V_{\text{horn}} = 2.57205487 \times 10^{-13}\ \text{J}. \tag{709}$$

At the canonical Compton angular frequency

$$\omega_c = 7.76344066 \times 10^{20}\ \text{s}^{-1},$$

we have

$$\hbar\omega_c = 8.18710572 \times 10^{-14}\ \text{J} = 510998.946\ \text{eV}. \tag{710}$$

Therefore,

$$\Pi_{\text{horn}} = \frac{P_{\max} V_{\text{horn}}}{\hbar\omega_c} = 3.14159235 \approx \pi. \tag{711}$$

Interpretation

Equation (711) is not yet a derivation of a particle mass or coupling constant. It is a useful dimensional coincidence produced by canonical SST scales and the horn-torus active-volume choice. It should be labelled:

$$\boxed{\text{[CALIBRATED / RESEARCH-TRACK NUMERICAL STRUCTURE]}}$$

until derived from the SST action or from a topological minimization principle.

Falsifiability

If Π_K controls linewidth, then families of analog swirl-string excitations should satisfy

$$\gamma_K \propto \omega_K \Pi_K^{-\eta}.$$

A measured absence of correlation between active-volume rigidity and linewidth would falsify Eq. (706).

P.5 Layered Relaxation and Multi-Mode Decay

Status and warning

Status: [RESEARCH-TRACK]. This appendix preserves the useful part of the delayed-decay idea: a multi-layer swirl-string system may relax through multiple modes. It explicitly rejects the naive identification

$$\tau_i = \frac{r_i}{\|\mathbf{v}_{\circlearrowleft}\|}$$

with observed delayed-neutron group lifetimes. For nuclear-scale radii, this gives $\tau_i \sim 10^{-21}$ s, whereas delayed-neutron groups are macroscopic nuclear decay times. Therefore direct radial-transit mapping is dimensionally possible but physically wrong in scale.

General multi-mode relaxation

If a perturbed swirl-string state has several metastable channels, its observable relaxation can be expanded as

$$\boxed{A(t) = \sum_{i=1}^{N} A_i e^{-t/\tau_i} \cos(\omega_i t + \phi_i)} \qquad \text{[ORTHODOX form; RESEARCH-TRACK SST interpretation]}. \tag{712}$$

The non-oscillatory limit is

$$A(t) = \sum_{i=1}^{N} A_i e^{-t/\tau_i}. \tag{713}$$

Pole interpretation

Equation (712) corresponds to a response function with poles

$$\omega = \omega_i - \frac{i}{\tau_i}. \tag{714}$$

Thus, each decay time τ_i is a lifetime of a response pole, not necessarily a physical radial crossing time.

Allowed SST interpretation

In SST, a layered or nested swirl-string configuration may contain:

- a core-scale response,
- a sheath-scale response,
- a linking/helicity response,
- a global Swirl-Clock phase response.

The decay constants τ_i should therefore be treated as *effective metastable amplitude-relaxation times*. With the linewidth convention of Eq. (705), γ_i is the half-width at half maximum in angular frequency and the time-domain amplitude is $e^{-\gamma_i t}$. Hence

$$\boxed{\tau_i = \frac{1}{\gamma_i} = \frac{2Q_i}{\omega_i}}, \tag{715}$$

where Q_i is the quality factor of the i-th mode. The earlier $\tau_i = Q_i/\omega_i$ convention is rejected because it is inconsistent by a factor of two with Eq. (705).

Critical scale check

For the canonical circulation scale,

$$\tau_{\text{cross,c}} = \frac{r_c}{\|\mathbf{v}_{\circlearrowleft}\|} = 1.28808868 \times 10^{-21} \text{ s}. \tag{716}$$

This is a single radial crossing time, distinct from the full circulation period $\tau_{\text{circ,c}} = 2\pi r_c/v_{\circlearrowleft}^*$ of Eq. (189) in the main canon. Therefore, an observed amplitude lifetime of 1 s at the core-scale angular frequency would require

$$Q = \frac{\omega_i \tau_i}{2} \sim 3.88 \times 10^{20}. \tag{717}$$

Such a huge Q-factor is not impossible as a formal metastability parameter, but it cannot be interpreted as a simple geometric shell crossing.

Experimental analogy

Multi-exponential decay models are common in nuclear delayed-neutron kinetics and in linear response theory. SST may use this only as an analogy unless a topological barrier calculation derives the Q_i or τ_i values independently.

Falsifiability

The layered relaxation ansatz predicts that perturbing a structured target should produce reproducible decay constants τ_i independent of total pulse amplitude, provided the response remains linear. If the apparent τ_i vary strongly with drive amplitude, heating, or detector threshold, the model is likely observing instrumental or thermal relaxation rather than swirl-string metastability.

P.6 Research Track: Quasinormal Swirl Spectroscopy and Spectral-Robustness Gate

Status, scope, and terminology

Status: [RESEARCH-TRACK / NOT CANONICAL]. This subsection transfers the mathematically useful parts of quasinormal-mode spectroscopy to perturbations of resolved SST vortex states [75]. It does *not* identify an SST knot with a Kerr black hole, does not assume an event horizon, and does not import the Einstein equations. The transferable content is the analysis of open, non-normal, or coarse-grained evolution through response poles, adjoint modes, resolvents, pseudospectra, nonlinear combination frequencies, and controlled mode-identification tests.

To avoid an ontological overstatement, this track uses the term *quasinormal swirl mode* (QSM). A QSM is an SST response resonance with a complex frequency. The term *QNM* remains reserved for the orthodox black-hole or analogue-gravity comparison.

Three objects must remain distinct:

1. the real normal-mode tower of a closed conservative carrier, as in Section 6.4;
2. the phenomenological Lorentzian response model of Section P.1;
3. QSMs defined as poles or isolated spectral resonances of a specified linearized SST evolution problem.

A fitted damped sinusoid or an FFT peak alone is not a QSM certification.

Relative-equilibrium background and linearized Euler operator

Let $\mathbf{v}_0(\mathbf{x})$ and $p_0(\mathbf{x})$ be a resolved, divergence-free SST background in a frame in which the chosen vortex ring or knot is stationary:

$$\nabla \cdot \mathbf{v}_0 = 0, \tag{718}$$

$$(\mathbf{v}_0 \cdot \nabla)\mathbf{v}_0 = -\frac{1}{\rho_f}\nabla p_0. \tag{719}$$

A translating or rotating vortex is therefore treated as a *relative equilibrium*; the translation and rotation generators belong to the frame definition rather than to the internal mode spectrum. Failure to remove those rigid motions produces spurious low-frequency modes.

Write

$$\mathbf{v} = \mathbf{v}_0 + \epsilon\mathbf{u} + O(\epsilon^2), \qquad p = p_0 + \epsilon\pi + O(\epsilon^2), \tag{720}$$

with $\nabla \cdot \mathbf{u} = 0$. The first-order incompressible Euler system is

$$\partial_t\mathbf{u} + (\mathbf{v}_0 \cdot \nabla)\mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{v}_0 = -\frac{1}{\rho_f}\nabla\pi. \tag{721}$$

Introducing the Leray projector $\mathbb{P}$ onto the admissible divergence-free subspace gives

$$\partial_t\mathbf{u} = \mathcal{L}_K\mathbf{u}, \qquad \mathcal{L}_K\mathbf{u} = -\mathbb{P}\left[(\mathbf{v}_0 \cdot \nabla)\mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{v}_0\right] + \mathcal{L}_{\text{frame}}\mathbf{u}. \tag{722}$$

Here $\mathcal{L}_{\text{frame}}$ contains only the explicitly declared co-translating or co-rotating frame generators and vanishes for a stationary laboratory-frame background. Equations (719)–(722) are the orthodox incompressible-flow linearization used as the starting point for this SST track [76].

The domain, core profile, regularization, topological sector, outer boundary, and allowed flux conditions are part of the operator definition. A spectrum quoted without this metadata is non-reproducible.

Complex frequencies and admissible damping mechanisms

A modal ansatz

$$\mathbf{u}_j(\mathbf{x},t) = \mathbf{q}_j(\mathbf{x})e^{s_j t} = \mathbf{q}_j(\mathbf{x})e^{-i\omega_j t}, \qquad s_j = -i\omega_j, \tag{723}$$

yields

$$\mathcal{L}_K \mathbf{q}_j = s_j \mathbf{q}_j. \tag{724}$$

With

$$\omega_j = \omega_{R,j} - i\gamma_j, \qquad \gamma_j > 0, \tag{725}$$

the amplitude behaves as $e^{-\gamma_j t}\cos(\omega_{R,j}t + \phi_j)$.

[INVISCID CONSISTENCY GATE]. In an exactly closed, fully resolved, inviscid system, damping may not be inserted as an unexplained friction coefficient. A negative imaginary part must be traced to at least one declared mechanism:

1. advective energy or vorticity flux through an open boundary;
2. phase mixing or transfer into a continuous spectrum;
3. transfer into unresolved internal degrees of freedom under an explicitly declared coarse graining;
4. projection onto a local observable while the total conservative state continues to evolve.

A numerical sponge or absorbing layer is a computational boundary treatment, not a physical SST damping mechanism, unless the extracted pole converges as the layer is moved outward and weakened.

The linewidth convention is fixed throughout this track by

$$\boxed{\gamma_j = \frac{1}{\tau_{A,j}} = \frac{\omega_{R,j}}{2Q_j}, \qquad \Delta\omega_{\mathrm{FWHM},j} = 2\gamma_j}. \tag{726}$$

Thus $\tau_{A,j}$ is the amplitude-decay time. An energy-decay time would be $\tau_{E,j} = 1/(2\gamma_j)$, and these two conventions must not be mixed.

Pole definition, residues, and continuous spectrum

Define the retarded resolvent

$$\mathcal{R}_K(s) = (sI - \mathcal{L}_K)^{-1}. \tag{727}$$

A certified QSM is an isolated pole s_j of the analytically continued response, or an isolated eigenvalue of a rigorously specified time-evolution generator on the chosen function space. Near a simple pole,

$$\chi_K(\omega) = \frac{\mathcal{B}_j}{\omega - \omega_j} + \chi_{\mathrm{reg}}(\omega), \tag{728}$$

where $\mathcal{B}_j$ is a residue fixed by the forcing and observable. Consequently, the Lorentzian forms in Eqs. (690) and (698) are local isolated-pole approximations, not primary definitions of the spectrum.

In general the response may contain both discrete resonances and a continuous contribution:

$$A(t) = \sum_j C_j e^{-i\omega_j t} + \int_{\mathcal{C}} C(\omega)e^{-i\omega t}\, d\omega. \tag{729}$$

The second term may produce phase-mixed transients or algebraic tails. Therefore a stable multi-exponential fit over one short time interval does not establish completeness of the pole sum [75].

Adjoint modes, biorthogonality, and first-order sensitivity

For a non-normal evolution generator, right and adjoint modes satisfy

$$\mathcal{L}_K \mathbf{q}_j = s_j \mathbf{q}_j, \tag{730}$$

$$\mathcal{L}_K^\dagger \mathbf{q}_j^\dagger = s_j^* \mathbf{q}_j^\dagger. \tag{731}$$

With an explicitly declared energy or Sobolev inner product, the modes may be normalized biorthogonally:

$$\left\langle \mathbf{q}_i^\dagger, \mathbf{q}_j \right\rangle_E = \delta_{ij}. \tag{732}$$

For a small structured perturbation $\delta\mathcal{L}_K$, the simple-eigenvalue shift is

$$\delta s_j = \frac{\left\langle \mathbf{q}_j^\dagger, \delta\mathcal{L}_K \mathbf{q}_j \right\rangle_E}{\left\langle \mathbf{q}_j^\dagger, \mathbf{q}_j \right\rangle_E} + O(\|\delta\mathcal{L}_K\|^2). \tag{733}$$

A useful conditioning diagnostic is

$$\kappa_j = \frac{\|\mathbf{q}_j^\dagger\|_E \|\mathbf{q}_j\|_E}{\left|\left\langle \mathbf{q}_j^\dagger, \mathbf{q}_j \right\rangle_E\right|}. \tag{734}$$

Large κ_j means that the pole can move strongly under a small change in core profile, embedding, boundary, or discretization. Reporting only ω_j without κ_j can therefore misclassify a fragile numerical overtone as a robust physical mode.

Pseudospectral robustness gate

The ε-pseudospectrum is

$$\boxed{\sigma_\varepsilon(\mathcal{L}_K) = \left\{ s \in \mathbb{C} : \left\| (sI - \mathcal{L}_K)^{-1} \right\|_E > \varepsilon^{-1} \right\}}. \tag{735}$$

It measures where small operator perturbations can create large responses or large spectral shifts. This is required because open gravitational spectra provide explicit examples in which very small potential perturbations strongly displace overtones [77, 75].

For SST, the perturbation classes must be separated:

$$\delta\mathcal{L}_{\text{geom}} : \text{embedding, reach, core-profile, and contact-map changes,} \tag{736}$$

$$\delta\mathcal{L}_{\text{num}} : \text{mesh, quadrature, regularization, and timestep changes,} \tag{737}$$

$$\delta\mathcal{L}_{\text{env}} : \text{outer-flow, wall, background-vorticity, and probe changes.} \tag{738}$$

A QSM is *spectrally robust* only if the relevant pseudospectral contours remain localized on the perturbation scale actually admitted by the benchmark. Broad contours are not automatically a falsification; they require the mode to be labelled *fragile* and prevent its promotion as a precision observable.

Conditional trapped-characteristic/eikonal route

A light-ring formula may not be copied into SST. The required starting point is the principal symbol of the actual SST perturbation equations,

$$D_K(\mathbf{x}, \mathbf{k}, \omega) = 0. \tag{739}$$

Its characteristic rays satisfy

$$\dot{\mathbf{x}} = \frac{\partial D_K}{\partial \mathbf{k}}, \qquad \dot{\mathbf{k}} = -\frac{\partial D_K}{\partial \mathbf{x}}. \tag{740}$$

Only if a specific perturbation branch possesses an unstable periodic characteristic with orbital frequency $\Omega_\star$ and transverse Lyapunov exponent $\Lambda_\star$ may SST test the asymptotic hypothesis

$$\boxed{\omega_{mn}^{\mathrm{QSM}} \simeq m\Omega_\star - i\left(n + \frac{1}{2}\right)\Lambda_\star} \qquad (m \gg 1). \tag{741}$$

This structure is motivated by the orthodox eikonal QNM relation [78, 75], but Eq. (741) is only an SST research hypothesis.

[CRITICAL GATE]. Incompressible Euler flow does not generically supply an acoustic null cone or a scalar wave branch. The characteristic branch, localization, WKB ordering, and large-m limit must therefore be derived before $\Omega_\star$ or $\Lambda_\star$ is assigned. Failure of this derivation blocks the entire trapped-characteristic route but does not block the resolvent-based QSM programme.

Second-order combination modes

Expand a perturbation variable as

$$q = \epsilon q^{(1)} + \epsilon^2 q^{(2)} + O(\epsilon^3). \tag{742}$$

The second-order equation has the schematic form

$$(\partial_t - \mathcal{L}_K)\, q^{(2)} = \mathcal{N}_2\left[q^{(1)}, q^{(1)}\right]. \tag{743}$$

If the first-order field contains modes a and b, the quadratic source contains candidate frequencies

$$\omega_{ab}^{(+)} = \omega_a + \omega_b, \qquad \omega_{ab}^{(-)} = \omega_a - \omega_b^*. \tag{744}$$

Their decay rates add. For an azimuthally resolved background the corresponding selection rules include $m_{ab} = m_a + m_b$ for the sum channel and $m_{ab} = m_a - m_b$ for the difference channel, subject to the actual parity and topology constraints of the chosen state.

In the perturbative regime,

$$A_{ab}^{(2)} \propto A_a^{(1)} A_b^{(1)} \propto \epsilon^2. \tag{745}$$

A peak near a sum or difference frequency is not sufficient: its amplitude must exhibit the quadratic scaling and its symmetry label must satisfy the selection rule. This provides a direct nonlinear benchmark inspired by quadratic black-hole modes, without importing their dynamics [75].

Mode avoidance and exceptional-point scan

For two interacting modes, a local reduced model is

$$H_{\mathrm{eff}} = \begin{pmatrix} s_1 & g_{12} \\ g_{21} & s_2 \end{pmatrix}, \tag{746}$$

with eigenvalues

$$s_\pm = \frac{s_1 + s_2}{2} \pm \sqrt{\left(\frac{s_1 - s_2}{2}\right)^2 + g_{12} g_{21}}. \tag{747}$$

An exceptional point requires the square-root discriminant to vanish and the eigenvectors to coalesce. Because this is one complex condition, two independent real control parameters are generically required. Candidate SST scans include

$$\left(\frac{a_{\rm core}}{L_K}, \frac{\Gamma_K}{v_{\rm ref} L_K}\right), \qquad \left(\mathrm{Tw}, \frac{\mathrm{reach}(K)}{L_K}\right). \tag{748}$$

A credible identification requires eigenvector exchange, adjoint-norm growth, and the expected square-root parameter dependence; an avoided crossing in a frequency plot alone is insufficient [75].

Dimensionless scaling and canonical numerical values

For any mode family define

$$\widehat{\omega}_j = \omega_j \frac{L_{\rm ref}}{v_{\rm ref}}. \tag{749}$$

The pair $(L_{\rm ref}, v_{\rm ref})$ must match the localization scale of the mode. The core reference choice $L_{\rm ref} = r_c$ and $v_{\rm ref} = \|\mathbf{v}_{\circlearrowleft}\|$ gives

$$t_c = \frac{r_c}{\|\mathbf{v}_{\circlearrowleft}\|} = 1.28808868 \times 10^{-21}\ \mathrm{s}, \tag{750}$$

$$\Omega_c = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c} = 7.76344066 \times 10^{20}\ \mathrm{s}^{-1}, \tag{751}$$

$$f_c = \frac{\Omega_c}{2\pi} = 1.23558996 \times 10^{20}\ \mathrm{Hz}. \tag{752}$$

These values are reference scales, not predictions that every global knot mode lies at Ω_c. A trefoil-scale mode must instead use a resolved global length such as L_K, $R_{\rm horn}$, or a certified localization length.

Numerical certification ladder

The minimum numerical programme is:

1. **Background certification.** Construct a vortex ring first and then a trefoil relative equilibrium. Record Euler residual, divergence residual, impulse, circulation, helicity, topology, reach, and core profile.

2. **Consistent tangent operator.** Linearize the same discretized evolution used for the nonlinear run or validate the continuum linearization against finite differences. Enforce the divergence-free projection and remove rigid translation/rotation modes.

3. **Dual extraction.** Compute leading eigenvalues with an Arnoldi or Jacobi–Davidson method and independently extract damped components from small-amplitude time-domain data using matrix-pencil, Prony, or validated dynamic-mode decomposition.

4. **Resolution and boundary sequence.** Vary spatial resolution, timestep, domain size, core regularization, and absorbing-boundary placement independently. Extrapolate before assigning a physical frequency.

5. **Adjoint and pseudospectral audit.** Compute left modes, κ_j, resolvent norms, and structured perturbations from $\delta\mathcal{L}_{\rm geom}$, $\delta\mathcal{L}_{\rm num}$, and $\delta\mathcal{L}_{\rm env}$.

6. **Nonlinear amplitude sweep.** Repeat at several perturbation amplitudes. Linear poles must be amplitude-independent over a declared interval, while quadratic combination amplitudes follow Eq. (745).

7. **Model hierarchy.** Validate the axisymmetric ring benchmark before interpreting a knotted spectrum. The default first knot is the resolved trefoil 3_1; extended knot families enter only after the ring and trefoil pass the same gates.

Passive tracer particles should be disabled during the eigensolve and restored afterward. They are visualization degrees of freedom, not part of $\mathcal{L}_K$, unless a benchmark explicitly promotes them to an active coupled field.

Preregistered pass/fail criteria

A mode may be labelled a *certified numerical QSM candidate* only if:

1. the complex frequency converges to within a preregistered tolerance, initially 1% for the fundamental ring and trefoil modes;
2. time-domain and operator-based estimates agree within their combined uncertainty;
3. divergence and background-residual errors decrease under refinement;
4. the knot type and resolved tube do not undergo reconnection during the linear measurement window;
5. boundary dependence is either negligible at the tolerance level or explicitly identified as the physical open-system mechanism;
6. the linewidth, Q, and decay-time convention satisfies Eq. (726);
7. the residual after subtracting the pole model is tested for a continuous-spectrum or tail contribution.

Promotion beyond a numerical candidate additionally requires robustness under physically admissible geometry perturbations and a reproducible observable coupling coefficient or residue.

Analogue-experiment discipline

Hydrodynamic analogue experiments demonstrate that complex vortex-response spectra can be measured and that many counter-rotating resonances may be excited in one nonequilibrium experiment [79, 75]. This supports the feasibility of a spectroscopy workflow, but it is not evidence for SST ontology. Any proposed analogue must independently vary and report dispersion, background vorticity, asymmetry, finite-size effects, and boundary reflections.

Falsifiers and promotion rule

This track is falsified or blocked for a stated configuration if:

1. the apparent poles do not converge under resolution and boundary refinement;
2. the damping disappears while no admissible inviscid transfer or open-flux mechanism remains;
3. the pole locations are dominated by numerical regularization or tracer settings;
4. Eq. (741) fails after its principal- symbol, localization, and asymptotic prerequisites have been satisfied;
5. predicted quadratic peaks fail the frequency, symmetry, or amplitude-scaling tests;
6. a claimed exceptional point lacks eigenvector coalescence and adjoint-conditioning growth.

No QSM formula is promoted to the main canon until at least one axisymmetric ring and one resolved trefoil benchmark pass the full certification ladder. Even after such promotion, the result would establish an SST spectral property, not a black-hole identity.

Summary of Canon Placement

The five additions should be placed as follows:

Appendix	**Canon status**	**Recommended placement**
Resonance Selection Principle	[RESEARCH-TRACK] strong candidate	CANON-v0.8.17 companion appendix; possible future main-canon principle after validation.
Time-Domain Kernel Response	[RESEARCH-TRACK] strong candidate	Near Swirl Clock, delay-mode selection, or quantum-response sections.
Swirl Impedance and Linewidth	[RESEARCH-TRACK] with corrected dimensions	Use as a response/linewidth ansatz; do not canonise the older dimensionally incorrect formula.
Layered Relaxation	[RESEARCH-TRACK] analogy only	Keep as metastable multi-mode response; reject direct delayed-neutron shell-crossing interpretation.
Quasinormal Swirl Spectroscopy	[RESEARCH-TRACK] certification framework	Keep in the companion track until a vortex ring and resolved trefoil pass the operator, boundary, adjoint, pseudospectral, and nonlinear gates.

Q Extended Vortex, Helicity, and Kairos Research Tracks

Q.1 Research Track: Nested Vortex–Antivortex States as Entropy-Minimizing Swirl Structures

Q.2 Motivation

Consider a coherent swirl structure consisting of an outer circulation region with vorticity $\omega_{\text{out}} > 0$ and an embedded inner core with opposite vorticity

$$\omega_{\text{in}} < 0.$$

Such configurations appear naturally in fluid dynamics, magnetic flux ropes, and vortex-ring interactions.

Within SST, opposite swirl chirality corresponds to opposite temporal orientation of the Swirl Clock:

$$S_t^{\circlearrowleft} \quad \leftrightarrow \quad \text{matter},$$

$$S_t^{\circlearrowright} \quad \leftrightarrow \quad \text{antimatter}.$$

Q.3 Helicity Cancellation

The total helicity may be decomposed as

$$H_{\text{tot}} = H_{\text{out}} + H_{\text{in}}.$$

Because the embedded structure carries opposite chirality,

$$H_{\text{in}} < 0.$$

Hence

$$|H_{\text{tot}}| < |H_{\text{out}}|.$$

The configuration therefore possesses reduced net topological complexity.

Q.4 Entropy Hypothesis

We introduce the conjecture

$$S_{\text{top}} \propto |H|,$$

where S_{top} denotes topological entropy.
Consequently,

$$S_{\text{top}}^{\text{nested}} < S_{\text{top}}^{\text{single}},$$

suggesting that nested vortex–antivortex states may represent preferred low-energy attractors of the swirl medium.

Q.5 Status

Research Track.

Not yet derived from the canonical SST Lagrangian. Proposed as a candidate mechanism for stability enhancement and matter–antimatter coexistence.

Q.6 Research Track: Helicity Conversion and Torsional Radiation

Q.7 Motivation

Experiments and simulations of knotted vortex structures indicate that reconnection events frequently preserve a substantial fraction of helicity.

Rather than disappearing, helicity is redistributed between knotting, linking, writhe, and twist.

Q.8 Helicity Partition

Write total helicity as

$$H = H_{\text{knot}} + H_{\text{twist}} + H_{\text{link}}.$$

During reconnection:

$$H_{\text{knot}} \downarrow$$

while

$$H_{\text{twist}} \uparrow .$$

Thus

$$H_{\text{tot}} \approx \text{const.}$$

Q.9 SST Interpretation

Within SST, mass-bearing T-phase structures correspond to localized knot helicity.

Radiative R-phase excitations correspond to propagating torsional modes.

We therefore conjecture

$$H_{\text{knot}} \to H_{\text{twist}} \to \gamma_R,$$

where γ_R denotes an R-phase torsional pulse.

Q.10 Mass–Radiation Conversion Chain

The proposed conversion pathway is

$$M \leftrightarrow H_{\text{knot}} \leftrightarrow H_{\text{twist}} \leftrightarrow \gamma_R.$$

In this picture:

- stable particles store helicity as topology,
- reconnections release helicity as torsional twist,
- torsional twist propagates as photon-like R-phase radiation.

Q.11 Connection to Photon Ontology

This hypothesis is consistent with the SST identification

$$\text{photon} = \text{torsional R-phase pulse},$$

and provides a possible mechanism for

- particle decay,
- annihilation,
- radiation emission,
- energy transport.

Q.12 Falsifiable Prediction

If the hypothesis is correct, vortex reconnection experiments should show

$$\Delta H_{\text{knot}} \approx -\Delta H_{\text{twist}}$$

with emitted torsional wave packets carrying the missing helicity.

Q.13 Status

Research Track.

Motivated by helicity-conservation studies in knotted vortex systems. Requires derivation from the SST radiation sector.

Q.14 Research Track: Kairos Surfaces as Chirality Transition Layers

Q.15 Definition

The Temporal Ontology introduces the Kairos event

$$\kappa,$$

representing an irreversible topological transition.
We extend this concept spatially.
Let

$$\omega(\mathbf{x}) = 0$$

define a surface separating opposite chiral domains:

$$\omega > 0 \quad \leftrightarrow \quad S_t^{\circlearrowleft},$$

$$\omega < 0 \quad \leftrightarrow \quad S_t^{\circlearrowright}.$$

We define the locus

$$\Sigma_\kappa = \{\mathbf{x} \mid \omega(\mathbf{x}) = 0\}$$

as a *Kairos Surface*.

Q.16 Physical Interpretation

Across Σ_κ the swirl orientation reverses sign.
The surface therefore acts as a boundary between distinct temporal sectors.

$$S_t^{\circlearrowleft} \longleftrightarrow S_t^{\circlearrowright}.$$

Such regions may act as:

- annihilation interfaces,
- vortex reconnection sites,
- phase-transition boundaries,
- chirality-conversion layers.

Q.17 Kairos Criterion

A local Kairos transition is conjectured whenever

$$\omega \to 0$$

and

$$\nabla\omega \neq 0.$$

The gradient supplies a preferred transition direction.

Q.18 Status

Research Track.
Consistent with the Temporal Ontology but not yet derived from the SST field equations.

R EM, Gravity, Diagnostics, Relativity, and Inertia Research Tracks

R.1 Euler-Decomposed Swirl Gravity and Probe Transport

R.2 Purpose and status

This appendix separates the canonical scalar pressure acceleration from orientation-dependent transverse transport effects. The pressure law is retained as the canon-level bridge, while Magnus-type terms are treated as probe-level corrections. The motivation is the orthodox transverse-force/Magnus result for moving vortices in acoustic-geometry and superfluid settings [31, 30].

[GUARDRAIL] The transverse term is not promoted here to the primary cause of universal free fall. In the present canon, universal fall and the Newtonian limit remain attached to the scalar pressure/clock sector. Magnus-type contributions are orientation-, circulation-, and chirality-sensitive probe transport corrections until a universal macroscopic coupling law is derived.

R.3 Canonical pressure acceleration

$$\mathbf{a}_P = -\frac{1}{\rho_f}\nabla p_{\mathrm{swirl}}. \tag{753}$$

R.4 Euler decomposition

For an incompressible inviscid background flow,

$$\frac{\partial \mathbf{u}}{\partial t} + \boldsymbol{\omega}\times\mathbf{u} + \nabla\left(\frac{u^2}{2}\right) = -\frac{1}{\rho_f}\nabla p_{\mathrm{swirl}}. \tag{754}$$

R.5 Filament Magnus transport

For a circulation-carrying swirl-string probe K,

$$\mathbf{F}_{\mathrm{M}}^{(K)} = \int_K \rho_f \Gamma_K \mathbf{T}(s)\times[\mathbf{U}_K(s) - \mathbf{u}(\mathbf{X}(s))]\,ds. \tag{755}$$

R.6 Curvature self-transport

In the local-induction approximation,

$$\mathbf{f}_\kappa \simeq -\frac{\rho_f\Gamma_K^2}{4\pi}\Lambda_K \kappa\,\mathbf{N}. \tag{756}$$

R.7 Dimensional and density check

Equation (755) is a force on a closed filament. The corresponding local density

$$\mathbf{f}_{\mathrm{M}} = \rho_f\Gamma_K\,\mathbf{T}\times(\mathbf{U}_K - \mathbf{u}) \tag{757}$$

has units

$$[\rho_f\Gamma_K\Delta U] = \mathrm{kg\,m^{-3}\,m^2\,s^{-1}\,m\,s^{-1}} = \mathrm{N\,m^{-1}}. \tag{758}$$

With the canonical values

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}^* = 9.6836192 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}, \tag{759}$$

$$\rho_f \Gamma_0 = 6.7785334 \times 10^{-15} \frac{\mathrm{N\,m^{-1}}}{\mathrm{m\,s^{-1}}}, \tag{760}$$

$$\rho_f \Gamma_0 v_{\circlearrowleft}^* = 7.4146692 \times 10^{-9}\ \mathrm{N\,m^{-1}}. \tag{761}$$

Thus, with ρ_f, the Magnus layer is a weak background-transport term rather than a replacement for the saturated core force scale $F_{\mathrm{swirl}}^{\max} = 29.053507$ N. Replacing ρ_f by ρ_{core} would change the physical sector and is not allowed in this transport estimate.

R.8 Axis/off-axis swirl interpretation

For an axisymmetric galactic swirl field

$$\mathbf{u} = u_\theta(r)\mathbf{e}_\theta, \qquad \boldsymbol{\omega} = \omega_z(r)\mathbf{e}_z, \tag{762}$$

an exactly axial probe suppresses the transverse Magnus correction by symmetry. Off-axis motion samples the azimuthal flow. For a flat-tail branch $u_\theta(r) \to v_0$,

$$\omega_z = \frac{1}{r}\frac{d}{dr}(r v_0) = \frac{v_0}{r}, \qquad \boldsymbol{\omega} \times \mathbf{u} = -\frac{v_0^2}{r}\mathbf{e}_r. \tag{763}$$

The inward sign is therefore already contained in the Euler decomposition of the background swirl. This observation should be used as an interpretation of probe drift and centripetal balance, not as an independent gravity postulate.

R.9 Status

The scalar pressure term is canon-level. The Magnus and curvature terms are research-track transport corrections until a universal coupling law for macroscopic matter is derived. Their role is to describe path curvature, chiral drift, and frame-dragging-like probe effects on top of the scalar pressure/clock bridge.

R.10 Research Track: Hybrid Density-Source Swirl-Clock Benchmark

Status. [RESEARCH TRACK / BENCHMARK CANDIDATE]. This subsection records a dimensionally corrected benchmark ansatz for coupling local Swirl-Clock dilation to density-source gravitation across microscopic and macroscopic regimes. It is not a canonical master equation. It may be promoted only if the transition function, source density, and rotational-length scale are independently derived from SST medium dynamics and pass the recovery tests listed below.

Hybrid clock ansatz. For a localized swirl-string configuration K, define the candidate clock factor

$$\boxed{S_{(t)\mathrm{hyb}}^{\circlearrowleft}(r;K) = \left[1 - \frac{2G_{\mathrm{hyb}}(r)M_{\mathrm{hyb}}(r;K)}{rc^2} - \frac{\|\mathbf{u}_{\mathrm{rel}} + \mathbf{v}_\theta(r;K)\|^2}{c^2} - \frac{\ell_\Omega^2(r;K)\Omega_{\mathrm{int}}^2(r;K)}{c^2}\right]^{1/2}} \tag{764}$$

where $\mathbf{u}_{\mathrm{rel}}$ is the relative velocity of the probe or localized configuration with respect to the background foliation, $\mathbf{v}_\theta$ is the resolved circular swirl velocity, and $\ell_\Omega \Omega_{\mathrm{int}}$ is the linear speed associated with an unresolved internal rotational mode.

Hybrid interpolation. The benchmark interpolation is written as

$$\mu(r) = \exp\left[-\left(\frac{r}{R_0}\right)^2\right], \tag{765}$$

$$G_{\rm hyb}(r) = \mu(r)G_{\rm swirl} + [1-\mu(r)]G_N, \tag{766}$$

$$M_{\rm hyb}(r;K) = \mu(r)M_{\rm eff}^{\rm SST}(r;K) + [1-\mu(r)]M_{\rm bulk}. \tag{767}$$

Here R_0 is a research-track transition length, not a canonical constant, and $M_{\rm bulk}$ denotes the independently measured macroscopic source mass used in the Newtonian recovery regime.

Density-source mass closure. The SST-compatible source is the mass-equivalent swirl-energy density, not a bare geometric density:

$$\boxed{M_{\rm eff}^{\rm SST}(r;K) = \int_{B_r} \rho_m^{\rm eff}(\mathbf{x};K)\, d^3x, \qquad \rho_m^{\rm eff} = \frac{\rho_E}{c^2} = \frac{\rho_f \,\|\mathbf{v}_{\circlearrowleft}\|^2}{2c^2}.} \tag{768}$$

This form keeps the source term aligned with the canonical density hierarchy $\rho_E = \frac{1}{2}\rho_f\|\mathbf{v}_{\circlearrowleft}\|^2$ and $\rho_m = \rho_E/c^2$.

Phenomenological spherical test profile. For numerical benchmarks one may use the non-canonical profile

$$\rho_m^{\rm eff}(r) = \rho_0 e^{-r/r_*}, \tag{769}$$

which gives the enclosed source mass

$$\boxed{M_{\rm eff}^{\rm exp}(r) = 8\pi\rho_0 r_*^3\left[1-\left(1+\frac{r}{r_*}+\frac{1}{2}\frac{r^2}{r_*^2}\right)e^{-r/r_*}\right].} \tag{770}$$

The small-radius limit is

$$M_{\rm eff}^{\rm exp}(r) = \frac{4\pi}{3}\rho_0 r^3 + O(r^4), \qquad r \ll r_*, \tag{771}$$

so the enclosed source remains regular at the origin. Omitting the quadratic term in Eq. (770) breaks this recovery limit and is therefore inadmissible.

Irrotational compression and boundary response. The exterior circular-flow branch used for bubble-boundary benchmarks is

$$\mathbf{v}_\theta(r;K) = \frac{\Gamma_K}{2\pi r}\,\hat{\boldsymbol{\theta}}, \qquad r \geq r_{\rm core}, \tag{772}$$

with pressure response inherited from Euler radial balance,

$$\frac{1}{\rho_f}\frac{dp_{\rm swirl}}{dr} = \frac{v_\theta^2}{r}. \tag{773}$$

Thus compression of the boundary at fixed circulation increases $v_\theta \propto r^{-1}$, raises the local kinetic energy density, and lowers the Swirl-Clock factor through Eq. (764). Energy addition may be represented by increasing Γ_K, by increasing the internal mode speed $\ell_\Omega\Omega_{\rm int}$, or by changing the boundary radius; these alternatives must be reported separately in benchmarks.

Dimensional guard for internal rotation. A bare factor Ω^2/c^2 is dimensionally inadmissible. The only admissible clock contribution from an angular rate is

$$\boxed{\frac{v_\Omega^2}{c^2} = \frac{\ell_\Omega^2(r;K)\Omega_{\text{int}}^2(r;K)}{c^2},} \tag{774}$$

where ℓ_Ω is the length scale on which the internal angular mode is realized. If $\ell_\Omega = r_c$ and $\Omega_{\text{int}} = \|\mathbf{v}_\circlearrowleft\|/r_c$, this term reduces to $\|\mathbf{v}_\circlearrowleft\|^2/c^2$ and must not be counted again as an independent correction.

Required benchmark reporting. Any numerical use of Eq. (764) must report the four dimensionless contributions separately:

$$\mathcal{B}_G = \frac{2G_{\text{hyb}}M_{\text{hyb}}}{rc^2}, \qquad \mathcal{B}_\theta = \frac{\|\mathbf{u}_{\text{rel}} + \mathbf{v}_\theta\|^2}{c^2}, \qquad \mathcal{B}_\Omega = \frac{\ell_\Omega^2\Omega_{\text{int}}^2}{c^2}, \qquad \mu(r). \tag{775}$$

At minimum the benchmark set must include electron-core, proton-core, Earth, Sun, and neutron-star limits. The candidate is rejected if it fails the weak field recovery $S^{\circlearrowleft}_{(t)\text{hyb}} \to \sqrt{1 - 2G_N M_{\text{bulk}}/(rc^2)}$ for $\mu(r) \to 0$, or if any contribution is dimensionally non-scalar.

R.11 Research Track: Swirl–Electromagnetic Normalization by Vorticity

R.12 Purpose and Status

This appendix records a research-track extension of Swirl–String Theory (SST) in which electromagnetic field variables are interpreted as SI-normalized coarse-grained measures of swirl kinematics. The result is not proposed as a replacement of Maxwell's equations at the level of observed electrodynamics. Instead, it is a constitutive bridge between the SST swirl-medium variables and the ordinary electromagnetic variables **E**, **B**, **A**, **D**, and **H**.

The central observation is that a dimensionally correct identification of the vector potential is obtained by scaling the local swirl velocity by the electron mass-to-charge ratio:

$$\boxed{\mathbf{A}_{\text{SI}} = \frac{m_e}{e}\,\mathbf{u}_\circlearrowleft(\mathbf{x},t)}. \tag{776}$$

Since $[\mathbf{A}] = \text{kg}\,\text{m}\,\text{C}^{-1}\,\text{s}^{-1}$ and $[(m_e/e)\mathbf{u}_\circlearrowleft] = \text{kg}\,\text{C}^{-1}\cdot\text{m}\,\text{s}^{-1}$, this equation has the correct SI units. This use of $\mathbf{u}_\circlearrowleft(\mathbf{x},t)$ is intentional: the EM bridge is a local-field bridge, not a curl of the calibrated scalar speed $v_\circlearrowleft^*$.

[CRITICAL NOTE] Eq. (776) is a constitutive normalization, not a proof that the full nonlinear Euler vorticity dynamics are isomorphic to vacuum Maxwell theory. The bridge is admissible only in a coarse-grained, transverse, weak-response regime where advective and vortex-stretching terms are either projected out, averaged away, or shown to be subleading by an explicit perturbation estimate.

The associated magnetic field is then

$$\boxed{\mathbf{B}_{\text{SST}} = \nabla\times\mathbf{A}_{\text{SI}} = \frac{m_e}{e}\,\boldsymbol{\omega}}, \qquad \boldsymbol{\omega} = \nabla\times\mathbf{u}_\circlearrowleft(\mathbf{x},t). \tag{777}$$

This is the cleanest form of the statement that magnetic flux arises from structured vorticity, not merely from charge flow. The charge-current description remains the effective macroscopic description; the SST interpretation places the microscopic origin of the magnetic field in quantized circulation and vorticity.

R.13 Why Direct Vorticity Source Terms Are Rejected

Earlier trial equations attempted to write vorticity as an additional source term directly inside Maxwell's equations, for example terms of the schematic form

$$\nabla \cdot \mathbf{E} \sim \frac{\rho}{\varepsilon_0} + \frac{F_{\max}\omega}{\|\mathbf{v}_{\circlearrowleft}\| {r_c}^2}, \qquad \nabla \times \mathbf{B} \sim \mu_0 \mathbf{J} + \frac{\|\mathbf{v}_{\circlearrowleft}\|\hbar}{q {r_c}^2}\boldsymbol{\Omega}. \tag{778}$$

These forms are not retained as canonical equations because the added terms do not generally have the SI dimensions required by Maxwell's source equations. In particular, $\nabla \cdot \mathbf{E}$ has units $\mathrm{V\,m^{-2}}$, while the proposed vorticity-force expression does not reduce to this unit without an additional calibrated constitutive coefficient. Likewise, the Ampere–Maxwell equation requires units of $\mathrm{T\,m^{-1}}$, not a force-per-charge expression multiplied by an unfixed vorticity vector.

Therefore, the research-track correction is:

> Vorticity should not be inserted as an uncalibrated extra Maxwell source. It should enter through a constitutive identification of the potential and field variables.

The retained bridge is thus Eq. (776), not Eq. (778).

R.14 Electric Field from Time-Dependent Swirl

With the vector potential bridge established, the electric field follows from the usual potential relation:

$$\boxed{\mathbf{E}_{\mathrm{SST}} = -\frac{\partial \mathbf{A}_{\mathrm{SI}}}{\partial t} - \nabla \Phi_{\mathrm{SST}} = -\frac{m_e}{e}\frac{\partial \mathbf{u}_{\circlearrowleft}(\mathbf{x},t)}{\partial t} - \nabla \Phi_{\mathrm{SST}}}. \tag{779}$$

Here Φ_{SST} is the SI-normalized scalar potential associated with compressive, pressure-like, or boundary-induced swirl response. In strictly incompressible regions, Φ_{SST} should be interpreted as an effective potential imposed by boundary constraints, topology, and coarse-grained pressure balance rather than by local volume compression.

Equation (779) gives the SST analogue of Faraday induction: time-varying swirl produces an electric field. Taking the curl gives

$$\nabla \times \mathbf{E}_{\mathrm{SST}} = -\frac{\partial}{\partial t}\left(\nabla \times \mathbf{A}_{\mathrm{SI}}\right) = -\frac{\partial \mathbf{B}_{\mathrm{SST}}}{\partial t}, \tag{780}$$

since $\nabla \times \nabla \Phi_{\mathrm{SST}} = 0$. Thus the ordinary Faraday law is preserved exactly.

R.15 Maxwell Equations as Effective Medium Equations

The correct research-track formulation is to retain Maxwell's equations in their macroscopic form while adding swirl-induced polarization and magnetization terms:

$$\nabla \cdot \mathbf{D}_{\mathrm{SST}} = \rho_{\mathrm{free}}, \tag{781}$$

$$\nabla \cdot \mathbf{B}_{\mathrm{SST}} = 0, \tag{782}$$

$$\nabla \times \mathbf{E}_{\mathrm{SST}} = -\frac{\partial \mathbf{B}_{\mathrm{SST}}}{\partial t}, \tag{783}$$

$$\nabla \times \mathbf{H}_{\mathrm{SST}} = \mathbf{J}_{\mathrm{free}} + \frac{\partial \mathbf{D}_{\mathrm{SST}}}{\partial t}. \tag{784}$$

The constitutive relations are

$$\mathbf{D}_{\mathrm{SST}} = \varepsilon_0 \mathbf{E}_{\mathrm{SST}} + \mathbf{P}_{\mathrm{swirl}}, \tag{785}$$

$$\mathbf{H}_{\mathrm{SST}} = \frac{1}{\mu_0}\mathbf{B}_{\mathrm{SST}} - \mathbf{M}_{\mathrm{swirl}}. \tag{786}$$

Here $\mathbf{P}_{\rm swirl}$ and $\mathbf{M}_{\rm swirl}$ encode the coarse-grained response of bound swirl structures, knot cores, circulating R-phase modes, and boundary-induced polarization.

This formulation has an important advantage: it does not violate the observed Maxwell structure. It instead interprets charge and current as effective descriptions of deeper circulation and vorticity organization.

R.16 Canonical Circulation-Scale Calibration and the QED Critical Field

At the SST canonical circulation scale the characteristic angular frequency scale is

$$\Omega_{\rm core} = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}. \tag{787}$$

Using the canonical SST values

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\ \mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15}\ \mathrm{m}, \tag{788}$$

one obtains

$$\Omega_{\rm core} = 7.76344066 \times 10^{20}\ \mathrm{s^{-1}}. \tag{789}$$

The corresponding SST magnetic field scale is

$$B^{\rm SST}_{\rm core} = \frac{m_e}{e}\Omega_{\rm core}. \tag{790}$$

Numerically,

$$\frac{m_e}{e} = 5.68563010 \times 10^{-12}\ \mathrm{kg\,C^{-1}}, \tag{791}$$

and therefore

$$\boxed{B^{\rm SST}_{\rm core} = 4.41400519 \times 10^{9}\ \mathrm{T}}. \tag{792}$$

This is not an arbitrary scale. The standard QED critical magnetic field is

$$B_{\rm QED} = \frac{m_e^2 c^2}{e\hbar} = 4.41400522 \times 10^{9}\ \mathrm{T}. \tag{793}$$

Thus

$$\frac{B^{\rm SST}_{\rm core}}{B_{\rm QED}} = 0.999999993. \tag{794}$$

The equality follows analytically from the SST Compton–horn-radius relation

$$\lambda_c = \frac{2\pi c r_c}{\|\mathbf{v}_{\circlearrowleft}\|}, \tag{795}$$

together with the standard identity

$$\lambda_c = \frac{2\pi\hbar}{m_e c}. \tag{796}$$

Equating these two expressions gives

$$\frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c} = \frac{m_e c^2}{\hbar}. \tag{797}$$

Substitution into Eq. (790) yields

$$B^{\rm SST}_{\rm core} = \frac{m_e}{e}\frac{m_e c^2}{\hbar} = \frac{m_e^2 c^2}{e\hbar} = B_{\rm QED}. \tag{798}$$

This is the strongest numerical result of this appendix: the SI-normalized vorticity bridge naturally reproduces the Schwinger/QED magnetic field scale at the SST core.

The associated electric field scale is

$$E^{\rm SST}_{\rm core} = cB^{\rm SST}_{\rm core} = 1.32328547 \times 10^{18}\ \mathrm{V\,m^{-1}}, \tag{799}$$

which matches the usual Schwinger electric field

$$E_{\rm Schwinger} = \frac{m_e^2 c^3}{e\hbar}. \tag{800}$$

R.17 Corrected Maximum Force Relation

The same calibration fixes the correct factor in the SST maximum-force relation. The expression

$$\frac{\|\mathbf{v}_{\circlearrowleft}\|\hbar}{{r_c}^2} \tag{801}$$

has units of force, but it is twice the canonical force scale. The consistent relation is

$$\boxed{F_{\max}^{\mathrm{swirl}} = \frac{\|\mathbf{v}_{\circlearrowleft}\|\hbar}{2{r_c}^2}}. \tag{802}$$

Using the canonical constants gives

$$F_{\max}^{\mathrm{swirl}} = 29.0535098 \text{ N}, \tag{803}$$

in agreement with the canonical value

$$F_{\max}^{\mathrm{swirl}} \simeq 29.053507 \text{ N}. \tag{804}$$

The electron rest energy relation is then

$$\boxed{m_e c^2 = 2F_{\max}^{\mathrm{swirl}} r_c}. \tag{805}$$

Equivalently,

$$m_e = \frac{2F_{\max}^{\mathrm{swirl}} r_c}{c^2}. \tag{806}$$

Numerically,

$$\frac{2F_{\max}^{\mathrm{swirl}} r_c}{c^2} = 9.10938 \times 10^{-31} \text{ kg}, \tag{807}$$

which matches the electron mass within the stated calibration precision.

R.18 Photon and R-Phase Normal Modes

A second research-track result is that photon frequency should not be introduced through an ad hoc energy-density law of the form

$$u(r,\omega,T) \propto \frac{F_{\max}\omega^3}{\|\mathbf{v}_{\circlearrowleft}\| r^2} \frac{1}{\exp(\hbar\omega/k_B T) - 1}. \tag{808}$$

This expression is not retained as a fundamental energy density because its bare prefactor does not have the units of energy per volume without additional geometric or constitutive factors.

Instead, the research-track photon sector is formulated as a normal-mode condition on a closed or quasi-closed R-phase path:

$$\boxed{k_n = \frac{2\pi n}{L_{\mathrm{eff}}(K)}, \qquad \omega_n = c_\gamma k_n = \frac{2\pi n c_\gamma}{L_{\mathrm{eff}}(K)}}. \tag{809}$$

Here $L_{\mathrm{eff}}(K)$ is the effective optical/swirl path length of the supporting configuration K, and c_γ is the propagation speed of the R-phase excitation. In the vacuum limit one expects $c_\gamma = c$. For a circular ring of radius R,

$$L_{\mathrm{eff}} = 2\pi R, \tag{810}$$

so that

$$\boxed{\omega_n = \frac{n c_\gamma}{R}}. \tag{811}$$

This gives a clean formulation of the statement:

Photon frequency depends on the supporting swirl-ring or path structure.

For knotted, linked, or boundary-deformed paths, $L_{\text{eff}}(K)$ may include writhe, torsion, self-linking, and medium-response corrections:

$$L_{\text{eff}}(K) = L(K) + \Delta L_{\text{writhe}} + \Delta L_{\text{torsion}} + \Delta L_{\text{boundary}} + \Delta L_{\text{medium}}. \tag{812}$$

This places the photon sector in the same mathematical family as the SST delay-loop and mode-selection principles.

R.19 Kelvin Impulse Bridge

For a thin circular swirl ring with circulation Γ, major radius R, and core radius a, the classical ring scalings are

$$E_{\text{ring}} = \frac{1}{2}\rho_f \Gamma^2 R \left[\ln\left(\frac{8R}{a}\right) - \alpha_{\text{ring}}\right], \tag{813}$$

$$P_{\text{ring}} = \rho_f \Gamma \pi R^2, \tag{814}$$

$$V_{\text{ring}} = \frac{\Gamma}{4\pi R}\left[\ln\left(\frac{8R}{a}\right) - \beta_{\text{ring}}\right]. \tag{815}$$

These relations provide a bridge between Kelvin's impulse theory and an SST momentum assignment. A research-track identification may be written as

$$\boxed{\hbar k \leftrightarrow \eta_I P_{\text{swirl}}}, \qquad P_{\text{swirl}} = \rho_f \Gamma \pi R^2. \tag{816}$$

The coefficient η_I is not a free fit parameter in the final theory. It must be determined from the same SI-normalization that fixes Eqs. (776)–(798), or from an independently specified R-phase normalization.

This suggests a useful research direction:

Planck momentum may correspond to normalized Kelvin impulse of a circulating R-phase excitation.

R.20 Topology, Helicity, and Reconnection

The ideal SST sector assumes conservation of circulation, linkage, and helicity in the absence of reconnection. This corresponds to the Helmholtz–Kelvin regime:

$$\frac{D\Gamma}{Dt} = 0, \qquad \frac{D\mathcal{H}}{Dt} = 0, \qquad \mathcal{H} = \int \mathbf{u}_{\circlearrowleft} \cdot \boldsymbol{\omega}\, d^3x. \tag{817}$$

In this ideal sector, knot class is preserved.

However, laboratory and numerical studies of knotted and linked fluid structures show that reconnection can transfer helicity between linking, twist, and writhe. Therefore, SST should distinguish two regimes:

$$\text{Ideal sector:} \quad K(t) = K(0), \qquad \Gamma = \text{constant}, \qquad \mathcal{H} = \text{constant}, \tag{818}$$

$$\text{Transition sector:} \quad K_i \to K_j, \qquad \Delta\mathcal{H} = \Delta\mathcal{L} + \Delta\mathcal{T} + \Delta\mathcal{W}, \tag{819}$$

where $\mathcal{L}$, $\mathcal{T}$, and $\mathcal{W}$ denote linking, twist, and writhe contributions respectively.

This is relevant for the SST interpretation of measurement, decay, and mode conversion. A stable particle-like excitation belongs to the ideal sector. A decay or transition corresponds to a controlled failure of ideal topological conservation, usually through reconnection or boundary interaction.

R.21 Pressure Envelopes and Spherical Equilibrium

The electromagnetic bridge above should be kept distinct from the pressure-envelope sector of SST. The pressure sector is governed by the Bernoulli-type swirl balance

$$p + \frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2 = p_0, \tag{820}$$

or, for a perturbation,

$$\Delta p = -\frac{1}{2}\rho_f \Delta \left(\|\mathbf{v}_{\circlearrowleft}\|^2\right). \tag{821}$$

This pressure defect can generate attraction-like behavior in the gravitational/swirl-clock sector. It should not be confused with the magnetic vorticity bridge

$$\mathbf{B}_{\mathrm{SST}} = \frac{m_e}{e}\boldsymbol{\omega}. \tag{822}$$

Thus the research-track separation is:

$$\text{EM-like sector:} \quad \mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\mathbf{u}_{\circlearrowleft}(\mathbf{x},t), \qquad \mathbf{B}_{\mathrm{SST}} = \frac{m_e}{e}\boldsymbol{\omega}, \tag{823}$$

$$\text{pressure/gravity sector:} \quad \Delta p = -\frac{1}{2}\rho_f \Delta(\|\mathbf{v}_{\circlearrowleft}\|^2), \qquad d\tau_{\mathrm{loc}}/dt_\infty = \sqrt{1 - \|\mathbf{v}_{\circlearrowleft}\|^2/c^2}. \tag{824}$$

R.22 Negative-Temperature and Entropy Analogy

A further research-track idea is that bounded swirl ensembles may possess entropy-ordering behavior analogous to negative-temperature vortex states. This is not required for the electromagnetic normalization, but it may be useful in the classification of high-energy knot ensembles.

A possible entropy functional is

$$S_{\mathrm{swirl}} = k_B \ln \Omega_{\mathrm{topo}}(E, \Gamma, \mathcal{H}), \tag{825}$$

where Ω_{topo} counts accessible configurations at fixed energy, circulation, and helicity. A negative-temperature-like regime would satisfy

$$\frac{\partial S_{\mathrm{swirl}}}{\partial E} < 0. \tag{826}$$

Such a regime would describe increasing order with increasing energy, for example through clustering, alignment, or coherent large-scale swirl organization. This idea should remain in the research track until it is connected to a concrete SST Hamiltonian and a well-defined finite phase space.

R.23 Research Claims Retained

The following claims are retained for research-track development:

1. Magnetic field strength can be SI-normalized as vorticity:

$$\mathbf{B}_{\mathrm{SST}} = \frac{m_e}{e}\boldsymbol{\omega}.$$

2. The vector potential is the SI-normalized swirl velocity:

$$\mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\mathbf{u}_{\circlearrowleft}(\mathbf{x},t).$$

3. The core-scale SST magnetic field equals the QED critical field:

$$B_{\text{core}}^{\text{SST}} = B_{\text{QED}} = \frac{m_e^2 c^2}{e\hbar}.$$

4. The associated electric field equals the Schwinger scale:

$$E_{\text{core}}^{\text{SST}} = cB_{\text{core}}^{\text{SST}} = \frac{m_e^2 c^3}{e\hbar}.$$

5. Photon frequency should be modeled by normal-mode closure:

$$\omega_n = \frac{2\pi n c_\gamma}{L_{\text{eff}}(K)}.$$

6. Kelvin impulse may provide the bridge to photon momentum:

$$\hbar k \leftrightarrow \eta_I \rho_f \Gamma \pi R^2.$$

7. Reconnection belongs to a separate transition sector and should not be mixed with ideal knot conservation.

R.24 Claims Not Retained as Canonical

The following statements are not retained as canonical equations:

1. Direct insertion of uncalibrated vorticity terms into Maxwell source equations.
2. The statement that charge is unnecessary in all electromagnetic phenomena. In this appendix charge remains the effective macroscopic source variable.
3. The claim that Podkletnov-type gravitational shielding is established evidence for SST. Such material may be mentioned only as speculative historical motivation, not as support for the canon.
4. Photon energy-density laws containing $F_{\max}\omega^3/(\|\mathbf{v}_{\circlearrowleft}\| r^2)$ without a complete dimensional and geometric derivation.

R.25 Minimal Experimental Tests

The vorticity-normalization bridge suggests several falsifiable directions.

1. Structured plasmonic or polaritonic media. If electromagnetic fields are coupled to structured circulation-like degrees of freedom, then anisotropic or hyperbolic media should show geometry-dependent emission and propagation consistent with an effective path length $L_{\text{eff}}(K)$.

2. Superfluid or quantum-fluid circulation. Quantized circulation structures should generate measurable effective magnetization only when the system contains charged, spinful, or electromagnetically coupled degrees of freedom. A neutral inviscid fluid alone should not emit ordinary electromagnetic fields unless an additional coupling channel exists.

3. Ring-resonator delay tests. The photon/R-phase normal-mode law predicts that frequency selection should scale as

$$\omega_n \propto \frac{1}{L_{\text{eff}}}.$$

Boundary deformation, writhe-like path changes, or torsional loading should shift the selected resonances.

4. Core-scale extrapolation. The equality

$$B_{\text{core}}^{\text{SST}} = B_{\text{QED}}$$

is not directly reachable in ordinary laboratory magnetic fields, but it can be tested indirectly by checking whether the same normalization reproduces Compton, Schwinger, and electron-rest-energy scales without additional tuning.

R.26 Numerical Verification Table

Using the canonical SST constants

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\ \mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15}\ \mathrm{m},$$

one obtains:

Quantity	Expression	Numerical value
Core angular scale	$\Omega_{\text{core}} = \|\mathbf{v}_{\circlearrowleft}\|/r_c$	$7.76344066 \times 10^{20}\ \mathrm{s^{-1}}$
Mass-charge ratio	m_e/e	$5.68563010 \times 10^{-12}\ \mathrm{kg\,C^{-1}}$
SST core magnetic field	$(m_e/e)\Omega_{\text{core}}$	$4.41400519 \times 10^{9}\ \mathrm{T}$
QED critical magnetic field	$m_e^2c^2/(e\hbar)$	$4.41400522 \times 10^{9}\ \mathrm{T}$
Ratio	$B_{\text{core}}^{\text{SST}}/B_{\text{QED}}$	0.999999993
SST electric field scale	$cB_{\text{core}}^{\text{SST}}$	$1.32328547 \times 10^{18}\ \mathrm{V\,m^{-1}}$
Circulation quantum	$2\pi r_c\|\mathbf{v}_{\circlearrowleft}\|$	$9.68361920 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}$
Corrected force ceiling	$\|\mathbf{v}_{\circlearrowleft}\|\hbar/(2{r_c}^2)$	29.0535098 N
Electron mass from force	$2F_{\max}^{\text{swirl}}r_c/c^2$	9.10938×10^{-31} kg

R.27 Conclusion

The strongest result of this appendix is the SI-normalized vorticity bridge

$$\mathbf{A}_{\text{SI}} = \frac{m_e}{e}\mathbf{u}_{\circlearrowleft}(\mathbf{x},t), \qquad \mathbf{B}_{\text{SST}} = \frac{m_e}{e}\boldsymbol{\omega}.$$

It preserves Maxwell's equations as effective field equations, avoids dimensionally invalid source terms, and reproduces the QED critical field at the SST core scale. For this reason it is suitable for inclusion in the research-track sections of CANON-v0.8.x.

The proposed canonical status is:

Result	Status
$\mathbf{A}_{\text{SI}} = (m_e/e)\mathbf{u}_{\circlearrowleft}(\mathbf{x},t)$	Research Track, strong candidate
$\mathbf{B}_{\text{SST}} = (m_e/e)\boldsymbol{\omega}$	Research Track, strong candidate
$B_{\text{core}}^{\text{SST}} = B_{\text{QED}}$	Numerically verified calibration result
Direct vorticity source terms in Maxwell equations	Rejected / not canonical
Photon frequency from $L_{\text{eff}}(K)$	Research Track, compatible with delay-loop sector
Kelvin impulse to $\hbar k$ bridge	Research Track, unresolved normalization
Negative-temperature swirl entropy	Speculative Research Track

R.28 Research Track: Flame-Lock, Photonless-Caustic, and Elastic-Shell Diagnostics

Status. [RESEARCH TRACK / DIAGNOSTIC ONLY]. This section records a three-channel diagnostic motivated by anomalous-field witness reports. The reports are not used as evidence. They serve only as a catalogue of separable experimental signatures compatible with the canonical sector-projection law of Section 11.15.

R.29 Mode I: flame-lock / plume stabilization

A flame or thermal plume may be stabilized without significant optical ray deflection if the dominant SST response is local impedance or damping rather than photon-index curvature. A minimal diagnostic equation is

$$\frac{\partial \mathbf{u}_{\text{plume}}}{\partial t} + (\mathbf{u}_{\text{plume}} \cdot \nabla)\mathbf{u}_{\text{plume}} = -\frac{1}{\rho_{\text{plume}}}\nabla p + \nu \nabla^2 \mathbf{u}_{\text{plume}} - \Gamma_{\circlearrowright} \mathbf{u}_{\text{plume}}. \tag{827}$$

The diagnostic signature is

$$P_{\text{plume}}(f) \downarrow, \qquad \Delta\phi_\gamma \approx 0, \qquad \nabla_\perp n_\gamma \approx 0. \tag{828}$$

This channel is a boundary/impedance response, not a standalone proof of a gravitational bulk force.

Primary orthodox confounders: ion wind, acoustic forcing, thermal convection suppression, electric-field stabilization of charged flame species, camera exposure artifacts, and ordinary airflow.

R.30 Mode II: photonless sphere / optical caustic

A dark or photon-depleted region is interpreted, if real, as an optical caustic, shadow, or phase-gradient effect, not as a literal astrophysical black hole. The minimal ray equation is

$$\frac{d}{ds}\left(n_\gamma \frac{d\mathbf{x}}{ds}\right) = \nabla n_\gamma, \qquad n_\gamma = S^{\circlearrowright}_{(t)}{}^{-1}. \tag{829}$$

The diagnostic signature is

$$\nabla_\perp n_\gamma \neq 0, \qquad \Delta\phi_\gamma = \frac{2\pi}{\lambda}\int (n_\gamma - 1)\, ds \neq 0, \qquad \mathbf{f}_{\text{matter}} \approx 0. \tag{830}$$

Macroscopic photon depletion or black-ball-like appearance over laboratory path lengths requires a resonant enhancement factor:

$$n_\gamma - 1 = Q_\gamma \frac{\|\mathbf{u}_{\circlearrowright}\|^2}{2c^2}, \qquad Q_\gamma \gg 1. \tag{831}$$

The gain Q_γ is apparatus-specific and is not a canonical constant.

Primary orthodox confounders: thermal lensing, plasma absorption, smoke or aerosol scattering, detector saturation, schlieren artifacts, camera auto-exposure, and ordinary optical shadowing.

R.31 Mode III: elastic shell / boundary-stress response

A localized repulsive shell is represented as a boundary-stress potential,

$$U_{\text{shell}}(r) = U_0 \exp\left[-\frac{(r - R_s)^2}{2\sigma_s^2}\right], \tag{832}$$

with radial force

$$F_r(r) = -\frac{dU_{\text{shell}}}{dr} = \frac{U_0(r - R_s)}{\sigma_s^2} \exp\left[-\frac{(r - R_s)^2}{2\sigma_s^2}\right]. \tag{833}$$

The diagnostic signature is

$$\nabla \cdot \boldsymbol{\sigma}^{\text{swirl}} \neq 0, \qquad \Delta\phi_\gamma \approx 0 \quad \text{or independently measurable.} \tag{834}$$

This mode is a stress/boundary response. It must not be rebranded as gravity shielding unless the bulk acceleration sector is independently measured.

Primary orthodox confounders: electrostatic repulsion, magnetic force, acoustic radiation pressure, airflow, vibration, thermal expansion, hidden mechanical coupling, and contact charging.

R.32 Required measurement vector

A valid test must record the full sectoral observable set

$$\boxed{\left\{F(t), \Delta\phi_\gamma(t), \frac{\Delta f}{f}(t), P_{\text{plume}}(f,t), \mathbf{E}(t), \mathbf{B}(t), T(t), p_{\text{air}}(t)\right\}.} \tag{835}$$

Any reported force must first be decomposed as

$$\mathbf{F}_{\text{meas}} = \mathbf{F}_{\text{EM}} + \mathbf{F}_{\text{EHD}} + \mathbf{F}_{\text{thermal}} + \mathbf{F}_{\text{acoustic}} + \mathbf{F}_{\text{mechanical}} + \mathbf{F}_\omega + \mathbf{F}_{\text{res}}. \tag{836}$$

Only $\mathbf{F}_{\text{res}}$, after uncertainty propagation and null tests, is admissible as an SST research-track candidate.

R.33 Null hierarchy

The minimum null hierarchy is:

1. reproduce the measurement with all high-voltage and high-current channels disabled;
2. randomize phase order at fixed RMS power;
3. replace the structured geometry with a symmetry-preserving dummy load of matched impedance;
4. repeat in still air, shielded enclosure, and reduced pressure where safe;
5. compare optical, force, and plume observables under the same drive waveform.

A positive SST candidate requires a correlated sectoral signature that survives these nulls and scales with the predicted mode-selection or stress-closure parameter.

R.34 Research Track: No-Monopole Audit of the Transport-Stress Gravity Candidate

Status. [RESEARCH TRACK / DERIVED CONSTRAINT]. This section records a constraint on the SST-73 transport-stress gravity candidate. It does not modify the canon-level weak-field closure $\nabla^2\Phi_\circlearrowleft = 4\pi G_{\text{swirl}}\rho_m$. It shows that a double-divergence stress source belongs to the local pressure/stress channel unless an additional non-divergence density closure or boundary source is specified.

R.35 Pressure equivalence lemma

For an incompressible inviscid medium with constant ρ_f,

$$\partial_i u_i = 0, \tag{837}$$

and no external body force, the Euler equation is

$$\rho_f\left(\partial_t u_i + u_j \partial_j u_i\right) = -\partial_i p. \tag{838}$$

Taking the divergence and using $\partial_i u_i = 0$ gives

$$\partial_i \partial_j\left(\rho_f u_i u_j\right) = -\nabla^2 p. \tag{839}$$

Therefore the SST-73 transport-stress candidate

$$\nabla^2\Phi_{\text{tr}} = \lambda\, \partial_i \partial_j\left(\rho_f u_i u_j\right) \tag{840}$$

is equivalent to

$$\nabla^2 (\Phi_{\rm tr} + \lambda p) = 0. \tag{841}$$

For decaying exterior boundary conditions,

$$\boxed{\Phi_{\rm tr} = -\lambda (p - p_\infty)} \tag{842}$$

up to an irrelevant harmonic constant. Candidate C is therefore not an independent long-range gravity mechanism in the smooth incompressible bulk; it is the Euler pressure channel written in Poisson form.

R.36 No-monopole theorem for smooth compact stress sources

Let

$$S(\mathbf{x}) = \partial_i \partial_j \Sigma_{ij}, \qquad \Sigma_{ij} = \rho_f u_i u_j, \tag{843}$$

with Σ_{ij} smooth and compactly supported, or decaying sufficiently fast at infinity. The monopole moment is

$$Q_0 = \int_{\mathbb{R}^3} S(\mathbf{x})\, dV = \oint_{\partial\mathbb{R}^3} \partial_j \Sigma_{ij}\, dA_i = 0. \tag{844}$$

The dipole moment also vanishes after integration by parts:

$$Q_k = \int_{\mathbb{R}^3} x_k S(\mathbf{x})\, dV = 0. \tag{845}$$

Hence the first possible nonzero exterior multipole is quadrupolar:

$$\Phi_{\rm tr}(r) = O(r^{-3}), \qquad \|\nabla \Phi_{\rm tr}\| = O(r^{-4}). \tag{846}$$

This cannot supply a Newtonian $1/r^2$ acceleration by itself.

R.37 Boundary and singular-core exception

The no-monopole result assumes a smooth regularized field on all of $\mathbb{R}^3$. If vortex tubes are excised, singular cores are retained, or material boundaries/shells are present, the surface term

$$\oint_{\partial V} \partial_j \Sigma_{ij}\, dA_i \tag{847}$$

need not vanish. Such terms are boundary-stress or shell-response physics $\mathcal{R}_\Pi$, not bulk Poisson gravity. A nonzero Newtonian monopole therefore requires either a genuine scalar density source, such as $\rho_m = \rho_E/c^2$, or an explicitly modeled boundary source.

R.38 Equivalence-principle consequence of the density source

If the gravitational source density is the same effective mass density that defines inertial mass,

$$\rho_m^{\rm eff} = \frac{\rho_E}{c^2}, \qquad \rho_E = \frac{1}{2}\rho_f \|\mathbf{u}_\circlearrowleft\|^2, \tag{848}$$

then

$$M_{\rm grav} = \int \rho_m^{\rm eff}\, dV = \frac{1}{c^2} \int \rho_E\, dV = M_{\rm inert}. \tag{849}$$

For full topological SST states this statement must be applied to the same coarse-grained mass functional used for inertial mass, including topology, geometric gate, and clock-impedance factors. The numerical value of $G_{\rm swirl}$ remains **[CALIBRATED]** until derived independently of orthodox G or t_p.

R.39 Minimal numerical tests

A verification script for this sector must include:

1. a smooth compact divergence-free test field for which $\int S\,dV \approx 0$ and $\int x_k S\,dV \approx 0$;
2. a far-field multipole fit verifying $\Phi_{\rm tr} \sim r^{-3}$, not r^{-1};
3. a pressure-equivalence test verifying $\Phi_{\rm tr}+\lambda p$ is harmonic/constant under the stated boundary conditions;
4. a separate density-source Poisson test verifying $\Phi \sim -G_{\rm swirl}M/r$ and $\|\mathbf{g}\| \sim r^{-2}$ for compact positive $\rho_m^{\rm eff}$.

R.40 Research Track: Atomic Gravity Closure and Condensate Coherence

Status. [CANON-CANDIDATE / RESEARCH BRIDGE]. This section records the electron-envelope and condensate-coherence clue for future gravity-sector tests. It does not claim gravitational shielding.

R.41 Electron envelope as atomic closure layer

In a neutral atom, the electron envelope cancels the far electromagnetic charge sector but does not cancel positive swirl-energy density. Therefore the electron is not the dominant mass reservoir, but it is the outer phase and linking boundary through which the neutral atom closes its electromagnetic projection:

$$Q_{\rm atom} = 0, \qquad \rho_m = \frac{\rho_E}{c^2} > 0. \tag{850}$$

The canon-safe interpretation is

> *The nucleus supplies most mass-energy, while the electron envelope supplies the neutral phase boundary and long-range matching layer.*

R.42 Condensate coherence extension

For coherent electronic or atomic condensates, a macroscopic order parameter

$$\Psi_{\rm cond}(\mathbf{x}) = \sqrt{n_s(\mathbf{x})}\, e^{i\theta(\mathbf{x})} \tag{851}$$

may coherently reweight boundary-stress and optical/clock projections:

$$\boldsymbol{\sigma}^{\rm swirl} \to \boldsymbol{\sigma}^{\rm swirl}_{\rm coh}[n_s, \theta, \nabla\theta], \tag{852}$$

$$n_\gamma \to n_{\gamma,\rm coh}. \tag{853}$$

The proposed clue from Meissner/BEC physics is therefore not mass cancellation, but collective phase closure: many microscopic carriers can impose one macroscopic boundary condition.

[CRITICAL NOTE]. This does not imply gravitational shielding or mass cancellation. The testable claim is only a possible phase-dependent change in boundary-stress, optical phase, or clock response across a coherence transition such as T_c.

R.43 Condensate test signature

A minimally admissible superconducting or BEC test compares the same apparatus above and below the coherence transition:

$$\Delta\mathcal{O}_{\rm coh} = \mathcal{O}(T<T_c) - \mathcal{O}(T>T_c), \qquad \mathcal{O} \in \{F, \Delta\phi_\gamma, \Delta f/f, \Delta\Phi\}\,. \tag{854}$$

A positive result must track coherence and vanish under matched non-coherent thermal, electromagnetic, and mechanical controls.

R.44 Relativity Emergence Ladder in SST

R.45 Purpose and status

This section records the precise epistemic status of special-relativistic and general-relativistic structures inside Swirl–String Theory (SST). The goal is not to overclaim that general relativity has been derived from incompressible Euler dynamics. The goal is to separate:

1. kinematic consequences of a common propagation cone;
2. clock and metric bridge structures;
3. weak-field gravitational closures;
4. the still-open problem of deriving the Einstein field equations.

Status summary. SST presently supports a conditional reconstruction of relativistic kinematics for its excitation sector, a conditional weak equivalence principle, and a conditional Newtonian limit. The full Einstein–Hilbert dynamics remain an open research target.

R.46 Primitive structure

The minimal SST substrate is taken to be an incompressible, inviscid medium on a preferred foliation,

$$\mathcal{M}_{\text{SST}} = \mathbb{R}^3 \times \mathbb{R}, \qquad \nabla \cdot \mathbf{v} = 0, \tag{855}$$

with structured vorticity, quantized circulation, an operationally canonical transverse torsion speed c, and a local Swirl–Clock factor

$$S_{(t)}(x) = \sqrt{1 - \frac{u(x)^2}{c^2}}. \tag{856}$$

[**ORTHODOX**] The functional form of Eq. (856) is the standard Lorentz time-dilation factor.

[**SST INTERPRETATION**] The velocity $u(x)$ is interpreted as a local medium-kinematic state rather than as an abstract spacetime postulate.

[**FOUNDATIONAL STATUS**] The equality between the locally measured R-phase torsion-pulse speed and the observed constant c is a canonical operational postulate of SST. If c is inserted as calibration input, the existence and universality of the limiting cone are not thereby derived. What must be derived is the common torsional constitutive speed, the dynamical response of clocks and rods, and the absence of species- or orientation-dependent retuning.

R.47 Operational vortex-atom postulate

Let A denote a stable vortex atom equipped with an internal cycle counter and an independently equilibrated length standard ℓ_A. For an out-and-back R-phase torsion-pulse exchange, define the measured two-way speed

$$c_A^{(2\text{w})} := \frac{2\ell_A}{\tau_A^{\text{recv}} - \tau_A^{\text{emit}}}. \tag{857}$$

The foundational SST target is

$$\boxed{c_A^{(2\text{w})} = c} \qquad \text{for every stable low-energy carrier } A. \tag{858}$$

The bare ratio dX/dT in substrate coordinates is not the observable in Eq. (857). It may vary under a change of substrate coordinates or through the clock/ruler response. The physical claim is that each vortex atom reconstructs the same local null cone.

After the two-way speed has been established relative to an independent ruler, Alice assigns radar coordinates

$$t_A(E) = \frac{\tau_A^{\rm recv} + \tau_A^{\rm emit}}{2}, \tag{859}$$

$$\ell_A(E) = \frac{c}{2}\left(\tau_A^{\rm recv} - \tau_A^{\rm emit}\right). \tag{860}$$

The corresponding local interval is

$$d\sigma_A^2 = -c^2 dt_A^2 + d\ell_A^2, \qquad d\sigma_A^2 = 0 \quad \text{for an R-phase torsion pulse.} \tag{861}$$

R.48 Alice–Bob operational Lorentz map

Let Bob move at relative speed u_{AB} along Alice's x-axis. Define

$$\beta_{AB} = \frac{u_{AB}}{c}, \qquad \gamma_{AB} = \frac{1}{\sqrt{1-\beta_{AB}^2}}. \tag{862}$$

If both observers' clocks, rulers, and synchronization signals are controlled by the same torsion cone, their measured coordinates obey

$$ct_B = \gamma_{AB}\left(ct_A - \beta_{AB} x_A\right), \tag{863}$$

$$x_B = \gamma_{AB}\left(x_A - \beta_{AB} ct_A\right), \tag{864}$$

and

$$-c^2 dt_A^2 + dx_A^2 = -c^2 dt_B^2 + dx_B^2. \tag{865}$$

[ORTHODOX] Equations (864)– (865) are standard special-relativistic kinematics [45].

[SST THEOREM TARGET] Although the substrate contains a preferred foliation, every low-energy Alice–Bob observable must depend only on u_{AB}, not separately on their velocities relative to that foliation. A residual dependence on the individual substrate velocities is a preferred-frame falsifier.

R.49 Non-circular emergence ladder

The non-circular derivation order is

$$\begin{aligned}\mathcal{L}_{\rm micro} &\longrightarrow (\rho_T, K_T) \longrightarrow c_T = \sqrt{K_T/\rho_T} \longrightarrow \text{clock and ruler response}\\ &\longrightarrow c_A^{(2\rm w)} = c_T \longrightarrow c_T \overset{\rm experiment}{=} c.\end{aligned} \tag{866}$$

Using measured c to set $K_T = \rho_T c^2$ is a valid calibrated bridge, but it does not constitute the first-principles derivation represented by Eq. (866).

R.50 Light-clock derivation of the Lorentz factor

Consider a clock whose tick is defined by a transverse R-phase torsion pulse with constitutive speed $c_T = c$ in the substrate frame. If the clock moves with speed u relative to the substrate, then during one half-cycle

$$c^2\Delta t^2 = u^2\Delta t^2 + c^2\Delta\tau^2. \tag{867}$$

Therefore

$$\Delta\tau = \Delta t\sqrt{1-\frac{u^2}{c^2}}. \tag{868}$$

[DERIVED, conditional] Equation (868) derives Lorentz time dilation for clocks whose internal communication is entirely mediated by substrate excitations with speed c [45].

[OPEN CANON GAP] Universal Lorentz kinematics for all matter species requires the stronger monometricity condition below.

R.51 Monometricity theorem target

Let A label low-energy SST excitation species. The principal symbol of the linearized equations for species A defines an effective characteristic surface

$$P_A(k) = 0. \tag{869}$$

Relativistic universality requires that, at low energy,

$$P_A(k) = 0 \quad \Longleftrightarrow \quad g^{\mu\nu}_{\text{eff}} k_\mu k_\nu = 0 \qquad \text{for all } A, \tag{870}$$

up to irrelevant conformal factors:

$$g^{\mu\nu}_{\text{eff},A} = \Omega^2_A(x)\, g^{\mu\nu}_{\text{eff}}. \tag{871}$$

[OPEN CANON GAP: MONOMETRICITY] SST currently contains at least two distinguished speeds,

$$c, \qquad \mathbf{v}_{\circlearrowleft} = \frac{\alpha c}{2}, \tag{872}$$

so exact low-energy Lorentz universality is not automatic. The required theorem is that matter knots, gauge/torsion pulses, and clock excitations share the same effective cone in the infrared. In condensed-matter language this is the Fermi-point universality problem [42].

Falsifier. Failure of Eq. (870) predicts species-dependent Lorentz violation and preferred-frame effects.

R.52 Operational universality benchmarks

A valid theorem-level closure must pass all of the following tests:

1. **Knot-species universality:**
$$\frac{c_A^{(2\text{w})}}{c} - 1 \longrightarrow 0 \qquad \text{for } A \in \{3_1, 4_1, 5_2, 6_1, \ldots\}. \tag{873}$$

2. **Orientation and polarization isotropy:**
$$\partial_{\hat{\mathbf{n}}} c_A^{(2\text{w})} = 0, \qquad \partial_{\text{pol}} c_A^{(2\text{w})} = 0 \tag{874}$$
in the infrared regime.

3. **Low-energy nondispersion:**
$$\omega_A^2(k) = c^2k^2\left[1 + O\Big((kr_c)^2\Big)\right], \qquad kr_c \ll 1. \tag{875}$$

4. **Relative-motion closure:** clock comparison, ruler comparison, and pulse propagation between Alice and Bob must be functions of u_{AB} only, within the stated approximation.

5. **Helmholtz-sector preservation:** ordinary R-phase propagation must satisfy $\Delta\Gamma_C = 0$ for every unaffected material loop and must not leave permanent vorticity in an initially irrotational material element.

R.53 Analogue-metric bridge

For a hyperbolic perturbation sector with common characteristic speed c, the linearized excitation equation can be written schematically as

$$\frac{1}{\sqrt{-g_{\text{eff}}}}\partial_\mu \left(\sqrt{-g_{\text{eff}}}\, g_{\text{eff}}^{\mu\nu}\partial_\nu\phi\right) = 0. \tag{876}$$

[ORTHODOX] Equations of the form (876) are standard in analogue-gravity systems [17, 47, 48].

[SST STATUS: CONDITIONAL BRIDGE] In SST this bridge applies only to the explicit hyperbolic excitation sector. It does not follow from incompressibility alone, because incompressible Euler pressure is a constraint field rather than an ordinary compressional sound mode. Exact incompressibility therefore removes the standard longitudinal acoustic branch from the SST foundation. The correct conditional statement is: if the independent torsion/transverse excitation sector has a hyperbolic principal operator with constitutive speed c_T, and if clocks and rulers couple universally to that sector, then an effective Lorentz cone follows. The operational canon identifies $c_T = c$; the microphysical derivation remains open.

[CRITICAL NOTE] Analogue gravity reproduces curved-spacetime kinematics for perturbations. It does not, by itself, derive the Einstein field equations.

R.54 Weak equivalence principle

There are two independent conditional routes to the weak equivalence principle.

Energy-source route. If both inertial mass and gravitational source strength are determined by the same localized energy functional,

$$m_i(T)c^2 = E_T, \qquad m_g(T)c^2 = E_T, \tag{877}$$

then

$$m_i(T) = m_g(T). \tag{878}$$

[DERIVED, conditional] This route is valid only if the gravitational clock/pressure field couples universally to the total localized SST energy functional.

Metric-coupling route. If matter is minimally coupled to a Lorentzian metric and $\nabla_\mu T^{\mu\nu} = 0$, then the point-particle limit follows geodesic motion,

$$u^\mu \nabla_\mu u^\nu = 0. \tag{879}$$

[ORTHODOX CONDITIONAL THEOREM] This is the standard GR result. In SST it is not yet a primitive-substrate derivation, because the Lorentzian metric and minimal coupling are assumed at the effective-field-theory level.

R.55 Newtonian limit

A Newtonian limit requires a scalar potential satisfying

$$\nabla^2\Phi = 4\pi G_{\text{eff}}\rho_m. \tag{880}$$

SST can motivate $\rho_m = \rho_E/c^2$ from the local energy accounting, but Eq. (880) remains a closure law unless derived from the coarse-grained vortex ensemble.

[CONDITIONAL] If Eq. (880) is admitted as the long-wavelength closure, the inverse-square force and Newtonian limit follow.

[OPEN CANON GAP] Derive Eq. (880) directly from the statistical mechanics or coarse-grained pressure response of many swirl strings.

R.56 Research Track: Compton–Horn Balance Angle

[RESEARCH-TRACK GEOMETRIC DIAGNOSTIC]. This subsection records a dimensionless visualization of the calibrated Compton–horn closure used in the core canon. It does not introduce a new gravity law and it must not be used as an independent derivation of Newton's constant.

Define ϕ_G by

$$G = \frac{v_{\circlearrowleft}^{*} c^3 t_p^2}{M_e r_c} \tan^2 \phi_G. \tag{881}$$

Equivalently,

$$\tan^2 \phi_G = \frac{G M_e r_c}{v_{\circlearrowleft}^{*} c^3 t_p^2}. \tag{882}$$

Using the v0.8.18 calibration chain

$$r_c = \frac{v_{\circlearrowleft}^{*}}{\omega_c}, \qquad \omega_c = \frac{M_e c^2}{\hbar}, \qquad t_p^2 = \frac{\hbar G}{c^5}, \tag{883}$$

one obtains

$$\tan^2 \phi_G = \frac{M_e c^2}{\hbar \omega_c} = 1, \qquad \phi_G = \frac{\pi}{4}. \tag{884}$$

Thus the 45° value means that the calibrated electron benchmark weights radial horn confinement and tangential swirl transport equally in this particular dimensionless projection. It does *not* mean that gravity acts at a spatial angle of 45°, and it does *not* add a new coupling constant.

[CRITICAL NOTE] Because $t_p^2 = \hbar G/c^5$ already contains G, Eq. (884) is a closure diagnostic, not a prediction of G from non-gravitational inputs. A future promotion would require deriving the effective long-range coupling or the Planck-time scale from SST medium dynamics without importing orthodox G.

R.56.1 Rejected microscopic Kelvin-angle ansatz

The dimensional ansatz

$$\tan \phi = \sqrt{\frac{I^{3/2}}{N \mu K^{1/2} \sqrt{\pi}}} \tag{885}$$

is not canon-ready. In SI units $[I] = \mathrm{kg\,m^2}$, while the present canon uses $[K] = \mathrm{kg\,m^{-3}\,s}$ for the density–rotation bridge. Unless μ and K are replaced by explicitly normalized quantities, Eq. (885) is not dimensionless, whereas $\tan\phi$ must be dimensionless.

A permissible research-track replacement must use only dimensionless knot functionals:

$$\tan^2 \phi_K = \frac{\mathcal{I}_K^{3/2}}{N_K \mathcal{M}_K \mathcal{K}_K^{1/2} \sqrt{\pi}}, \tag{886}$$

with, for example,

$$\mathcal{I}_K = \frac{I_K}{M_K {r_c}^2}, \qquad \mathcal{K}_K = \frac{K_K}{K_0}, \qquad \mathcal{M}_K = \frac{\mu_K}{\mu_0^*}. \tag{887}$$

For the electron/trefoil benchmark the admissible normalization condition is

$$\tan^2 \phi_{3_1} = 1, \tag{888}$$

but this is a benchmark closure condition, not yet a microscopic theorem.

R.57 Einstein dynamics: three candidate routes

Route I: thermodynamic gravity. Jacobson's route derives the Einstein equation as an equation of state from $\delta Q = T\,dS$ applied to local Rindler horizons, assuming an entropy-area law and the Unruh temperature [49]. In SST this requires:

$$S_{\rm defect} \propto A, \qquad T_{\rm SST} = \frac{\hbar a}{2\pi c k_B}. \tag{889}$$

[RESEARCH TRACK] This is the most promising route because it connects directly to horizon, boundary, and holographic SST sectors.

Source-paper integration for Route I. The thermodynamic-gravity route should be linked explicitly to three source papers. SST-63 supplies the constraint-holographic boundary reconstruction layer. SST-23 supplies the accelerated torsion / Unruh-response candidate. SST-56 supplies the superfluid topology diagnostics—Hall angle, writhe barrier, and lifetime hierarchy—needed to test whether a line-tangle or fabric sector has stable protected response. Thus the route is not a single paper claim but a three-asset programme:

$$\boxed{\text{SST-63} + \text{SST-23} + \text{SST-56} \;\longrightarrow\; (S = \eta A,\ T_{\rm SST},\ \delta Q = T\,dS)}. \tag{890}$$

The next open lemma is the Unruh-response derivation in the hyperbolic torsion sector. The area law may be attacked through a line-piercing entropy model, but its coefficient remains calibrated unless the vacuum line density is derived independently.

Route II: Sakharov induced gravity. With ultraviolet cutoff $\ell_{\rm UV} \sim r_c$, a schematic induced gravity estimate gives [50]

$$G_{\rm ind} \sim \frac{r_c^2 c^3}{\hbar N}. \tag{891}$$

Matching Newton's constant requires

$$N_{\rm req} \sim \frac{r_c^2 c^3}{\hbar G} = \left(\frac{r_c}{L_p}\right)^2 = 7.60 \times 10^{39}. \tag{892}$$

[CRITICAL NOTE] Unless $N_{\rm req}$ is independently derived from SST mode counting, this route only restates the hierarchy r_c/L_p.

Route III: infrared EFT universality. If coarse-grained SST organizes into a diffeomorphism-invariant metric EFT, then the leading two-derivative action has the form

$$S_{\rm IR} = \frac{c^3}{16\pi G_{\rm eff}} \int d^4x\sqrt{-g}\,(R - 2\Lambda) + S_{\rm matter} + S_{\rm clock}. \tag{893}$$

[RESEARCH TRACK] This route explains why Einstein–Hilbert appears as the leading IR term, but it does not explain why a preferred-foliation incompressible medium becomes diffeomorphism invariant in the infrared.

[CRITICAL NOTE] The cosmological-constant problem is severe [55]:

$$\frac{\rho_f}{\rho_\Lambda} \approx 1.2 \times 10^{20}, \qquad \frac{\rho_{\rm horn}^{\rm eff}}{\rho_\Lambda} \approx 6.6 \times 10^{44}. \tag{894}$$

The first ratio is not arbitrary: it is close to one power of the core–Planck impedance ratio r_c/ℓ_P. This observation motivates the research-track density route below, without changing the canonical status of ρ_f in the main text.

R.58 Cosmological Impedance Route for the Background Density

Status. **[RESEARCH TRACK / NOT CANON-DERIVED].** This subsection records a candidate route for reducing the calibrated background density ρ_f to a cosmological vacuum input plus a UV–IR impedance lift. It does not promote ρ_f out of the primitive calibration set in the main canon.

The orthodox vacuum mass-equivalent density associated with a cosmological constant is

$$\rho_\Lambda = \frac{\Lambda c^2}{8\pi G}. \tag{895}$$

For a spatially flat ΛCDM background,

$$\Lambda = \frac{3\Omega_\Lambda H_0^2}{c^2}, \qquad \rho_\Lambda = \Omega_\Lambda \frac{3H_0^2}{8\pi G}. \tag{896}$$

Using the Planck-2018 baseline values $H_0 \simeq 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_\Lambda \simeq 0.685$, this gives

$$\Lambda \simeq 1.09 \times 10^{-52}\,\mathrm{m^{-2}}, \tag{897}$$

$$\rho_\Lambda \simeq 5.85 \times 10^{-27}\,\mathrm{kg\,m^{-3}}. \tag{898}$$

The ratio required to obtain the canonical SST background density is

$$\frac{\rho_f}{\rho_\Lambda} \simeq 1.20 \times 10^{20}. \tag{899}$$

This is close to the one-power core–Planck ratio

$$\frac{r_c}{\ell_P} = 8.72 \times 10^{19}, \tag{900}$$

which suggests the line-impedance ansatz

$$\rho_f = \chi_\Lambda \left(\frac{r_c}{\ell_P}\right) \rho_\Lambda, \qquad \chi_\Lambda \simeq 1.37. \tag{901}$$

Two non-equivalent simple closures are numerically relevant:

$$\rho_f^{(c_s)} := \sqrt{2}\left(\frac{r_c}{\ell_P}\right) \rho_\Lambda \simeq 7.21 \times 10^{-7}\,\mathrm{kg\,m^{-3}}, \tag{902}$$

$$\rho_f^{(\Omega)} := 2\Omega_\Lambda \left(\frac{r_c}{\ell_P}\right) \rho_\Lambda \simeq 6.98 \times 10^{-7}\,\mathrm{kg\,m^{-3}}. \tag{903}$$

The first closure uses the internal core-acoustic factor already present in $c_s = \mathbf{v}_{\circlearrowleft}/\sqrt{2}$, but does not yet derive why the cosmological vacuum sector should couple through that impedance. The second closure is numerically sharper for Planck-2018 parameters, but it introduces an explicitly epoch-dependent projection Ω_Λ and is therefore not acceptable as a fundamental constant derivation unless SST supplies a time-independent interpretation of that factor.

Dimensional interpretation. The power of r_c/ℓ_P must be one, not two or three. A volume lift $(r_c/\ell_P)^3$ or area lift $(r_c/\ell_P)^2$ overshoots catastrophically. The only viable route is therefore a one-dimensional UV–IR mode-count or line-impedance theorem. In physical terms, the cosmological vacuum density would have to be lifted into a local inertial background density by counting Planck-normal modes along one core/horn length, not across an area or volume.

Canonical conclusion. Equation (901) is a promising falsifier route, but not a completed derivation. The main canon should continue to label ρ_f as a calibrated effective background density. Promotion to **[DERIVED]** requires an SST theorem that derives either $\chi_\Lambda = \sqrt{2}$, or an epoch-invariant replacement for $2\Omega_\Lambda$, from the coarse-graining map between the cosmological vacuum sector and the local incompressible swirl medium.

R.59 Tensor-speed naturalness

R.60 Separatrix Surface-Density Lift via the Spherical Monopole DtN Sector

Status. [RESEARCH TRACK / SPHERICAL MONOPOLE DtN NORMALIZATION PROVEN / SOURCE COUPLING OPEN]. This subsection records a candidate boundary-response route for the Separatrix Surface-Density Lift (SSDL). It proves only the geometric R_∂-normalization of the spherical exterior monopole Dirichlet-to-Neumann (DtN) sector. It does not prove the constitutive coupling of the cosmological vacuum density ρ_Λ to an SST separatrix source, and it does not replace the calibrated status of ρ_f.

Exterior boundary problem. Let $\partial\mathcal{B}$ be a spherical externally resolved separatrix of radius R_∂. In the exterior region $r \geq R_\partial$, let an isotropic response potential satisfy

$$\nabla^2\Phi = 0, \qquad \Phi(R_\partial) = \Phi_0, \qquad \Phi(\infty) = 0. \tag{904}$$

The unique spherical solution is

$$\Phi(r) = \Phi_0 \frac{R_\partial}{r}. \tag{905}$$

With the normal convention

$$\Lambda_\partial[\Phi_0] := -\partial_r\Phi\big|_{r=R_\partial}, \tag{906}$$

the spherical monopole DtN eigenvalue is

$$\Lambda_{\partial,0} = \frac{1}{R_\partial}, \qquad \Lambda_{\partial,0}^{-1} = R_\partial. \tag{907}$$

For a spherical exterior harmonic mode one has more generally

$$\Lambda_{\partial,\ell} = \frac{\ell+1}{R_\partial}, \qquad \ell = 0, 1, 2, \ldots. \tag{908}$$

Thus the R_∂ lift belongs specifically to the isotropic monopole sector, not to arbitrary tangential boundary data.

Monopole projection. Let Π_0 denote projection onto the spherical monopole sector,

$$\Pi_0 f = \frac{1}{4\pi}\int_{S^2} f(\theta,\varphi)\, d\Omega. \tag{909}$$

Then

$$\Pi_0\Lambda_\partial^{-1}\Pi_0[1] = R_\partial. \tag{910}$$

If the cosmological vacuum density enters the SST boundary-response problem as an isotropic normal source, the proposed surface-density projection is

$$\sigma_{\Lambda,\partial} := \Pi_0\Lambda_\partial^{-1}\Pi_0[\rho_\Lambda] = \rho_\Lambda R_\partial. \tag{911}$$

Resolving this surface density across a Planck-normal boundary thickness L_p gives

$$\rho_{\text{eff}} = \frac{\sigma_{\Lambda,\partial}}{L_p} = \left(\frac{R_\partial}{L_p}\right)\rho_\Lambda. \tag{912}$$

Including a present-epoch vacuum projection factor gives the compact SSDL candidate

$$\rho_f^{\text{SSDL}} = \frac{\Omega_{\Lambda,0}}{L_p}\Pi_0\Lambda_\partial^{-1}\Pi_0[\rho_\Lambda]. \tag{913}$$

Electron separatrix candidate. For the charged electron separatrix, take the external response radius to be the classical electron radius

$$R_\partial = R_e = 2r_c = \frac{e^2}{4\pi\varepsilon_0 m_e c^2}. \tag{914}$$

The spherical-monopole reduction of Eq. (913) then becomes

$$\rho_f^{\mathrm{SSDL}} = \Omega_{\Lambda,0}\left(\frac{R_e}{L_p}\right)\rho_\Lambda. \tag{915}$$

Using the audit constants

$$R_e = 2.8179403262\times10^{-15}\,\mathrm{m}, \qquad L_p = 1.6162550\times10^{-35}\,\mathrm{m}, \tag{916}$$

$$\rho_\Lambda = 5.8450\times10^{-27}\,\mathrm{kg\,m^{-3}}, \qquad \Omega_{\Lambda,0} = 0.685, \tag{917}$$

one obtains

$$\rho_f^{\mathrm{SSDL}} = 6.9806682\times10^{-7}\,\mathrm{kg\,m^{-3}}, \tag{918}$$

which differs from the calibrated canon value $\rho_f = 7.0\times10^{-7}\,\mathrm{kg\,m^{-3}}$ by -0.2762%. This numerical agreement is not a derivation of ρ_f; it is a falsifiable research-track target.

Planck-normal mode-count equivalent. The discrete normal-stack version of the same route is obtained by restricting the active response space to the isotropic sector,

$$\mathcal{H}_{\mathrm{active}} = \Pi_0\Big(L^2([0,R_\partial])\otimes L^2(S^2)\Big) = L^2([0,R_\partial]). \tag{919}$$

Tangential modes are not claimed to be absent; they are projected out of the isotropic density response by Π_0. If L_p is the minimal resolved normal thickness, the active normal cell count is

$$N_\perp^{\mathrm{cell}} = \frac{R_\partial}{L_p} + O(1). \tag{920}$$

Equivalently, a one-dimensional spectral convention with half-wavelength resolution uses

$$k_{\max} = \frac{\pi}{L_p}, \tag{921}$$

so that Weyl counting gives

$$N_\perp^{\mathrm{spec}} = \mathrm{Tr}_\perp\,\Theta\left[\left(\frac{\pi}{L_p}\right)^2 + \partial_n^2\right] = \frac{R_\partial}{L_p} + O(1). \tag{922}$$

Using $k_{\max} = L_p^{-1}$ instead would introduce an unwanted factor $1/\pi$; the cutoff convention is therefore part of the research-track claim.

Audit status and open lemmas. A v0.2 audit harness supports the operator and counting consistency of this route: the BEM cross-check recovers R_e through $\Pi_0\Lambda_\partial^{-1}\Pi_0$ with relative error 2.68×10^{-4}, and the analytic Planck-normal count gives $N_\perp = R_e/L_p = 1.743499835236\times10^{20}$ up to integer boundary convention. These checks are numerical and algebraic consistency tests, not constitutive proofs. Promotion to [**DERIVED**] requires:

1. proof that ρ_Λ couples to the SST separatrix as an isotropic normal source;
2. derivation of $\Omega_{\Lambda,0}$, or its replacement by an epoch-invariant SST projection functional;

3. derivation of L_p as the correct normal resolution thickness for the separatrix boundary layer.

In the Einstein–Æther comparison layer,

$$c_T^2 = \frac{c^2}{1 - c_{13}}, \qquad c_{13} = c_1 + c_3. \tag{923}$$

If SST tensor modes are identified with transverse substrate waves, and if the canonical transverse wave speed is c, then

$$c_T = c \quad \Rightarrow \quad c_{13} = 0. \tag{924}$$

[DERIVED, conditional] The luminal spin-2 speed is then not a tuning but an internal consistency condition of the mode identification [52, 53]. This mirrors the main-canon proposition of Section 11.12.

[LIMIT] This does not prove the Einstein field equations. It only fixes the spin-2 propagation speed inside the effective comparison theory.

R.61 Final status statement

The accurate SST statement is:

> SST conditionally reconstructs relativistic kinematics for its low-energy excitation sector, provided monometricity holds. It provides conditional routes to the weak equivalence principle and to the Newtonian limit. It naturally explains luminal tensor-mode speed if gravitational tensor modes are transverse substrate waves. The full Einstein field equations remain an open research target, with thermodynamic, induced-gravity, and infrared-EFT routes under active investigation.

R.62 Observational falsifier ladder

The relativity-emergence programme has three immediate observational gates:

1. **Tensor-speed gate.** Multimessenger propagation requires $c_T = c$, equivalently $c_{13} = 0$ in the Einstein–Æther comparison layer. Quantitatively, GW170817/GRB170817A gives

$$-3 \times 10^{-15} < \frac{c_T}{c} - 1 < 7 \times 10^{-16}, \qquad |c_{13}| \lesssim 10^{-15} \tag{925}$$

   in the Einstein–Æther comparison convention [66, 67]. This is the spin-2 version of the transverse-mode identification.

2. **Preferred-frame gate.** Any residual preferred-foliation coupling must satisfy the post-Newtonian preferred-frame constraints of the comparison theory. In practice this means that the effective α_1 and α_2 parameters cannot be freely chosen; they must be derived from the same substrate couplings that set the clock sector. Strong-field pulsar timing gives representative bounds

$$\hat{\alpha}_1 = (-0.4^{+3.7}_{-3.1}) \times 10^{-5} \quad (95\%\ \mathrm{CL};\ \mathrm{PSR}\ J1738 + 0333), \tag{926}$$

$$|\hat{\alpha}_2| < 1.6 \times 10^{-9} \quad (95\%\ \mathrm{CL};\ \mathrm{PSR}\ B1937 + 21, J1744 - 1134). \tag{927}$$

   Binary-pulsar damping and PSR J0337+1715 further reduce the post-GW170817 viable Einstein–Æther parameter region by about one order of magnitude [68, 69, 70].

3. **Species-cone gate.** Matter knots, gauge/torsion pulses, and clock excitations must share the same infrared characteristic cone. Failure of this gate predicts species-dependent Lorentz violation and directly falsifies exact monometricity.

[**CRITICAL NOTE**] In the orthodox Einstein–Æther comparison theory, matter is usually coupled only to the metric, not directly to the Æther field. SST cannot inherit this protection automatically, because matter is modeled as a substrate excitation. The species-cone gate is therefore stricter for SST than for a purely metric matter-coupled Einstein–Æther EFT.

[**STATUS**] This is not an additional derivation. It is the observational test hierarchy attached to the monometricity and tensor-speed theorem targets.

R.63 Research Track: Minimal Relational Link–Field Action

R.63.1 Status and objective

[RESEARCH-TRACK / CANON CANDIDATE]. The purpose of this section is to replace the ambiguous statement that a photon is "local rotation of the medium" by a sharper microscopic hypothesis: material rotation belongs to the site or vortex sector, whereas radiative torsion belongs to oriented connections between neighbouring Æther elements. The action below is the minimal local, parity-even, conservative, Abelian model that realizes this distinction. It is a model-selection hypothesis, not yet a derivation from a completed Æther microphysics.

R.63.2 Discrete relational variables

Let the microscopic adjacency structure be an oriented cellular graph

$$\mathcal{G} = (V, E, P), \tag{928}$$

with sites $i \in V$, oriented links $e = (i \to j) \in E$, elementary plaquettes $p \in P$, and characteristic link spacing a_ℓ. Assign to each oriented link a compact relative torsion phase

$$U_e = e^{i\vartheta_e} \in U(1), \qquad \vartheta_{\bar{e}} = -\vartheta_e \pmod{2\pi}. \tag{929}$$

The compact $U(1)$ choice is the minimal hypothesis when relative torsion is a periodic phase. A noncompact real link field is recovered by linearizing around one phase branch.

Introduce a site variable $\phi_i(T)$ only as a temporal connection or Lagrange multiplier. It is not the physical angular velocity of the Æther element and carries no independent site-rotation kinetic term. Under a local change of phase zero,

$$\vartheta_{ij} \mapsto \vartheta_{ij} + \chi_j - \chi_i, \tag{930}$$

$$\phi_i \mapsto \phi_i - \dot{\chi}_i, \tag{931}$$

the gauge-invariant link twist rate and plaquette holonomy are

$$\mathscr{E}_{ij} := -\dot{\vartheta}_{ij} - (\phi_j - \phi_i), \tag{932}$$

$$\mathscr{B}_p := \textstyle\sum_{e \in \partial p} s_{pe} \vartheta_e \pmod{2\pi}, \tag{933}$$

where $s_{pe} = \pm 1$ records relative orientation.

R.63.3 Minimal compact link action

The proposed source-free microscopic action is

$$S_{\text{link}}^{\text{lat}} = \int dT \left[\frac{I_\ell}{2} \sum_{e \in E} \mathscr{E}_e^2 - K_\ell \sum_{p \in P} (1 - \cos \mathscr{B}_p) \right], \tag{934}$$

with

$$I_\ell > 0, \qquad K_\ell > 0, \qquad [I_\ell] = \mathrm{kg\,m}^2, \qquad [K_\ell] = \mathrm{J}. \tag{935}$$

Here I_ℓ is a link inertia and K_ℓ is a plaquette torsional energy scale. No value of c, e, m_e, h, ε_0, or μ_0 occurs in Eq. (934).

Variation with respect to ϕ_i gives the source-free discrete Gauss constraint

$$\sum_{e \ni i} s_{ie}\, I_\ell \mathscr{E}_e = 0. \tag{936}$$

Variation with respect to an oriented link gives, up to the declared orientation convention,

$$I_\ell \dot{\mathscr{E}}_e = K_\ell \sum_{p \supset e} s_{pe} \sin \mathscr{B}_p. \tag{937}$$

Equations (936)– (937) evolve link torsion without introducing a net site-spin degree of freedom.

R.63.4 Linear spectrum and emergent causal speed

For $|\mathscr{B}_p| \ll 1$, the compact potential becomes quadratic. On a regular isotropic cubic complex, after gauge fixing and projection onto either transverse polarization, the linear branch satisfies

$$\omega^2(\mathbf{k}) = \frac{4K_\ell}{I_\ell} \sum_{a=1}^{3} \sin^2\left(\frac{k_a a_\ell}{2}\right). \tag{938}$$

Therefore, for $|\mathbf{k}| a_\ell \ll 1$,

$$\omega^2 = c_T^2 |\mathbf{k}|^2 + O\left(a_\ell^4 |\mathbf{k}|^4\right), \qquad \boxed{c_T = a_\ell \sqrt{\frac{K_\ell}{I_\ell}}}. \tag{939}$$

Equation (939) is the first zero-legacy speed target: c_T is generated by a relation stiffness divided by a relation inertia. The operational identification $c_T = c$ is tested only after a_ℓ, I_ℓ, and K_ℓ have been determined independently.

R.63.5 Continuum limit and Maxwell–Faraday structure

Write, on a regular branch,

$$\vartheta_{i,a} = a_\ell \mathcal{A}_{\ell,a}(\mathbf{x}_i, T) + O(a_\ell^2), \qquad \phi_i = \Phi_\ell(\mathbf{x}_i, T). \tag{940}$$

Define

$$\boldsymbol{\mathcal{E}}_\ell := -\partial_T \boldsymbol{\mathcal{A}}_\ell - \nabla \Phi_\ell, \tag{941}$$

$$\boldsymbol{\mathcal{B}}_\ell := \nabla \times \boldsymbol{\mathcal{A}}_\ell. \tag{942}$$

Up to cell-shape and coordination factors, the infrared action is

$$S^{\mathrm{IR}}_{\mathrm{link}} = \int dT\, d^3x \left[\frac{\epsilon_\ell}{2} |\boldsymbol{\mathcal{E}}_\ell|^2 - \frac{1}{2\mu_\ell} |\boldsymbol{\mathcal{B}}_\ell|^2\right], \tag{943}$$

where

$$\epsilon_\ell \simeq \frac{I_\ell}{a_\ell}, \qquad \mu_\ell^{-1} \simeq K_\ell a_\ell, \qquad \frac{1}{\epsilon_\ell \mu_\ell} = c_T^2. \tag{944}$$

The symbols ϵ_ℓ and μ_ℓ are SST link-response coefficients; they are not inserted as the measured electromagnetic constants ε_0 and μ_0.

With

$$\boldsymbol{\mathcal{D}}_\ell := \epsilon_\ell \boldsymbol{\mathcal{E}}_\ell, \qquad \boldsymbol{\mathcal{H}}_\ell := \mu_\ell^{-1} \boldsymbol{\mathcal{B}}_\ell, \tag{945}$$

variation and the definitions in Eq. (942) give

$$\nabla \cdot \boldsymbol{\mathcal{D}}_\ell = 0, \qquad \nabla \times \boldsymbol{\mathcal{H}}_\ell - \partial_T \boldsymbol{\mathcal{D}}_\ell = 0, \tag{946}$$

$$\nabla \cdot \boldsymbol{\mathcal{B}}_\ell = 0, \qquad \nabla \times \boldsymbol{\mathcal{E}}_\ell + \partial_T \boldsymbol{\mathcal{B}}_\ell = 0. \tag{947}$$

Thus the source-free Maxwell–Faraday form is an infrared consequence of the minimal link action. This is a structural isomorphism, not yet a proof that the SST fields equal the experimentally normalized electromagnetic fields.

Gauge invariance removes a physical longitudinal connection mode and forbids a bare mass term $m_A^2 |\boldsymbol{\mathcal{A}}_\ell|^2$. The remaining linear vacuum sector has two transverse polarizations. A parity-odd term proportional to $\boldsymbol{\mathcal{E}}_\ell \cdot \boldsymbol{\mathcal{B}}_\ell$, nonlinear terms, higher derivatives, and source couplings are deliberately excluded from the minimal parity-even action and belong to later research stages.

R.63.6 Relation to the existing continuum radiation bridge

The existing field $\mathbf{A}_{\text{eff}}$ may be treated as a field-rescaled representative of the continuum link connection,

$$\mathbf{A}_{\text{eff}} = \Lambda_\ell \boldsymbol{\mathcal{A}}_\ell. \tag{948}$$

In radiation gauge this gives the identifications

$$\rho_f = \frac{\epsilon_\ell}{\Lambda_\ell^2}, \qquad K_\gamma = \frac{\mu_\ell^{-1}}{\Lambda_\ell^2}, \qquad \frac{K_\gamma}{\rho_f} = \frac{1}{\epsilon_\ell \mu_\ell} = c_T^2. \tag{949}$$

The speed is invariant under the field normalization Λ_ℓ. Therefore the current quadratic continuum bridge need not be discarded; it is supplied with a microscopic relational interpretation.

R.63.7 Energy, flux, and link momentum

The positive quadratic energy density and energy flux are

$$u_\ell = \frac{1}{2}\left(\epsilon_\ell |\boldsymbol{\mathcal{E}}_\ell|^2 + \mu_\ell^{-1} |\boldsymbol{\mathcal{B}}_\ell|^2\right), \tag{950}$$

$$\mathbf{S}_\ell = \boldsymbol{\mathcal{E}}_\ell \times \boldsymbol{\mathcal{H}}_\ell, \qquad \partial_T u_\ell + \nabla \cdot \mathbf{S}_\ell = 0. \tag{951}$$

The associated momentum density in the isotropic linear branch is

$$\mathbf{g}_\ell = \frac{\mathbf{S}_\ell}{c_T^2} = \boldsymbol{\mathcal{D}}_\ell \times \boldsymbol{\mathcal{B}}_\ell. \tag{952}$$

These quantities reside in the link field. They do not require nonzero net material angular momentum of the endpoint sites.

R.63.8 Helmholtz-sector separation

At leading vacuum order the total action is required to separate as

$$S^{(0)} = S_{\text{Euler}}[\mathbf{v}, p] + S_{\text{link}}[\vartheta, \phi]. \tag{953}$$

The material vorticity

$$\boldsymbol{\zeta} = \nabla \times \mathbf{v} \tag{954}$$

is not the link curvature $\boldsymbol{\mathcal{B}}_\ell$. Ordinary source-free link propagation must therefore satisfy

$$\Delta\Gamma_C = 0 \tag{955}$$

for every unaffected material loop $C(T)$ and must leave the site sector label “rotating” versus “nonrotating” unchanged. Any later matter–link coupling must preserve this gate except at an explicitly declared source, boundary, reconnection, singularity, or Kairos event.

A gauge-invariant source extension may later be written schematically as

$$\mathcal{L}_{\mathrm{src}} = \mathbf{J}_\ell \cdot \boldsymbol{\mathcal{A}}_\ell - q_\ell \Phi_\ell, \qquad \partial_T q_\ell + \nabla \cdot \mathbf{J}_\ell = 0, \tag{956}$$

but neither q_ℓ nor its normalization is identified with the elementary charge at this stage.

R.63.9 Zero-legacy dimensional closure

If the only dimensional microscopic inputs allowed for the link layer are a mass density ρ_f, a circulation scale Γ_0, and the link spacing a_ℓ, dimensional analysis permits

$$I_\ell = \iota_\ell \rho_f a_\ell^5, \tag{957}$$
$$K_\ell = \kappa_\ell \rho_f \Gamma_0^2 a_\ell, \tag{958}$$

where ι_ℓ and κ_ℓ are dimensionless microstructural response coefficients. Equation (939) then becomes

$$\boxed{c_T = \frac{\Gamma_0}{a_\ell}\sqrt{\frac{\kappa_\ell}{\iota_\ell}}}. \tag{959}$$

Using $\Gamma_0 = 2\pi r_c v_{\circlearrowleft}^{*}$ only after the circulation convention has been fixed gives the dimensionless relation

$$2\frac{v_{\circlearrowleft}^{*}}{c_T} = \frac{a_\ell}{\pi r_c}\sqrt{\frac{\iota_\ell}{\kappa_\ell}}. \tag{960}$$

For the special identification $a_\ell = r_c$, the original SST kinematic target $\alpha = 2v_{\circlearrowleft}^{*}/c_T$ reduces to

$$\alpha_{\mathrm{SST}} = \frac{1}{\pi}\sqrt{\frac{\iota_\ell}{\kappa_\ell}}. \tag{961}$$

Equations (959)– (961) use no electromagnetic or quantum constant as an algebraic input. They become predictions only if a_ℓ, ι_ℓ, and κ_ℓ are obtained from an independently specified microstructure rather than fitted to c or α. Agreement with the historical capacitance/impedance identity would then be a closure test; the capacitance formula and its impedance rewriting are not two independent routes.

R.63.10 Required certification programme

The link-field hypothesis is not promoted beyond research-track status until all of the following are completed:

1. specify the microscopic graph or continuum adjacency law without choosing a_ℓ from a target value of c;

2. calculate I_ℓ from the quadratic energy cost of link twist rate and K_ℓ from the quadratic energy cost of plaquette holonomy;
3. recover two transverse modes and no physical longitudinal radiative mode after constraint reduction;
4. verify Eq. (938) and the isotropic infrared limit under lattice rotation and refinement;
5. demonstrate positive energy, local energy flux, and stable evolution;
6. verify Helmholtz-sector preservation and zero residual material vorticity after a source-free pulse passes;
7. determine whether the microscopic dynamics has an exact front-speed cone or only an infrared group-velocity cone;
8. add conserved knot sources and derive the static force law without inserting e, ε_0, or μ_0;
9. compare the independently predicted c_T and dimensionless coupling ratios with experiment only after the above steps are frozen.

Falsifiers. The minimal hypothesis fails if the link action produces a ghost or negative energy branch, a species- or direction-dependent infrared speed without a controlled symmetry-breaking mechanism, an unavoidable longitudinal photon mode, permanent generation of material vorticity by a source-free pulse, or a value of c_T that can be matched only by reusing the observed value of c in I_ℓ, K_ℓ, or a_ℓ.

R.64 Research Track: Core–Torsion Impedance Matching for Inertia Closure

R.65 Purpose and status

[RESEARCH-TRACK]. This section records the canon-safe form of the inertia-closure programme. It separates the internal NLSE/Gross–Pitaevskii core layer from the transverse torsion/shear radiation layer. The separation is mandatory because the canonical swirl speed $v_{\circlearrowleft}^*$ is much smaller than the observed causal speed c:

$$\frac{v_{\circlearrowleft}^*}{c} = 3.64867628 \times 10^{-3}. \tag{962}$$

A derivation that produces $u^2/v_{\circlearrowleft}^{*2}$ or u^2/c_s^2 therefore does not yet derive the Lorentz-compatible factor u^2/c^2.

R.66 NLSE/Gross–Pitaevskii core diagnostic

For a defocusing NLSE/Gross–Pitaevskii core with Madelung phase velocity,

$$\mathbf{u}_{\rm NLS} = \frac{\hbar_*}{m_*}\nabla S, \tag{963}$$

the circulation quantum and healing length may be written

$$\Gamma_q = \frac{h_*}{m_*}N_\Gamma, \qquad \xi = \frac{\hbar_*}{\sqrt{2}\, m_* c_s}. \tag{964}$$

For the single-winding convention $N_\Gamma = 1$, imposing

$$\Gamma_q = \Gamma_0 = 2\pi r_c v_{\circlearrowleft}^*, \qquad \xi = r_c \tag{965}$$

gives

$$c_s = \frac{\Gamma_0}{2\sqrt{2}\pi r_c} = \frac{v_{\circlearrowleft}^*}{\sqrt{2}} = 7.73465663 \times 10^5 \ \mathrm{m\,s^{-1}}. \tag{966}$$

[DERIVED-CONDITIONAL]. Equation (966) is a useful internal diagnostic of the core layer. It is not a derivation of the Lorentz signal speed.

R.67 Transverse torsion/shear causal layer

A minimal transverse torsion displacement field $\mathbf{A}$, with $\nabla \cdot \mathbf{A} = 0$, may be assigned the quadratic bridge Lagrangian

$$\mathcal{L}_{\text{torsion}} = \frac{1}{2}\rho_f \|\partial_t \mathbf{A}\|^2 - \frac{1}{2}K_T \|\nabla \times \mathbf{A}\|^2. \tag{967}$$

The transverse propagation speed is

$$c_T^2 = \frac{K_T}{\rho_f}. \tag{968}$$

The canonical operational light-speed identification belongs only to this transverse layer:

$$c_T = c. \tag{969}$$

With the canonical fluid density this corresponds to the calibrated stiffness target

$$K_T^{\text{target}} = \rho_f c^2 = 6.29128625115772348 \times 10^{10} \ \text{Pa}. \tag{970}$$

Equation (970) fixes the value required by the operational postulate. A first-principles closure must obtain this value from the substrate microphysics without using c as the same-step input. Its intensive impedance is

$$Z_T = \rho_f c_T = \sqrt{\rho_f K_T}, \tag{971}$$

which has the units of acoustic impedance, $\mathrm{kg\,m^{-2}\,s^{-1}}$.

R.68 Core–Torsion Impedance Matching Lemma

Let K be a localized closed swirl-string configuration with rest-swirl energy $E_0[K]$. Define the torsion-dressed inertial tensor by the quadratic stored-energy response

$$\Delta E_{\text{torsion}}[K;\mathbf{u}] = \frac{1}{2}\mathbf{u}^{\mathsf{T}} \mathsf{M}_{\text{torsion}}[K]\mathbf{u} + O(\|\mathbf{u}\|^4). \tag{972}$$

The required Lorentz-compatible closure is

$$\boxed{\mathsf{M}_{\text{torsion}}[K] \stackrel{?}{=} \frac{2E_0[K]}{c_T^2}\,\mathsf{I}} \tag{973}$$

or equivalently

$$\Delta E_{\text{torsion}}[K;u] \stackrel{?}{=} E_0[K]\frac{u^2}{c_T^2} + O(u^4). \tag{974}$$

If $c_T = c$, Eq. (973) supplies the missing bridge from core inertia to the Lorentz-compatible clock factor.

Define the diagnostic residual

$$\chi_K^{(T)} = \frac{c_T^2}{2E_0[K]} \lambda_{\mathrm{iso}}(\mathsf{M}_{\mathrm{torsion}}[K]) \,. \tag{975}$$

The bridge closes only if

$$\chi_K^{(T)} = 1 \tag{976}$$

within numerical and topological tolerance.

[OPEN LEMMA]. The equality $\chi_K^{(T)} = 1$ is the falsifiable theorem target. If numerical torsion-dressing returns a stable non-unity value, the Lorentz clock factor remains a constitutive bridge rather than a derived theorem from the core model.

R.69 Required numerical tests

A canon-safe numerical programme must report:

1. the NLSE boost coefficient $M_{\mathrm{core}}[K]$ and its comparison to $2E_0[K]/c_s^2$, not to $2E_0[K]/c^2$;
2. the torsion-dressed tensor $\mathsf{M}_{\mathrm{torsion}}[K]$ and the residual $\chi_K^{(T)}$;
3. a stiffness sweep K_T verifying whether $\chi_K^{(T)} = 1$ occurs naturally at $c_T = c$ or only after retuning;
4. the circulation-normalization ratio $\mathcal{R}_\Gamma = (h_*/m_*)/\Gamma_0$, which must equal unity under the canonical single-winding convention;
5. the trefoil-locked source test comparing $T_{2,3}$ against generic helical sources.

[CANON INTERPRETATION]. The operational statement that stable vortex matter measures R-phase torsion pulses locally at c, together with the Lorentz-form Swirl-Clock, is a founding canon postulate. The open problem is not the intended meaning of c, but the non-circular derivation of K_T, universal clock-and-ruler dressing, and the impedance identity from the substrate. The NLSE core alone must not be presented as supplying that derivation.

R.69.1 Research Track: EM-to-Swirl Force-Density Correspondence

Status labels. The correspondence proposed in this section is [**SPECULATIVE**] and [**PENDING DERIVATION**]. It is not a canonical replacement of Maxwell–Lorentz electrodynamics. It is a Rosetta-level research bridge between electromagnetic force density and SST substrate stress.

Structural correspondence. The electromagnetic Lorentz force density is

$$\mathbf{f}_{\mathrm{EM}} = \rho_e \mathbf{E} + \mathbf{J} \times \mathbf{B}.$$

The magnetic part has the same force-density dimension as the SST swirl-force density,

$$[\mathbf{J} \times \mathbf{B}] = \mathrm{N\,m^{-3}}, \qquad [\rho_f \mathbf{v} \times \boldsymbol{\omega}] = \mathrm{N\,m^{-3}}.$$

Hence the most conservative Rosetta correspondence is

$$\boxed{\mathbf{J} \times \mathbf{B} \;\longleftrightarrow\; \lambda_{\mathrm{EM}\to\circlearrowleft}\, \rho_f \mathbf{v} \times \boldsymbol{\omega}.}$$

Since both sides already carry units of force density,

$$\boxed{[\lambda_{\mathrm{EM}\to\circlearrowleft}] = 1.}$$

The coefficient $\lambda_{\mathrm{EM}\to\circlearrowleft}$ is therefore dimensionless. It is not fixed by dimensional analysis and remains [**SPECULATIVE / PENDING DERIVATION**] until a field-level dictionary or normalization condition is specified.

Relation to the canonical flux-impulse channel. The canonical SST electromagnetic bridge proceeds through phase, flux, and flux-piercing variables, including the effective Faraday/flux-impulse channel and the flux quantum

$$\Phi_0 = \frac{h}{2e}.$$

The bulk correspondence

$$\mathbf{J} \times \mathbf{B} \;\longleftrightarrow\; \lambda_{\mathrm{EM}\to\circlearrowleft}\rho_f \mathbf{v} \times \boldsymbol{\omega}$$

is therefore not an independent canonical EM bridge. It is a candidate continuum-limit or bulk-stress projection of the already canonical phase/flux channel. A canonical upgrade requires showing how the flux observable, phase winding, and current density reduce to a definite $\lambda_{\mathrm{EM}\to\circlearrowleft}$ in the continuum limit.

Possible field-level dictionary. A possible but non-canonical dictionary is the classical hydrodynamic analogy

$$\mathbf{A} \;\longleftrightarrow\; \alpha_A \mathbf{v}, \qquad \mathbf{B} = \nabla \times \mathbf{A} \;\longleftrightarrow\; \alpha_A \boldsymbol{\omega},$$

and

$$\mu_0 \mathbf{J} = \nabla \times \mathbf{B} \;\longleftrightarrow\; \alpha_A \nabla \times \boldsymbol{\omega}.$$

The constants and normalization implied by α_A are not fixed here. This dictionary is retained only as a research-track route toward deriving $\lambda_{\mathrm{EM}\to\circlearrowleft}$.

Stress closure and SST-44 tensor. The canonical swirl-stress candidate is the symmetric Cauchy-type stress used in the SST-44 appendix,

$$\sigma_{ij}^{(44)} = \rho_f v_i v_j + \rho_f {r_c}^2 \left(\omega_i \omega_j - \frac{1}{2}\delta_{ij}|\boldsymbol{\omega}|^2\right) - \delta_{ij}\frac{1}{2}\rho_f |\mathbf{v}|^2.$$

Because this tensor already contains the isotropic dynamic-pressure term

$$-\delta_{ij}\frac{1}{2}\rho_f |\mathbf{v}|^2,$$

one must not also add an independent $-\nabla p_\circlearrowleft$ term with

$$p_\circlearrowleft = \frac{1}{2}\rho_f |\mathbf{v}|^2.$$

Otherwise the Bernoulli contribution is counted twice.

Two admissible decompositions. The full-stress convention is

$$\boxed{\mathbf{f}_{\mathrm{SST}} = \nabla \cdot \boldsymbol{\sigma}^{(44)} + \mathbf{f}_{\mathrm{link}}.}$$

The split-stress convention is

$$\boxed{\mathbf{f}_{\mathrm{SST}} = -\nabla p_\circlearrowleft + \nabla \cdot \boldsymbol{\sigma}_\circlearrowleft^{\mathrm{dev}} + \mathbf{f}_{\mathrm{link}},}$$

where

$$\boldsymbol{\sigma}_\circlearrowleft^{\mathrm{dev}} \equiv \rho_f \mathbf{v}\mathbf{v} + \rho_f {r_c}^2 \left(\boldsymbol{\omega}\boldsymbol{\omega} - \frac{1}{2}\mathbf{I}|\boldsymbol{\omega}|^2\right)$$

contains the advective and torsional anisotropic parts but excludes the isotropic dynamic-pressure term. These two conventions must not be mixed.

Topological link force. The link-force term is not yet canonical. The intended structure is a topological potential functional

$$U_{\text{link}} = U_{\text{link}}[Lk(C_a, C_b), \Gamma_a, \Gamma_b, H]\,,$$

with force density obtained schematically by variation,

$$\boxed{\mathbf{f}_{\text{link}} = -\frac{\delta U_{\text{link}}}{\delta \mathbf{x}}}$$

or, for a collective coordinate q,

$$\boxed{F_{\text{link},q} = -\frac{\partial U_{\text{link}}}{\partial q}.}$$

The linking number is

$$Lk(C_a, C_b) = \frac{1}{4\pi}\oint_{C_a}\oint_{C_b}\frac{(d\mathbf{x}_a \times d\mathbf{x}_b)\cdot(\mathbf{x}_a - \mathbf{x}_b)}{|\mathbf{x}_a - \mathbf{x}_b|^3}.$$

Until U_{link} is specified and normalized, $\mathbf{f}_{\text{link}}$ remains **[PENDING DERIVATION]**.

Research-track conclusion. The candidate bulk EM-to-swirl mapping is therefore

$$\boxed{\mathbf{f}_{\text{EM}} = \rho_e \mathbf{E} + \mathbf{J}\times\mathbf{B} \quad \rightsquigarrow \quad \mathbf{f}_{\text{SST}} = \nabla\cdot\boldsymbol{\sigma}^{(44)} + \mathbf{f}_{\text{link}}}$$

under the full-stress convention, or equivalently

$$\boxed{\mathbf{f}_{\text{SST}} = -\nabla p_{\circlearrowleft} + \nabla\cdot\boldsymbol{\sigma}_{\circlearrowleft}^{\text{dev}} + \mathbf{f}_{\text{link}}}$$

under the split-stress convention. The correspondence becomes canonizable only after $\lambda_{\text{EM}\to\circlearrowleft}$, the field-level dictionary, and U_{link} are derived or experimentally fixed.

R.70 Research Track: Taylor-Column Analogues for Finite-Thickness Swirl Strings

Status. This subsection is **Research Track**. The Taylor–Proudman, Hill-vortex, relative-helicity, inertial-wave, and Feynman–Onsager statements are orthodox rotating-fluid or superfluid mechanics [8, 7, 6, 10, 9, 11, 12]. Their SST use is restricted to analogue benchmarking, scale-separation diagnostics, and falsification tests. No macroscopic rotating-tank parameter is identified with the SST canonical core scale.

R.70.1 Background kinematics and resolved scale separation

Consider a rotating cylindrical background of height $H = 1\,\text{m}$, radius $R = 0.25\,\text{m}$, and constant angular velocity

$$\boldsymbol{\Omega} = \Omega\,\hat{\mathbf{z}}, \qquad \Omega = 1\,\text{s}^{-1}.$$

The unperturbed velocity and vorticity fields are

$$\mathbf{v}_b = \Omega r\,\hat{\boldsymbol{\theta}}, \qquad \boldsymbol{\omega}_b = \nabla\times\mathbf{v}_b = 2\Omega\,\hat{\mathbf{z}}.$$

The background has zero local kinetic helicity density:

$$\mathbf{v}_b\cdot\boldsymbol{\omega}_b = (\Omega r\,\hat{\boldsymbol{\theta}})\cdot(2\Omega\,\hat{\mathbf{z}}) = 0.$$

Let a denote the resolved separatrix radius of a macroscopic vortex structure, such as a vortex ring, Hill-type vortex, or trefoil vortex tube. Let r_c denote the SST canonical core radius. The resolved-scale guardrail is

$$\boxed{a \neq r_c.}$$

Using

$$r_c = 1.40897017 \times 10^{-15}\,\mathrm{m}, \qquad \mathbf{v}_{\circlearrowleft} = 1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}},$$

the canonical core-scale circulation is

$$\Gamma_0 = 2\pi r_c \mathbf{v}_{\circlearrowleft} = 9.68361920 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}.$$

For a macroscopic separatrix radius

$$a = \frac{R}{4} = 0.0625\,\mathrm{m},$$

the background circulation around the same radius is

$$\Gamma_b(a) = \oint_{r=a} \mathbf{v}_b \cdot d\boldsymbol{\ell} = 2\pi\Omega a^2.$$

This defines the dimensionless synchronization parameter

$$\boxed{\chi_\Omega = \frac{\Gamma_K}{2\pi\Omega a^2}.}$$

For $\Gamma_K = \Gamma_0$ and $\Omega = 1\,\mathrm{s^{-1}}$,

$$2\pi\Omega a^2 = 2.45436926 \times 10^{-2}\,\mathrm{m^2\,s^{-1}},$$

so that

$$\boxed{\chi_\Omega = 3.94546141 \times 10^{-7}.}$$

Thus a macroscopic separatrix equipped only with the canonical core circulation is kinematically far below synchronization with a $1\,\mathrm{s^{-1}}$ tank-scale rotation.

Dimensional check.

$$[\Gamma_K] = \mathrm{m^2\,s^{-1}}, \qquad [\Omega a^2] = \mathrm{s^{-1}m^2} = \mathrm{m^2\,s^{-1}},$$

so χ_Ω is dimensionless.

R.70.2 Separatrix response and relative helicity

For a slowly translating finite-thickness structure with axial speed U, the Rossby number is

$$Ro = \frac{U}{2\Omega a}.$$

In the low-Rossby regime $Ro \ll 1$, the leading-order Taylor–Proudman constraint gives

$$\frac{\partial \mathbf{u}}{\partial z} \approx 0.$$

Hence, to leading order, a no-through-flow separatrix behaves externally as a columnar obstruction. This does not imply that the internal topology is sphere-like; it only states that the exterior flow sees a columnar separatrix constraint.

For an incompressible closed cross-section, the axial flux balance is

$$\boxed{\int_0^R 2\pi r\, u_z(r)\, dr = 0.}$$

A simple two-zone model is

$$u_z(r) = \begin{cases} U, & 0 \le r < a, \\ -U\dfrac{a^2}{R^2 - a^2}, & a < r \le R. \end{cases}$$

Indeed,

$$\int_0^a 2\pi r U\, dr + \int_a^R 2\pi r \left(-U\frac{a^2}{R^2 - a^2}\right) dr = \pi a^2 U - \pi a^2 U = 0.$$

For a vortex ring with circulation Γ_K and ring surface S_K, a useful diagnostic is the relative or mutual helicity with the rotating background [36]:

$$\boxed{\mathcal{H}_\times = 2\Gamma_K \int_{S_K} \boldsymbol{\omega}_b \cdot d\mathbf{S}.}$$

Because

$$\boldsymbol{\omega}_b = 2\Omega\hat{\mathbf{z}}, \qquad \int_{S_K} \hat{\mathbf{z}} \cdot d\mathbf{S} = \pi a^2,$$

one obtains

$$\boxed{\mathcal{H}_\times = 4\pi\Omega a^2 \Gamma_K.}$$

For $\Gamma_K = \Gamma_0$,

$$\boxed{\mathcal{H}_{\times,0} = 4.75343546 \times 10^{-10}\,\mathrm{m}^4\,\mathrm{s}^{-2}.}$$

Dimensional check.

$$[\mathcal{H}_\times] = [\Omega a^2 \Gamma_K] = \mathrm{s}^{-1}\mathrm{m}^2\,\mathrm{m}^2\mathrm{s}^{-1} = \mathrm{m}^4\,\mathrm{s}^{-2} = [\Gamma_K]^2.$$

R.70.3 Hill-vortex sphere analogue

A vortex-ring dipole with a closed separatrix may be approximated, in the continuous Euler limit, by Hill's spherical vortex [9]. In the frame where the undisturbed fluid at infinity is at rest, the kinetic energy splits into an external irrotational contribution and an internal rotational contribution:

$$E_{\mathrm{out}} = \frac{1}{3}\pi\rho_f a^3 U^2, \qquad E_{\mathrm{in}} = \frac{23}{21}\pi\rho_f a^3 U^2.$$

Therefore

$$\boxed{E_{\mathrm{tot}} = E_{\mathrm{out}} + E_{\mathrm{in}} = \frac{10}{7}\pi\rho_f a^3 U^2.}$$

This result is used only as a continuum benchmark for finite-thickness separatrix energetics. It does not identify a trefoil vortex with a featureless sphere; the equivalence is external and leading-order only.

Numerical reference for $U = 1\,\mathrm{m\,s}^{-1}$. With

$$\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m}^{-3}, \qquad a = 0.0625\,\mathrm{m},$$

one obtains

$$E_{\mathrm{out}} = 1.78964425 \times 10^{-10}\,\mathrm{J},$$
$$E_{\mathrm{in}} = 5.88025969 \times 10^{-10}\,\mathrm{J},$$

and

$$E_{\mathrm{tot}} = 7.66990394 \times 10^{-10}\,\mathrm{J}.$$

R.70.4 Axial force density from vorticity-pressure gradients

If the local background rotation varies along the axis, $\Omega = \Omega(z)$, ordinary Euler/Bernoulli balance gives

$$\frac{1}{\rho_f}\frac{\partial p}{\partial r} = \Omega(z)^2 r.$$

Integrating from the axis to the wall yields

$$p(0,z) = p(R,z) - \frac{1}{2}\rho_f \Omega(z)^2 R^2.$$

If $p(R,z)$ is fixed or externally controlled, the axial force density on the central region is

$$\boxed{f_z = -\frac{\partial p(0,z)}{\partial z} = \frac{1}{2}\rho_f R^2 \frac{\partial}{\partial z}\left[\Omega(z)^2\right].}$$

This force is part of the ordinary vorticity-pressure impulse budget:

$$\boxed{\mathbf{F}_z \subset \mathbf{F}_\omega.}$$

It must be analytically subtracted before any residual SST-specific force claim is considered.

Dimensional check.

$$[\rho_f R^2 \partial_z \Omega^2] = \mathrm{kg\,m^{-3}\,m^2\,m^{-1}s^{-2}} = \mathrm{kg\,m^{-2}s^{-2}} = \mathrm{N\,m^{-3}}.$$

Thus f_z has the unit of force density.

R.70.5 Trefoil dynamics and He-II falsification

A macroscopic trefoil vortex is not a sphere internally. Its internal geometry carries writhe, twist, finite-thickness contact structure, and self-induced Biot–Savart dynamics. In water-like classical fluids, hydrofoil-generated trefoil vortices are expected to deform strongly, redistribute helicity, and undergo reconnection [13, 33].

In orthodox rotating superfluid helium, the background rotation is carried by a Feynman–Onsager lattice of quantized vortex lines [11, 12]. For ^{4}He,

$$\kappa_4 = \frac{h}{m_4} = 9.9693 \times 10^{-8}\,\mathrm{m^2\,s^{-1}}.$$

The vortex-line areal density is

$$n_v = \frac{2\Omega}{\kappa_4}.$$

For $\Omega = 1\,\mathrm{s^{-1}}$,

$$\boxed{n_v = 2.0062 \times 10^7\,\mathrm{m^{-2}}.}$$

The number of quantized vortex lines crossing the resolved separatrix area πa^2 is

$$N_v = n_v \pi a^2.$$

For $a = 0.0625\,\mathrm{m}$,

$$\boxed{N_v = 2.4619 \times 10^5.}$$

The orthodox He-II expectation is therefore a high reconnection rate, Kelvin-wave cascade, partial helicity redistribution, and likely unknotting or decay of the trefoil. By contrast, the ideal SST continuum limit assumes inviscid, incompressible flow with no reconnecting cuts, so topology is preserved unless an explicit reconnection or defect-bridge sector is added. The disagreement defines a falsification domain:

$$\boxed{\text{He-II quantized vortex lattice: reconnection-enabled decay,}}$$

$$\boxed{\text{ideal SST continuum: topology-preserving transport.}}$$

R.71 Research Track: Rotating-Fluid Delay and Swirl-Clock Parity

Status. This subsection is **Research Track**. It uses orthodox inertial-wave and rotating-vorticity relations as analogue benchmarks and then records the SST-compatible parity split between odd tracer observables and even clock-rate observables. No dimensionally invalid fine-structure normalization is used.

R.71.1 Arrival delay of the surface signal: inertial-wave delay

For a small perturbation in a uniformly rotating inviscid incompressible fluid, the inertial-wave dispersion relation is

$$\omega = 2\Omega\frac{k_z}{k}, \qquad k^2 = k_r^2 + k_z^2.$$

The vertical group velocity is

$$c_{g,z} = \frac{\partial\omega}{\partial k_z} = 2\Omega\frac{k_r^2}{k^3}.$$

Therefore the arrival time of a disturbance over height H is

$$\boxed{t_{\rm arr} = \frac{H}{c_{g,z}} = \frac{Hk^3}{2\Omega k_r^2}.}$$

Thus

$$\boxed{t_{\rm arr} \propto \Omega^{-1}.}$$

This is a propagation delay of the rotating-fluid disturbance. It is not a swirl-clock time-dilation effect.

Dimensional check.

$$\left[\frac{Hk^3}{2\Omega k_r^2}\right] = \frac{\mathrm{m\,m^{-3}}}{\mathrm{s^{-1}m^{-2}}} = \mathrm{s}.$$

R.71.2 Clock delay from local swirl speed: even-in-direction effect

In the SST swirl-clock interpretation, the local proper-time rate associated with a resolved azimuthal swirl speed u_θ is taken in the dimensionally consistent form

$$\boxed{\frac{d\tau}{dt} = \sqrt{1 - \frac{u_\theta^2}{c^2}}.}$$

For $|u_\theta| \ll c$,

$$\frac{d\tau}{dt} = 1 - \frac{1}{2}\frac{u_\theta^2}{c^2} + \mathcal{O}\left(\frac{u_\theta^4}{c^4}\right).$$

The accumulated proper-time deficit over a laboratory interval T is therefore

$$\Delta\tau - T = \int_0^T \left(\sqrt{1 - \frac{u_\theta(t)^2}{c^2}} - 1\right) dt \approx -\frac{1}{2c^2}\int_0^T u_\theta(t)^2\, dt.$$

Because the dependence is quadratic,

$$u_\theta \to -u_\theta \quad \Longrightarrow \quad u_\theta^2 \to u_\theta^2,$$

so

$$\boxed{\Delta\tau(+u_\theta) = \Delta\tau(-u_\theta) < T.}$$

Both swirl directions therefore produce clock slowing. Directional reversal changes the sign of tracer rotation, but not the sign of the clock-rate deficit.

R.71.3 Vorticity response to vertical displacement

In the low-Rossby rotating-fluid limit, the linearized vorticity equation gives

$$\partial_t \boldsymbol{\omega} \approx 2(\boldsymbol{\Omega} \cdot \nabla)\mathbf{u}.$$

For an axisymmetric vertical displacement field

$$\mathbf{u} = w(r, z, t)\, \hat{\mathbf{z}}, \qquad w = \partial_t \xi,$$

the vertical vorticity response is

$$\partial_t \omega_z = 2\Omega\, \partial_z w = 2\Omega\, \partial_z \partial_t \xi.$$

Integrating in time with zero initial perturbation vorticity gives

$$\boxed{\omega_z(r, z, t) = 2\Omega\, \partial_z \xi(r, z, t).}$$

Thus column stretching and compression generate opposite-signed vertical vorticity.

For a Gaussian displacement profile,

$$\xi(r, z, t) = Z(t) \exp\left[-\frac{r^2 + z^2}{a^2}\right],$$

one obtains

$$\omega_z(r, z, t) = -\frac{4\Omega Z(t) z}{a^2} \exp\left[-\frac{r^2 + z^2}{a^2}\right].$$

Hence the induced vorticity is odd in z:

$$\boxed{\omega_z(r, -z, t) = -\omega_z(r, z, t).}$$

R.71.4 Resolved swirl velocity from induced vorticity

For an axisymmetric azimuthal velocity $u_\theta(r, z, t)$,

$$\omega_z = \frac{1}{r} \frac{\partial}{\partial r} (r u_\theta).$$

Using the Gaussian vorticity source above gives

$$\boxed{u_\theta(r, z, t) = -\frac{2\Omega Z(t) z}{a^2} e^{-z^2/a^2} \frac{1 - e^{-r^2/a^2}}{r}.}$$

Therefore

$$u_\theta(r, -z, t) = -u_\theta(r, z, t), \qquad u_\theta(r, -z, t)^2 = u_\theta(r, z, t)^2.$$

This establishes the parity split:

$$\boxed{\text{tracer angle is odd in swirl direction,} \qquad \text{clock deficit is even in swirl direction.}}$$

R.71.5 C1–C4 conclusions

C1. Inertial-wave benchmark. The delayed push–pull surface response in a rotating tank is explained, to leading order, as an inertial-wave arrival delay:

$$\boxed{t_{\text{arr}} = \frac{H k^3}{2\Omega k_r^2}.}$$

It scales as $t_{\text{arr}} \propto \Omega^{-1}$, and therefore disappears as a rotating-fluid effect when the background rotation is removed.

C2. Vorticity generation by vertical forcing. A vertical displacement in a rotating incompressible fluid produces vertical vorticity according to

$$\boxed{\omega_z = 2\Omega\, \partial_z \xi.}$$

Stretching and compression of the rotating column generate opposite signs of relative vorticity above and below the symmetry plane.

C3. Odd tracer response and cancellation over symmetric cycles. The induced azimuthal velocity satisfies

$$u_\theta(r, -z, t) = -u_\theta(r, z, t).$$

Hence tracer angles reverse under reflection or reversed forcing. For an ideal symmetric up–down cycle, the leading-order angular displacement cancels:

$$\boxed{\Delta\theta_{\text{cycle}} \simeq 0 \qquad \text{to leading order in } Ro.}$$

C4. Even clock-rate response and accumulated delay. The swirl-clock deficit depends on u_θ^2, not on u_θ. Therefore

$$\boxed{\Delta\tau(+u_\theta) = \Delta\tau(-u_\theta) < T.}$$

Equivalently,

$$\boxed{\Delta\tau - T \approx -\frac{1}{2c^2} \int_0^T u_\theta(t)^2 \, dt.}$$

Thus a symmetric cycle can have zero net tracer rotation while still producing a nonzero accumulated clock deficit.

Core-scale guardrail. The laboratory rotation rate Ω must not be identified with the SST canonical core turnover rate. The latter is

$$\Omega_0 = \frac{\mathbf{v}_{\circlearrowleft}}{r_c}.$$

Using the canonical constants gives

$$\boxed{\Omega_0 = 7.76344066 \times 10^{20}\,\mathrm{s}^{-1}.}$$

Therefore

$$\boxed{\Omega_{\text{lab}} \neq \Omega_0.}$$

The rotating-tank experiment is a resolved analogue benchmark, not a direct identification of macroscopic rotation with the canonical SST core scale.

Summary. The experiment separates two parities:

$$\boxed{\theta(+u_\theta) = -\theta(-u_\theta),}$$

but

$$\boxed{\Delta\tau(+u_\theta) = \Delta\tau(-u_\theta).}$$

This provides a falsifiable benchmark: angular tracer motion is direction sensitive, whereas the clock-rate or energy-density deficit is direction independent at leading quadratic order.

Child-level analogy. A knotted vortex in a rotating fluid is like a small spinning propeller inside a slowly rotating elevator shaft. From outside, it may push fluid like a round object moving up and down, but inside the object the strings are tied into a knot. Turning the knot the other way reverses the visible swirl, but the amount of busy motion stored in the fluid still slows the clock in both directions.

References

[1] J. K. Hale and S. M. Verduyn Lunel, *Introduction to Functional Differential Equations*, Springer (1993).

[2] CMS Collaboration, High-precision measurement of the W boson mass with the CMS experiment, Nature **652** (2026), 321–337, DOI: 10.1038/s41586-026-10168-5.

[3] S. Navas et al. (Particle Data Group), Review of Particle Physics, Phys. Rev. D **110** (2024), 030001, DOI: 10.1103/PhysRevD.110.030001.

[4] J. de Blas et al., Global analysis of electroweak data in the Standard Model, Phys. Rev. D **106** (2022), 033003, DOI: 10.1103/PhysRevD.106.033003.

[5] CDF Collaboration, High-precision measurement of the W boson mass with the CDF II detector, Science **376** (2022), 170–176, DOI: 10.1126/science.abk1781.

[6] G. K. Batchelor, *An Introduction to Fluid Dynamics*, Cambridge University Press (1967).

[7] G. I. Taylor, The motion of a sphere in a rotating liquid, *Proceedings of the Royal Society of London A* **102** (1922), 180–189, DOI: 10.1098/rspa.1922.0079.

[8] J. Proudman, On the motion of solids in a liquid possessing vorticity, *Proceedings of the Royal Society of London A* **92** (1916), 408–424, DOI: 10.1098/rspa.1916.0026.

[9] M. J. M. Hill, On a spherical vortex, *Philosophical Transactions of the Royal Society of London A* **185** (1894), 213–245, DOI: 10.1098/rsta.1894.0006.

[10] H. P. Greenspan, *The Theory of Rotating Fluids*, Cambridge University Press (1968), ISBN: 978-0521051477.

[11] L. Onsager, Statistical hydrodynamics, *Il Nuovo Cimento* **6**(S2) (1949), 279–287, DOI: 10.1007/BF02780991.

[12] R. P. Feynman, Application of quantum mechanics to liquid helium, *Progress in Low Temperature Physics* **1** (1955), 17–53, DOI: 10.1016/S0079-6417(08)60077-3.

[13] D. Kleckner and W. T. M. Irvine, Creation and dynamics of knotted vortices, *Nature Physics* **9** (2013), 253–258, DOI: 10.1038/nphys2560.

[14] T. Jacobson, Einstein-Æther gravity: a status report, PoS QG-PH (2007) 020, DOI: 10.22323/1.043.0020.

[15] D. Blas, O. Pujolàs, and S. Sibiryakov, Models of non-relativistic quantum gravity: the good, the bad and the healthy, JHEP **04** (2011) 018, DOI: 10.1007/JHEP04(2011)018.

[16] B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), GW170817: Observation of gravitational waves from a binary neutron star inspiral, Phys. Rev. Lett. **119** (2017) 161101, DOI: 10.1103/PhysRevLett.119.161101.

[17] W. G. Unruh, Experimental black-hole evaporation?, Phys. Rev. Lett. **46** (1981), 1351–1353, DOI: 10.1103/PhysRevLett.46.1351.

[18] M. Visser, Acoustic black holes: horizons, ergospheres and Hawking radiation, Class. Quantum Grav. **15** (1998), 1767–1791, DOI: 10.1088/0264-9381/15/6/024.

[19] P. Painlevé, La mécanique classique et la théorie de la relativité, C. R. Acad. Sci. (Paris) **173** (1921), 677–680.

[20] A. Gullstrand, Allgemeine Lösung des statischen Einkörperproblems in der Einsteinschen Gravitationstheorie, Arkiv. Mat. Astron. Fys. **16**(8) (1922), 1–15.

[21] E. Schrödinger, Quantisierung als Eigenwertproblem, Annalen der Physik **384** (1926), 361–376, DOI: 10.1002/andp.19263840404.

[22] E. Madelung, Quantentheorie in hydrodynamischer Form, Z. Phys. **40** (1927), 322–326, DOI: 10.1007/BF01400372.

[23] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley (1999).

[24] M. Planck, Ueber das Gesetz der Energieverteilung im Normalspectrum, Annalen der Physik **309** (1901), 553–563, DOI: 10.1002/andp.19013090310.

[25] D. Rolfsen, *Knots and Links*, Publish or Perish (1976); AMS Chelsea reprint (2003).

[26] J. H. Conway, An enumeration of knots and links, and some of their algebraic properties, in *Computational Problems in Abstract Algebra*, Pergamon Press (1970), pp. 329–358, DOI: 10.1016/B978-0-08-012975-4.50034-5.

[27] W. Wien, Ueber die Energieverteilung im Emissionsspectrum eines schwarzen Körpers, Annalen der Physik **294** (1896), 662–669, DOI: 10.1002/andp.18962940310.

[28] G. Călugăreanu, Sur les classes d'isotopie des noeuds tridimensionnels et leurs invariants, Czech. Math. J. **11** (1961), 588–625.

[29] J. H. White, Self-linking and the Gauss integral in higher dimensions, Amer. J. Math. **91** (1969), 693–728, DOI: 10.2307/2373348.

[30] P. G. Saffman, *Vortex Dynamics*, Cambridge University Press (1992).

[31] P.-M. Zhang, L.-M. Cao, Y.-S. Duan, and C.-K. Zhong, Transverse force on a moving vortex with the acoustic geometry, arXiv:hep-th/0501073 (2005). Permalink: `https://arxiv.org/abs/hep-th/0501073`.

[32] S. Kida and M. Takaoka, Vortex reconnection, Annual Review of Fluid Mechanics **26** (1994), 169–189, DOI: 10.1146/annurev.fl.26.010194.001125.

[33] J. Koplik and H. Levine, Vortex reconnection in superfluid helium, Physical Review Letters **71** (1993), 1375–1378, DOI: 10.1103/PhysRevLett.71.1375.

[34] G. P. Bewley, M. S. Paoletti, K. R. Sreenivasan, and D. P. Lathrop, Characterization of reconnecting vortices in superfluid helium, Proceedings of the National Academy of Sciences **105** (2008), 13707–13710, DOI: 10.1073/pnas.0806002105.

[35] E. Varoquaux, Anderson's considerations on the flow of superfluid helium: Some offshoots, Reviews of Modern Physics **87** (2015), 803–854, DOI: 10.1103/RevModPhys.87.803.

[36] H. K. Moffatt, The degree of knottedness of tangled vortex lines, J. Fluid Mech. **35** (1969), 117–129, DOI: 10.1017/S0022112069000991.

[37] J. Cantarella and J. Parsley, A new cohomological formula for helicity in $\mathbb{R}^{2k+1}$ reveals the effect of a diffeomorphism on helicity, *Journal of Geometry and Physics* **60** (2010), 1127–1155, DOI: 10.1016/j.geomphys.2010.04.001, arXiv: 0903.1465.

[38] W. H. Zurek, Decoherence, einselection, and the quantum origins of the classical, Rev. Mod. Phys. **75** (2003), 715–775, DOI: 10.1103/RevModPhys.75.715.

[39] S. Weinberg, A model of leptons, Phys. Rev. Lett. **19** (1967), 1264–1266, DOI: 10.1103/PhysRevLett.19.1264.

[40] G. W. Gibbons, The maximum tension principle in general relativity, Found. Phys. **32** (2002), 1891–1901, DOI: 10.1023/A:1022370717626.

[41] F. Wilczek and A. Zee, "Linking Numbers, Spin, and Statistics of Solitons," *Physical Review Letters* **51**, 2250–2252 (1983), doi:10.1103/PhysRevLett.51.2250.

[42] G. E. Volovik, *The Universe in a Helium Droplet*, International Series of Monographs on Physics, Vol. 117, Clarendon Press, Oxford (2003).

[43] S. Autti, V. V. Dmitriev, J. T. Mäkinen, A. A. Soldatov, G. E. Volovik, A. N. Yudin, V. V. Zavjalov, and V. B. Eltsov, "Observation of Half-Quantum Vortices in Topological Superfluid ^{3}He," *Physical Review Letters* **117**, 255301 (2016), doi:10.1103/PhysRevLett.117.255301, arXiv:1508.02197.

[44] S. Chen and Y. Tanizaki, "Solitonic Symmetry beyond Homotopy: Invertibility from Bordism and Noninvertibility from Topological Quantum Field Theory," *Physical Review Letters* **131**, 011602 (2023), doi:10.1103/PhysRevLett.131.011602, arXiv:2210.13780.

[45] A. Einstein, Zur Elektrodynamik bewegter Körper, Annalen der Physik **322** (1905), 891–921, DOI: 10.1002/andp.19053221004.

[46] A. Einstein, Die Feldgleichungen der Gravitation, Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften (1915), 844–847, permalink: https://einsteinpapers.press.princeton.edu/vol6-trans/129.

[47] C. Barceló, S. Liberati, and M. Visser, Analogue Gravity, Living Reviews in Relativity **14** (2011), 3, DOI: 10.12942/lrr-2011-3.

[48] C. Barceló, S. Liberati, and M. Visser, "Analogue Gravity," *Living Reviews in Relativity* **29**, 2 (2026), DOI: 10.1007/s41114-026-00064-9, permalink: https://doi.org/10.1007/s41114-026-00064-9.

[49] T. Jacobson, Thermodynamics of spacetime: The Einstein equation of state, Phys. Rev. Lett. **75** (1995), 1260–1263, DOI: 10.1103/PhysRevLett.75.1260.

[50] A. D. Sakharov, Vacuum quantum fluctuations in curved space and the theory of gravitation, Dokl. Akad. Nauk SSSR **177** (1967), 70–71; Sov. Phys. Dokl. **12** (1968), 1040–1041.

[51] T. Jacobson and D. Mattingly, Gravity with a dynamical preferred frame, Phys. Rev. D **64** (2001), 024028, DOI: 10.1103/PhysRevD.64.024028.

[52] B. Z. Foster and T. Jacobson, Post-Newtonian parameters and constraints on Einstein-Æther theory, Phys. Rev. D **73** (2006), 064015, DOI: 10.1103/PhysRevD.73.064015.

[53] B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), GW170817: Observation of gravitational waves from a binary neutron star inspiral, Phys. Rev. Lett. **119** (2017), 161101, DOI: 10.1103/PhysRevLett.119.161101.

[54] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, CODATA recommended values of the fundamental physical constants: 2022, Rev. Mod. Phys. **97** (2025), 025002, DOI: 10.1103/RevModPhys.97.025002.

[55] N. Aghanim et al. (Planck Collaboration), Planck 2018 results. VI. Cosmological parameters, Astronomy & Astrophysics **641** (2020), A6, DOI: 10.1051/0004-6361/201833910.

[56] J. Cantarella, D. DeTurck, H. Gluck, and M. Teytel, "The Spectrum of the Curl Operator on Spherically Symmetric Domains," *Physics of Plasmas* **7** (2000), 2766–2775, DOI: 10.1063/1.874127.

[57] J. Cantarella, D. DeTurck, and H. Gluck, "Upper Bounds for the Writhing of Knots and the Helicity of Vector Fields," in *Knots, Braids, and Mapping Class Groups—Papers Dedicated to Joan S. Birman*, AMS/IP Studies in Advanced Mathematics **24** (2001), 1–21, Permalink: author PDF.

[58] J. Cantarella, J. H. G. Fu, R. B. Kusner, J. M. Sullivan, and N. C. Wrinkle, "Criticality for the Gehring Link Problem," *Geometry & Topology* **10** (2006), 2055–2115, DOI: 10.2140/gt.2006.10.2055, arXiv: math/0402212.

[59] J. Cantarella, D. DeTurck, and H. Gluck, "The Biot–Savart Operator for Application to Knot Theory, Fluid Dynamics, and Plasma Physics," *Journal of Mathematical Physics* **42** (2001), 876–905, DOI: 10.1063/1.1329659.

[60] J. Cantarella, D. DeTurck, and H. Gluck, "Vector Calculus and the Topology of Domains in 3-Space," *The American Mathematical Monthly* **109** (2002), 409–442, DOI: 10.1080/00029890.2002.11919870.

[61] J. Cantarella, R. B. Kusner, and J. M. Sullivan, On the Minimum Ropelength of Knots and Links, *Inventiones Mathematicae* **150** (2002), 257–286, DOI: 10.1007/s00222-002-0234-y, arXiv: math/0103224.

[62] J. Cantarella, D. DeTurck, H. Gluck, and M. Teytel, Isoperimetric problems for the helicity of vector fields and the Biot–Savart and curl operators, *Journal of Mathematical Physics* **41** (2000), 5615–5641, DOI: 10.1063/1.533429.

[63] T. Ashton, J. Cantarella, M. Piatek, and E. J. Rawdon, Knot Tightening by Constrained Gradient Descent, *Experimental Mathematics* **20**(1) (2011), 57–90, DOI: 10.1080/10586458.2011.544581, arXiv: 1002.1723.

[64] E. Denne, Y. Diao, and J. M. Sullivan, Quadrisecants Give New Lower Bounds for the Ropelength of a Knot, *Geometry & Topology* **10** (2006), 1–26, DOI: 10.2140/gt.2006.10.1, arXiv: math/0408026.

[65] R. V. Buniy and T. W. Kephart, A Model of Glueballs, *Physics Letters B* **576** (2003), 127–134, DOI: 10.1016/j.physletb.2003.09.081, arXiv: hep-ph/0209339.

[66] B. P. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration, Fermi Gamma-ray Burst Monitor, and INTEGRAL), Gravitational Waves and Gamma-rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A, *Astrophysical Journal Letters* **848**, L13 (2017), DOI: 10.3847/2041-8213/aa920c.

[67] J. Oost, S. Mukohyama, and A. Wang, Constraints on Einstein-aether theory after GW170817, *Physical Review D* **97**, 124023 (2018), DOI: 10.1103/PhysRevD.97.124023.

[68] L. Shao and N. Wex, New tests of local Lorentz invariance of gravity with small-eccentricity binary pulsars, *Classical and Quantum Gravity* **29**, 215018 (2012), DOI: 10.1088/0264-9381/29/21/215018.

[69] L. Shao, R. N. Caballero, M. Kramer, N. Wex, D. J. Champion, and A. Jessner, A new limit on local Lorentz invariance violation of gravity from solitary pulsars, *Classical and Quantum Gravity* **30**, 165019 (2013), DOI: 10.1088/0264-9381/30/16/165019.

[70] T. Gupta, M. Herrero-Valea, D. Blas, E. Barausse, N. Cornish, K. Yagi, and N. Yunes, New binary pulsar constraints on Einstein-æther theory after GW170817, *Classical and Quantum Gravity* **38**, 195003 (2021), DOI: 10.1088/1361-6382/ac1a69.

[71] T. Annala, R. Zamora-Zamora, and M. Möttönen, Topologically protected vortex knots and links, *Communications Physics* **5** (2022), 309, DOI: 10.1038/s42005-022-01071-2.

[72] D. Kleckner, L. H. Kauffman, and W. T. M. Irvine, How superfluid vortex knots untie, *Nature Physics* **12** (2016), 650–655, DOI: 10.1038/nphys3679.

[73] E. Pisanty, G. Jiménez, V. Vicuña-Hernández, A. Picón, A. Celi, J. P. Torres, and M. Lewenstein, Knotting fractional-order knots with the polarization state of light, *arXiv:1808.05193* (2018), arXiv: 1808.05193.

[74] R. L. Ricca, Applications of knot theory in fluid mechanics, *Banach Center Publications* **42** (1998), 321–346, Permalink: EuDML:252224.

[75] E. Berti et al., Black hole spectroscopy: from theory to experiment, *Classical and Quantum Gravity* **43** (2026), 123001, DOI: 10.1088/1361-6382/ae59e2.

[76] A. J. Majda and A. L. Bertozzi, *Vorticity and Incompressible Flow*, Cambridge Texts in Applied Mathematics, Cambridge University Press (2002), DOI: 10.1017/CBO9780511613203.

[77] J. L. Jaramillo, R. Panosso Macedo, and L. Al Sheikh, Pseudospectrum and black hole quasinormal mode instability, *Physical Review X* **11** (2021), 031003, DOI: 10.1103/PhysRevX.11.031003.

[78] V. Cardoso, A. S. Miranda, E. Berti, H. Witek, and V. T. Zanchin, Geodesic stability, Lyapunov exponents, and quasinormal modes, *Physical Review D* **79** (2009), 064016, DOI: 10.1103/PhysRevD.79.064016.

[79] T. Torres, S. Patrick, M. Richartz, and S. Weinfurtner, Quasinormal mode oscillations in an analogue black hole experiment, *Physical Review Letters* **125** (2020), 011301, DOI: 10.1103/PhysRevLett.125.011301.